THE CORE OF A RANDOM THREE-SIDED ASSIGNMENT MARKET IS EMPTY

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Abstract. Take three sectors of q agents each. Every pair of agents from different sectors has an independent value, uniform on $[0,1]$, and a team of one agent from each sector is worth the sum of its three pair values. We show that the core of the resulting three-sided assignment game is empty with probability tending to 1 as $q \to \infty$. The core is nonempty exactly when the gap D_q between the optimum of the fractional relaxation of the assignment problem and the value of the best assignment is zero, and we prove that $\mathbb{E}D_q \geq (0.050 - o(1))\, q^{1/3}$ and that $D_q/\mathbb{E}D_q \to 1$ in probability. The proof compares two constants on the scale $q^{1/3}$. With high probability every assignment falls short of the trivial bound $3q$ by at least $\beta_0 q^{1/3}$, $\beta_0 = 2.302$, by a first moment conditioned on the local stars of all agents. The fractional optimum falls short in expectation by at most $(\alpha_0 + o(1))q^{1/3}$, $\alpha_0 = 2.2519$, because a random perfect matching on the limiting Poisson hypertree, built from a solution of a recursive distributional equation, can be carried to the finite market as a fractional solution. The solution exists by a monotonicity property of the recursion in the dispersive order. Both constants rest on computations in interval arithmetic. All statements are asymptotic and say nothing at any size that can be simulated.

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1. Introduction

1.1. **The problem.** In the two-sided assignment game of buyers and sellers the core is never empty: the linear programme behind the optimal assignment has an integral optimal solution, and its dual solutions are the core payoffs [25]. With three sides this fails. A three-sided assignment game can have an empty core, and the core is nonempty exactly when the fractional relaxation of the three-dimensional assignment problem has an integral optimal solution. How often this happens in a large random market has been an open question; the only figures found in the sources we searched concern very small markets, or markets with one small side, with independent team values [20].

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We study the following market. There are three sectors of q agents. Every pair of agents from different sectors has a value, and the $3q^2$ pair values are independent and uniform on $[0,1]$. A team consists of one agent from each sector and is worth the sum of its three pair values. A coalition is worth the best total value of disjoint teams that it can form. This is the model of Frieze, Pegden and Tkocz [9], read as a market: they showed that the best assignment falls short of the trivial upper bound $3q$ by an amount of order $q^{1/3}$.

Let T_q be the value of the best perfect assignment, LP_q^* the optimum of the fractional relaxation, and $D_q = \mathrm{LP}_q^* - T_q \geq 0$ the gap. The core is nonempty if and only if $D_q = 0$ (Proposition 2.3). Full definitions are in Section 2.

1.2. **The main theorem.** Throughout, $\beta_0 = 2.302$ and $\alpha_0 = 2.2519$.

Theorem 1.1. *In the random three-sided market with independent uniform pair values, as $q \to \infty$:*

(i) $\mathbb{E}D_q \;\geq\; (\beta_0 - \alpha_0 - o(1))\, q^{1/3} \;\geq\; (0.050 - o(1))\, q^{1/3}$*; in particular* $\mathbb{E}D_q \to \infty$.
(ii) $D_q/\mathbb{E}D_q \to 1$ *in probability.*
(iii) $\mathbb{P}(\text{the core is nonempty}) = \mathbb{P}(D_q = 0) \to 0$.

So the core of the random three-sided assignment market is empty with probability tending to 1. The theorem is asymptotic. The proof gives no bound at any size that can be simulated, and the thresholds implicit in it are astronomically large (Section 12). Two steps of the proof are computer-assisted (Sections 1.6 and 11).

1.3. **The four ingredients.** Write $\Phi_q = 3q - T_q$ and $\Phi_q^{\mathrm{LP}} = 3q - \mathrm{LP}_q^*$ for the integer and the fractional deficit, so that $D_q = \Phi_q - \Phi_q^{\mathrm{LP}}$. Both are of order $q^{1/3}$. The theorem follows from four statements.

The gap is asymptotically deterministic (Theorem 3.3, Section 3). For every $\varepsilon \in (0, \frac{1}{2})$, with high probability the optimal assignment uses no pair of deficit above $C_\varepsilon q^{-1/2+\varepsilon}$, and no team in the support of an optimal fractional solution has deficit above $C_\varepsilon q^{-1/2+\varepsilon}$. With a Poincaré inequality this gives $\operatorname{Var} D_q = O(q^{-1/6+\varepsilon})$. Hence $\mathbb{P}(D_q = 0) \to 0$ as soon as $\mathbb{E}D_q$ stays bounded away from zero; far less than (i) is needed for (iii).

An integer lower bound (Theorem 5.1, Sections 4 and 5). With high probability every perfect assignment has total deficit at least $\beta_0 q^{1/3}$, that is, $\Phi_q \geq \beta_0 q^{1/3}$. The bound is first proved in the model with independent team deficits (Theorem 4.1) and then carried to the market by a comparison inequality (Theorem 5.2).

A fractional upper bound from the tree (Theorem 9.1, Sections 8 and 9). Let $\mathbb{T}$ be the Poisson hypertree, the local limit of the hypergraph of near-tight teams, and let m be any invariant random perfect matching on $\mathbb{T}$, with $c(m)$ the expected cost of the root's team. Then $\limsup_q \mathbb{E}\Phi_q^{\mathrm{LP}}/q^{1/3} \leq c(m)$.

A cheap matching on the tree (Theorems 6.12 and 7.1, Sections 6 and 7). Let (ξ_k) be a Poisson process of intensity $\frac{1}{2}\xi^2\,\mathrm{d}\xi$ on $[0,\infty)$. A law μ on $\mathbb{R}$ is a fixed point of the map Θ if

$$X \stackrel{d}{=} \min_k\bigl(\xi_k - X'_k - X''_k\bigr), \tag{1.1}$$

where X and all X'_k, X''_k have law μ and are independent of each other and of (ξ_k). A fixed point defines an invariant perfect matching on $\mathbb{T}$ of cost $9\,\mathbb{E}X$. A fixed point exists, and every fixed point has $9\,\mathbb{E}X \le \alpha_0$.

Given these, the proof of Theorem 1.1 takes a few lines (Section 10). The last two ingredients give $\mathbb{E}\Phi_q^{\mathrm{LP}} \le (\alpha_0 + o(1))q^{1/3}$. The second, with $\Phi_q \ge 0$, gives $\mathbb{E}\Phi_q \ge (\beta_0 - o(1))q^{1/3}$. Subtracting gives (i). Chebyshev's inequality with the variance bound gives $\mathbb{P}(|D_q - \mathbb{E}D_q| > t\,\mathbb{E}D_q) \le \operatorname{Var} D_q/(t\,\mathbb{E}D_q)^2 \to 0$ for every fixed $t > 0$, which is (ii), and (iii) is the case $t = 1/2$.

1.4. **Overview of the proof.** The difficulty is that Φ_q and Φ_q^{LP} have the same order $q^{1/3}$ and, as far as the numerical evidence goes, constants that differ by a few per cent. Three natural approaches fail, each for a reason that shaped the proof.

The plain first moment is too weak. There are $(q!)^2$ perfect assignments, and the total deficit of a fixed one is a sum of $3q$ independent uniform variables. The expected number of assignments with deficit at most $\beta q^{1/3}$ is therefore at most $(q!)^2(\beta q^{1/3})^{3q}/(3q)! = (\beta^3 e/27 + o(1))^q$, which tends to zero only for $\beta < 3e^{-1/3} = 2.1496\ldots$ This is a valid lower bound for $\Phi_q/q^{1/3}$, but it lies below 2.2495, the cost of the matching on the tree constructed below, and below the fractional optimum as measured numerically in the independent-team model (about 2.20, floating point). So it cannot separate the two optima.

No bounded structure carries the gap. One might look for a local obstruction: a short odd cycle in the support of the fractional optimum, a block of agents with a positive gap, a local repair that rounds the fractional solution. But the neighbourhood of a typical agent in the hypergraph of near-tight teams is a tree, and on the limiting tree the two problems have the same value: every invariant fractional perfect matching on $\mathbb{T}$ is the marginal of an invariant random integral perfect matching of the same cost (Theorem 6.7). The difference between LP_q^* and T_q is that the finite market cannot realise the cheap integral matchings of the tree, and this is not visible in any neighbourhood of bounded size. In simulations the fractional optimum is fractional at most agents, so rounding it would need of order q repairs (Section 12).

The recursion is unstable under shifts. The fixed-point equation (1.1) is the three-sided analogue of the recursion behind the limit of the random two-sided assignment problem [2]. Here, shifting the input law by a constant c shifts the output by $-2c$. A perturbation in the shift direction therefore

doubles at every step, plain iteration is unstable, and no narrow order interval of laws is invariant under Θ. Existence of a fixed point is not routine, and a numerical solution is not a proof: the difference $\beta_0 - \alpha_0$ is about 0.05, and the cost $9\,\mathbb{E}X$ moves by $9c$ under a shift by c.

The proof gets round these obstacles with three ideas.

A first moment conditioned on local stars (Section 4). In the independent-team model, fix a truncation level K and a partition of $[0, K]$ into bins, and condition on the *star* of every agent: the number of its teams in each bin. Given the stars, the hypergraph of near-tight teams is uniform among simple hypergraphs with those stars, which is bounded through the configuration model, and the expected number of perfect assignments with a prescribed number of teams in each bin can be written down exactly. When the stars are typical, its exponential rate is a concave functional of the usage profile, and a tangent bound reduces the question to the sign of one explicit number $\theta\beta + \Psi_0$. That number is negative for $\beta = 2.30224$; this is the first computer-assisted step. Conditioning matters because the unconditioned count is taken over all star configurations at once, while the conditioned count is needed only for typical ones. The bound is carried to the market by a comparison inequality (Section 5): the expected number of cheap assignments in the market, restricted to a good event of the near-tight hypergraph, is at most $e^{o(q)}$ times the same expectation in the independent-team model on a slightly larger event, plus a negligible term. The comparison is made for the conditional law of the near-tight teams outside a candidate assignment, given the pair values of that assignment, on the event that few near-tight teams share a pair. The probability that a family of teams contains no near-tight team is bounded by the argument of Boppana and Spencer [6] for Janson's inequality, which uses Harris's inequality [12] once.

Fractional solutions as conditional expectations of a tree process, with patching (Sections 6, 8, 9). A fixed point of (1.1) gives a family of messages on $\mathbb{T}$, and with them an invariant random perfect matching m whose cost per team is $9\,\mathbb{E}X$ (Theorem 6.12). The matching is not a function of the tree and its costs; it uses further randomness. What can be computed from a finite neighbourhood is the conditional probability, given the ball of radius r around an agent, that a given team at that agent belongs to m. These conditional probabilities are a fractional solution with small slack: their cost is $c(m)$ by the tower property, and by martingale convergence the slack tends to zero as $r \to \infty$. Convergence of marked balls in total variation (Theorem 8.4) carries the rule to the finite market. The remaining slack is patched with an independent part of the randomness, obtained by writing each pair deficit as the minimum of two independent variables; this splitting device is Aldous's, from his work on the two-sided problem [1, 2]. The cost of the patch is controlled by the upper bound of [9], Theorem 2(b), for the integer problem on the set of unserved agents. Averaging over the patches gives an exact fractional assignment of cost $(c(m) + o(1))q^{1/3}$. Fractionality enters at one place only, the conditional expectation. The same construction

would give an *integral* assignment if m were a measurable function of the marked tree, and by the integer lower bound it cannot be one when $c(m) < \beta_0$ (Section 12).

The dispersive order (Section 7). Write a law as the law of $Q(E)$, with E a standard exponential variable and Q nondecreasing, and say that $Q_0 \preceq_{\mathrm{disp}} Q_1$ if $Q_1 - Q_0$ is nondecreasing. This order ignores shifts, which removes the unstable direction. On laws with increasing failure rate, the map Θ is monotone in this order (Theorem 7.7); the proof uses the theorem of Prékopa [22] that integrals of log-concave functions are log-concave, and every image of Θ has increasing failure rate. The iterates $\Theta^n(\delta_0)$ increase in the dispersive order, which ignores shifts; normalised at a fixed level they converge, and the limit is a fixed point of Θ after a shift that the shift law determines (Theorem 7.1). A functional that equals the cost at a fixed point is antitone, so every fixed point costs at most $c_n = 3\big(\mathbb{E}\,\Theta(Q_n) + 2\,\mathbb{E}\,Q_n\big)$ for every n, where Q_n is the nth iterate and $\mathbb{E}\,Q$ denotes the mean of the law Q. Monotonicity also makes a one-sided numerical evaluation legitimate: replacing an iterate by a less dispersed law can only increase the bound. Ten rounded iterations in interval arithmetic give $9\,\mathbb{E}X \le \alpha_0 = 2.2519$; this is the second computer-assisted step. In floating point the cost of the fixed point is 2.2495456.

1.5. **Related work.** The statements in this subsection are limited to sources that were opened in full text during the project, and say so where a source was not opened. The literature search was done by AI agents; its queries and gaps are recorded with the project files. It did not cover Google Scholar, SSRN or dissertation databases, and a paper with an unusual title could have been missed.

The model. Frieze, Pegden and Tkocz [9] study exactly the cost array $W_{ijk} = a_{ij} + b_{ik} + c_{jk}$ with independent uniform entries, in minimisation form. Their Theorem 2 shows that the expected optimum is at least $c_1 n^{1/3}$ and that a greedy algorithm achieves expected cost at most $c_2 n^{1/3}$. The paper contains nothing on the fractional relaxation, on duality or on the core. We use its Theorem 2(b) in the form: independent uniform pair costs, bound in expectation. Several agents read this from the arXiv version; the final text should quote the published statement. **[TODO: quote Theorem 2(b) of the published version of FPT20 verbatim]**

Random cores. Lucas [20] reports results of Crowley and Graham for three-sided games with independent uniform values on triples, not pair-additive values: the probability of a nonempty core is 209/240 for two agents per side and about 73% for three, and it tends to 1 when one side stays fixed and the others grow. The balanced limit is not treated there. The primary sources behind these figures were not opened. For two sides, Hassidim and Romm [13] prove an approximate law of one price in random assignment games with bounded values; the core is always nonempty there, and the question is its size. A related paper of Kanoria, Saban and Sethuraman

on the size of the core in two-sided markets [18] was opened in part. The papers [17] and [23] on cores of partitioning and multi-sided games were not opened (the first at abstract level only); no statement here rests on them.

Multi-dimensional assignment with independent costs. Martin, Mézard and Rivoire [21] treat costs that are independent over hyperedges by the cavity method. They report that the replica-symmetric equations are inconsistent beyond two dimensions and do not discuss a linear-programming relaxation. Frieze and Sorkin [10] bound the optimum for independent exponential costs and remark that even its scaling is not known. The search found no source that compares the fractional and the integer optimum for a random three-dimensional assignment problem.

Fractional and integral matchings in random hypergraphs. At the level of thresholds the two notions are close in the random r-uniform hypergraph. Kahn [15] shows that a perfect matching appears once the number of edges exceeds $(1+\varepsilon)(n/r)\log n$. A result of Krivelevich, known to us only as restated in [8] (the original was not opened), says that at the first time without isolated vertices the random hypergraph has a perfect fractional matching with high probability. Devlin and Kahn [8] study perfect fractional matchings in k-out hypergraphs, including the r-partite case. These are first-order statements about existence; our result concerns the constant on the scale $q^{1/3}$ in a weighted problem.

The two-dimensional analogue. For minimum matching on the complete graph with exponential costs, Hessler and Wästlund [14] show that a relaxed problem and the integer problem have expected costs within $O(\log n/n)$ of each other, less than the standard deviation of either. The search found no result on the probability that the fractional matching optimum is integral in that setting.

Methods. The use of a limiting tree, recursive distributional equations and invariant (unimodular) random networks follows the objective method [1, 2, 4, 3]. The dispersive order is treated in [24], and the variance bound uses the Poincaré inequality for the exponential measure [5]; see also [7]. Balanced matrices [11] appear only in remarks. None of the works cited in this paragraph, nor [6], [12] and [22], was opened during the project: the attributions are from the agents' recollection, and the statements used from them are either proved in the text or are standard. They must be checked against the sources before any submission.

1.6. **What is computer-assisted.** Two inequalities between explicit real numbers are established by computation.

The first is $\Psi_0 \le -3.84751328322$ for an explicit functional Ψ_0 of finitely many parameters, which gives the constant $2.30224586 \ge \beta_0$ in Theorem 4.1 and, with an explicit error budget, in Theorem 5.1. The second is the bound $9\,\mathbb{E}X \le 2.2519$ of Theorem 7.1, obtained by ten rounded iterations of the map Θ.

Both computations use interval arithmetic with outward rounding, and the second also uses floating-point sums with a proved error bound. Their trusted base is the four basic operations of the interval library and, for the second, IEEE arithmetic in the summations. During the work the library's interval exponential and logarithm were found not to be strict enclosures; neither certificate uses them. Section 11 states exactly what each program certifies, how to reproduce it, and which independent recomputations were made. Everything else in the paper is a written proof. All numerical values labelled as floating point, in remarks and in Section 12, are evidence and are not used in any proof.

2. The model

This section fixes the market, the two optimisation problems, the core, the scaling and the comparison model. A table of notation is at the end.

2.1. **The market.** There are three sectors A, B, C, each with q agents, written a_i, b_j, c_k for $1 \le i, j, k \le q$; when no index is needed we write a, b, c for agents of the three sectors. A *pair* is a set of two agents from different sectors. There are $3q^2$ pairs. Each pair p has a *value* $w(p)$, and the $3q^2$ values are independent and uniform on $[0, 1]$. The *deficit* of a pair is

$$d(p) = 1 - w(p), \tag{2.1}$$

so the pair deficits are again independent and uniform on $[0, 1]$.

A *team* is a triple $e = (a, b, c)$ with one agent from each sector. It contains the three pairs $\{a, b\}$, $\{b, c\}$, $\{a, c\}$. Its *value* and its *deficit* are

$$\begin{aligned} v(e) &= w(a, b) + w(b, c) + w(a, c), \\ \Delta(e) &= d(a, b) + d(b, c) + d(a, c) = 3 - v(e). \end{aligned} \tag{2.2}$$

There are q^3 teams. Two teams that share a pair have dependent values; two teams that share no pair have independent values. Since $\Delta(e)$ is a sum of three independent uniform variables,

$$\mathbb{P}(\Delta(e) \le t) = \frac{t^3}{6} \qquad (0 \le t \le 1), \tag{2.3}$$

and the density of $\Delta(e)$ is $t^2/2$ on $[0, 1]$ and at most $t^2/2$ on $[0, 3]$.

2.2. **Assignments and the fractional relaxation.** A *perfect assignment* is a set of q pairwise disjoint teams; it covers every agent exactly once. Its value is the sum of the values of its teams. We write

$$T_q = \max_N \sum_{e \in N} v(e), \qquad \Phi_q = 3q - T_q = \min_N \sum_{e \in N} \Delta(e), \tag{2.4}$$

where N ranges over the perfect assignments. A perfect assignment is determined by its set of $3q$ pairs, so two different assignments have equal value only on a null set, and the maximiser is almost surely unique. We call it the *optimal assignment* and write M for it.

A *fractional assignment* is a vector $x = (x_e)$ of nonnegative weights on the teams with

$$\sum_{e \ni i} x_e = 1 \qquad \text{for every agent } i. \tag{2.5}$$

The *fractional relaxation* is the linear programme

$$\mathrm{LP}_q^* = \max_x \sum_e v(e)\, x_e, \qquad \Phi_q^{\mathrm{LP}} = 3q - \mathrm{LP}_q^* = \min_x \sum_e \Delta(e)\, x_e, \tag{2.6}$$

where x ranges over the fractional assignments and the second form uses $\sum_e x_e = q$. The indicator vector of a perfect assignment is a fractional assignment, so $\mathrm{LP}_q^* \geq T_q$. The *gap* is

$$D_q = \mathrm{LP}_q^* - T_q = \Phi_q - \Phi_q^{\mathrm{LP}} \;\geq\; 0. \tag{2.7}$$

The *support* of a fractional assignment x is the set of teams with $x_e > 0$.

The dual of (2.6) has one variable u_i for each agent, unrestricted in sign because the constraints (2.5) are equations. Writing $u(e) = u_a + u_b + u_c$ for $e = (a, b, c)$, it is

$$\min\Big\{ \sum_i u_i : \; u(e) \geq v(e) \text{ for every team } e \Big\}. \tag{2.8}$$

The programme (2.6) is feasible and bounded. By the duality theorem of linear programming the minimum in (2.8) is attained and equals LP_q^*, and for every optimal x and every optimal u,

$$x_e > 0 \quad \Longrightarrow \quad u(e) = v(e). \tag{2.9}$$

2.3. **The core.** The market defines a cooperative game with transferable utility on the set I of the $3q$ agents. A coalition $S \subseteq I$ can form disjoint teams from its own members, and its worth $v(S)$ is the largest total value of a set of pairwise disjoint teams contained in S (zero if S contains no team). In particular $v(I) = T_q$, because all values are nonnegative and the sectors have equal size, so that a set of disjoint teams extends to a perfect assignment without loss of value.

Definition 2.1. The *core* is the set of payoff vectors $p \in \mathbb{R}^I$ with $\sum_{i \in I} p_i = v(I)$ and $\sum_{i \in S} p_i \geq v(S)$ for every coalition $S \subseteq I$.

Lemma 2.2. *A vector $p \in \mathbb{R}^I$ is in the core if and only if*

$$p_i \geq 0 \;\text{ for every agent } i, \quad p(e) \geq v(e) \;\text{ for every team } e, \quad \sum_{i \in I} p_i = T_q, \tag{2.10}$$

where $p(e) = p_a + p_b + p_c$ for $e = (a, b, c)$.

Proof. If p is in the core, apply the definition to $S = \{i\}$, whose worth is 0, to $S = \{a, b, c\}$, whose worth is $v(e)$, and to $S = I$. Conversely, let p satisfy (2.10) and let S be a coalition. Choose disjoint teams $e_1, \dots, e_k \subseteq S$ with $v(S) = \sum_j v(e_j)$. Since $p \geq 0$, $\sum_{i \in S} p_i \geq \sum_j p(e_j) \geq \sum_j v(e_j) = v(S)$. $\square$

Proposition 2.3. *The core is nonempty if and only if $D_q = 0$.*

Proof. Let p be in the core. By Lemma 2.2, p is feasible for the dual (2.8) and has total T_q. By weak duality $\mathrm{LP}_q^* \le \sum_i p_i = T_q \le \mathrm{LP}_q^*$, so $D_q = 0$.

Conversely, let $D_q = 0$ and let u be an optimal solution of (2.8), so that $\sum_i u_i = \mathrm{LP}_q^* = T_q$ and $u(e) \ge v(e) \ge 0$ for every team. The vector u may have negative entries, and we shift it sector by sector. Put $m_A = \min_{a\in A} u_a$ and define m_B, m_C in the same way. Every triple is a team, so $s := m_A + m_B + m_C = \min_e u(e) \ge 0$. Define

$$p_a = u_a - m_A + \tfrac{s}{3}, \qquad p_b = u_b - m_B + \tfrac{s}{3}, \qquad p_c = u_c - m_C + \tfrac{s}{3}.$$

Then $p_i \ge s/3 \ge 0$ for every agent, $p(e) = u(e) - s + s = u(e) \ge v(e)$ for every team, and, since each sector has q agents, $\sum_i p_i = \sum_i u_i - qs + qs = T_q$. By Lemma 2.2, p is in the core. □

Thus the statement "the core is empty" is the statement $D_q > 0$: the optimal assignment is not an optimal solution of the fractional relaxation. From here on the game plays no role, and the paper is about the random variable D_q.

2.4. **Deficits and the scaling.** A fixed agent lies in q^2 teams. By (2.3) the expected number of these with deficit at most t is $q^2t^3/6$ for $t \le 1$, which is of order one when t is of order $q^{-2/3}$. We therefore measure team deficits on that scale and define the *rescaled cost*

$$\xi_e = q^{2/3}\,\Delta(e). \tag{2.11}$$

For $K \le q^{2/3}$ the expected number of teams e at a given agent with $\xi_e \le K$ is exactly $K^3/6 = \Lambda([0,K])$, where

$$\Lambda(\,\mathrm{d}\xi) = \frac{\xi^2}{2}\,\mathrm{d}\xi \qquad \text{on } [0,\infty). \tag{2.12}$$

A perfect assignment whose teams have rescaled costs of order one has total deficit of order $q \cdot q^{-2/3} = q^{1/3}$. This is the order of Φ_q: by [9], Theorem 2, $\mathbb{E}\Phi_q$ lies between two constant multiples of $q^{1/3}$. All the constants of this paper are constants in front of $q^{1/3}$:

$$\frac{\Phi_q}{q^{1/3}} = \frac{1}{q}\sum_{e\in M}\xi_e, \qquad \frac{\Phi_q^{\mathrm{LP}}}{q^{1/3}} = \frac{1}{q}\sum_e \xi_e\, x_e \quad \text{for an optimal } x.$$

Each is an average cost per team, equivalently per agent of one sector.

For $K > 0$ the *near-tight hypergraph* $H_q(K)$ has the $3q$ agents as vertices and the teams with $\xi_e \le K$ as hyperedges, each marked by its cost ξ_e. Its local limit is the Poisson hypertree $\mathbb{T}$ of Section 6 restricted to teams of cost at most K. In $\mathbb{T}$ every agent has countably many teams whose costs form a Poisson process of intensity Λ, every team has three agents, and the incidence graph is a tree; after the restriction an agent has a Poisson number of teams with mean $K^3/6$. The precise statement is Theorem 8.4.

2.5. **The independent-team model.** In the market, teams that share a pair have dependent deficits. The *independent-team model* removes this dependence and keeps everything else. It has the same $3q$ agents and q^3 teams, and the team deficits $\Delta(e)$ are independent, each distributed as the sum of three independent uniform variables on $[0,1]$. So a single team has the same law (2.3) in both models, and there are no pair values. Perfect assignments, rescaled costs and the near-tight hypergraph are defined as in the market. Where both models appear together, $\mathbb{P}$ and $\mathbb{E}$ refer to the market and $\mathbb{P}'$ and $\mathbb{E}'$ to the independent-team model.

The integer lower bound is first proved in the independent-team model (Theorem 4.1) and then carried to the market by a comparison inequality (Theorem 5.2). In the market, a team with $\xi_e \le K$ shares a pair with about $\frac{9}{10}K^2q^{-1/3}$ other such teams on average. This tends to zero, but slowly, and it is the reason why the two models agree only at very large q (Section 12).

2.6. **Notation.** Table 1 lists the symbols used throughout. Logarithms are natural. "With high probability" means with probability tending to 1 as $q \to \infty$.

Table 1. Notation.

Symbol	Meaning
A, B, C; a_i, b_j, c_k	the three sectors; their agents
$w(p)$, $d(p) = 1 - w(p)$	value and deficit of a pair p
$v(e)$, $\Delta(e) = 3 - v(e)$	value and deficit of a team e
T_q	best total value of a perfect assignment
M	the optimal assignment (almost surely unique)
$\Phi_q = 3q - T_q$	integer deficit
LP_q^*	optimum of the fractional relaxation
$\Phi_q^{\mathrm{LP}} = 3q - \mathrm{LP}_q^*$	fractional deficit
$D_q = \mathrm{LP}_q^* - T_q$	the gap; the core is nonempty iff $D_q = 0$
u	a solution of the dual (2.8)
$\xi_e = q^{2/3}\Delta(e)$	rescaled cost of a team
$H_q(K)$	near-tight hypergraph: teams with $\xi_e \le K$
$\mathbb{P}'$, $\mathbb{E}'$	probability, expectation in the independent-team model
Λ	the measure $\frac{1}{2}\xi^2\,\mathrm{d}\xi$ on $[0,\infty)$
$\mathbb{T}$	the Poisson hypertree (Section 6)
Θ	$\Theta(\mu)$ is the law of $\min_k(\xi_k - X_k' - X_k'')$; see (1.1)
$\preceq_{\mathrm{disp}}$	dispersive order (Section 7)
IFR	increasing failure rate
$\beta_0 = 2.302$	integer constant (Theorems 4.1, 5.1)
$\alpha_0 = 2.2519$	bound on the tree cost (Theorem 7.1)
c_2	the constant of [9], Theorem 2(b)
C_0	$\sup_n \mathbb{E}\Phi_n/n^{1/3}$

3. The gap is asymptotically deterministic

This section shows that the gap $D_q = \mathrm{LP}_q^* - T_q$ has variance $O(q^{-1/6+\varepsilon})$ for every $\varepsilon > 0$. The proof has three parts. First, with high probability every optimal assignment uses only pairs of deficit $O(q^{-1/2+\varepsilon})$, and every optimal solution of the fractional relaxation is supported on teams of deficit $O(q^{-1/2+\varepsilon})$ (Propositions 3.8 and 3.7); both rest on an expansion property of the market that holds uniformly over all sets of positions (Lemmas 3.5 and 3.6). Second, when the pair deficits are written as functions of independent exponential variables, the gradient of T_q is the vector of deficits of the pairs used by the optimal assignment, and the gradient of LP_q^* is the vector of deficits weighted by the pair loads (Lemma 3.11). Third, the Poincaré inequality for the exponential measure (Lemma 3.10) bounds the variance by the expected squared gradient, which is at most the largest deficit in use times the total deficit; the total deficit is $O(q^{1/3})$ in expectation by a theorem of Frieze, Pegden and Tkocz (Proposition 3.1).

Throughout the section logarithms are natural, and a team $e = (a, b, c)$ always has $a \in A$, $b \in B$, $c \in C$. A *pair* is two agents from different sectors. The value of a team is $v(e) = 3 - \Delta(e)$, where $\Delta(e) = d(a, b) + d(b, c) + d(a, c)$. For a set $\mathcal{M}$ of disjoint teams we write $\Delta(\mathcal{M}) = \sum_{e \in \mathcal{M}} \Delta(e)$ and call it the deficit of $\mathcal{M}$; it is the sum of the deficits of the pairs *used* by $\mathcal{M}$, a pair being used if some team of $\mathcal{M}$ contains both of its agents. A perfect assignment is determined by the set of $3q$ pairs it uses. An *optimal assignment* is a perfect assignment of least deficit, so that $\Phi_q = 3q - T_q$ is the deficit of an optimal assignment. When the size of the market has to be shown we write T_n and Φ_n for the market with n agents in each sector.

3.1. Statement.

The only input that is not elementary is the following.

Proposition 3.1 (Frieze, Pegden and Tkocz [9, Theorem 2(b)])**.** *Consider the market with n agents in each sector, the cost of a team being its deficit, that is, the sum of three pair costs, where the $3n^2$ pair costs are independent and uniform on $[0, 1]$. There is a constant c_2 such that the greedy algorithm of* [9]*, run until $n - n^{1/4}$ disjoint teams have been chosen, finds a solution of expected cost at most $c_2 n^{1/3}$; the last step of the algorithm, which completes the teams chosen greedily to a perfect assignment, may be carried out by an arbitrary completion. Consequently*

$$C_0 := \sup_{n \geq 1} \frac{\mathbb{E}\Phi_n}{n^{1/3}} < \infty. \tag{3.1}$$

Proof of (3.1) *from the cited statement.* The completed output of the greedy algorithm is a perfect assignment, so Φ_n is at most its deficit. Whether or not the bound $c_2 n^{1/3}$ is read as including the completion, the completion adds at most $n^{1/4} + 1$ teams of deficit at most 3 each, so

$$\mathbb{E}\Phi_n \leq c_2 n^{1/3} + 3n^{1/4} + 3 \leq (c_2 + 6)\, n^{1/3}.$$

If the cited statement is read as holding only for large n, then C_0 is still finite, because $\Phi_n \le 3n$ always. $\square$

[TODO: Proposition 3.1 was transcribed from version 1 of the arXiv preprint of [9]; compare with the published version before circulation.]

Remark 3.2. Only the finiteness of C_0 is used, and only in Theorem 3.3(ii) and Corollary 3.13. The bound does not depend on the pair costs being uniform rather than exponential: if the pair costs are independent exponential variables E_p of rate 1, then $d(p) := 1 - e^{-E_p}$ is uniform on $[0,1]$ and $d(p) \le E_p$, so every assignment is cheaper for the uniform deficits, and a bound of the form $c_2 n^{1/3}$ for exponential costs implies the same bound for uniform costs.

For $\varepsilon \in (0, 1/2)$ put

(3.2)

$$r_A := \Big\lceil \frac{1}{2\varepsilon} \Big\rceil - 1, \qquad r_B := \Big\lceil \frac{1}{2\varepsilon} \Big\rceil + 1, \qquad \kappa := \frac{0.3^{\,r_A}}{8 \cdot 4^{\,r_A+1}}, \qquad C_\varepsilon := 3\Big(\Big\lceil \frac{1}{2\varepsilon} \Big\rceil + 2 \Big).$$

Note that $r_A \ge 1$, because $1/(2\varepsilon) > 1$. For $K \ge 1$ let $q_0(\varepsilon, K)$ be the least integer $q_0 \ge 2$ such that every $q \ge q_0$ satisfies

$$\text{(Q1)} \qquad q^{2\varepsilon} \ge 4 \max\big(13(K+5),\, 600\big) \log q,$$

$$\text{(Q2)} \qquad q^{2\varepsilon} \ge (20/3)^{\lceil 1/(2\varepsilon) \rceil},$$

$$\text{(Q3)} \qquad q^{-K} \ge 3q^3 \Big[3e^{-0.009(q-1)} + e^{-q^{2\varepsilon}/32} + e^{-\kappa q^{2\varepsilon}} \Big] + 3e^{-0.009q}.$$

Each of the three conditions holds for all large q, so $q_0(\varepsilon, K)$ is finite.

Theorem 3.3. *Fix $\varepsilon \in (0, 1/2)$ and put $s = q^{-1/2+\varepsilon}$, with C_ε as in* (3.2).

(i) *For every $K \ge 1$ and every $q \ge q_0(\varepsilon, K)$, with probability at least $1 - 150 \cdot 2^K q^{-K}$ the following two statements hold: no optimal assignment uses a pair of deficit at least $C_\varepsilon s$; and every team in the support of every optimal solution of the fractional relaxation has deficit less than $C_\varepsilon s$.*

(ii) *With C_0 as in* (3.1)*, for every $q \ge q_0(\varepsilon, 2)$,*

$$\operatorname{Var} T_q \le 4\, C_\varepsilon C_0\, q^{-1/6+\varepsilon} + \frac{7000}{q},$$

$$\operatorname{Var} \mathrm{LP}_q^* \le 4\, C_\varepsilon C_0\, q^{-1/6+\varepsilon} + \frac{7000}{q},$$

$$\operatorname{Var} D_q \le 16\, C_\varepsilon C_0\, q^{-1/6+\varepsilon} + \frac{28000}{q}.$$

The consequence for the core is Corollary 3.13 at the end of the section.

Remark 3.4. For $\varepsilon \ge 1/2$ part (i) is trivial, since then $C_\varepsilon s \ge 9$ while pair deficits are at most 1 and team deficits at most 3; and the bounds of part (ii) for $\varepsilon \ge 1/2$ follow from the case $\varepsilon = 1/4$, because $C_{1/4}\, q^{1/12} = 12\, q^{1/12} \le$

$9\,q^{1/3} \le C_\varepsilon\, q^{-1/6+\varepsilon}$ for $q \ge 4$. The thresholds $q_0(\varepsilon, K)$ are astronomically large (for $\varepsilon = 1/4$, condition (Q1) alone requires $q > 10^9$); the theorem is asymptotic only.

3.2. **Preliminaries.**

Loads. Let $\mathcal{P}$ be the polytope of fractional assignments: the vectors $x \ge 0$ indexed by the q^3 teams with $\sum_{e \ni i} x_e = 1$ for each agent i. Thus $\mathrm{LP}_q^* = \max_{x \in \mathcal{P}} \sum_e v(e) x_e$, and the *support* of x is $\{e : x_e > 0\}$. The *load* of a pair p under x is

$$\ell_p(x) := \sum_{e \supset p} x_e,$$

the sum over the teams containing both agents of p. Since every team contains three pairs and $\sum_e x_e = q$,

(3.3)

$$0 \le \ell_p(x) \le 1, \qquad \sum_p \ell_p(x) = 3q,$$

$$\sum_e v(e) x_e = \sum_p \ell_p(x)\big(1 - d(p)\big), \qquad \sum_p \ell_p(x)\, d(p) = 3q - \sum_e v(e) x_e.$$

Positions. A *position* of type $\{X, Y\}$, where $X \ne Y$ are two of the sectors, is a pair (x, y) with $x \in X$ and $y \in Y$; its *third sector* is the remaining one. Fix $0 < s \le 1$. An agent z of the third sector is *s-close* to the position (x, y) if $d(x, z) + d(y, z) < s$. This event has probability

$$\pi := s^2/2 \le 1/2,$$

the probability that the sum of two independent uniform variables is below s. Positions are *vertex-disjoint* if no two of them share an agent. For a set S of positions of one type, $N(S)$ denotes the set of agents of the third sector that are s-close to at least one position of S. The map N is monotone: $S' \subseteq S$ implies $N(S') \subseteq N(S)$.

Standard facts. The following are used without proof.

- (C1) *Chernoff bound, lower tail.* If Y is binomial with parameters n and ϱ, and $0 \le \delta \le 1$, then $\mathbb{P}(Y \le (1-\delta) n\varrho) \le \exp(-\delta^2 n\varrho/2)$; see [7].
- (C2) *Chernoff–Hoeffding bound.* If Y is binomial with parameters n and ϱ, and $0 < a < \varrho$, then $\mathbb{P}(Y \le an) \le \exp\big(-n\,\mathrm{KL}(a \,\|\, \varrho)\big)$, where $\mathrm{KL}(a \,\|\, \varrho) = a \log(a/\varrho) + (1-a) \log\big((1-a)/(1-\varrho)\big)$; see [7].
- (C3) *Monotonicity.* If $\varrho \ge \varrho_0$ then $\mathbb{P}(\mathrm{Bin}(n, \varrho) \le t) \le \mathbb{P}(\mathrm{Bin}(n, \varrho_0) \le t)$ for every t.
- (M) *Bipartite matchings.* If $y \ge 0$ is a weighting of the edges of a bipartite graph in which the total weight of the edges at each vertex is at most 1, then the graph has a matching with at least $\sum_{\text{edges}} y$ edges, hence, the size of a matching being an integer, with at least $\lceil \sum_{\text{edges}} y \rceil$ edges. (By Kőnig's theorem there is a vertex cover with as many

vertices as the largest matching has edges, and the total weight is at most the number of vertices in any vertex cover.)

(LP) *Duality.* The fractional relaxation is feasible and bounded, so its dual,

$$\min \sum_i u_i \quad \text{subject to} \quad u(e) := u_a + u_b + u_c \geq v(e) \text{ for every team } e = (a, b, c),$$

with u unrestricted in sign, has an optimal solution, and for every optimal x and every optimal u, $x_e > 0$ implies $u(e) = v(e)$ (complementary slackness).

(ES) *Efron–Stein inequality.* If f is a square-integrable function of independent random variables $E_1, \dots, E_m$, then $\operatorname{Var} f \leq \sum_{i=1}^{m} \mathbb{E}[\operatorname{Var}_i f]$, where Var_i denotes the variance in E_i with the other coordinates held fixed; see [7].

3.3. **Uniform expansion.**

Lemma 3.5 (Uniform expansion)**.** *Consider the market with $n \geq 2$ agents in each sector and independent uniform pair deficits. Let $K \geq 1$, $0 < s \leq 1$ and $\pi = s^2/2$, and assume*

$$n\pi \geq 13(K+2)\log n, \tag{U1}$$

$$n\pi \geq 600 \log n. \tag{U2}$$

Let $\mathcal{U}(n, s)$ be the event that, for each of the three position types and every *nonempty set S of vertex-disjoint positions of that type,*

(a) $|N(S)| \geq 0.3\, n\pi\, |S|$ *if* $|S| \leq 1/\pi$*;*
(b) $|N(S)| \geq 0.9\, n$ *if* $|S| \geq 3/\pi$*.*

Then

$$\mathbb{P}\big(\mathcal{U}(n, s) \text{ fails}\big) \leq 6\, n^{-K} + 3\, e^{-0.009\, n}.$$

Proof. Fix a type, say $\{A, C\}$ with third sector B, and a set

$$S = \{(x_1, y_1), \dots, (x_m, y_m)\}$$

of m vertex-disjoint positions of this type. An agent $z \in B$ belongs to $N(S)$ if and only if $d(x_i, z) + d(y_i, z) < s$ for some i. Because the positions are vertex-disjoint, the $2m$ pairs (x_i, z), (y_i, z) are distinct for fixed z, and pairs with different z are distinct. So the indicators $\mathbf{1}\{z \in N(S)\}$, over the n agents $z \in B$, are independent with probability

$$\pi_m := 1 - (1 - \pi)^m,$$

and $|N(S)|$ is binomial with parameters n and π_m. The number of sets of m vertex-disjoint positions of one type is $\binom{n}{m}^2 m! \leq n^{2m}$.

Part (a). Let $1 \leq m \leq 1/\pi$. Since $1 - \pi \leq e^{-\pi}$ and $t \mapsto (1 - e^{-t})/t$ is decreasing on $(0, 1]$,

$$\pi_m \geq 1 - e^{-\pi m} \geq (1 - e^{-1})\, \pi m =: \varrho_0, \qquad 1 - e^{-1} = 0.63212\ldots$$

Apply (C3) and then (C1) with $\varrho = \varrho_0$ and with δ defined by $(1-\delta)(1-e^{-1}) = 0.3$, that is, $\delta = 0.52540\ldots$:

$$\begin{aligned}\mathbb{P}\big(|N(S)| < 0.3\,n\pi m\big) &\le \exp\big(-\delta^2(1-e^{-1})\,n\pi m/2\big)\\ &= \exp(-0.08724\ldots\,n\pi m) \le \exp(-0.079\,n\pi m).\end{aligned}$$

By (U1), $0.079\,n\pi \ge 1.027\,(K+2)\log n \ge (K+2)\log n$, so the last bound is at most $n^{-(K+2)m}$. The union bound over all sets S of this type with $1 \le |S| \le 1/\pi$ gives at most

$$\sum_{m\ge 1} n^{2m}\, n^{-(K+2)m} = \sum_{m\ge 1} n^{-Km} \le 2\,n^{-K},$$

the last step because $n^K \ge 2$.

Part (b). Put $m_b := \lceil 3/\pi \rceil$. A set with $|S| \ge 3/\pi$ has at least m_b elements, so by monotonicity of N it is enough to treat the sets with $|S| = m_b$ (if there are none, (b) is vacuous). For such a set, $\pi_{m_b} \ge 1 - e^{-\pi m_b} \ge 1 - e^{-3} = 0.95021\ldots$, and by (C3) and (C2),

$$\begin{aligned}\mathbb{P}\big(|N(S)| < 0.9\,n\big) &\le \exp\big(-n\,\mathrm{KL}(0.9 \,\|\, 1-e^{-3})\big)\\ &= \exp(-0.020879\ldots\,n) \le \exp(-0.0208\,n).\end{aligned}$$

The number of such sets is at most n^{2m_b}, and $2m_b \log n \le (6/\pi + 2)\log n$. By (U2), $(6/\pi)\log n \le n/100$; and, using $\pi \le 1/2$, $\log n \le n\pi/600 \le n/1200$, so $2\log n \le n/600$. The union bound over the sets of this type with $|S| = m_b$ gives at most

$$\exp\big((0.01 + 0.001667 - 0.0208)\,n\big) \le \exp(-0.009\,n).$$

Adding the bounds of parts (a) and (b) and multiplying by 3 for the three position types gives the claim. □

The point of Lemma 3.5 is that the event $\mathcal{U}(n,s)$ concerns all sets S simultaneously. The set S may therefore be chosen after looking at the instance (at an optimal assignment, or at an optimal solution of the fractional relaxation), and no conditioning is involved.

Part (a) of the event stops at $|S| = 1/\pi$. The next lemma, which is deterministic, carries the growth on to linear size.

Lemma 3.6 (Growth through the cutoff)**.** *Assume the hypotheses of Lemma 3.5 and that the event $\mathcal{U}(n,s)$ holds, and put $\lambda := 0.3\,n\pi$. Then $\lambda > 1$, and for every nonempty set S of vertex-disjoint positions of one type, with $m = |S|$:*

(G1) $|N(S)| \ge \min(\lambda m,\, 0.15\,n)$*;*
(G2) *if $m \ge 0.15\,n$, then $|N(S)| \ge 0.9\,n$.*

Moreover:

(G3) *if real numbers $x_0, x_1, \ldots$ satisfy $x_{i+1} \ge \min(\lambda x_i,\, 0.15\,n)$ for all $i \ge 0$, then $x_i \ge \min(\lambda^i x_0,\, 0.15\,n)$ for all $i \ge 0$.*

Proof. By (U2), $n\pi \geq 600 \log 2 > 415$, so $\lambda > 1$.

(G1) If $m \leq 1/\pi$ this is part (a) of $\mathcal{U}(n, s)$. If $m > 1/\pi$, let $m_0 := \lfloor 1/\pi \rfloor$. Then $2 \leq m_0 \leq m$, because $\pi \leq 1/2$, so S has a subset S' with exactly m_0 positions. The set S' consists of vertex-disjoint positions and $|S'| \leq 1/\pi$, so by monotonicity and part (a),

$$|N(S)| \geq |N(S')| \geq 0.3\, n\pi\, \lfloor 1/\pi \rfloor \geq 0.3\, n\pi\, (1/\pi - 1) = 0.3\, n\, (1 - \pi) \geq 0.15\, n,$$

the last step because $\pi \leq 1/2$.

(G2) Since $n\pi \geq 20$, we have $0.15\, n \geq 3/\pi$, so $m \geq 3/\pi$ and part (b) of $\mathcal{U}(n, s)$ applies.

(G3) By induction on i. The case $i = 0$ is trivial. If $x_i \geq \min(\lambda^i x_0,\, 0.15\, n)$, then, using $\lambda \geq 1$,

$$x_{i+1} \geq \min(\lambda x_i,\, 0.15\, n) \geq \min(\lambda^{i+1} x_0,\, 0.15\, \lambda n,\, 0.15\, n) = \min(\lambda^{i+1} x_0,\, 0.15\, n).$$

□

There is no rounding issue in Lemmas 3.5 and 3.6: (a), (G1) and (G2) are real lower bounds on integer cardinalities, and the only integer parts taken are $\lfloor 1/\pi \rfloor$ in (G1) and $\lceil 3/\pi \rceil$ in part (b) of Lemma 3.5.

3.4. **The support of the fractional relaxation.**

Proposition 3.7 (Support of the fractional relaxation)**.** *Fix $\varepsilon \in (0, 1/2)$ and $K \geq 1$, put $s = q^{-1/2+\varepsilon}$ and $\pi = s^2/2 = q^{-1+2\varepsilon}/2$, and let $r_B = \lceil 1/(2\varepsilon) \rceil + 1$ as in* (3.2)*. Let $q \geq 2$ satisfy* (Q1) *and* (Q2)*. Then, on the event $\mathcal{U}(q, s)$ of Lemma 3.5 for the whole market ($n = q$), every optimal solution x of the fractional relaxation and every team f with $x_f > 0$ satisfy*

$$\Delta(f) < 3\, (r_B + 1)\, s.$$

Consequently this holds with probability at least $1 - 6\, q^{-K} - 3\, e^{-0.009\, q}$.

Proof. By (Q1), $n\pi = q^{2\varepsilon}/2 \geq \max(13(K + 2), 600) \log q$, so Lemma 3.5 applies with $n = q$, and Lemma 3.6 holds on $\mathcal{U}(q, s)$ with $\lambda = 0.15\, q^{2\varepsilon}$. Assume $\mathcal{U}(q, s)$ from now on.

Fix an optimal x, an optimal dual solution u as in (LP), and a support team $f = (a, b, c)$. Recall that $u(g) \geq v(g) = 3 - \Delta(g)$ for every team g, with equality on the support of x.

Step 1 (start). For every $b' \in B$ we have $u(a, b', c) \geq v(a, b', c)$, and $u(f) = v(f)$. Subtracting,

$$u_{b'} - u_b \geq d(a, b) + d(b, c) - \big[d(a, b') + d(b', c)\big].$$

Let $\mathcal{B}_0 := N(\{(a, c)\})$. For $b' \in \mathcal{B}_0$ the bracket is below s, so

$$u_{b'} > u_b + d(a, b) + d(b, c) - s \qquad \text{for all } b' \in \mathcal{B}_0. \tag{3.4}$$

Step 2 (one link). If $g = (a', b', c')$ is a support team and $b'' \in B$ is s-close to the position (a', c'), then $u(a', b'', c') \geq v(a', b'', c')$ and $u(g) = v(g)$ give

$$u_{b''} - u_{b'} \geq d(a', b') + d(b', c') - \big[d(a', b'') + d(b'', c')\big] > -s, \tag{3.5}$$

because deficits are nonnegative and the bracket is below s.

Step 3 (growth). Suppose that $\mathcal{B}_i \subseteq B$ is nonempty. Let H_i be the bipartite graph with vertex classes A and C whose edges are the positions (a', c') for which some support team (a', b', c') has $b' \in \mathcal{B}_i$, and put

$$y(a', c') := \sum_{b' \in \mathcal{B}_i} x_{(a', b', c')}.$$

Each team has exactly one agent in B and each agent has total weight 1, so $\sum_{\text{edges}} y = |\mathcal{B}_i|$; and for each $a' \in A$, $\sum_{c'} y(a', c') \le \sum_{e \ni a'} x_e = 1$, and likewise at each $c' \in C$. By (M), H_i contains a matching S_i with $|S_i| \ge |\mathcal{B}_i|$. The set S_i is a set of vertex-disjoint positions of type $\{A, C\}$, each of which is the (A, C) part of a support team whose agent in B lies in $\mathcal{B}_i$. Define

$$\mathcal{B}_{i+1} := N(S_i).$$

By (3.5), every $b'' \in \mathcal{B}_{i+1}$ satisfies $u_{b''} > u_{b'} - s$ for some $b' \in \mathcal{B}_i$. With (3.4) and induction on i,

$$u_{b''} > u_b + d(a, b) + d(b, c) - (i+1)\, s \qquad \text{for all } b'' \in \mathcal{B}_i \text{ and all } i \ge 0. \tag{3.6}$$

Step 4 (sizes). Part (a) of $\mathcal{U}(q, s)$ with $|S| = 1$ gives $|\mathcal{B}_0| \ge \lambda$. By (G1) and $|S_i| \ge |\mathcal{B}_i|$, $|\mathcal{B}_{i+1}| \ge \min(\lambda |\mathcal{B}_i|,\, 0.15\, q)$, so by (G3)

$$|\mathcal{B}_i| \ge \min(\lambda^{i+1},\, 0.15\, q) \qquad \text{for all } i \ge 0;$$

in particular every $\mathcal{B}_i$ is nonempty and the construction of Step 3 can be continued. Put $h := \lceil 1/(2\varepsilon) \rceil = r_B - 1$. Then $2\varepsilon(h+1) \ge 1 + 2\varepsilon$, so

$$\lambda^{h+1} = 0.15^{\,h+1}\, q^{2\varepsilon(h+1)} \ge 0.15^{\,h+1}\, q^{1+2\varepsilon} \ge 0.15\, q,$$

the last inequality being (Q2), that is, $q^{2\varepsilon} \ge (20/3)^h$. Hence $|\mathcal{B}_h| \ge 0.15\, q$, so $|S_h| \ge 0.15\, q$, and (G2) gives

$$|\mathcal{B}_{r_B}| = |N(S_h)| \ge 0.9\, q.$$

(The count is: $h = \lceil 1/(2\varepsilon) \rceil$ applications of (G1) after $\mathcal{B}_0$, then one application of (G2); in total $r_B = h + 1$ steps after $\mathcal{B}_0$.)

Step 5 (symmetry). Lemma 3.5 covers all three position types, so the same argument with the roles of the sectors permuted gives sets $\mathcal{A}_r \subseteq A$, $\mathcal{B}_r \subseteq B$, $\mathcal{C}_r \subseteq C$, each of size at least $0.9\, q$, with $r = r_B$ and

$$\begin{aligned} u_{a''} &> u_a + d(a, b) + d(a, c) - (r+1)\, s && \text{for } a'' \in \mathcal{A}_r, \\ u_{b''} &> u_b + d(a, b) + d(b, c) - (r+1)\, s && \text{for } b'' \in \mathcal{B}_r, \\ u_{c''} &> u_c + d(a, c) + d(b, c) - (r+1)\, s && \text{for } c'' \in \mathcal{C}_r. \end{aligned}$$

Step 6 (close). The teams whose agent in A lies outside $\mathcal{A}_r$ carry total weight $q - |\mathcal{A}_r| \le 0.1\, q$; likewise for $\mathcal{B}_r$ and $\mathcal{C}_r$. Since the total weight of all teams is $q > 0.3\, q$, some support team $t = (a'', b'', c'')$ lies in $\mathcal{A}_r \times \mathcal{B}_r \times \mathcal{C}_r$. Adding the three inequalities of Step 5 and using $u(t) = v(t) \le 3$ and $u(f) = v(f) = 3 - \Delta(f)$,

$$3 \ge v(t) = u(t) > u(f) + 2\,\Delta(f) - 3(r+1)\, s = 3 + \Delta(f) - 3(r+1)\, s,$$

so $\Delta(f) < 3(r_B + 1)\, s$.

The probability statement follows from Lemma 3.5 with $n = q$. □

The argument uses complementary slackness, so it says nothing about the optimal assignment when $D_q > 0$. That case needs a different argument.

3.5. **The optimal assignment.**

Proposition 3.8 (Pairs used by an optimal assignment)**.** *Fix $\varepsilon \in (0, 1/2)$ and $K \geq 1$, put $s = q^{-1/2+\varepsilon}$ and $\pi = s^2/2$, and let r_A and κ be as in* (3.2)*. Let $q \geq 2$ satisfy* (Q1)*. Then the probability that some optimal assignment uses a pair of deficit at least $(r_A + 2)\, s$ is at most*

$$\eta_A := 3q^3\Big[6\,(q-1)^{-(K+3)} + 3\,e^{-0.009\,(q-1)} + e^{-q^{2\varepsilon}/32} + e^{-\kappa q^{2\varepsilon}}\Big]. \tag{3.7}$$

Since $(q-1)^{-(K+3)} \leq 2^{K+3}q^{-(K+3)}$, the first term of η_A is at most $144 \cdot 2^K q^{-K}$.

Proof. The law of the market is invariant under permutations of the three sectors, so it is enough to bound the probability for pairs with one agent in A and one in B and to multiply by 3. Fix such a pair $p = (a, b)$, and write $\delta_A := (r_A + 2)\, s$. Note that (Q1) and $q \geq 2$ force $q^{2\varepsilon} \geq 2400 \log 2$, so $q \geq 3$.

Step 1 (threshold; deterministic). Let W_0 be the least value of $\Delta(\mathcal{M}) - d(p)$ over the perfect assignments $\mathcal{M}$ that use p, and W_1 the least value of $\Delta(\mathcal{M})$ over the perfect assignments that do not use p; both classes are nonempty. Neither W_0 nor W_1 depends on $d(p)$. The least deficit of a perfect assignment is $\min(W_0 + d(p),\, W_1)$. If an optimal assignment uses p, then $W_0 + d(p) \leq W_1$, that is, $d(p) \leq \rho_p := W_1 - W_0$. Hence

$$\{\text{some optimal assignment uses } p \text{ and } d(p) \geq \delta_A\} \subseteq \{\rho_p \geq \delta_A\}. \tag{3.8}$$

Step 2 (decoupling; deterministic). For $c \in C$ let I_c be the market obtained by removing the agents a, b and c (it has $n := q - 1$ agents in each sector), and let M_c be an optimal assignment of I_c, chosen by a fixed rule (say the lexicographically first optimum), so that M_c is a function of the pair deficits of I_c only. A perfect assignment that uses p contains the team (a, b, c) for exactly one c, and the rest of it is a perfect assignment of I_c. Hence

$$W_0 = \min_{c \in C}\big[d(a, c) + d(b, c) + \Delta(M_c)\big]. \tag{3.9}$$

Step 3 (rotations; deterministic). Write $M_c = \{(a_j, b_j, c_j) : j = 1, \ldots, n\}$. For $k \geq 1$ and distinct indices $j_1, \ldots, j_k$, the *rotation* of M_c along $(j_1, \ldots, j_k)$ is the following set of q teams: the teams of M_c with index outside $\{j_1, \ldots, j_k\}$; the team (a_{j_1}, b, c_{j_1}); the teams $(a_{j_{i+1}}, b_{j_i}, c_{j_{i+1}})$ for $i = 1, \ldots, k-1$; and the team (a, b_{j_k}, c). Every agent appears exactly once (in A: all a_j and a; in C: all c_j and c; in B: b in the team with index j_1, b_{j_i} in the team with index j_{i+1}, b_{j_k} with a and c, and the others unmoved), so the rotation is a perfect assignment, and it does not use p, because b is teamed with $a_{j_1} \neq a$. The

pairs it uses are: the pairs (a_j, c_j) for all j, and (a, c); the pairs containing an agent of B in the unmoved teams; and the $2(k+1)$ *new pairs*

$$(a_{j_1}, b),\ (b, c_{j_1}); \qquad (a_{j_{i+1}}, b_{j_i}),\ (b_{j_i}, c_{j_{i+1}}) \quad (1 \le i < k); \qquad (a, b_{j_k}),\ (b_{j_k}, c).$$

Let ν be the sum of the deficits of the new pairs. The deficit of the rotation is at most $d(a,c) + \Delta(M_c) + \nu \le d(a,c) + d(b,c) + \Delta(M_c) + \nu$, so

$$W_1 \le d(a,c) + d(b,c) + \Delta(M_c) + \nu \qquad \text{for every } c \text{ and every rotation of } M_c.$$

Taking c to be a minimiser in (3.9) shows that $\rho_p \le \nu$ for every rotation of that M_c. Therefore, with

$$\mathcal{R}(p,c) := \{\text{some rotation of } M_c \text{ has } \nu < \delta_A\},$$

we have

$$\{\rho_p \ge \delta_A\} \subseteq \bigcup_{c \in C} \mathcal{R}(p,c)^{\mathrm{c}}. \tag{3.10}$$

(The minimiser depends on the pairs at a and at b; the union over all q fixed values of c removes that dependence.)

Step 4 (three independent pieces; c fixed). On the index set $\{1, \dots, n\}$ of M_c write $j \to l$ if b_j is s-close to the position (a_l, c_l) (the case $j = l$ is allowed), and define

$$L := \{l :\ b_l \text{ is } s\text{-close to } (a,c)\} = \{l :\ d(a,b_l) + d(b_l,c) < s\},$$
$$J := \{j :\ b \text{ is } s\text{-close to } (a_j, c_j)\} = \{j :\ d(a_j,b) + d(b,c_j) < s\}.$$

Three disjoint families of pairs are involved: the pairs of I_c, which determine M_c and the relation $\to$; the pairs (a, b_l), (b_l, c), which together with M_c determine L; and the pairs (a_j, b), (b, c_j), which together with M_c determine J. Hence, conditionally on I_c, the sets L and J are independent, and each of them contains each index independently with probability π. In particular, J is independent of (I_c, L), with that law.

Let $\Gamma^0 := L$ and $\Gamma^{i+1} := \Gamma^i \cup \{j :\ j \to l \text{ for some } l \in \Gamma^i\}$. Thus Γ^i is the set of indices from which L is reached by a path of the relation $\to$ with at most i steps.

Claim. If $J \cap \Gamma^{r_A} \ne \emptyset$ then $\mathcal{R}(p,c)$ holds. Indeed, let j be in the intersection and take a path $j = j_1 \to j_2 \to \cdots \to j_k \in L$ with the fewest steps; its vertices are distinct and $k - 1 \le r_A$. In the rotation along $(j_1, \dots, j_k)$, the two new pairs at b have deficits summing to less than s (because $j_1 \in J$), the two new pairs at b_{j_i} have deficits summing to less than s for each $i < k$ (because $j_i \to j_{i+1}$), and the two new pairs at b_{j_k} have deficits summing to less than s (because $j_k \in L$). So $\nu < (k+1)\, s \le (r_A + 2)\, s = \delta_A$.

Step 5 (expansion on I_c). The market I_c has $n = q - 1 \ge q/2$ agents in each sector and independent uniform pair deficits. By (Q1), $n\pi \ge q^{2\varepsilon}/4 \ge \max(13(K+5), 600) \log q$, so Lemma 3.5 applies to I_c with $K+3$ in place of K:

$$\mathbb{P}\big(\mathcal{U}(I_c) \text{ fails}\big) \le 6\,(q-1)^{-(K+3)} + 3\, e^{-0.009\,(q-1)},$$

where $\mathcal{U}(I_c)$ denotes the event $\mathcal{U}(n,s)$ for the market I_c. On $\mathcal{U}(I_c)$, for any nonempty index set R the positions $\{(a_l,c_l) : l \in R\}$ are vertex-disjoint and of type $\{A,C\}$, and, identifying the agent b_j of I_c with its index j,

$$N\big(\{(a_l,c_l) : l \in R\}\big) = \{j : \ j \to l \text{ for some } l \in R\}.$$

So by (G1), whenever $L \neq \emptyset$,

$$|\Gamma^{i+1}| \geq \min(\lambda\,|\Gamma^i|,\, 0.15\,n), \qquad \lambda = 0.3\,n\pi,$$

and by (G3), $|\Gamma^{r_A}| \geq \min(\lambda^{r_A}|L|,\, 0.15\,n)$. (Only part (a) of the event and the cutoff step (G1) are used here; part (b) is not needed for Proposition 3.8.)

Step 6 (probability for fixed p and c). The cardinality $|L|$ is binomial with parameters n and π, so by (C1) with $\delta = 1/2$,

$$\mathbb{P}\big(|L| < n\pi/2\big) \leq e^{-n\pi/8} \leq e^{-q^{2\varepsilon}/32}.$$

Let

$$\mathcal{G} := \mathcal{U}(I_c) \cap \{|L| \geq n\pi/2\}, \qquad G := \Gamma^{r_A}.$$

Both $\mathcal{G}$ and G are functions of (I_c, L). On $\mathcal{G}$, by Step 5,

$$|G| \geq \min\Big(\frac{0.3^{\,r_A}(n\pi)^{r_A+1}}{2},\ 0.15\,n\Big), \qquad \text{hence}$$
$$\pi\,|G| \geq \min\Big(\frac{0.3^{\,r_A}(n\pi)^{r_A+2}}{2n},\ 0.15\,n\pi\Big).$$

Here $n\pi \geq q^{2\varepsilon}/4$ and $2\varepsilon(r_A{+}1) = 2\varepsilon\lceil 1/(2\varepsilon)\rceil \geq 1$, so $(n\pi)^{r_A+1} \geq q^{2\varepsilon(r_A+1)}/4^{r_A+1} \geq q/4^{r_A+1}$. With $n \leq q$ this gives

$$\frac{0.3^{\,r_A}(n\pi)^{r_A+2}}{2n} = \frac{0.3^{\,r_A}}{2}\,(n\pi)\,\frac{(n\pi)^{r_A+1}}{n} \geq \frac{0.3^{\,r_A}}{2}\cdot\frac{q^{2\varepsilon}}{4}\cdot\frac{1}{4^{r_A+1}} = \kappa\,q^{2\varepsilon}.$$

Also $0.15\,n\pi \geq 0.0375\,q^{2\varepsilon} \geq \kappa\,q^{2\varepsilon}$, since $\kappa \leq 0.3/128$ for $r_A \geq 1$. Thus $\pi\,|G| \geq \kappa\,q^{2\varepsilon}$ on $\mathcal{G}$. Since J is independent of (I_c, L) and contains each index independently with probability π,

$$\mathbb{P}\big(J \cap G = \emptyset \;\big|\; I_c, L\big) = (1-\pi)^{|G|} \leq e^{-\pi|G|}.$$

By the Claim, $\mathcal{R}(p,c)^{\mathrm{c}} \subseteq \{J \cap G = \emptyset\}$, so

$$\begin{aligned}\mathbb{P}\big(\mathcal{R}(p,c)^{\mathrm{c}}\big) &\leq \mathbb{P}(\mathcal{G}^{\mathrm{c}}) + \mathbb{E}\big[\mathbf{1}_{\mathcal{G}}\,\mathbb{P}\big(J \cap G = \emptyset \;\big|\; I_c, L\big)\big] \\ &\leq \mathbb{P}\big(\mathcal{U}(I_c) \text{ fails}\big) + \mathbb{P}\big(|L| < n\pi/2\big) + e^{-\kappa q^{2\varepsilon}} \\ &\leq 6\,(q-1)^{-(K+3)} + 3\,e^{-0.009\,(q-1)} + e^{-q^{2\varepsilon}/32} + e^{-\kappa q^{2\varepsilon}}. \qquad (3.11)\end{aligned}$$

Step 7 (union bounds). By (3.8) and (3.10), for each of the q^2 pairs $p = (a,b)$ with $a \in A$ and $b \in B$,

$$\mathbb{P}\big(\text{some optimal assignment uses } p \text{ and } d(p) \geq \delta_A\big) \leq \sum_{c \in C} \mathbb{P}\big(\mathcal{R}(p,c)^{\mathrm{c}}\big),$$

a sum of q terms, each bounded by (3.11). Summing over the q^2 pairs of this type and multiplying by 3 for the three types of pair gives $3q^3$ times the bound (3.11), which is η_A.

Finally, $3q^3 \cdot 6\,(q-1)^{-(K+3)} \le 18 \cdot 2^{K+3} q^{-K} = 144 \cdot 2^K q^{-K}$. $\square$

Remark 3.9. (1) Uniqueness of the optimal assignment is not used in Proposition 3.8. (2) When $1/(2\varepsilon)$ is an integer, the set Γ^{r_A} is only guaranteed to have size of order q with a small constant factor, not $0.15\,n$; this is enough, as Step 6 shows. (3) Lemma 3.5 is applied to the $3q^3$ markets I_c, one for each pair p and each third agent c; this is why it is invoked with $K+3$ in place of K.

3.6. **A Poincaré inequality for the exponential measure.** The inequality of the next lemma, with the constant 4, is classical [5]. The short proof is included to keep the section self-contained and to make the regularity hypotheses explicit.

Lemma 3.10 (Poincaré inequality)**.** *Let μ be the law of m independent exponential variables of rate 1 on $[0,\infty)^m$. Let $f\colon [0,\infty)^m \to \mathbb{R}$ be continuous, bounded, and Lipschitz in each coordinate separately. Then each partial derivative $\partial_i f$ exists μ-almost everywhere, and*

$$\operatorname{Var}_\mu(f) \le 4\,\mathbb{E}_\mu\Big[\sum_{i=1}^{m} (\partial_i f)^2\Big].$$

Proof. Consider one dimension first. Let $g\colon [0,\infty) \to \mathbb{R}$ be bounded and Lipschitz. It is absolutely continuous, and g' exists almost everywhere and is bounded. Put $h := g - g(0)$. Since h^2 is absolutely continuous and $h(0) = 0$, integration by parts on $[0,R]$ gives

$$\int_0^R h(x)^2 e^{-x}\,\mathrm{d}x = -h(R)^2 e^{-R} + 2\int_0^R h(x)h'(x)e^{-x}\,\mathrm{d}x.$$

Let $R \to \infty$. Since h is bounded the boundary term vanishes, and the Cauchy–Schwarz inequality gives

$$\mathbb{E}h^2 = 2\,\mathbb{E}[hh'] \le 2\,(\mathbb{E}h^2)^{1/2}(\mathbb{E}h'^2)^{1/2},$$

that is, $\mathbb{E}h^2 \le 4\,\mathbb{E}h'^2$, expectations being with respect to the exponential law of rate 1. Hence

$$\operatorname{Var} g \le \mathbb{E}\big(g - g(0)\big)^2 \le 4\,\mathbb{E}g'^2.$$

Now let f be as in the statement. For every value of the other coordinates, the function $t \mapsto f(\dots,t,\dots)$ of the i-th coordinate is bounded and Lipschitz, so by the one-dimensional case $\operatorname{Var}_i f \le 4\,\mathbb{E}_i(\partial_i f)^2$, where $\mathbb{E}_i$ is the expectation in the i-th coordinate alone and the derivative exists for almost every t. The set where $\partial_i f$ fails to exist is measurable and meets every line parallel to the i-th axis in a null set, so it is μ-null by Fubini's theorem. Taking expectations, summing over i and applying (ES) gives the claim. $\square$

3.7. **Gradients.** Write each pair deficit as

$$d(p) = e^{-E_p},$$

with (E_p) independent exponential variables of rate 1 indexed by the $m := 3q^2$ pairs. Then the deficits are independent and uniform on $[0,1]$, as required, and the value of the pair p is $1 - e^{-E_p}$. Regard T_q, LP_q^* and D_q as functions of $E = (E_p) \in [0,\infty)^m$, and write ∇ for the gradient in these coordinates. For a perfect assignment $\mathcal{M}$ we write $p \in \mathcal{M}$ when $\mathcal{M}$ uses the pair p.

Lemma 3.11 (Gradients and uniqueness of the loads)**.** (1) *The functions T_q, LP_q^* and D_q are continuous and bounded by $3q$. Each is Lipschitz in each coordinate, with constant 1 for T_q and LP_q^* and constant 2 for D_q.*

(2) *Almost surely the optimal assignment M is unique, and for every pair p,*

$$\partial T_q / \partial E_p = \mathbf{1}\{p \in M\}\, d(p).$$

(3) *Optimal solutions x of the fractional relaxation need not be unique, but almost surely all of them have the same load vector $\ell^* = (\ell_p^*)$, and for every pair p,*

$$\partial \mathrm{LP}_q^* / \partial E_p = \ell_p^*\, d(p).$$

(4) *Consequently, almost surely,*

$$|\nabla T_q|^2 = \sum_{p \in M} d(p)^2 \le 3q,$$

$$|\nabla \mathrm{LP}_q^*|^2 = \sum_p (\ell_p^*)^2\, d(p)^2 \le \sum_p \ell_p^* = 3q,$$

$$|\nabla D_q|^2 \le 2\,|\nabla T_q|^2 + 2\,|\nabla \mathrm{LP}_q^*|^2.$$

Proof. For a point $\ell \in \mathbb{R}^m$ write

$$\varphi_\ell(E) := \sum_p \ell_p \big(1 - e^{-E_p}\big);$$

it is smooth, with $\partial \varphi_\ell / \partial E_p = \ell_p e^{-E_p} = \ell_p\, d(p)$. The function T_q is the maximum of φ_ℓ over the finitely many indicator vectors ℓ of the sets of pairs used by perfect assignments. By (3.3), LP_q^* is the maximum of $\sum_e v(e) x_e = \varphi_{\ell(x)}(E)$ over $x \in \mathcal{P}$, that is, the maximum of φ_ℓ over the *load polytope* $\mathcal{L} := \{\ell(x) : x \in \mathcal{P}\}$, a polytope in $[0,1]^m$ (the linear image of $\mathcal{P}$). Since $\ell \mapsto \varphi_\ell(E)$ is linear, LP_q^* is the maximum of φ_ℓ over the finitely many vertices of $\mathcal{L}$.

(1) If $0 \le \ell_p \le 1$ for all p and $\sum_p \ell_p = 3q$, then φ_ℓ is 1-Lipschitz in each coordinate and takes values in $[0, 3q]$. A maximum of finitely many such functions is continuous and has the same two properties; and D_q is the difference of two of them, with $0 \le D_q \le 3q$.

(2), (3) Let V be a finite set of points of $\mathbb{R}^m$ (the indicator vectors of the assignments, or the vertices of $\mathcal{L}$). In the coordinates $X_p := 1 - d(p)$ we have $\varphi_\ell = \langle X, \ell \rangle$. If the maximiser of φ_ℓ over V is not unique at X, then $\langle X, \ell - \ell' \rangle = 0$ for two distinct $\ell, \ell' \in V$; such X form a finite union of hyperplanes, which is Lebesgue-null. The map $E \mapsto X$ is a diffeomorphism of $(0,\infty)^m$ onto $(0,1)^m$, so the exceptional set is null in the coordinates E as well, hence of probability zero. Where the maximiser ℓ^* over V is unique, it remains the unique maximiser on a neighbourhood (finitely many strict inequalities), so the maximum equals φ_{ℓ^*} there, and its partial derivatives are $\ell^*_p\, d(p)$. For T_q, the vector ℓ^* is the indicator of the pairs used by M, and distinct perfect assignments use distinct sets of pairs, so M is unique. For LP^*_q, uniqueness of the maximising vertex of $\mathcal{L}$ means that the set of optimal loads is the single point ℓ^* (the optimal face of $\mathcal{L}$ is the convex hull of the optimal vertices), that is, every optimal x has $\ell(x) = \ell^*$. Non-uniqueness of x itself is possible, because the linear map $x \mapsto \ell(x)$ from $\mathbb{R}^{q^3}$ to $\mathbb{R}^{3q^2}$ is not injective; it is harmless, since the objective and the gradient depend on x only through $\ell(x)$.

(4) follows from (2) and (3), from $d(p) \le 1$, $0 \le \ell^*_p \le 1$ and $\sum_p \ell^*_p = 3q$, and from $(u-w)^2 \le 2u^2 + 2w^2$. □

Proposition 3.12 (Variance bound with the bad events)**.** *Let $\delta_A, \delta_B > 0$. Let $\mathcal{G}_A$ be the event that no optimal assignment uses a pair of deficit at least δ_A, and let $\mathcal{G}_B$ be the event that every team in the support of every optimal solution of the fractional relaxation has deficit less than δ_B. Suppose that $\mathbb{P}(\mathcal{G}_A \text{ fails}) \le \eta_A$ and $\mathbb{P}(\mathcal{G}_B \text{ fails}) \le \eta_B$. Then*

$$\begin{aligned}
\operatorname{Var} T_q &\le 4\big[\delta_A\, \mathbb{E}\Phi_q + 3q\,\eta_A\big],\\
\operatorname{Var} \mathrm{LP}^*_q &\le 4\big[\delta_B\, \mathbb{E}\Phi_q + 3q\,\eta_B\big],\\
\operatorname{Var} D_q &\le 8\big[(\delta_A + \delta_B)\, \mathbb{E}\Phi_q + 3q\,(\eta_A + \eta_B)\big].
\end{aligned}$$

Proof. By Lemma 3.11(1), the functions T_q, LP^*_q and D_q satisfy the hypotheses of Lemma 3.10. That lemma, together with parts (2)–(4) of Lemma 3.11, gives

$$\operatorname{Var} T_q \le 4\,\mathbb{E}\sum_{p\in M} d(p)^2, \qquad \operatorname{Var} \mathrm{LP}^*_q \le 4\,\mathbb{E}\sum_p (\ell^*_p)^2 d(p)^2,$$

$$\operatorname{Var} D_q \le 8\,\mathbb{E}\Big[\sum_{p\in M} d(p)^2 + \sum_p (\ell^*_p)^2 d(p)^2\Big].$$

On $\mathcal{G}_A$, every pair used by M has deficit less than δ_A, so

$$\sum_{p\in M} d(p)^2 \le \delta_A \sum_{p\in M} d(p) = \delta_A\, \Phi_q.$$

Off $\mathcal{G}_A$ the sum is at most $3q$, by Lemma 3.11(4). So

$$\mathbb{E}\sum_{p\in M} d(p)^2 \le \delta_A\, \mathbb{E}\Phi_q + 3q\,\eta_A.$$

On $\mathcal{G}_B$: if $\ell_p^* > 0$, take any optimal x; then $\ell_p(x) = \ell_p^* > 0$, so some support team contains the pair p, and $d(p)$ is at most the deficit of that team, which is less than δ_B. With $\ell_p^* \le 1$ and (3.3),

$$\sum_p (\ell_p^*)^2 d(p)^2 \le \delta_B \sum_p \ell_p^*\, d(p) = \delta_B\,(3q - \mathrm{LP}_q^*) \le \delta_B\,(3q - T_q) = \delta_B\,\Phi_q,$$

using $\mathrm{LP}_q^* \ge T_q$. Off $\mathcal{G}_B$ the sum is at most $3q$. So

$$\mathbb{E}\sum_p (\ell_p^*)^2 d(p)^2 \le \delta_B\,\mathbb{E}\Phi_q + 3q\,\eta_B.$$

The three bounds follow. □

3.8. **Proof of Theorem 3.3 and the consequence for the core.**

Proof of Theorem 3.3. Let $\varepsilon \in (0, 1/2)$, $s = q^{-1/2+\varepsilon}$ and $q \ge q_0(\varepsilon, K)$.

(i) Proposition 3.8, whose hypothesis is (Q1), gives: off an event of probability at most η_A, no optimal assignment uses a pair of deficit at least

$$\delta_A := (r_A + 2)\,s = \big(\lceil 1/(2\varepsilon)\rceil + 1\big)\,s \le C_\varepsilon\, s.$$

Proposition 3.7, whose hypotheses are (Q1) and (Q2), gives: off an event of probability at most $\eta_B := 6\,q^{-K} + 3\,e^{-0.009\,q}$, every support team of every optimal solution of the fractional relaxation has deficit less than

$$\delta_B := 3\,(r_B + 1)\,s = 3\big(\lceil 1/(2\varepsilon)\rceil + 2\big)\,s = C_\varepsilon\, s.$$

The total failure probability is at most

$$\eta_A + \eta_B \le (144\cdot 2^K + 6)\,q^{-K} + 3q^3\Big[3e^{-0.009(q-1)} + e^{-q^{2\varepsilon}/32} + e^{-\kappa q^{2\varepsilon}}\Big] + 3e^{-0.009q}.$$

By (Q3) the exponentially small terms are together at most q^{-K}, so

$$\eta_A + \eta_B \le (144\cdot 2^K + 7)\,q^{-K} \le 150\cdot 2^K q^{-K}.$$

(ii) Take $K = 2$ and apply Proposition 3.12 with the events and the values δ_A, δ_B, η_A, η_B of part (i). By Proposition 3.1, $\mathbb{E}\Phi_q \le C_0\, q^{1/3}$, so for $\delta \in \{\delta_A, \delta_B\}$,

$$\delta\,\mathbb{E}\Phi_q \le C_\varepsilon\, q^{-1/2+\varepsilon}\cdot C_0\, q^{1/3} = C_\varepsilon C_0\, q^{-1/6+\varepsilon}.$$

By (Q3) with $K = 2$, the exponentially small part of η_A and that of η_B are each at most q^{-2}, so

$$3q\,\eta_A \le 3q\,(576 + 1)\,q^{-2} = \frac{1731}{q}, \qquad 3q\,\eta_B \le 3q\cdot 7\,q^{-2} = \frac{21}{q}.$$

Hence

$$\begin{aligned}
\operatorname{Var} T_q &\le 4\,C_\varepsilon C_0\, q^{-1/6+\varepsilon} + \frac{6924}{q},\\
\operatorname{Var} \mathrm{LP}_q^* &\le 4\,C_\varepsilon C_0\, q^{-1/6+\varepsilon} + \frac{84}{q},\\
\operatorname{Var} D_q &\le 16\,C_\varepsilon C_0\, q^{-1/6+\varepsilon} + \frac{14016}{q},
\end{aligned}$$

which are within the stated bounds. □

Recall from Section 2 that the core is nonempty exactly when $D_q = 0$.

Corollary 3.13. *Let $\varepsilon \in (0, 1/2)$. If $\mathbb{E}D_q \geq q^{-1/12+\varepsilon}$ for all large q, then for all large q*

$$\mathbb{P}(\textit{the core is nonempty}) = \mathbb{P}(D_q = 0) \leq \frac{\operatorname{Var} D_q}{(\mathbb{E}D_q)^2} \leq 16\, C_\varepsilon C_0\, q^{-\varepsilon} + 28000\, q^{-5/6-2\varepsilon},$$

which tends to 0. In particular, $\mathbb{P}(D_q = 0) \to 0$ whenever $\liminf_q \mathbb{E}D_q > 0$.

Proof. Since $D_q \geq 0$ and $\mathbb{E}D_q > 0$, we have $\{D_q = 0\} \subseteq \{|D_q - \mathbb{E}D_q| \geq \mathbb{E}D_q\}$, and Chebyshev's inequality gives $\mathbb{P}(D_q = 0) \leq \operatorname{Var} D_q/(\mathbb{E}D_q)^2$. If $\mathbb{E}D_q \geq q^{-1/12+\varepsilon}$, then $(\mathbb{E}D_q)^2 \geq q^{-1/6+2\varepsilon}$, and Theorem 3.3(ii) with the same ε gives, for $q \geq q_0(\varepsilon, 2)$,

$$\mathbb{P}(D_q = 0) \leq 16\, C_\varepsilon C_0\, q^{-\varepsilon} + 28000\, q^{-1}\, q^{1/6-2\varepsilon} = 16\, C_\varepsilon C_0\, q^{-\varepsilon} + 28000\, q^{-5/6-2\varepsilon}.$$

If $\liminf_q \mathbb{E}D_q > 0$, then $\mathbb{E}D_q \geq q^{-1/12+1/24}$ for all large q, and the first part applies with $\varepsilon = 1/24$. □

Remark 3.14. The restriction to $\varepsilon < 1/2$ in Corollary 3.13 loses nothing: since $\mathbb{E}D_q \leq \mathbb{E}\Phi_q \leq C_0\, q^{1/3}$, the hypothesis $\mathbb{E}D_q \geq q^{-1/12+\varepsilon}$ cannot hold for all large q when $\varepsilon > 5/12$, so for $\varepsilon \geq 1/2$ it is vacuous.

4. The integer optimum in the independent-team model

This section treats the independent-team model, in which the q^3 teams have independent deficits, each with the law of a sum of three independent $U[0, 1]$ variables. It is the market of Section 2 with the dependence between teams that share a pair removed; Section 5 carries the conclusion back to the market.

Theorem 4.1. *In the independent-team model, with probability tending to 1 as $q \to \infty$, every perfect assignment has deficit at least $\beta_0\, q^{1/3}$, where $\beta_0 = 2.302$.*

The proof is a first-moment computation for the number of perfect assignments of small deficit, carried out conditionally on the stars of the agents (the numbers of cheap teams at each agent, recorded in bins of cost). The steps are: an exact formula for the conditional first moment in a configuration model (Subsection 4.2); exponential bounds and the passage to a limit functional on typical stars (Subsection 4.3); a concavity property of that functional, which bounds its supremum by its value on a tangent (Subsection 4.4); and the evaluation of one explicit number in interval arithmetic (Subsection 4.5). The analytic part is stated as a criterion, Proposition 4.3, valid for every choice of the parameters; the computer-assisted part is Proposition 4.4.

4.1. **The model, the parameters and the criterion.**

The model. In this section the three sectors are written V_1, V_2, V_3; each has q agents. A team is a triple $e \in V_1 \times V_2 \times V_3$. The q^3 team deficits $\Delta(e)$ are independent, and each is distributed as $U + U' + U''$ with U, U', U'' independent and uniform on $[0,1]$. Two facts about the density f of $\Delta(e)$ are used, and nothing else about the one-team law:

$$f(t) = \tfrac{1}{2}t^2 \quad \text{for } 0 \le t \le 1, \tag{F1}$$

$$f(t) \le \tfrac{1}{2}t^2 \quad \text{for all } t \ge 0. \tag{F2}$$

Indeed $2f(t) = t^2 - 3(t-1)^2 \le t^2$ on $[1,2]$, $2f(t) = (3-t)^2 \le 1 \le t^2$ on $[2,3]$, and $f = 0$ on $(3,\infty)$.

As in Section 2, the rescaled cost of a team is $\xi_e = q^{2/3}\Delta(e)$. For a perfect assignment N (a set of q pairwise disjoint teams) put $\xi(N) = \sum_{e \in N} \xi_e$. The deficit of N is $q^{-2/3}\xi(N)$, so N has deficit below $\beta q^{1/3}$ if and only if $\xi(N) < \beta q$. For $\beta \in \mathbb{R}$ let

$$Z_\beta = \#\{N : \ N \text{ a perfect assignment with } \xi(N) < \beta q\}.$$

Recall $\Lambda(\mathrm{d}x) = \frac{1}{2}x^2\,\mathrm{d}x$. By (F1), for every interval $I \subset [0, q^{2/3}]$,

$$\mathbb{P}(\xi_e \in I) = \Lambda(I)/q^2. \tag{4.1}$$

Each agent lies in q^2 teams, so the number of teams at a given agent with cost in I is Binomial$(q^2, \Lambda(I)/q^2)$, which converges to Poisson$(\Lambda(I))$. This is the origin of the reference star below.

Parameters. Fix a truncation level $K > 0$, a number of bins $B \ge 1$, and a tilt $\theta > 0$. Put $\eta = K/B$ and $I_b = ((b-1)\eta, b\eta]$ for $1 \le b \le B$. A team e is *low of bin* b if $\xi_e \in I_b$, and *high* if $\xi_e > K$ (the event $\xi_e = 0$ has probability zero and is ignored). Put

$$\lambda_b = \Lambda(I_b) = \frac{\eta^3(3b^2 - 3b + 1)}{6}, \qquad \sum_{b=1}^{B} \lambda_b = \frac{K^3}{6}, \qquad \lambda_{\max} = \max_b \lambda_b = \lambda_B,$$

$$\kappa_b = \int_{I_b} \mathrm{e}^{-\theta x}\, \tfrac{1}{2}x^2\,\mathrm{d}x, \qquad J = \int_K^\infty \mathrm{e}^{-\theta x}\, \tfrac{1}{2}x^2\,\mathrm{d}x = \mathrm{e}^{-\theta K}\Big(\frac{K^2}{2\theta} + \frac{K}{\theta^2} + \frac{1}{\theta^3}\Big).$$

Vectors in $\mathbb{R}^{B+1}$ have coordinates indexed by $1, \dots, B, \infty$; for $x, y \in \mathbb{R}^{B+1}$ write $\langle x, y\rangle = \sum_b x_b y_b + x_\infty y_\infty$. The symbol $\mathbf{1}$ also denotes the vector $(1, \dots, 1) \in \mathbb{R}^{B+1}$. A *weight vector* is any $u = (u_1, \dots, u_B, u_\infty) \in \mathbb{R}^{B+1}$; put $w_b = \mathrm{e}^{u_b}$ and $w_\infty = \mathrm{e}^{u_\infty}$. The *reference star* is a random vector $D = (D_1, \dots, D_B)$ with independent coordinates, $D_b \sim$ Poisson(λ_b); its law on $\mathbb{N}^B$ is denoted γ. Put

$$W = W(u) = w_\infty + \sum_{b=1}^{B} D_b w_b.$$

Since $w_\infty \le W \le w_\infty + |D| \max_b w_b$, where $|d| = \sum_b d_b$, the variable $\ln W$ has finite moments of all orders.

Definition 4.2. For $u \in \mathbb{R}^{B+1}$ put

$$\Phi(u) = \mathbb{E}\ln W, \qquad \rho_b(u) = \mathbb{E}\Big[\frac{w_b}{W+w_b}\Big]\ (1 \le b \le B), \qquad \varepsilon_0(u) = \mathbb{E}\Big[\frac{w_\infty}{W}\Big],$$

$$C_b(u) = \mathrm{e}^{-3u_b}\,\frac{\kappa_b}{\lambda_b}, \qquad C_\infty(u) = \mathrm{e}^{-3u_\infty} J,$$

and

$$\Psi_0(K,B,\theta,u) = 3\,\Phi(u) + \sum_{b=1}^{B} \lambda_b \ln\big((1-\rho_b)^3 + C_b\rho_b^3\big) - 3\,\varepsilon_0 + C_\infty \varepsilon_0^3, \tag{4.2}$$

with $\rho_b = \rho_b(u)$, $\varepsilon_0 = \varepsilon_0(u)$, $C_b = C_b(u)$, $C_\infty = C_\infty(u)$.

Proposition 4.3 (The criterion)**.** *Let $K > 0$, $B \ge 1$, $\theta > 0$, $u \in \mathbb{R}^{B+1}$ and $\beta \in \mathbb{R}$ satisfy*

$$\theta\beta + \Psi_0(K,B,\theta,u) < 0. \tag{4.3}$$

Then, in the independent-team model, $\mathbb{P}(Z_\beta \ge 1) \to 0$ as $q \to \infty$: with probability tending to 1 every perfect assignment has deficit at least $\beta q^{1/3}$. More precisely, there are $\delta > 0$ and $c > 0$, depending only on (K,B,θ,u,β), such that for all large q

$$\mathbb{P}(Z_\beta \ge 1) \le \mathbb{P}\big(D(G) \notin \mathcal{T}_\delta\big) + \mathrm{e}^{-cq}, \qquad \mathbb{P}\big(D(G) \notin \mathcal{T}_\delta\big) \to 0,$$

where $D(G)$ is the array of binned stars defined in Subsection 4.2 and $\mathcal{T}_\delta$ is the set of typical arrays defined in (4.16)*. No rate is tracked for the first term.*

The proof occupies Subsections 4.2–4.4 and is concluded in Subsection 4.5. The following statement is the only computer-assisted step of this section.

Proposition 4.4 (What the computation certifies)**.** *Let $K = 5$, $B = 50$ (so $\eta = 1/10$), $\theta = 1.6712$, and let $u^* \in \mathbb{R}^{51}$ be the vector whose coordinates $u_1^*, \dots, u_{50}^*$ are, in this order,*

0.71406, 0.58839, 0.44812, 0.30536, 0.16226, 0.01936, −0.12321,
−0.26544, −0.40739, −0.54916, −0.69086, −0.83260, −0.97452,
−1.11672, −1.25934, −1.40250, −1.54629, −1.69084, −1.83624,
−1.98257, −2.12991, −2.27832, −2.42785, −2.57854, −2.73041,
−2.88345, −3.03766, −3.19302, −3.34949, −3.50701, −3.66554,
−3.82501, −3.98535, −4.14649, −4.30837, −4.47092, −4.63405,
−4.79770, −4.96183, −5.12636, −5.29125, −5.45647, −5.62197,
−5.78770, −5.95361, −6.11975, −6.28597, −6.45224, −6.61880,
−6.78546

and $u_\infty^* = -4.61708$. Then

$$\Psi_0(5,50,1.6712,u^*) \le -3.84751328322. \tag{4.4}$$

Consequently $\Psi_0 \le -3.84751328$, $-\Psi_0/\theta \ge 2.30224586$, and $\theta\beta + \Psi_0 \le -0.0004108832$ for $\beta = 2.302$.

Proof. Inequality (4.4) is the output of the interval-arithmetic script

`lemmaC/round3/final/certI/certify_rigorous.py`

run on the weights file `lemmaC/round3/firstmoment/w_K5_e10.json` (paths relative to the lane folder) with mesh parameter 0.005. The script encloses $\Phi(u^*)$, the $\rho_b(u^*)$ and $\varepsilon_0(u^*)$ by quadrature of convex integrands with rigorous error bounds, evaluates (4.2) in outward-rounded arithmetic, and returns the upper endpoint of its enclosure of Ψ_0 as an exact binary rational,

$$-\frac{116283953735057163395091 7}{2^{78}} \;\le\; -3.84751328322.$$

How the enclosures are obtained, and what is trusted, is described in Section 11. The remaining statements follow by exact arithmetic:

$$1.6712 \times 2.30224586 = 3.847513281232 \le 3.84751328322,$$
$$1.6712 \times 2.302 - 3.84751328322 = -0.00041088322. \qquad \square$$

Note that the bound $-\Psi_0/\theta \ge 2.30224586$ uses the eleven-decimal form (4.4); the eight-decimal form $\Psi_0 \le -3.84751328$ alone gives only $-\Psi_0/\theta \ge 2.30224585$.

Proof of Theorem 4.1, given Propositions 4.3 and 4.4. By Proposition 4.4, condition (4.3) holds for $K = 5$, $B = 50$, $\theta = 1.6712$, $u = u^*$ and $\beta = \beta_0 = 2.302$. By Proposition 4.3, $\mathbb{P}(Z_{\beta_0} \ge 1) \to 0$. On the event $\{Z_{\beta_0} = 0\}$ every perfect assignment N has $\xi(N) \ge \beta_0 q$, that is, deficit at least $\beta_0 q^{1/3}$. $\square$

Remark 4.5. The same argument gives the conclusion of Theorem 4.1 for every $\beta < -\Psi_0/\theta$, and so for every $\beta \le 2.30224586$. A second run of the same script (weights file `lemmaC/round3/firstmoment/w_K5_e25.json`: $K = 5$, $B = 20$, $\theta = 1.6671$) certifies $\Psi_0 \le -3.83621828$ and $-\Psi_0/\theta \ge 2.30113267$; it is not used. In floating point, the largest value of $-\Psi_0/\theta$ found over θ and u at $K = 5$ is about 2.30114 for $B = 20$ and 2.30225 for $B = 50$, and the corresponding continuum variational problem has value about 2.30246 at $K = 5$ and 2.30248 without truncation. These floating-point values are not used in any proof; they suggest, without proving it, that the method does not give a constant above about 2.30248.

Remark 4.6. The proof of Proposition 4.3 uses only (F1), (F2) and the independence of the teams.

4.2. **The conditional first moment.** Throughout this subsection and the next, K, B, θ and β are fixed and $q^{2/3} \ge K$. Put

$$p_b = \frac{\lambda_b}{q^2} \quad (1 \le b \le B), \qquad p = \sum_{b=1}^{B} p_b = \frac{K^3}{6q^2}.$$

By (4.1), $p_b = \mathbb{P}(\xi_e \in I_b)$ and $1 - p = \mathbb{P}(\xi_e > K)$; moreover $p \le 1/6$ because $q^2 \ge K^3$.

Tilting, and the split into low and high teams. Let G be the *labelled low hypergraph*: the set of pairs (e,b) with $\xi_e \in I_b$. It is the hypergraph of teams of cost at most K, each marked by the bin of its cost. Write $e \in G$ if e is low, and $b(e)$ for its bin. Conditionally on G the costs are independent. By (F1) the density of ξ_e is $x^2/(2q^2)$ on $[0, q^{2/3}]$, so a low team of bin b has conditional density $x^2/(2\lambda_b)$ on I_b. A high team has conditional density $q^{-2/3}f(q^{-2/3}x)/(1-p)$ on (K,∞), which is at most $x^2/(2q^2(1-p))$ by (F2). Hence

(4.5)
$$\mathbb{E}\big[\mathrm{e}^{-\theta\xi_e}\,\big|\,G\big] = \frac{\kappa_b}{\lambda_b} \ \text{ if } (e,b)\in G, \qquad \mathbb{E}\big[\mathrm{e}^{-\theta\xi_e}\,\big|\,G\big] \le \frac{J}{q^2(1-p)} \ \text{ if } e \text{ is high.}$$

Since $\mathbf{1}\{\xi(N) < \beta q\} \le \exp(\theta(\beta q - \xi(N)))$, conditional independence and (4.5) give

$$\mathbb{E}[Z_\beta \mid G] \le \mathrm{e}^{\theta\beta q} \sum_N \prod_{e\in N\cap G} \frac{\kappa_{b(e)}}{\lambda_{b(e)}} \cdot \Big(\frac{J}{q^2(1-p)}\Big)^{|N\setminus G|}, \tag{4.6}$$

the sum being over all perfect assignments N.

A *matching in* G is a set of pairwise disjoint low teams. A *profile* is a vector $m = (m_1,\dots,m_B) \in \mathbb{N}^B$ with $|m| = \sum_b m_b \le q$; a matching has profile m if it contains exactly m_b teams of bin b for every b. For a profile m put $r = r(m) = q - |m|$, and let $Y_m(G)$ be the number of matchings in G of profile m. There are at most $(q+1)^B$ profiles.

Group the assignments in (4.6) according to the matching $N\cap G$. If $N\cap G$ has profile m, then $N\setminus G$ is a perfect assignment of the $r+r+r$ agents not covered by $N\cap G$, and there are $(r!)^2$ perfect assignments of these agents. Dropping the requirement that the teams of $N\setminus G$ be high (a union bound over the teams above K) gives

$$\mathbb{E}[Z_\beta \mid G] \le \mathrm{e}^{\theta\beta q} \sum_m Y_m(G) \prod_{b=1}^{B} \Big(\frac{\kappa_b}{\lambda_b}\Big)^{m_b} (r!)^2 \Big(\frac{J}{q^2(1-p)}\Big)^{r}, \tag{4.7}$$

the sum being over all profiles. An agent not covered by the matching is said to take the *high option*. The high option cannot be omitted: an agent whose star is empty below K is covered by no low team, and every perfect assignment uses a high team at it.

Degree arrays and the configuration model. A *degree array* is an array $D = (D^s_{v,b})$ of nonnegative integers, indexed by $s \in \{1,2,3\}$, $v \in V_s$ and $1 \le b \le B$, such that for every b the column sum

$$n_b = n_b(D) = \sum_{v\in V_s} D^s_{v,b}$$

is the same for $s = 1,2,3$. Put $n = \sum_b n_b$ and $D^s = (D^s_{v,b})_{v\in V_s,\, b\le B}$. The degree array $D(G)$ of the labelled low hypergraph has $D^s_{v,b}(G)$ equal to the number of low teams of bin b that contain the agent $v \in V_s$; its row

$(D^s_{v,b}(G))_{b\le B}$ is the *binned star* of v. For a degree array D let $\mathcal{G}(D)$ be the set of labelled hypergraphs with degree array D: sets g of pairs (e,b), e a team and $1\le b\le B$, in which no team occurs twice (neither twice with the same bin, which is automatic for a set, nor with two different bins), and with $D(g)=D$.

(a) The teams are independent, each is low of bin b with probability p_b and high with probability $1-p$. So every $g\in\mathcal{G}(D)$ has

$$\mathbb{P}(G=g)=\prod_{b=1}^{B} p_b^{n_b}\,(1-p)^{q^3-n}, \tag{4.8}$$

which depends on g only through D: conditionally on $D(G)=D$, the hypergraph G is uniform on $\mathcal{G}(D)$.

(b) *Configurations.* For each s and b take half-edges (v,b,i) with $v\in V_s$ and $1\le i\le D^s_{v,b}$; there are n_b half-edges of bin b in each sector. A *configuration* is a family $c=(\sigma_b,\tau_b)_{b\le B}$ in which σ_b and τ_b are bijections from the half-edges of bin b in sector 1 to the half-edges of bin b in sectors 2 and 3 respectively. There are $\prod_b (n_b!)^2$ configurations. The *triples* of c are the n triples of half-edges $(x,\sigma_b(x),\tau_b(x))$, x a half-edge of bin b in sector 1; such a triple has bin b, and its *team* is the triple of agents carrying its three half-edges. A configuration is *simple* if the teams of its n triples are pairwise distinct; that is, no two of its triples have the same team, whether the two triples lie in the same bin or in two different bins. The *projection* of a simple configuration is the set of pairs (team, bin) of its triples, an element of $\mathcal{G}(D)$. A configuration that is not simple projects to no element of $\mathcal{G}(D)$.

Every $g\in\mathcal{G}(D)$ is the projection of exactly $\prod_{s,v,b} D^s_{v,b}!$ configurations. Indeed, a configuration projecting to g is obtained by choosing, for every s, $v\in V_s$ and b, a bijection between the $D^s_{v,b}$ teams of g of bin b through v and the half-edges $(v,b,1),\dots,(v,b,D^s_{v,b})$; since the teams of g are distinct, different choices give different sets of triples, hence different configurations.

(c) For a configuration c and a profile m let $\widetilde{Y}_m(c)$ be the number of sets of triples of c, containing exactly m_b triples of bin b for every b, whose teams are pairwise disjoint. If c is simple with projection g, the triples of c correspond bijectively to the elements of g, and $\widetilde{Y}_m(c)=Y_m(g)$.

For a profile m and $s\in\{1,2,3\}$, a *choice function* of sector s is a selection of m_b half-edges of bin b in sector s, for every b, such that the $|m|$ selected half-edges are at pairwise distinct agents. Let $N_s(m;D^s)$ be the number of choice functions. Each agent of V_s either has no selected half-edge or has exactly one, of some bin b, chosen among $D^s_{v,b}$; therefore

$$N_s(m;D^s)=\big[z_1^{m_1}\cdots z_B^{m_B} z_\infty^{q-|m|}\big]\prod_{v\in V_s}\Big(z_\infty+\sum_{b=1}^{B} D^s_{v,b} z_b\Big). \tag{4.9}$$

Lemma 4.7 (Exact count)**.** *For every degree array D and every profile m with $m_b \le n_b$ for all b, the sum over all configurations with degree array D is*

$$\sum_{c} \widetilde{Y}_m(c) = \prod_{s=1}^{3} N_s(m; D^s) \cdot \prod_{b=1}^{B} \big(m_b!\,(n_b - m_b)!\big)^2.$$

Equivalently, the average of $\widetilde{Y}_m$ over the $\prod_b (n_b!)^2$ configurations is

$$\prod_{s=1}^{3} N_s(m; D^s) \cdot \prod_{b=1}^{B} \binom{n_b}{m_b}^{-2}.$$

Proof. Count pairs (c, S) with S a set of triples of c as in (c). The set S determines, in each sector, the set of half-edges it uses; because the teams of S are pairwise disjoint, this is a choice function. Conversely, a triple of choice functions (one per sector) together with, for every b, a bijection from the selected half-edges of bin b in sector 1 to those in sector 2 and another to those in sector 3, is a set S of triples of half-edges with pairwise disjoint teams. So there are $\prod_s N_s(m; D^s) \cdot \prod_b (m_b!)^2$ possible sets S. A given S is a set of triples of exactly those configurations whose bijections extend the ones prescribed by S; the remaining $n_b - m_b$ half-edges of bin b in each sector are matched arbitrarily, in $\prod_b ((n_b - m_b)!)^2$ ways. □

Lemma 4.8 (Simplicity conditioning)**.** *For a degree array D put*

$$Q(D) = \prod_{b=1}^{B} p_b^{n_b}\,(1-p)^{q^3 - n} \cdot \frac{\prod_b (n_b!)^2}{\prod_{s,v,b} D^s_{v,b}!}.$$

Then for every degree array D and every profile m with $m_b \le n_b$ for all b,

$$\mathbb{E}\big[Y_m(G);\, D(G) = D\big] \le Q(D) \cdot \prod_{s=1}^{3} N_s(m; D^s) \cdot \prod_{b=1}^{B} \binom{n_b}{m_b}^{-2}, \tag{4.10}$$

while $Y_m(G) = 0$ on $\{D(G) = D\}$ if $m_b > n_b$ for some b. Moreover

$$\sum_{D} Q(D) \le \exp\Big(\frac{K^6}{36\, q\,(1-p)}\Big), \tag{4.11}$$

the sum being over all degree arrays.

Proof. A matching of profile m contains m_b distinct low teams of bin b, so $Y_m(G) = 0$ if $m_b > n_b(D(G))$ for some b. Let $m_b \le n_b$ for all b. By (4.8), (b) and (c),

$$\begin{aligned}\mathbb{E}\big[Y_m(G);\, D(G) = D\big] &= \prod_b p_b^{n_b} (1-p)^{q^3 - n} \sum_{g \in \mathcal{G}(D)} Y_m(g) \\ &= \frac{\prod_b p_b^{n_b} (1-p)^{q^3-n}}{\prod_{s,v,b} D^s_{v,b}!} \sum_{c \text{ simple}} \widetilde{Y}_m(c).\end{aligned}$$

Since $\widetilde{Y}_m \geq 0$, the sum over simple configurations is at most the sum over all configurations, which is given by Lemma 4.7; this is (4.10).

For (4.11), fix $(n_b)_{b\leq B} \in \mathbb{N}^B$. By the multinomial theorem, for each s and b the sum of $1/\prod_{v\in V_s} D^s_{v,b}!$ over all $(D^s_{v,b})_{v\in V_s}$ with sum n_b equals $q^{n_b}/n_b!$. Hence the sum of $Q(D)$ over the degree arrays with column sums (n_b) is

$$(1-p)^{q^3-n}\prod_b p_b^{n_b}(n_b!)^2\Big(\frac{q^{n_b}}{n_b!}\Big)^3 = (1-p)^{q^3}\prod_b \frac{1}{n_b!}\Big(\frac{q^3 p_b}{1-p}\Big)^{n_b}.$$

Summing over $(n_b) \in \mathbb{N}^B$ gives at most

$$(1-p)^{q^3}\exp\Big(\frac{q^3 p}{1-p}\Big) \leq \exp\Big(-q^3 p + \frac{q^3 p}{1-p}\Big) = \exp\Big(\frac{q^3 p^2}{1-p}\Big),$$

and $q^3p^2 = K^6/(36q)$. □

Remark 4.9. If $\mathcal{G}(D) \neq \emptyset$ then $Q(D)$ is the ratio of $\mathbb{P}(D(G) = D)$ to the probability that a uniform configuration with degree array D is simple. Lemma 4.8 says that the conditional first moment given the stars is at most the configuration-model value divided by the probability of simplicity, and that this division costs a factor $1 + o(1)$ after averaging over D. No lower bound on the probability of simplicity is needed.

Lemma 4.10 (Conditional first moment)**.** *Let $q^{2/3} \geq K$. For every set $\mathcal{T}$ of degree arrays,*

$$\begin{aligned}(4.12)\quad \mathbb{E}\big[Z_\beta;\, D(G)\in\mathcal{T}\big] \leq \exp\Big(\frac{K^6}{36\,q\,(1-p)}\Big)\,\mathrm{e}^{\theta\beta q} \\ \times \sum_m \sup_{D\in\mathcal{T}:\ n(D)\geq m} \Big[\prod_{s=1}^{3} N_s(m; D^s)\cdot \prod_{b=1}^{B}\binom{n_b}{m_b}^{-2}\Big] \\ \times \prod_{b=1}^{B}\Big(\frac{\kappa_b}{\lambda_b}\Big)^{m_b}(r!)^2\Big(\frac{J}{q^2(1-p)}\Big)^r,\end{aligned}$$

where the sum is over all profiles m, $r = q - |m|$, the condition $n(D) \geq m$ means $n_b(D) \geq m_b$ for all b, and the supremum of the empty set is 0.

Proof. Multiply (4.7) by $\mathbf{1}\{D(G) = D\}$, take expectations, and sum over $D \in \mathcal{T}$. For each profile m this gives the factor $\sum_{D\in\mathcal{T}} \mathbb{E}[Y_m(G); D(G) = D]$, which by (4.10) is at most $\sum_D Q(D)$ times the supremum, over $D \in \mathcal{T}$ with $n(D) \geq m$, of the bracket. Apply (4.11). □

4.3. **Exponential bounds and the limit functional.**

Choice functions. For a degree array D, $s \in \{1, 2, 3\}$ and $u \in \mathbb{R}^{B+1}$ put

$$\widehat{\Phi}_s(u) = \widehat{\Phi}_s(u; D) = \frac{1}{q}\sum_{v\in V_s} \ln\Big(\mathrm{e}^{u_\infty} + \sum_{b=1}^{B} D^s_{v,b}\,\mathrm{e}^{u_b}\Big).$$

For a profile m write $\mu_b = m_b/q$, $\varepsilon = r/q = 1 - \sum_b \mu_b$, and $\langle \mu, u\rangle = \sum_b \mu_b u_b + \varepsilon u_\infty$. The polynomial in (4.9) has nonnegative coefficients, so each coefficient is at most the value of the polynomial at a positive point divided by the corresponding monomial. Taking $z_b = \mathrm{e}^{u_b}$, $z_\infty = \mathrm{e}^{u_\infty}$ gives, for every $u \in \mathbb{R}^{B+1}$ and every profile m,

$$N_s(m; D^s) \le \exp\Big(q\big[\widehat{\Phi}_s(u) - \langle \mu, u\rangle\big]\Big). \tag{4.13}$$

Binomials and factorials. Let $h(x) = -x\ln x - (1-x)\ln(1-x)$ be the binary entropy, and $g(x,y) = y\,h(x/y)$ for $0 \le x \le y$, $y > 0$, with $g(0,0) = 0$; then $0 \le g(x,y) \le y\ln 2$ and g is continuous on $\{0 \le x \le y\}$. The following elementary bounds are used: $\binom{n}{m} \ge \exp(n\,h(m/n))/(n+1)$ for $0 \le m \le n$, and $r! \le \mathrm{e}\,(r+1)(r/\mathrm{e})^r$ for $r \ge 0$ (with $0^0 = 1$).

Lemma 4.11 (The exponent)**.** *Let $q^{2/3} \ge K$. For a degree array D and a profile m with $m_b \le n_b(D)$ for all b, put $\nu_b = n_b/q$ and*

$$\Xi(\mu; D) = \theta\beta + \sum_{s=1}^{3} \inf_{u \in \mathbb{R}^{B+1}} \big[\widehat{\Phi}_s(u; D) - \langle \mu, u\rangle\big] - 2\sum_{b=1}^{B} g(\mu_b, \nu_b) + \sum_{b=1}^{B} \mu_b \ln\frac{\kappa_b}{\lambda_b} + \varepsilon\ln J + 2\varepsilon\ln\varepsilon - 2\varepsilon \tag{4.14}$$

(with $0\ln 0 = 0$). Then for every set $\mathcal{T}$ of degree arrays

$$\mathbb{E}\big[Z_\beta;\, D(G) \in \mathcal{T}\big] \le c_q\,(q+1)^{B+2}\ \sup\Big[\prod_{b=1}^{B}(n_b+1)^2 \cdot \exp\big(q\,\Xi(\mu; D)\big)\Big], \tag{4.15}$$

where the supremum is over all $D \in \mathcal{T}$ and all profiles m with $m_b \le n_b(D)$ for all b, and

$$c_q = \mathrm{e}^2 \exp\Big(\frac{K^6}{36\,q\,(1-p)}\Big)(1-p)^{-q} \le \mathrm{e}^2 \exp\Big(\frac{K^{9/2}}{30} + \frac{K^{3/2}}{5}\Big).$$

The factor $\prod_b (n_b+1)^2$ depends on the degree array and stays inside the supremum.

Proof. Fix D and m with $m_b \le n_b$. By (4.13), optimised over u separately for each s, $\prod_s N_s(m; D^s) \le \exp(q\sum_s \inf_u[\widehat{\Phi}_s(u) - \langle \mu, u\rangle])$. By the bound on binomial coefficients, $\prod_b \binom{n_b}{m_b}^{-2} \le \prod_b (n_b+1)^2 \exp(-2n_b h(m_b/n_b)) = \prod_b (n_b+1)^2 \cdot \exp(-2q\sum_b g(\mu_b, \nu_b))$. Next, $\prod_b (\kappa_b/\lambda_b)^{m_b} = \exp(q\sum_b \mu_b \ln(\kappa_b/\lambda_b))$. Finally, by the bound on $r!$ and $r = \varepsilon q$,

$$\begin{aligned}(r!)^2\Big(\frac{J}{q^2(1-p)}\Big)^r &\le \mathrm{e}^2(r+1)^2(1-p)^{-r}\Big(\frac{r}{\mathrm{e}q}\Big)^{2r} J^r \\ &\le \mathrm{e}^2(q+1)^2(1-p)^{-q}\exp\big(q[2\varepsilon\ln\varepsilon - 2\varepsilon + \varepsilon\ln J]\big).\end{aligned}$$

So the term of (4.12) belonging to m and D, times $\mathrm{e}^{\theta\beta q}$, is at most $\mathrm{e}^2(q+1)^2(1-p)^{-q}\prod_b(n_b+1)^2\exp(q\,\Xi(\mu;D))$. There are at most $(q+1)^B$ profiles, which gives (4.15). For the bound on c_q use $p \le 1/6$ and $q \ge K^{3/2}$: then $K^6/(36q(1-p)) \le K^{9/2}/30$ and $(1-p)^{-q} \le \exp(qp/(1-p)) \le \exp(K^3/(5q)) \le \exp(K^{3/2}/5)$. □

Typical stars. For $s \in \{1,2,3\}$ let $\widehat{\gamma}_s$ be the empirical law on $\mathbb{N}^B$ of the q binned stars of sector s: $\widehat{\gamma}_s(d) = q^{-1}\#\{v \in V_s : (D^s_{v,b})_{b\le B} = d\}$. Recall that γ is the law of the reference star and $|d| = \sum_b d_b$. For $\delta > 0$ let $\mathcal{T}_\delta$ be the set of degree arrays D with

$$\sum_{d\in\mathbb{N}^B} \bigl|\widehat{\gamma}_s(d) - \gamma(d)\bigr|\,(1+|d|) \le \delta \qquad \text{for } s = 1,2,3. \tag{4.16}$$

Lemma 4.12 (Stars are typical)**.** *For every $\delta > 0$, $\mathbb{P}(D(G) \in \mathcal{T}_\delta) \to 1$ as $q \to \infty$.*

Proof. Fix s. The binned star of $v \in V_s$ is a function of the costs of the q^2 teams containing v, and these sets of teams are disjoint for distinct agents of V_s. So the q stars of sector s are independent, with common law γ_q, the law of $\sum_{i\le q^2}\zeta_i$ where the ζ_i are independent, $\zeta_i = \mathbf{e}_b$ (the bth unit vector of $\mathbb{N}^B$) with probability p_b and $\zeta_i = 0$ with probability $1-p$.

Total variation. Let ζ' have independent Poisson(p_b) coordinates. Then $\mathbb{P}(\zeta' = 0) = \mathrm{e}^{-p} \ge 1-p$ and $\mathbb{P}(\zeta' = \mathbf{e}_b) = p_b\mathrm{e}^{-p} \le p_b$, so the total variation distance between the laws of ζ_1 and ζ' is $\sum_b(p_b - p_b\mathrm{e}^{-p}) = p(1-\mathrm{e}^{-p}) \le p^2$. The sum of q^2 independent copies of ζ' has law γ, because $q^2p_b = \lambda_b$. Coupling term by term,

$$d_{\mathrm{TV}}(\gamma_q,\gamma) \le q^2p^2 = \frac{K^6}{36q^2}.$$

Small stars. Let $R \ge 1$. For each d, $q\widehat{\gamma}_s(d)$ is Binomial$(q,\gamma_q(d))$, so $\mathbb{E}|\widehat{\gamma}_s(d) - \gamma_q(d)| \le (\gamma_q(d)/q)^{1/2}$. Hence

$$\begin{aligned}\mathbb{E}\sum_{|d|\le R}\bigl|\widehat{\gamma}_s(d)-\gamma(d)\bigr|(1+|d|) &\le (1+R)\Bigl[\sum_{|d|\le R}\Bigl(\frac{\gamma_q(d)}{q}\Bigr)^{1/2} + 2\,d_{\mathrm{TV}}(\gamma_q,\gamma)\Bigr]\\ &\le (1+R)\Bigl[\frac{(R+1)^B}{q^{1/2}} + \frac{K^6}{18q^2}\Bigr],\end{aligned}$$

because there are at most $(R+1)^B$ vectors $d \in \mathbb{N}^B$ with $|d| \le R$ and $\sum_d|\gamma_q(d)-\gamma(d)| = 2\,d_{\mathrm{TV}}(\gamma_q,\gamma)$. The right side tends to 0 as $q\to\infty$ for fixed R.

Large stars. Under γ the total $|D|$ is Poisson$(K^3/6)$, and under γ_q it is Binomial(q^2,p), with the same mean $K^3/6$ and variance at most $K^3/6$. In both cases $\mathbb{E}(1+|D|)^2 \le C := (1+K^3/6)^2 + K^3/6$, and so $\mathbb{E}[(1+|D|);|D| >$

$R] \le C/(1+R)$. Therefore

$$\mathbb{E}\sum_{|d|>R}\big|\widehat{\gamma}_s(d)-\gamma(d)\big|(1+|d|)$$
$$\le \mathbb{E}_{\gamma_q}\big[(1+|D|);|D|>R\big]+\mathbb{E}_{\gamma}\big[(1+|D|);|D|>R\big]\le\frac{2C}{1+R},$$

uniformly in q.

Given $\delta>0$ and $\omega>0$, choose R with $2C/(1+R)\le\delta\omega/12$ and then q_0 so large that for $q\ge q_0$ the bound for small stars is at most $\delta\omega/12$. For $q\ge q_0$ the left side of (4.16) has expectation at most $\delta\omega/6$, so by Markov's inequality it exceeds δ with probability at most $\omega/6$, and a union bound over $s=1,2,3$ gives $\mathbb{P}(D(G)\notin\mathcal{T}_\delta)\le\omega/2$. □

Two consequences of (4.16) are used. Let $D\in\mathcal{T}_\delta$.

(T1) $|\nu_b-\lambda_b|\le\delta$ for every b, because $\nu_b=\sum_d\widehat{\gamma}_s(d)\,d_b$ and $\lambda_b=\sum_d\gamma(d)\,d_b$. In particular $n_b+1\le 1+(\lambda_{\max}+\delta)q$.

(T2) Let $A>0$ and $u\in[-A,0]^{B+1}$. For every $d\in\mathbb{N}^B$, $\mathrm{e}^{-A}\le\mathrm{e}^{u_\infty}+\sum_b d_b\mathrm{e}^{u_b}\le 1+|d|$, so the logarithm of the middle term is at most $A+|d|\le(A+1)(1+|d|)$ in absolute value. Since $\widehat{\Phi}_s(u;D)$ and $\Phi(u)$ are the integrals of $d\mapsto\ln(\mathrm{e}^{u_\infty}+\sum_b d_b\mathrm{e}^{u_b})$ against $\widehat{\gamma}_s$ and γ respectively,

$$\big|\widehat{\Phi}_s(u;D)-\Phi(u)\big|\le(A+1)\,\delta\qquad\text{for all } u\in[-A,0]^{B+1},\ s=1,2,3. \tag{4.17}$$

The limit functional. Let

$$\mathcal{P}=\Big\{\mu=(\mu_1,\dots,\mu_B,\varepsilon)\in\mathbb{R}^{B+1}:\ 0\le\mu_b\le\lambda_b,\ \varepsilon\ge0,\ \sum_{b=1}^{B}\mu_b+\varepsilon=1\Big\}.$$

For $\mu\in\mathbb{R}^{B+1}$ with coordinate sum 1 (the last coordinate being written ε) and $A>0$ define

$$H(\mu)=\inf_{u\in\mathbb{R}^{B+1}}\big[\Phi(u)-\langle\mu,u\rangle\big],\quad H_A(\mu)=\min_{u\in[-A,0]^{B+1}}\big[\Phi(u)-\langle\mu,u\rangle\big],$$

and for $\mu\in\mathcal{P}$

$$S_0(\mu)=\sum_{b=1}^{B}\lambda_b\,h\Big(\frac{\mu_b}{\lambda_b}\Big)-\varepsilon\ln\varepsilon+\varepsilon,$$
$$L(\mu)=\theta\beta+3H(\mu)-2S_0(\mu)+\langle\ell,\mu\rangle,\qquad \ell_b=\ln\frac{\kappa_b}{\lambda_b},\quad \ell_\infty=\ln J.$$

Put $L^*=\sup_{\mu\in\mathcal{P}}L(\mu)$. Here $H(\mu)\in[-\infty,\infty)$, and H is concave, being an infimum of affine functions. Since $\Phi(u+t\mathbf{1})=\Phi(u)+t$ and $\langle\mu,\mathbf{1}\rangle=1$, the bracket is invariant under $u\mapsto u+t\mathbf{1}$; one may therefore restrict the infimum to u with $\max u=0$, and consequently

$$H_A(\mu)\downarrow H(\mu)\quad\text{as } A\uparrow\infty. \tag{4.18}$$

Lemma 4.13 (Passage to the limit functional)**.** *For $\delta > 0$ let $\mathcal{K}_\delta$ be the set of pairs (μ, ν) with $\mu = (\mu_1, \dots, \mu_B, \varepsilon)$ a probability vector, $\nu \in \mathbb{R}^B$, $|\nu_b - \lambda_b| \le \delta$ and $0 \le \mu_b \le \nu_b$ for all b; and for $A > 0$ put*

$$L_A(\mu, \nu) = \theta\beta + 3H_A(\mu) - 2\sum_{b=1}^{B} g(\mu_b, \nu_b) + \langle \ell, \mu \rangle + 2\varepsilon \ln \varepsilon - 2\varepsilon.$$

Then for all $A > 0$ and $\delta > 0$,

$$\limsup_{q \to \infty} \frac{1}{q} \ln \mathbb{E}\big[Z_\beta;\, D(G) \in \mathcal{T}_\delta\big] \le \sup_{\mathcal{K}_\delta} L_A + 3(A+1)\,\delta, \tag{4.19}$$

and

$$\inf_{A>0} \inf_{\delta>0} \Big[\sup_{\mathcal{K}_\delta} L_A + 3(A+1)\,\delta\Big] \le L^*. \tag{4.20}$$

In particular, if $L^* < 0$ there are $\delta > 0$ and $c > 0$ such that $\mathbb{E}[Z_\beta; D(G) \in \mathcal{T}_\delta] \le \mathrm{e}^{-cq}$ for all large q.

Proof. The function H_A is a minimum of affine functions over a compact set; it is finite and concave on $\mathbb{R}^{B+1}$, hence continuous. The functions g and $\varepsilon \mapsto \varepsilon \ln \varepsilon$ are continuous. So L_A is continuous on the compact set $\mathcal{K}_\delta$, and its supremum is finite.

Proof of (4.19). Let $D \in \mathcal{T}_\delta$ and let m be a profile with $m_b \le n_b$ for all b. Then $(\mu, \nu) \in \mathcal{K}_\delta$ by (T1). By (4.17),

$$\inf_{u \in \mathbb{R}^{B+1}} \big[\widehat{\Phi}_s(u; D) - \langle \mu, u \rangle\big] \le \min_{u \in [-A,0]^{B+1}} \big[\widehat{\Phi}_s(u; D) - \langle \mu, u \rangle\big] \le H_A(\mu) + (A+1)\,\delta$$

for each s, so that $\Xi(\mu; D) \le L_A(\mu, \nu) + 3(A+1)\delta$. By (T1), $\prod_b (n_b + 1)^2 \le (1 + (\lambda_{\max} + \delta)q)^{2B}$, uniformly in $D \in \mathcal{T}_\delta$. Lemma 4.11 with $\mathcal{T} = \mathcal{T}_\delta$ now gives

$$\begin{aligned} &\mathbb{E}\big[Z_\beta;\, D(G) \in \mathcal{T}_\delta\big] \\ &\qquad \le c_q\,(q+1)^{B+2}\big(1 + (\lambda_{\max} + \delta)q\big)^{2B} \exp\Big(q\Big[\sup_{\mathcal{K}_\delta} L_A + 3(A+1)\,\delta\Big]\Big), \end{aligned}$$

and the factor in front of the exponential is polynomial in q.

Proof of (4.20). Let $c > L^*$. On $\mathcal{P}$, $-2\sum_b g(\mu_b, \lambda_b) + 2\varepsilon \ln \varepsilon - 2\varepsilon = -2S_0(\mu)$, so $L_A(\mu, \lambda) = \theta\beta + 3H_A(\mu) - 2S_0(\mu) + \langle \ell, \mu \rangle$, which by (4.18) decreases to $L(\mu)$ as $A \uparrow \infty$, for every $\mu \in \mathcal{P}$. The sets $\{\mu \in \mathcal{P} : L_A(\mu, \lambda) \ge c\}$ are compact and decrease as A increases; a common point μ would have $L(\mu) \ge c > L^*$, which is impossible. So one of them is empty: there is A with $\sup_{\mu \in \mathcal{P}} L_A(\mu, \lambda) < c$. Fix this A. The compact sets $\mathcal{K}_\delta$ decrease, as $\delta \downarrow 0$, to $\mathcal{K}_0 = \mathcal{P} \times \{\lambda\}$. The compact sets $\{(\mu, \nu) \in \mathcal{K}_\delta : L_A(\mu, \nu) \ge c\}$ decrease with δ and have empty intersection, so one of them is empty: there is $\delta_0 > 0$ with $\sup_{\mathcal{K}_\delta} L_A < c$ for $\delta \le \delta_0$. Letting $\delta \downarrow 0$, the left side of (4.20) is at most c. As $c > L^*$ was arbitrary, (4.20) follows. (If $L^* = -\infty$ the same argument applies to every real c.)

The last assertion follows from (4.19) and (4.20): choose A and δ with $\sup_{\mathcal{K}_\delta} L_A + 3(A+1)\delta < 0$, and let c be half the absolute value of this number. □

The argument is by compactness: it gives no explicit A and δ, and no rate. Lemmas 4.10 and 4.11 are stated for an arbitrary set $\mathcal{T}$ of degree arrays, so that they can be applied to sets of arrays other than $\mathcal{T}_\delta$.

4.4. **Concavity and the tangent inequality.** This subsection concerns only the reference star D (independent Poisson(λ_b) coordinates) and is exact for finite B. Its outcome is Proposition 4.18: $L^* \le \theta\beta + \Psi_0(K, B, \theta, u_0)$ for every weight vector u_0.

Notation. For $u \in \mathbb{R}^{B+1}$, with $w = \mathrm{e}^u$ and $W = w_\infty + \sum_b D_b w_b$ as before, define the random probability vector $\pi = \pi(u)$ on $\{1, \dots, B, \infty\}$ by

$$\pi_b = \frac{D_b w_b}{W} \quad (1 \le b \le B), \qquad \pi_\infty = \frac{w_\infty}{W}.$$

The function $u \mapsto \ln W$ is smooth, with gradient π and Hessian $\mathrm{diag}(\pi) - \pi\pi^{\mathsf{T}}$; all its partial derivatives of positive order are polynomials in the coordinates of π, hence bounded, so Φ is smooth and may be differentiated under the expectation:

$$\nabla\Phi(u) = \mathbb{E}\,\pi =: \mu(u), \qquad \nabla^2\Phi(u) = \Sigma(u) := \mathbb{E}\big[\mathrm{diag}(\pi) - \pi\pi^{\mathsf{T}}\big].$$

Let $\mathcal{H}_0 = \{x \in \mathbb{R}^{B+1} : \langle x, \mathbf{1}\rangle = 0\}$ and $\mathcal{H}_1 = \{x \in \mathbb{R}^{B+1} : \langle x, \mathbf{1}\rangle = 1\}$. Then $\mu(u) \in \mathcal{H}_1$, $\Sigma(u)\mathbf{1} = 0$, and for $x \in \mathbb{R}^{B+1}$

$$x^{\mathsf{T}}\Sigma(u)\,x = \mathbb{E}\Big[\sum_i \pi_i x_i^2 - \Big(\sum_i \pi_i x_i\Big)^2\Big] \ge 0,$$

the expected variance of x under π. With positive probability all $D_b \ge 1$, and then π charges every coordinate; so $x^{\mathsf{T}}\Sigma(u)x > 0$ unless x is a multiple of $\mathbf{1}$. Thus $\Sigma(u)$ maps $\mathcal{H}_0$ to itself and its restriction $\Sigma(u)|_{\mathcal{H}_0}$ is positive definite; Φ is convex on $\mathbb{R}^{B+1}$ and strictly convex on $\mathcal{H}_0$. Since $\Phi(u + t\mathbf{1}) = \Phi(u) + t$, the quantities $\mu(u)$, $\Sigma(u)$, $\rho_b(u)$ and $\varepsilon_0(u)$ are unchanged when u is replaced by $u + t\mathbf{1}$.

Lemma 4.14 (Poisson identities)**.** *For every $u \in \mathbb{R}^{B+1}$: $\mu_b(u) = \lambda_b\rho_b(u)$ with $\rho_b(u) \in (0,1)$ for $1 \le b \le B$; $\mu_\infty(u) = \varepsilon_0(u) \in (0,1]$; and*

(4.21)

$$\mathbb{E}\big[\pi\pi^{\mathsf{T}}\big] = \int_0^\infty t\,\mathbb{E}\big[\mathrm{e}^{-tW}\big]\, v_t v_t^{\mathsf{T}}\,\mathrm{d}t + \mathrm{diag}\Big(\lambda_b\,\mathbb{E}\Big[\Big(\frac{w_b}{W + w_b}\Big)^2\Big]\ (b \le B),\ 0\Big),$$

where $v_t \in \mathbb{R}^{B+1}$ has coordinates $(v_t)_b = \lambda_b w_b \mathrm{e}^{-tw_b}$ and $(v_t)_\infty = w_\infty$.

Proof. Let $\mathbf{e}_b$ be the bth unit vector of $\mathbb{N}^B$. For independent Poisson variables D_b with means λ_b and every bounded function χ on $\mathbb{N}^B$, the Mecke

formula gives

$$\begin{aligned}\mathbb{E}[D_b\,\chi(D)] &= \lambda_b\,\mathbb{E}\,\chi(D+\mathbf{e}_b),\\ \mathbb{E}[D_bD_c\,\chi(D)] &= \lambda_b\lambda_c\,\mathbb{E}\,\chi(D+\mathbf{e}_b+\mathbf{e}_c)\qquad (b\neq c),\\ \mathbb{E}[D_b^2\,\chi(D)] &= \lambda_b^2\,\mathbb{E}\,\chi(D+2\mathbf{e}_b)+\lambda_b\,\mathbb{E}\,\chi(D+\mathbf{e}_b).\end{aligned}$$

Adding $\mathbf{e}_b$ to D replaces W by $W+w_b$. The first formula with $\chi = w_b/W$ gives $\mu_b(u) = \mathbb{E}[D_bw_b/W] = \lambda_b\mathbb{E}[w_b/(W+w_b)]$, and $0 < w_b/(W+w_b) < 1$. By definition $\mu_\infty(u) = \mathbb{E}[w_\infty/W] = \varepsilon_0(u)$, and $0 < w_\infty/W \le 1$.

For (4.21) apply the formulas to $\chi = w_bw_c/W^2$, w_b^2/W^2 and w_bw_∞/W^2. Entry by entry, for $b \neq c$,

$$\begin{aligned}\mathbb{E}[\pi_b\pi_c] &= \lambda_b\lambda_cw_bw_c\,\mathbb{E}\big[(W+w_b+w_c)^{-2}\big],\\ \mathbb{E}[\pi_b^2] &= \lambda_b^2w_b^2\,\mathbb{E}\big[(W+2w_b)^{-2}\big]+\lambda_bw_b^2\,\mathbb{E}\big[(W+w_b)^{-2}\big],\\ \mathbb{E}[\pi_b\pi_\infty] &= \lambda_bw_bw_\infty\,\mathbb{E}\big[(W+w_b)^{-2}\big],\qquad \mathbb{E}[\pi_\infty^2] = w_\infty^2\,\mathbb{E}\big[W^{-2}\big].\end{aligned}$$

Now insert $x^{-2} = \int_0^\infty t\,\mathrm{e}^{-tx}\,\mathrm{d}t$ for $x>0$ into every term except the second term of $\mathbb{E}[\pi_b^2]$, and exchange expectation and integral (the integrands are nonnegative). For instance

$$\lambda_b\lambda_cw_bw_c\,\mathbb{E}\big[(W+w_b+w_c)^{-2}\big] = \int_0^\infty t\,\mathbb{E}\big[\mathrm{e}^{-tW}\big]\,(v_t)_b(v_t)_c\,\mathrm{d}t.$$

The four families of terms are the entries of $\int_0^\infty t\,\mathbb{E}[\mathrm{e}^{-tW}]\,v_tv_t^{\mathsf{T}}\,\mathrm{d}t$, and the second term of $\mathbb{E}[\pi_b^2]$ is the diagonal matrix in (4.21). The integral converges because $W \ge w_\infty > 0$. □

Lemma 4.15 (Covariance domination)**.** *For every $u \in \mathbb{R}^{B+1}$, in the positive semidefinite order,*

$$\Sigma(u) \preceq \Gamma(u) := \operatorname{diag}\big(\lambda_b\rho_b(1-\rho_b)\ (b\le B),\ \varepsilon_0\big),\qquad \rho_b = \rho_b(u),\ \varepsilon_0 = \varepsilon_0(u).$$

Proof. The integral in (4.21) is positive semidefinite, and $\mathbb{E}[(w_b/(W+w_b))^2] \ge \rho_b^2$ by Jensen's inequality. So $\mathbb{E}[\pi\pi^{\mathsf{T}}] \succeq \operatorname{diag}(\lambda_b\rho_b^2, 0)$. By Lemma 4.14, $\mathbb{E}\operatorname{diag}(\pi) = \operatorname{diag}(\lambda_b\rho_b, \varepsilon_0)$. Hence

$$\Sigma(u) = \operatorname{diag}(\lambda_b\rho_b,\varepsilon_0) - \mathbb{E}\big[\pi\pi^{\mathsf{T}}\big] \preceq \operatorname{diag}\big(\lambda_b\rho_b - \lambda_b\rho_b^2,\ \varepsilon_0\big).$$ □

Lemma 4.16 (Legendre facts)**.** *Let $\mathcal{D} = \nabla\Phi(\mathbb{R}^{B+1}) = \nabla\Phi(\mathcal{H}_0)$.*

(i) *$\nabla\Phi$ is a C^∞ diffeomorphism from $\mathcal{H}_0$ onto $\mathcal{D}$, and $\mathcal{D}$ is a relatively open subset of $\mathcal{H}_1$ contained in $\{\mu : 0 < \mu_b < \lambda_b\ (b\le B),\ \varepsilon > 0\}$. For $\mu = \mu(u) \in \mathcal{D}$,*

$$H(\mu(u)) = \Phi(u) - \langle\mu(u), u\rangle.$$

On $\mathcal{D}$ the function H is C^2; at $\mu = \mu(u)$ its derivative in a direction $x \in \mathcal{H}_0$ is $-\langle u, x\rangle$ and its second derivative in that direction is $-x^{\mathsf{T}}(\Sigma(u)|_{\mathcal{H}_0})^{-1}x$.

(ii) *If $\mu_0 \in \mathcal{D}$, $\mu \in \mathcal{H}_1$ and $H(\mu) > -\infty$, then $\mu_t = \mu_0 + t(\mu-\mu_0) \in \mathcal{D}$ for every $t \in [0,1)$.*

Proof. (i) The two descriptions of $\mathcal{D}$ agree because $\nabla\Phi(u+t\mathbf{1}) = \nabla\Phi(u)$. The restriction of Φ to $\mathcal{H}_0$ is smooth and strictly convex, so for $u \neq u'$ in $\mathcal{H}_0$, $\langle \nabla\Phi(u) - \nabla\Phi(u'), u-u'\rangle > 0$; thus $\nabla\Phi$ is injective on $\mathcal{H}_0$. Its derivative at u, as a map from $\mathcal{H}_0$ to the direction space $\mathcal{H}_0$ of $\mathcal{H}_1$, is $\Sigma(u)|_{\mathcal{H}_0}$, which is invertible. By the inverse function theorem $\nabla\Phi$ is a local diffeomorphism from $\mathcal{H}_0$ to $\mathcal{H}_1$; being injective, it is a diffeomorphism onto its image, which is open in $\mathcal{H}_1$. The inclusion is Lemma 4.14.

Let $\mu = \mu(u)$. The convex function $u' \mapsto \Phi(u') - \langle \mu, u'\rangle$ on $\mathbb{R}^{B+1}$ has gradient $\nabla\Phi(u) - \mu = 0$ at u, so it is minimised there, and $H(\mu) = \Phi(u) - \langle \mu, u\rangle$. Write $u = u(\mu) \in \mathcal{H}_0$ for the inverse map, which is smooth. Differentiating $H(\mu) = \Phi(u(\mu)) - \langle \mu, u(\mu)\rangle$ in a direction $x \in \mathcal{H}_0$ gives $\langle \nabla\Phi(u) - \mu, \partial_x u\rangle - \langle x, u\rangle = -\langle u, x\rangle$. Differentiating once more gives $-\langle \partial_x u, x\rangle = -x^{\mathsf{T}}(\Sigma(u)|_{\mathcal{H}_0})^{-1}x$, because the derivative of $\mu \mapsto u(\mu)$ is the inverse of $\Sigma(u)|_{\mathcal{H}_0}$.

(ii) Let $\varphi_0(u) = \Phi(u) - \langle \mu_0, u\rangle$ for $u \in \mathcal{H}_0$. It is strictly convex and, by (i), has a minimiser u_0, which is therefore unique. Let $a > 0$ be the minimum of $\varphi_0 - \varphi_0(u_0)$ on the unit sphere of $\mathcal{H}_0$ centred at u_0; by convexity, $\varphi_0(u) - \varphi_0(u_0) \geq a\,|u-u_0|$ whenever $|u-u_0| \geq 1$. Fix $t \in [0,1)$. For $u \in \mathcal{H}_0$,

$$\Phi(u) - \langle \mu_t, u\rangle = (1-t)\,\varphi_0(u) + t\big[\Phi(u) - \langle \mu, u\rangle\big] \geq (1-t)\,\varphi_0(u) + t\,H(\mu),$$

which tends to $+\infty$ as $|u| \to \infty$ in $\mathcal{H}_0$. So the continuous function $u \mapsto \Phi(u) - \langle \mu_t, u\rangle$ attains its minimum over $\mathcal{H}_0$ at some u_t. There the vector $\nabla\Phi(u_t) - \mu_t$ is orthogonal to $\mathcal{H}_0$; it also lies in $\mathcal{H}_0$, because both $\nabla\Phi(u_t)$ and μ_t lie in $\mathcal{H}_1$. Hence $\nabla\Phi(u_t) = \mu_t$ and $\mu_t \in \mathcal{D}$. □

Proposition 4.17 (Tangent inequality)**.** *Let $u_0 \in \mathbb{R}^{B+1}$ and write $\rho_b = \rho_b(u_0)$, $\varepsilon_0 = \varepsilon_0(u_0)$. For every $\mu = (\mu_1, \dots, \mu_B, \varepsilon) \in \mathcal{P}$,*

$$H(\mu) - S_0(\mu) \leq \Phi(u_0) + \sum_{b=1}^{B} \lambda_b \ln(1-\rho_b) - \varepsilon_0 - \sum_{b=1}^{B} \mu_b\Big[u_{0,b} + \ln\frac{1-\rho_b}{\rho_b}\Big] - \varepsilon\big[u_{0,\infty} - \ln\varepsilon_0\big]. \tag{4.22}$$

Proof. Both sides of (4.22) are unchanged when u_0 is replaced by $u_0 + t\mathbf{1}$ (the terms $\Phi(u_0)$ and $-\langle \mu, u_0\rangle$ change by t and $-t$), so one may assume $u_0 \in \mathcal{H}_0$. If $H(\mu) = -\infty$ there is nothing to prove. Otherwise let $\mu_0 = \mu(u_0) \in \mathcal{D}$, $x = \mu - \mu_0 \in \mathcal{H}_0$, $\mu_t = \mu_0 + tx$ and $\varphi(t) = H(\mu_t)$ for $t \in [0,1]$. The function φ is concave and finite at both ends, hence finite on $[0,1]$. By Lemma 4.16(ii), $\mu_t \in \mathcal{D}$ for $t \in [0,1)$; let $u_t \in \mathcal{H}_0$ be the point with $\mu(u_t) = \mu_t$. Since $\mathcal{D} \subset \{0 < \mu_b < \lambda_b,\ \varepsilon > 0\}$ and $\mathcal{P}$ is convex, $\mu_t \in \mathcal{P}$ for all $t \in [0,1]$, and both φ and $t \mapsto S_0(\mu_t)$ are C^2 on $[0,1)$.

Second derivatives. By Lemma 4.16(i), $\varphi''(t) = -x^{\mathsf{T}}(\Sigma(u_t)|_{\mathcal{H}_0})^{-1}x$. The function S_0 is a sum of functions of the single coordinates, with

$$\frac{\partial S_0}{\partial \mu_b} = \ln\frac{1-\mu_b/\lambda_b}{\mu_b/\lambda_b}, \qquad \frac{\partial^2 S_0}{\partial \mu_b^2} = -\frac{1}{\lambda_b\,(\mu_b/\lambda_b)(1-\mu_b/\lambda_b)},$$
$$\frac{\partial S_0}{\partial \varepsilon} = -\ln\varepsilon, \qquad \frac{\partial^2 S_0}{\partial \varepsilon^2} = -\frac{1}{\varepsilon}.$$

At $\mu_t = \mu(u_t)$ one has $\mu_{t,b}/\lambda_b = \rho_b(u_t)$ and $\varepsilon_t = \varepsilon_0(u_t)$ by Lemma 4.14, so the Hessian of S_0 at μ_t is $-\Gamma(u_t)^{-1}$ and $\frac{\mathrm{d}^2}{\mathrm{d}t^2}S_0(\mu_t) = -x^{\mathsf{T}}\Gamma(u_t)^{-1}x$.

The matrix inequality. Write $\Sigma = \Sigma(u_t)$, $\Gamma = \Gamma(u_t)$. Since $x \in \mathcal{H}_0$ and $\Sigma\mathbf{1} = 0$,

$$x^{\mathsf{T}}(\Sigma|_{\mathcal{H}_0})^{-1}x = \sup_{y\in\mathbb{R}^{B+1}}\big[2\langle x,y\rangle - y^{\mathsf{T}}\Sigma y\big] \geq \sup_{y\in\mathbb{R}^{B+1}}\big[2\langle x,y\rangle - y^{\mathsf{T}}\Gamma y\big] = x^{\mathsf{T}}\Gamma^{-1}x.$$

The first equality holds because the bracket depends on y only through its projection to $\mathcal{H}_0$, and on $\mathcal{H}_0$ the supremum of $2\langle x,y\rangle - y^{\mathsf{T}}\Sigma y$ is attained at $y = (\Sigma|_{\mathcal{H}_0})^{-1}x$. The inequality is Lemma 4.15. The last equality holds because Γ is diagonal with positive entries.

The tangent at $t = 0$. Consequently $t \mapsto \varphi(t) - S_0(\mu_t)$ has nonpositive second derivative on $[0,1)$, so it is concave there and lies below its tangent at $t = 0$. By Lemma 4.16(i), $\varphi'(0) = -\langle u_0, x\rangle$. Hence for $t \in [0,1)$

$$\varphi(t) - S_0(\mu_t) \leq H(\mu_0) - S_0(\mu_0) - t\,\big\langle u_0 + \nabla S_0(\mu_0),\, x\big\rangle, \tag{4.23}$$

where $\nabla S_0(\mu_0)$ has coordinates $\ln((1-\rho_b)/\rho_b)$ for $b \leq B$ and $-\ln\varepsilon_0$.

The boundary. Let $t \uparrow 1$ in (4.23). The function S_0 is continuous on $\mathcal{P}$, so $S_0(\mu_t) \to S_0(\mu)$. Since φ is concave and finite on $[0,1]$, $\varphi(t) \geq (1-t)\varphi(0) + t\varphi(1)$, so $\liminf_{t\uparrow 1}\varphi(t) \geq \varphi(1) = H(\mu)$. Therefore

$$H(\mu) - S_0(\mu) \leq H(\mu_0) - S_0(\mu_0) - \big\langle u_0 + \nabla S_0(\mu_0),\, \mu - \mu_0\big\rangle.$$

Evaluation. By Lemma 4.16(i), $H(\mu_0) + \langle u_0, \mu_0\rangle = \Phi(u_0)$. Using $h(r) - r\,h'(r) = -\ln(1-r)$ and $\mu_{0,b} = \lambda_b\rho_b$,

$$S_0(\mu_0) - \langle \nabla S_0(\mu_0), \mu_0\rangle = \sum_b \lambda_b\big[h(\rho_b) - \rho_b h'(\rho_b)\big] + \big[-\varepsilon_0\ln\varepsilon_0 + \varepsilon_0 + \varepsilon_0\ln\varepsilon_0\big]$$
$$= -\sum_b \lambda_b \ln(1-\rho_b) + \varepsilon_0.$$

Substituting these two identities gives (4.22). □

Proposition 4.18 (Tangent bound)**.** *For every $K > 0$, $B \geq 1$, $\theta > 0$, $\beta \in \mathbb{R}$ and $u_0 \in \mathbb{R}^{B+1}$,*

$$L^* = \sup_{\mu\in\mathcal{P}} L(\mu) \leq \theta\beta + \Psi_0(K, B, \theta, u_0).$$

Proof. Write $L(\mu) - \theta\beta = 3\,(H(\mu) - S_0(\mu)) + S_0(\mu) + \langle \ell, \mu\rangle$ and insert (4.22):

$$L(\mu) - \theta\beta \leq 3\Big[\Phi(u_0) + \sum_b \lambda_b \ln(1-\rho_b) - \varepsilon_0\Big] + S_0(\mu) + \langle a, \mu\rangle, \tag{4.24}$$

with

$$a_b = \ell_b - 3u_{0,b} - 3\ln\frac{1-\rho_b}{\rho_b} \quad (b \le B), \qquad a_\infty = \ell_\infty - 3u_{0,\infty} + 3\ln\varepsilon_0.$$

Now drop the constraint $\sum_b \mu_b + \varepsilon = 1$ and maximise coordinate by coordinate. For real a, $\sup_{0\le r\le 1}[h(r)+ar] = \ln(1+\mathrm{e}^a)$ and $\sup_{\varepsilon\ge 0}[-\varepsilon\ln\varepsilon+\varepsilon+a\varepsilon] = \mathrm{e}^a$. With $r = \mu_b/\lambda_b$ this gives

$$\begin{aligned} S_0(\mu) + \langle a, \mu\rangle &= \sum_b \lambda_b\Big[h\Big(\frac{\mu_b}{\lambda_b}\Big) + a_b\frac{\mu_b}{\lambda_b}\Big] + \big[-\varepsilon\ln\varepsilon + \varepsilon + a_\infty\varepsilon\big] \\ &\le \sum_b \lambda_b \ln\big(1+\mathrm{e}^{a_b}\big) + \mathrm{e}^{a_\infty}. \end{aligned}$$

Finally $\mathrm{e}^{a_b} = C_b(u_0)\,\rho_b^3/(1-\rho_b)^3$, so $(1-\rho_b)^3(1+\mathrm{e}^{a_b}) = (1-\rho_b)^3 + C_b\rho_b^3$, and $\mathrm{e}^{a_\infty} = C_\infty(u_0)\,\varepsilon_0^3$. The right side of (4.24) is therefore at most $3\Phi(u_0) + \sum_b \lambda_b \ln((1-\rho_b)^3 + C_b\rho_b^3) - 3\varepsilon_0 + C_\infty\varepsilon_0^3 = \Psi_0(K, B, \theta, u_0)$. □

Remark 4.19. The tilt uses the exact conditional Laplace transform κ_b/λ_b of the cost within a bin, see (4.5), so the binning causes no error in the accounting of costs; the only loss is that the conditioning is on coarser stars. Every $B \ge 1$ gives a valid bound. Likewise Proposition 4.18 holds for every weight vector u_0, so the certificate does not need an exact stationary point: the vector u^* of Proposition 4.4 was found by a floating-point search, and how it was found is irrelevant to the proof.

4.5. **Proof of the criterion.**

Proof of Proposition 4.3. Let K, B, θ, u, β satisfy (4.3). By Proposition 4.18 with $u_0 = u$, $L^* \le \theta\beta + \Psi_0(K, B, \theta, u) < 0$. By Lemma 4.13 there are $\delta > 0$ and $c > 0$, depending only on (K, B, θ, u, β), such that $\mathbb{E}[Z_\beta; D(G) \in \mathcal{T}_\delta] \le \mathrm{e}^{-cq}$ for all large q. By Markov's inequality,

$$\mathbb{P}(Z_\beta \ge 1) \le \mathbb{P}\big(D(G) \notin \mathcal{T}_\delta\big) + \mathbb{E}\big[Z_\beta;\, D(G) \in \mathcal{T}_\delta\big] \le \mathbb{P}\big(D(G) \notin \mathcal{T}_\delta\big) + \mathrm{e}^{-cq},$$

and the first term tends to 0 by Lemma 4.12. □

Remark 4.20 (The certificate)**.** For the parameters of Proposition 4.4 the script reports the following outward-rounded enclosures, which are intermediate quantities and are not used elsewhere:

$$\begin{aligned} \Phi(u^*) &\in [-0.522226313363,\ -0.522223198517], \\ \varepsilon_0(u^*) &\in [0.02081077388,\ 0.02081090331], \\ \rho_1(u^*) &\in [0.7564957391,\ 0.7565004438], \\ \textstyle\sum_b \lambda_b\rho_b(u^*) + \varepsilon_0(u^*) &\in [0.99999792704,\ 1.00000414596]. \end{aligned}$$

The last interval contains 1, as it must by Lemma 4.14. The certified upper bounds for $\theta\beta + \Psi_0$ are -0.0037532832 at $\beta = 2.30$, -0.0004108832 at $\beta = 2.302$, -0.0000766432 at $\beta = 2.3022$ and -0.0000097952 at $\beta = 2.30224$. The enclosure of Ψ_0 is rigorous given the correctness of the script, of exact integer and rational arithmetic, and of the four interval operations and the

comparisons of the interval library (mpmath 1.3.0). The library's interval exponential and logarithm are not used, because they are not strict enclosures; see Section 11.

Remark 4.21 (Checks that carry no weight in the proof)**.** Lemma 4.7 was compared with an exact enumeration of all 20,736 configurations of an instance with three agents per sector and two bins, for all ten profiles (`lemmaC/round3/firstmoment/check_partial.py`). Lemma 4.15 and inequality (4.22) were tested in floating point on random instances with one to three bins (`lemmaC/round3/firstmoment/check_concave.py`): the largest eigenvalue of $\Gamma^{-1/2}\Sigma\,\Gamma^{-1/2}$ found over 60 instances was 0.999999, and the largest value of the left side of (4.22) minus its right side over 360 pairs (u_0, μ) was -1.3×10^{-11}.

5. From independent teams to the pair-additive market

Theorem 4.1 is a statement about the independent-team model. This section carries it to the market and proves the integer lower bound that enters Theorem 1.1.

Theorem 5.1. *In the market, for every $\beta \le \beta_0 = 2.302$,*

$$\mathbb{P}\big(\Phi_q \le \beta\, q^{1/3}\big) \longrightarrow 0 \qquad (q \to \infty).$$

In particular $\Phi_q \ge \beta_0\, q^{1/3}$ with high probability. No rate is obtained. The method reaches $\beta = 2.30224$; see the note after Table 2.

The proof has three parts. The first is a comparison inequality (Theorem 5.2): the expected number of cheap perfect assignments in the market, restricted to a good event of the near-tight hypergraph, is at most $e^{o(q)}$ times the same expectation in the independent-team model on a slightly larger event, plus a negligible term. The second is that the good event is typical in the market (Proposition 5.8). The third is that the first-moment bound behind Theorem 4.1 survives the enlargement of the event, with explicit constants (Proposition 5.14). The parts are assembled in Section 5.10.

Two features of the comparison should be kept in mind. A cheap assignment uses some teams of rescaled cost above the truncation level K, and their costs are not a function of the near-tight hypergraph; so it is not enough to compare the laws of the near-tight hypergraph in the two models. The comparison is made for the conditional law of the near-tight teams outside a candidate assignment N, given the values of all pairs of N (Lemma 5.6). In that form the comparison factor contains a term that is exponential in the number of near-tight teams sharing a pair with N. This number is not controlled by any event of the near-tight hypergraph alone; the term is removed in expectation in the independent-team model (Lemma 5.7), and this is what forces the enlargement of the good event.

5.1. **Setting and notation.** Throughout the section $K \geq 1$ is a truncation level, $B \geq 1$ is the number of bins, $\eta = K/B$ is the bin width, and

$$I_b = \big((b-1)\eta,\, b\eta\big], \qquad \lambda_b = \Lambda(I_b) = \frac{\eta^3(3b^2-3b+1)}{6} \qquad (1 \leq b \leq B),$$

as in Section 4; $\lambda_{\max} = \max_b \lambda_b = \lambda_B$ and $\sum_b \lambda_b = K^3/6$. We always assume $q^{2/3} \geq K$ and put

$$t = Kq^{-2/3} \leq 1, \qquad p_b = \frac{\lambda_b}{q^2}, \qquad p = \sum_b p_b = \frac{K^3}{6q^2} \leq \frac{1}{6}.$$

By (2.3), in the market and in the independent-team model alike, $p_b = \mathbb{P}(\xi_e \in I_b)$ and $p = \mathbb{P}(\xi_e \leq K)$ for every team e, and $p_b \geq p_1 = \eta^3/(6q^2)$. A team with $\xi_e \leq K$ is called *near-tight*. As in Section 2.5, $\mathbb{P}$, $\mathbb{E}$ refer to the market and $\mathbb{P}'$, $\mathbb{E}'$ to the independent-team model; random objects of the independent-team model carry a prime.

Configurations. A *labelled team* is a pair (e, b) of a team and a bin. A *configuration* is a finite set h of labelled teams in which every team occurs at most once; $|h|$ is its number of teams, and we write $e \in h$ if e occurs in h and b_e for its label. The *binned near-tight hypergraph* is the random configuration

$$H = \big\{(e,b) :\ \xi_e \in I_b\big\},$$

and H' is the same object in the independent-team model. For a configuration h let $\mathrm{pairs}(h)$ be the set of pairs contained in some team of h, and for a pair π let $\mathrm{mult}_h(\pi)$ be the number of teams of h containing π. The *pair-sharing excess* of h is

$$s(h) = 3|h| - |\,\mathrm{pairs}(h)| = \sum_{\pi \in \mathrm{pairs}(h)} \big(\mathrm{mult}_h(\pi) - 1\big).$$

It is nondecreasing in h: a further team adds 3 to $3|h|$ and at most 3 to $|\,\mathrm{pairs}(h)|$. For a set $\mathcal{G}$ of configurations and $\tau > 0$, $\mathcal{G}^{(\tau)}$ denotes the set of configurations h' for which there is $h \in \mathcal{G}$ with $|h \triangle h'| \leq \tau q$, the symmetric difference being that of sets of labelled teams.

Cheap assignments. As in Section 4, for a perfect assignment N put $\xi(N) = \sum_{e \in N} \xi_e$, so that the deficit of N is $q^{-2/3}\xi(N)$, and for $\beta > 0$

$$Z_\beta = \#\big\{N \text{ a perfect assignment} :\ \xi(N) < \beta q\big\}.$$

By (2.4), $\{\Phi_q < \beta q^{1/3}\} = \{Z_\beta \geq 1\}$. The variable Φ_q is the minimum of finitely many variables with continuous laws, so $\mathbb{P}(\Phi_q = \beta q^{1/3}) = 0$ and

$$\mathbb{P}\big(\Phi_q \leq \beta q^{1/3}\big) = \mathbb{P}(Z_\beta \geq 1). \tag{5.1}$$

Geometry of a perfect assignment. Let N be a perfect assignment.

(G1) Two distinct teams share at most one pair: if they agree in two sectors they differ in the third, and then only the pair inside the two common sectors is shared.

(G2) A team $f \notin N$ contains at most one pair of N. If $f = (a, y, z)$ and both $\{a, y\}$ and $\{y, z\}$ are pairs of N, then the team of N at y contains a and z, so it is f; the other two cases are the same.

(G3) Let ∂N be the set of teams outside N that contain a pair of N. Each of the $3q$ pairs of N lies in q teams, one of which belongs to N, and by (G2) no team outside N is counted for two pairs. Hence $|\partial N| = 3q(q-1)$.

The q teams of N are disjoint, so they use $3q$ distinct pairs. We write d_N for the vector of the $3q$ deficits of these pairs and $\xi_N = (\xi_e)_{e \in N}$, a function of d_N. In the market the coordinates of ξ_N are independent with the law of $q^{2/3}$ times a sum of three independent uniform variables; in the independent-team model ξ'_N has the same law. Let

$$H^- = \{(e, b) \in H : \ e \notin N\}$$

be the binned set of near-tight teams outside N; H is a function of (ξ_N, H^-).

For a configuration h^- none of whose teams lies in N, the excess of h^- *relative to* N counts the pairs of N as already used:

$$s(h^-; N) = 3|h^-| - \big|\mathrm{pairs}(h^-) \setminus \mathrm{pairs}(N)\big|,$$

where $\mathrm{pairs}(N)$ is the set of the $3q$ pairs of N. By (G2) every pair of $\mathrm{pairs}(h^-) \cap \mathrm{pairs}(N)$ lies in a team of $h^- \cap \partial N$, and every such team contains exactly one pair of N. Therefore

$$s(h^-; N) = s(h^-) + \big|\mathrm{pairs}(h^-) \cap \mathrm{pairs}(N)\big| \le s(h^-) + |h^- \cap \partial N|. \tag{5.2}$$

We also allow $N = \emptyset$, with $\mathrm{pairs}(\emptyset) = \emptyset$, no conditioning, and $s(h; \emptyset) = s(h)$.

Constants of the comparison.

$$C_1 = \frac{36K^5}{\eta^6}, \qquad W_q = C_1\, q^{2/3}, \qquad \Gamma_q = \frac{3q^3(q-1)\,t^5}{40} \le \frac{3}{40} K^5 q^{2/3}. \tag{5.3}$$

Since $\eta \le K \le q^{2/3}$, $W_q \ge 36$.

5.2. **The comparison theorem.**

Theorem 5.2 (Comparison)**.** *Let $K \ge 1$, $B \ge 1$, $\beta > 0$, $\tau > 0$ and $\zeta \in (0, \frac{1}{3})$. Put $s_0 = q^{2/3+\zeta}$, and let $\mathcal{G} = \mathcal{G}_q$ be arbitrary sets of configurations. For every*

$$q \ \ge \ q_0(K, B, \tau) = \max\Big\{K^{3/2}, \ \frac{1}{\tau}, \ \Big(\frac{eK^3 C_1}{\tau}\Big)^{12}\Big\}, \tag{5.4}$$

$$\begin{aligned} &\mathbb{E}\big[Z_\beta; \ H \in \mathcal{G}, \ s(H) \le s_0, \ |H| \le K^3 q\big] \\ &\qquad \le e^{R_q}\Big(\mathbb{E}'\big[Z_\beta; \ H' \in \mathcal{G}^{(2\tau)}\big] + 3\,\mathbb{E}'[Z_\beta]\, e^{-(\tau/4)\, q \ln q}\Big), \end{aligned} \tag{5.5}$$

where

$$R_q = \Gamma_q + 2K^6 + s_0 \ln W_q + \tfrac{1}{2} K^3 W_q = O\big(q^{2/3+\zeta} \ln q\big) = o(q). \tag{5.6}$$

Consequently, for all sufficiently large q (depending on K, B, β, τ, ζ),

$$\mathbb{P}(Z_\beta \geq 1) \leq \mathbb{P}(H \notin \mathcal{G}) + \mathbb{P}\big(s(H) > s_0\big) + \mathbb{P}\big(|H| > K^3 q\big) + e^{R_q}\,\mathbb{E}'\big[Z_\beta;\ H' \in \mathcal{G}^{(2\tau)}\big] + e^{-(\tau/5)\,q\ln q}. \tag{5.7}$$

The four terms of R_q have separate origins: Γ_q is the correction in the bound for the probability that a family of teams has no near-tight member (Lemma 5.5); $2K^6$ accounts for the teams left out of that family; $s_0 \ln W_q$ is the price of the pair-sharings inside H (Lemma 5.4); and $\frac{1}{2}K^3 W_q$ is the price of the near-tight teams that share a pair with the candidate assignment (Lemma 5.7). The number of profiles, the binning and the truncation at K do not appear: they stay inside the first moment of the independent-team model. The proof of Theorem 5.2 is in Section 5.7, after four lemmas.

The plain first moment is the same in both models and is used only as a crude bound.

Lemma 5.3 (Plain first moment)**.** *For every $\beta > 0$ and $q \geq 1$,*

$$\mathbb{E}[Z_\beta] = \mathbb{E}'[Z_\beta] \leq \frac{(q!)^2\,(\beta q^{1/3})^{3q}}{(3q)!} \leq e^2(q+1)^2\Big(\frac{\beta^3 e}{27}\Big)^q.$$

For $\beta \leq 2.31$ the last bracket is at most $e^{0.22}$.

Proof. Fix a perfect assignment N. In the market $q^{-2/3}\xi(N)$ is the sum of the $3q$ independent uniform deficits of the pairs of N; in the independent-team model it is the sum of q independent team deficits, each a sum of three independent uniform variables. The two laws coincide. The set $\{x \in [0,1]^{3q} : \sum x_i \leq y\}$ is contained in the simplex $\{x \geq 0 : \sum x_i \leq y\}$, of volume $y^{3q}/(3q)!$; take $y = \beta q^{1/3}$. There are $(q!)^2$ perfect assignments. The second inequality follows from $q! \leq e(q+1)(q/e)^q$ and $(3q)! \geq (3q/e)^{3q}$. Finally $2.31^3 e/27 < 1.241 < e^{0.22}$. □

5.3. **The pair-sharing excess.**

Lemma 5.4 (Factor per unit of excess)**.** *Let N be a perfect assignment or $N = \emptyset$, and let h^- be a configuration none of whose teams lies in N. Then, almost surely,*

$$\mathbb{P}\big(\xi_f \in I_{b_f} \text{ for all } f \in h^- \;\big|\; d_N\big) \leq W_q^{\,s(h^-;N)} \prod_{f \in h^-} p_{b_f}.$$

Proof. Call $f \in h^-$ *isolated* if none of its three pairs is a pair of N or a pair of another team of h^-. Let I be the set of isolated teams and R the set of the others. The $3|I|$ pairs of the teams of I are distinct and belong neither to pairs(R) nor to pairs(N). Hence the events $\{\xi_f \in I_{b_f}\}$, $f \in I$, are independent of each other, of d_N and of the deficits of the pairs of R, and each has probability p_{b_f}. Let $Q_R = \text{pairs}(R) \setminus \text{pairs}(N)$. If $\xi_f \leq K$ for every

$f \in R$ then $d(\pi) \le t$ for every $\pi \in Q_R$; this event has probability $t^{|Q_R|}$ and is independent of d_N. Therefore the left side is at most

$$\prod_{f \in I} p_{b_f} \cdot t^{|Q_R|}.$$

Write $s = s(h^-; N)$. Isolated teams contribute nothing to s, so $s = 3|R| - |Q_R|$. Counting the $3|R|$ incidences between teams of R and their pairs,

$$s = \#\big\{(f,\pi) :\ f \in R,\ \pi \in \mathrm{pairs}(N),\ \pi \subset f\big\} + \sum_{\pi \in Q_R} \big(\mathrm{mult}_R(\pi) - 1\big).$$

Every team of R contains a pair of N or a pair of Q_R of multiplicity at least 2 (a team sharing a pair with a non-isolated team is itself not isolated). The number of teams of the first kind is at most the first term. The number of the second kind is at most $\sum_{\pi:\mathrm{mult}_R(\pi)\ge 2} \mathrm{mult}_R(\pi) \le 2\sum_{\pi \in Q_R}(\mathrm{mult}_R(\pi) - 1)$. Hence $|R| \le 2s$. Using $p_b \ge \eta^3/(6q^2)$ and $|Q_R| = 3|R| - s$,

$$\begin{aligned} \frac{t^{|Q_R|}}{\prod_{f\in R} p_{b_f}} &\le \big(Kq^{-2/3}\big)^{3|R|-s}\Big(\frac{6q^2}{\eta^3}\Big)^{|R|} = \Big(\frac{6K^3}{\eta^3}\Big)^{|R|} K^{-s} q^{2s/3} \\ &\le \Big(\frac{36K^5}{\eta^6}\, q^{2/3}\Big)^{s} = W_q^{\,s}, \end{aligned}$$

because $6K^3/\eta^3 \ge 1$ and $|R| \le 2s$. □

The bound is uniform in d_N; this is what allows conditioning on the exact values of the pairs of N. The power of q cannot be improved: for two teams sharing a pair, $\mathbb{P}(\text{both near-tight})/p^2 = (t^5/20)/(t^3/6)^2 = \frac{9}{5} q^{2/3}/K$.

5.4. **The void bound.** The next lemma bounds the probability that a family of teams contains no near-tight team. It is an inequality of Janson type, and the proof is the argument of Boppana and Spencer [6]: a union bound, exact independence of events on disjoint sets of pairs, and one application of Harris's inequality [12].

Lemma 5.5 (Void bound)**.** *In the market, for every family $\mathcal{F}$ of teams,*

$$\mathbb{P}\big(\xi_e > K \text{ for all } e \in \mathcal{F}\big) \le \exp\big(-|\mathcal{F}|\, p + \Gamma_q\big).$$

Proof. For a team e let $B_e = \{\Delta(e) \le t\}$, so $\mathbb{P}(B_e) = p$. As a subset of the cube $[0,1]^{3q^2}$ of pair deficits, B_e is decreasing (a down-set), and it depends only on the three pairs of e. Write $e \sim f$ if $e \ne f$ and e, f share a pair. Enumerate $\mathcal{F}$ as $e_1, \dots, e_n$ and write $B_i = B_{e_i}$. Fix i and put

$$\mathcal{D} = \bigcap_{j<i,\ e_j \sim e_i} B_j^c, \qquad \mathcal{I} = \bigcap_{j<i,\ e_j \not\sim e_i} B_j^c.$$

Since $B_i \cap \mathcal{I} \cap \mathcal{D}^c \subset \bigcup_{j<i,\, e_j \sim e_i} (B_i \cap B_j \cap \mathcal{I})$,

$$\begin{aligned}\mathbb{P}(B_i \cap \mathcal{D} \cap \mathcal{I}) &\geq \mathbb{P}(B_i \cap \mathcal{I}) - \sum_{j<i,\ e_j \sim e_i} \mathbb{P}(B_i \cap B_j \cap \mathcal{I}) \\ &\geq \mathbb{P}(B_i)\,\mathbb{P}(\mathcal{I}) - \sum_{j<i,\ e_j \sim e_i} \mathbb{P}(B_i \cap B_j)\,\mathbb{P}(\mathcal{I}).\end{aligned}$$

For the first term: the teams e_j entering $\mathcal{I}$ contain none of the three pairs of e_i, so $\mathcal{I}$ is independent of B_i. For the second: $B_i \cap B_j$ is a decreasing event and $\mathcal{I}$, an intersection of complements of decreasing events, is increasing, so $\mathbb{P}(B_i \cap B_j \cap \mathcal{I}) \leq \mathbb{P}(B_i \cap B_j)\mathbb{P}(\mathcal{I})$ by Harris's inequality for the product measure of the pair deficits. If $\mathbb{P}(\mathcal{D} \cap \mathcal{I}) > 0$, then, since $\mathbb{P}(\mathcal{D} \cap \mathcal{I}) \leq \mathbb{P}(\mathcal{I})$,

$$\mathbb{P}(B_i \mid \mathcal{D} \cap \mathcal{I}) \;\geq\; \frac{\mathbb{P}(B_i \cap \mathcal{D} \cap \mathcal{I})}{\mathbb{P}(\mathcal{I})} \;\geq\; p - \sum_{j<i,\ e_j \sim e_i} \mathbb{P}(B_i \cap B_j),$$

and therefore

$$\begin{aligned}\mathbb{P}\Big(B_i^c \,\Big|\, \bigcap_{j<i} B_j^c\Big) &\leq 1 - p + \sum_{j<i,\ e_j \sim e_i} \mathbb{P}(B_i \cap B_j) \\ &\leq \exp\Big(-p + \sum_{j<i,\ e_j \sim e_i} \mathbb{P}(B_i \cap B_j)\Big).\end{aligned}$$

Multiplying over i (if some conditioning event has probability 0, the probability in the lemma is 0) gives

$$\mathbb{P}\Big(\bigcap_{i \leq n} B_i^c\Big) \leq \exp\big(-np + \Gamma_{\mathcal{F}}\big), \qquad \Gamma_{\mathcal{F}} = \sum_{\{e,f\} \subset \mathcal{F},\ e \sim f} \mathbb{P}(B_e \cap B_f).$$

It remains to bound $\Gamma_{\mathcal{F}}$ by its value for the family of all teams. A team shares a pair with $3(q-1)$ others, by (G1), so there are $3q^3(q-1)/2$ unordered pairs $\{e, f\}$ with $e \sim f$. For such a pair let x be the deficit of the common pair; given $x \leq t$, each team is near-tight if its other two pair deficits have sum at most $t - x$, which has probability $(t-x)^2/2$, independently for the two teams. Hence

$$\mathbb{P}(B_e \cap B_f) = \int_0^t \Big(\frac{(t-x)^2}{2}\Big)^2 \,\mathrm{d}x = \frac{t^5}{20} \qquad (e \sim f), \tag{5.8}$$

and $\Gamma_{\mathcal{F}} \leq \frac{3}{2} q^3 (q-1) \cdot t^5/20 = \Gamma_q$. □

The proof never conditions on an event B_i. So no correlation inequality is needed for the conditional law of the pair deficits given $\Delta(e) \leq t$, which is not positively associated.

5.5. **The planted comparison.**

Lemma 5.6 (Planted comparison)**.** *Let N be a perfect assignment and h^- a configuration none of whose teams lies in N. Then, almost surely,*

$$\mathbb{P}\big(H^- = h^- \,\big|\, d_N\big) \leq \rho(h^-, N)\,\mathbb{P}'\big(H'^- = h^-\big),$$

where

$$\rho(h^-,N)=\exp\Big(\Gamma_q+\frac{K^3(|h^-|+q)}{2q}+\frac{K^6}{36q}\Big)\,W_q^{s(h^-;N)}.$$

The factor ρ does not depend on the values of the pairs of N.

Proof. Let $\mathcal{F}$ be the set of teams none of whose pairs lies in $\mathrm{pairs}(h^-)\cup\mathrm{pairs}(N)$. It contains no team of N and no team of h^-, so $\{H^-=h^-\}\subset E_1\cap E_2$ with

$$E_1=\{\xi_f\in I_{b_f}\text{ for all }f\in h^-\},\qquad E_2=\{\xi_f>K\text{ for all }f\in\mathcal{F}\}.$$

E_1 and d_N are functions of the deficits of the pairs in $\mathrm{pairs}(h^-)\cup\mathrm{pairs}(N)$, and E_2 is a function of the other pair deficits. Hence $\mathbb{P}(E_1\cap E_2\mid d_N)=\mathbb{P}(E_1\mid d_N)\,\mathbb{P}(E_2)$. Lemma 5.4 bounds the first factor and Lemma 5.5 the second. At most $3(|h^-|+q)$ pairs are excluded and each pair lies in q teams, so $|\mathcal{F}|\ge q^3-3q(|h^-|+q)$ and

$$|\mathcal{F}|\,p\;\ge\;q^3p-\frac{K^3(|h^-|+q)}{2q}.$$

Thus

$$\mathbb{P}\big(H^-=h^-\;\big|\;d_N\big)\le W_q^{s(h^-;N)}\prod_{f\in h^-}p_{b_f}\cdot\exp\Big(-q^3p+\frac{K^3(|h^-|+q)}{2q}+\Gamma_q\Big).$$

In the independent-team model the q^3-q teams outside N are independent, so

$$\mathbb{P}'\big(H'^-=h^-\big)=\prod_{f\in h^-}p_{b_f}\cdot(1-p)^{q^3-q-|h^-|}\ge\prod_{f\in h^-}p_{b_f}\cdot\exp\big(-q^3p-q^3p^2\big),$$

because $\ln(1-p)\ge-p-p^2$ for $0\le p\le\frac12$; and $q^3p^2=K^6/(36q)$. Dividing gives the claim. □

In the independent-team model ξ'_N and H'^- are independent, and ξ'_N has the law of ξ_N. So for every measurable $G\ge0$,

$$\begin{aligned}\mathbb{E}\big[G(\xi_N,H^-)\big]&=\mathbb{E}\Big[\sum_{h^-}G(\xi_N,h^-)\,\mathbb{P}\big(H^-=h^-\;\big|\;d_N\big)\Big]\\&\le\mathbb{E}'\big[G(\xi'_N,H'^-)\,\rho(H'^-,N)\big].\end{aligned}\tag{5.9}$$

The costs of all teams of N, below and above K, are carried along exactly and have the same law on both sides. No separate union bound over the teams above K is needed in the comparison.

5.6. **Teams sharing a pair with the candidate assignment.** Fix a perfect assignment N and let

$$A_N=\big\{\xi(N)<\beta q\big\},\qquad X_N=\big|H'^-\cap\partial N\big|,$$

the number of near-tight teams of ∂N in the independent-team model. Apply (5.9) to

$$G=\mathbf{1}_{A_N}\cdot\mathbf{1}\{H\in\mathcal{G}\}\cdot\mathbf{1}\{s(H)\le s_0\}\cdot\mathbf{1}\{|H|\le K^3q\},$$

a function of (ξ_N, H^-). On the support of G we have $|h^-| \le |H| \le K^3 q$ and, by (5.2) and the monotonicity of s, $s(h^-;N) \le s(H) + |h^- \cap \partial N| \le s_0 + |h^- \cap \partial N|$. Since $W_q \ge 1$ and $\frac{1}{2}K^3(K^3+1) + K^6/(36q) \le 2K^6$,

$$\mathbb{E}\big[\mathbf{1}_{A_N};\ H \in \mathcal{G},\ s(H) \le s_0,\ |H| \le K^3 q\big] \le e^{\Gamma_q + 2K^6 + s_0 \ln W_q}\ \mathbb{E}'\big[\mathbf{1}_{A_N}\,\mathbf{1}\{H' \in \mathcal{G}\}\,W_q^{X_N}\big]. \tag{5.10}$$

The weight $W_q^{X_N}$ cannot be dropped by restricting to an event of H. In the market, given that a team of N is near-tight, each of the $q-1$ other teams through one of its pairs is near-tight with probability of order $q^{-4/3}$, not q^{-2}; for a cheap assignment the number of near-tight teams in ∂N is of order $q^{2/3}$ and not of order 1. The weight is removed on the side of the independent-team model, where ∂N is a fixed family of independent teams.

Lemma 5.7 (Exponential moment)**.** *Let N be a perfect assignment, $\mathcal{G}$ a set of configurations and $\tau > 0$. For $q \ge q_0(K, B, \tau)$ as in* (5.4)*,*

$$\mathbb{E}'\big[\mathbf{1}_{A_N}\,\mathbf{1}\{H' \in \mathcal{G}\}\,W_q^{X_N}\big] \le e^{K^3 W_q/2}\Big(\mathbb{E}'\big[\mathbf{1}_{A_N}\,\mathbf{1}\{H' \in \mathcal{G}^{(2\tau)}\}\big] + 3\,\mathbb{P}'(A_N)\,e^{-(\tau/4)\,q\ln q}\Big).$$

Proof. In the independent-team model the three objects ξ'_N, $H'_\partial = H'^- \cap \partial N$ and $H'_{\mathrm{free}} = \{(e,b) \in H' : e \notin N \cup \partial N\}$ are independent, because N, ∂N and the remaining teams are fixed disjoint families. The variable $X_N = |H'_\partial|$ is binomial with parameters $3q(q-1)$ and p, by (G3). The configuration $H'_0 = H' \setminus H'_\partial$ is a function of $(\xi'_N, H'_{\mathrm{free}})$, hence independent of X_N, and $|H' \triangle H'_0| = X_N$. So if $H' \in \mathcal{G}$ and $X_N \le \tau q$ then $H'_0 \in \mathcal{G}^{(\tau)}$, and if $H'_0 \in \mathcal{G}^{(\tau)}$ and $X_N \le \tau q$ then $H' \in \mathcal{G}^{(2\tau)}$. Splitting according to $X_N \le \tau q$ or $X_N > \tau q$ and using the independence twice,

$$\begin{aligned}\mathbb{E}'\big[\mathbf{1}_{A_N}\mathbf{1}\{H' \in \mathcal{G}\}W_q^{X_N}\big] &\le \mathbb{E}'\big[\mathbf{1}_{A_N}\mathbf{1}\{H'_0 \in \mathcal{G}^{(\tau)}\}\big]\,\mathbb{E}'\big[W_q^{X_N}\big] \\ &\quad + \mathbb{P}'(A_N)\,\mathbb{E}'\big[W_q^{X_N};\,X_N > \tau q\big], \\ \mathbb{E}'\big[\mathbf{1}_{A_N}\mathbf{1}\{H'_0 \in \mathcal{G}^{(\tau)}\}\big] &\le \mathbb{E}'\big[\mathbf{1}_{A_N}\mathbf{1}\{H' \in \mathcal{G}^{(2\tau)}\}\big] + \mathbb{P}'(A_N)\,\mathbb{P}'(X_N > \tau q).\end{aligned}$$

Now $\mathbb{E}'[W_q^{X_N}] = (1 + (W_q - 1)p)^{3q(q-1)} \le \exp(3q^2 p W_q) = \exp(K^3 W_q/2)$. For the tail put $a = K^3 W_q/2 \ge 3q(q-1)pW_q$. Then

$$\mathbb{E}'\big[W_q^{X_N};\,X_N > \tau q\big] \le \sum_{x > \tau q} \binom{3q(q-1)}{x} (pW_q)^x \le \sum_{x > \tau q} \frac{a^x}{x!}.$$

Let $r_q = ea/(\tau q) = eK^3 C_1/(2\tau q^{1/3})$. By (5.4), $q^{1/12} \ge eK^3 C_1/\tau$, so $r_q \le \frac{1}{2}q^{-1/4} < 1$. For $x > \tau q$ the ratio of consecutive terms of the last sum is $a/(x+1) \le r_q/e \le 1/e$, and its first term, at $x_1 = \lfloor \tau q \rfloor + 1$, is $a^{x_1}/x_1! \le$

$(ea/x_1)^{x_1} \le r_q^{x_1} \le r_q^{\tau q}$. Hence

$$\begin{aligned}\mathbb{E}'\big[W_q^{X_N};\, X_N > \tau q\big] &\le \frac{1}{1-e^{-1}}\, r_q^{\tau q} \le 2\, r_q^{\tau q} \\ &\le 2\cdot 2^{-\tau q}\, e^{-(\tau/4)\, q \ln q} \le e^{-(\tau/4)\, q \ln q},\end{aligned}$$

the last step because $\tau q \ge 1$. Since $W_q \ge 1$, also $\mathbb{P}'(X_N > \tau q) \le e^{-(\tau/4) q \ln q}$. Inserting the second display into the first, the two error terms are at most $e^{K^3 W_q/2}\, \mathbb{P}'(A_N)\, e^{-(\tau/4) q \ln q}$ each, which gives the claim (with 2 in place of 3). □

5.7. **Proof of Theorem 5.2.** Sum (5.10) over all perfect assignments N and apply Lemma 5.7 to each term:

$$\begin{aligned}&\mathbb{E}\big[Z_\beta;\, H \in \mathcal{G},\ s(H) \le s_0,\ |H| \le K^3 q\big] \\ &\quad= \sum_N \mathbb{E}\big[\mathbf{1}_{A_N};\ H \in \mathcal{G},\ s(H) \le s_0,\ |H| \le K^3 q\big] \\ &\quad\le e^{\Gamma_q + 2K^6 + s_0 \ln W_q} \sum_N \mathbb{E}'\big[\mathbf{1}_{A_N} \mathbf{1}\{H' \in \mathcal{G}\} W_q^{X_N}\big] \\ &\quad\le e^{R_q} \sum_N \Big(\mathbb{E}'\big[\mathbf{1}_{A_N} \mathbf{1}\{H' \in \mathcal{G}^{(2\tau)}\}\big] + 3\, \mathbb{P}'(A_N)\, e^{-(\tau/4) q \ln q}\Big).\end{aligned}$$

Since $\sum_N \mathbf{1}_{A_N} = Z_\beta$, this is (5.5). The estimate of R_q in (5.6) follows from (5.3):

$$R_q \le \tfrac{3}{40} K^5 q^{2/3} + 2K^6 + q^{2/3+\zeta} \ln\big(C_1 q^{2/3}\big) + \tfrac{1}{2} K^3 C_1\, q^{2/3}, \tag{5.11}$$

and $\zeta < \frac{1}{3}$.

For (5.7), the event $\{H \in \mathcal{G},\ s(H) \le s_0,\ |H| \le K^3 q\}$ is a function of H, so

$$\begin{aligned}\mathbb{P}(Z_\beta \ge 1) \le \mathbb{P}(H \notin \mathcal{G}) + \mathbb{P}\big(s(H) > s_0\big) + \mathbb{P}\big(|H| > K^3 q\big) \\ + \mathbb{E}\big[Z_\beta;\ H \in \mathcal{G},\ s(H) \le s_0,\ |H| \le K^3 q\big],\end{aligned}$$

and by Lemma 5.3 and $R_q = o(q)$, $3e^{R_q}\mathbb{E}'[Z_\beta] e^{-(\tau/4) q \ln q} \le e^{-(\tau/5) q \ln q}$ for all large q. □

5.8. **Typicality in the market.** The *binned star* of an agent v in a configuration h is the vector $d_v(h) = (d_{v,1}, \dots, d_{v,B}) \in \mathbb{N}^B$, where $d_{v,b}$ is the number of teams of h with label b that contain v; $|d_v| = \sum_b d_{v,b}$ is the degree of v. As in Section 4, the *reference star* is a vector $D = (D_1, \dots, D_B)$ of independent Poisson variables with means $\lambda_1, \dots, \lambda_B$, and γ is its law on $\mathbb{N}^B$.

Proposition 5.8 (Typicality)**.** *In the market, for fixed $K \ge 1$, $B \ge 1$ and $\zeta > 0$:*

(a) $\mathbb{P}(|H| > K^3 q) \le 1/q$;
(b) $\mathbb{P}\big(s(H) > q^{2/3+\zeta}\big) \le \frac{3}{40} K^5 q^{-\zeta}$;

(c) *for each of the three sectors S and every function $\psi : \mathbb{N}^B \to \mathbb{R}$ with $|\psi(d)| \le c\,(1+|d|)^m$ for some constants c, m,*

$$\frac{1}{q}\sum_{v\in S}\psi\big(d_v(H)\big) \longrightarrow \mathbb{E}\,\psi(D) \qquad \textit{in probability, } D\sim\gamma.$$

Proof. (a) $\mathbb{E}|H| = q^3p = K^3q/6 =: \mu$. The indicators of $\{\xi_e \le K\}$ and $\{\xi_f \le K\}$ are independent unless $e \sim f$, there are $3q^3(q-1)$ ordered pairs with $e \sim f$, and for these the covariance is at most $t^5/20$ by (5.8). So $\operatorname{Var}|H| \le \mu + 2\Gamma_q$, and by Chebyshev's inequality

$$\mathbb{P}\big(|H| > 6\mu\big) \le \frac{\mu+2\Gamma_q}{25\mu^2} \le \frac{6}{25K^3q} + \frac{27}{125\,K\,q^{4/3}} \le \frac{1}{q}.$$

(b) $s(H) = \sum_\pi (\mathrm{mult}_H(\pi)-1)^+ \le \sum_\pi \binom{\mathrm{mult}_H(\pi)}{2} = Y$, where Y is the number of unordered pairs of near-tight teams that share a pair (by (G1) each such pair of teams is counted once). By (5.8), $\mathbb{E}Y = \Gamma_q$ exactly, and Markov's inequality gives the claim.

(c) The three sectors play symmetric roles; fix one of them, S, and an agent a of S. Its teams are the triples (a,y,z) with y and z in the other two sectors. Let $X_b = d_{a,b}(H)$ and let $k = (k_1,\dots,k_B) \in \mathbb{N}^B$, $|k| = \sum_b k_b$. The joint factorial moment $\mathbb{E}\prod_b (X_b)_{k_b}$ is the sum, over ordered $|k|$-tuples of distinct teams at a of which the first k_1 are assigned bin 1, the next k_2 bin 2, and so on, of the probability that every team of the tuple lies in its assigned bin. The pairs $\{y,z\}$ of the teams of a tuple are distinct, so the excess of the tuple, as a configuration, is

$$s = 2|k| - \#\{\text{values of } y\} - \#\{\text{values of } z\}.$$

There are $(q)_{|k|}^2$ tuples with $s=0$; their teams share no pair, each such tuple has probability exactly $\prod_i p_{b_i}$, and their total contribution is $\prod_b \lambda_b^{k_b}\,(1 - O(|k|^2/q))$. For $s \ge 1$ there are at most $|k|^{2|k|}q^{2|k|-s}$ tuples (choose which positions share a value of y and which share a value of z, then the $2|k|-s$ values), each of probability at most $W_q^s\prod_i p_{b_i}$ by Lemma 5.4 with $N=\emptyset$. Their total contribution is at most

$$|k|^{2|k|}\prod_b \lambda_b^{k_b}\sum_{s\ge1}\Big(\frac{W_q}{q}\Big)^s = O\big(q^{-1/3}\big),$$

since $W_q/q = C_1 q^{-1/3}$. Hence every joint factorial moment of (X_b) converges to $\prod_b \lambda_b^{k_b}$, the corresponding moment of γ.

Now let $a \ne a'$ be two agents of S, with bin counts (X_b) and (X'_b), and consider the mixed factorial moment $\mathbb{E}\prod_b (X_b)_{k_b}\prod_b (X'_b)_{k'_b}$. It is a sum over pairs of tuples, one at a and one at a'. A team at a and a team at a' share a pair exactly when they have the same (y,z). Write the excess of the union of the two tuples as $s_1+s_2+s_3$, where s_1, s_2 are the excesses of the two tuples and s_3 is the number of pairs $\{y,z\}$ used by both. Suppose $s_3 \ge 1$, and let the s_3 teams at a' whose pair $\{y,z\}$ is also used at a have r distinct

values of y and c distinct values of z. These $r+c$ values are determined by the tuple at a and the pattern of coincidences, so the number of pairs of tuples with given (s_1, s_2, s_3, r, c) is at most a constant (depending on $|k|$, $|k'|$) times

$$q^{2|k|-s_1} \cdot q^{2|k'|-s_2-(r+c)}.$$

The s_3 teams in question are distinct teams at a' on $r+c$ values, so their excess is $2s_3 - r - c$, which is at most s_2 by monotonicity; together with $r, c \geq 1$ this gives $r+c \geq \max(2,\, 2s_3 - s_2)$. Each pair of tuples has probability at most $W_q^{s_1+s_2+s_3} \prod p_{b_i} \prod p_{b'_i}$ by Lemma 5.4, so the contribution of a class (s_1, s_2, s_3, r, c) is $O(q^{\omega})$ with

$$\omega = -\frac{s_1+s_2}{3} - (r+c) + \frac{2s_3}{3}.$$

If $2s_3 - s_2 \leq 2$ then $\omega \leq -\frac{s_2}{3} - 2 + \frac{2}{3}\big(1 + \frac{s_2}{2}\big) = -\frac{4}{3}$; if $2s_3 - s_2 > 2$ then $\omega \leq -\frac{s_2}{3} - (2s_3 - s_2) + \frac{2s_3}{3} = -\frac{2}{3}(2s_3 - s_2) < -\frac{4}{3}$. If $s_3 = 0$ and $s_1 + s_2 \geq 1$ then $r = c = 0$ and $\omega \leq -\frac{1}{3}$. So the pairs of tuples of positive excess contribute $O(q^{-1/3})$ in total. The pairs of tuples of excess 0 number $(q)_{|k|}^2 (q)_{|k'|}^2 (1 - O(q^{-2}))$ and have product probability. Hence the mixed factorial moments converge to $\prod_b \lambda_b^{k_b} \prod_b \lambda_b^{k'_b}$.

A product of Poisson laws is determined by its moments. Therefore $(d_a(H), d_{a'}(H))$ converges in distribution to $\gamma \otimes \gamma$, and all moments of the degrees $|d_a(H)|$, $|d_{a'}(H)|$ are bounded uniformly in q. Let $|\psi(d)| \leq c(1+|d|)^m$. By uniform integrability, $\mathbb{E}\psi(d_a(H)) \to \mathbb{E}\psi(D)$, $\operatorname{Var}\psi(d_a(H))$ stays bounded, and $\operatorname{Cov}\big(\psi(d_a(H)), \psi(d_{a'}(H))\big) \to 0$. By exchangeability of the agents of S,

$$\operatorname{Var}\Big(\frac{1}{q}\sum_{v \in S} \psi\big(d_v(H)\big)\Big) \leq \frac{\operatorname{Var}\psi(d_a(H))}{q} + \operatorname{Cov}\big(\psi(d_a(H)), \psi(d_{a'}(H))\big) \longrightarrow 0,$$

and Chebyshev's inequality gives (c). □

Only stars of depth one and a law of large numbers over agents, without a rate, are used here; Theorem 8.4 is not needed for the integer bound.

Remark 5.9 (Numerical evidence, floating point)**.** The scripts `lemmaC/round3/sc/cmp.py` and `cmp2.py` (same folder) simulate both models for $16 \leq q \leq 48$ and $0.8 \leq K \leq 2$, without bins. The mean of Y agrees with Γ_q in all 16 cells within sampling error; the probability that H is empty lies between its value in the independent-team model and the bound of Lemma 5.5, and nearly attains the latter (for $q = 32$, $K = 0.8$: 0.0822 against 0.0829); the ratio of the probabilities of $\{|H| = n,\ s(H) = 0\}$ in the two models stays below the factor of Lemma 5.6 with $N = \emptyset$, after one apparent exceedance was resolved as sampling noise by a larger run. None of this is used in the proofs, and the sizes are far from the range in which the comparison is effective (Remark 5.16).

5.9. **The robust form of the first moment.** Theorem 5.2 asks for a bound on $\mathbb{E}'[Z_\beta;\, H' \in \mathcal{G}^{(2\tau)}]$, where $\mathcal{G}$ is typical *in the market* and the event has been enlarged by $2\tau q$ teams. The typical set $\mathcal{T}_\delta$ of Section 4 is defined through a distance between the empirical law of the stars and γ that carries the weight $1 + |d|$. It is not suited to this: adding one team at an agent of degree $|d|$ can move that distance by as much as $(3 + 2|d|)/q$, so the set is not stable under changes of τq teams. The first moment, however, depends on the stars only through two families of averages, and each of them changes by a bounded amount when one team is changed. The good set is defined through these. In addition, the compactness step of Section 4 (Lemma 4.13), which gives no explicit constants, is replaced by an explicit estimate (Lemma 5.13), so that the size of the admissible enlargement τ is explicit.

Notation and imported statements. The notation of Section 4 is used: the tilt $\theta > 0$; the integrals

$$\kappa_b = \int_{I_b} e^{-\theta x}\, \tfrac{1}{2}x^2 \,\mathrm{d}x, \qquad J = \int_K^\infty e^{-\theta x}\, \tfrac{1}{2}x^2 \,\mathrm{d}x = e^{-\theta K}\Big(\frac{K^2}{2\theta} + \frac{K}{\theta^2} + \frac{1}{\theta^3}\Big);$$

vectors $u = (u_1, \dots, u_B, u_\infty)$ and $\mu = (\mu_1, \dots, \mu_B, \varepsilon)$ in $\mathbb{R}^{B+1}$, indexed by the B bins and the *high option* ∞, with $\langle \mu, u\rangle = \sum_b \mu_b u_b + \varepsilon u_\infty$; the vector $\mathbf{1}$ of ones; and

$$\ell_b = \ln \frac{\kappa_b}{\lambda_b} \quad (1 \le b \le B), \qquad \ell_\infty = \ln J.$$

We add $\ell_{\max} = \max(\max_b |\ell_b|,\, |\ell_\infty|)$ and write $\|\cdot\|_1$ for the ℓ^1 norm on $\mathbb{R}^{B+1}$. Four points of notation differ from Section 4. The labelled low hypergraph G of that section is the binned near-tight hypergraph of the independent-team model, here H'. Its probability $\mathbb{P}$ is here $\mathbb{P}'$. The Legendre functional, written H there, is written H here, where H is the hypergraph. Degree arrays, written D there, are written $\mathbf{D}$ here, to keep them apart from the reference star D.

For a star $d \in \mathbb{N}^B$ let

$$L_u(d) = \ln\Big(e^{u_\infty} + \sum_{b=1}^{B} d_b\, e^{u_b}\Big),$$

that is, $\ln(w_\infty + \sum_b d_b w_b)$ with the weights $w = e^u$. The *functional mean* is $\Phi(u) = \mathbb{E}L_u(D)$, $D \sim \gamma$, as in Definition 4.2. For a configuration h, its *degree array* is $\mathbf{D}(h) = (d_v(h))_v$, the family of the binned stars of all $3q$ agents; n_b is the number of teams with label b and $\nu_b = n_b/q$ is the *bin density*. The three sectors are numbered $s = 1, 2, 3$, with agent sets V_1, V_2, V_3, and

$$\widehat{\Phi}_s(u; \mathbf{D}) = \frac{1}{q} \sum_{v \in V_s} L_u(d_v)$$

is the empirical mean of L_u over the stars of sector s. We write $\nu_b(h)$ and $\widehat{\Phi}_s(u; h)$ for the same quantities of $\mathbf{D}(h)$. A *profile* for $\mathbf{D}$ is a vector μ with

$q\mu_b \in \mathbb{N}$, $0 \le \mu_b \le \nu_b$ and $\varepsilon = 1 - \sum_b \mu_b \ge 0$: the assignment uses $q\mu_b$ teams of bin b and $q\varepsilon$ teams above K. With $h(x) = -x\ln x - (1-x)\ln(1-x)$ and $g(x,y) = y\,h(x/y)$ for $0 \le x \le y$ ($g(0,0) = 0$), the exponent (4.14) of the first moment is

$$\Xi(\mu;\mathbf{D}) = \theta\beta + \sum_{s=1}^{3} \inf_{u\in\mathbb{R}^{B+1}} \big[\widehat{\Phi}_s(u;\mathbf{D}) - \langle \mu, u\rangle\big] - 2\sum_b g(\mu_b,\nu_b) + \sum_b \mu_b\ell_b + \varepsilon\,\ell_\infty + 2\varepsilon\ln\varepsilon - 2\varepsilon. \tag{5.12}$$

Further,

$$\mathcal{P} = \Big\{\mu :\ 0 \le \mu_b \le \lambda_b,\ \varepsilon \ge 0,\ \sum_b \mu_b + \varepsilon = 1\Big\},$$

$$\mathsf{H}(\mu) = \inf_{u\in\mathbb{R}^{B+1}} \big[\Phi(u) - \langle \mu, u\rangle\big], \qquad S_0(\mu) = \sum_b \lambda_b\, h\Big(\frac{\mu_b}{\lambda_b}\Big) - \varepsilon\ln\varepsilon + \varepsilon,$$

and $\Psi_0 = \Psi_0(K,B,\theta,u_0)$ is the functional of Definition 4.2, defined for every weight vector $u_0 \in \mathbb{R}^{B+1}$.

Four statements are imported from Section 4. Each of them concerns the reference star or the independent-team model.

(I1) *First moment on a set of degree arrays* (Lemma 4.11). For every set $\mathcal{T}$ of degree arrays and $q^{2/3} \ge K$,

$$\mathbb{E}'\big[Z_\beta;\ \mathbf{D}(H') \in \mathcal{T}\big] \le c_q\,(q+1)^{B+2} \sup_{\mathbf{D}\in\mathcal{T},\ \mu} \Big[\prod_b (n_b+1)^2 \cdot e^{\,q\,\Xi(\mu;\mathbf{D})}\Big], \tag{5.13}$$

where the supremum is over $\mathbf{D} \in \mathcal{T}$ and profiles μ for $\mathbf{D}$, and $c_q = e^2 e^{K^6/(36q(1-p))}(1-p)^{-q} \le e^2\exp(K^{9/2}/30 + K^{3/2}/5)$. The factor $\prod_b (n_b+1)^2$ depends on the degree array and stays inside the supremum. This bound holds for an arbitrary set $\mathcal{T}$; it contains the tilting inequality $\mathbf{1}\{\xi(N) < \beta q\} \le e^{\theta(\beta q - \xi(N))}$, the union bound over the teams above K, the simplicity conditioning, the choice-function bound and the count of profiles.

(I2) *Legendre facts* (Subsection 4.4 and Lemma 4.16(i)). Φ is finite, convex and differentiable on $\mathbb{R}^{B+1}$; $\Phi(u + c\mathbf{1}) = \Phi(u) + c$; $\nabla\Phi(u) \in \mathcal{P}$ for every u; and if $\mu' = \nabla\Phi(\bar u)$ then $\mathsf{H}(\mu') = \Phi(\bar u) - \langle \mu', \bar u\rangle$.

(I3) *Tangent bound* (Proposition 4.18). For every $u_0 \in \mathbb{R}^{B+1}$ and every $\mu \in \mathcal{P}$,

$$3\mathsf{H}(\mu) - 2S_0(\mu) + \langle \ell, \mu\rangle \le \Psi_0(K,B,\theta,u_0).$$

(I4) *Certified value* (Proposition 4.4). For $K = 5$, $B = 50$, $\theta = 1.6712$ and the weight vector $u_0 = u^*$ listed there, $\Psi_0 \le -3.84751328322$. Hence $\Psi_0 \le -3.84751328$, the form used below, and $-\Psi_0/\theta \ge 2.30224586$ (this needs the eleven-decimal bound). The enclosure is certified in interval arithmetic by the script `lemmaC/round3/final/certI/certify_rigorous.py` (Section 11).

Not imported: the typical set $\mathcal{T}_\delta$ of (4.16), the statement that it is typical in the independent-team model (Lemma 4.12) and the compactness step (Lemma 4.13). In particular Theorem 4.1 and Proposition 4.3 are not used as statements; the argument below runs parallel to their proof, with a different typical set.

We record the size of the prefactor in (5.13). If $\nu_b(\mathbf{D}) \le M$ for all b and all $\mathbf{D} \in \mathcal{T}$, then $n_b + 1 \le q(1+M)$, and $q+1 \le 2q$, so

(5.14)
$$\mathbb{E}'\big[Z_\beta;\ \mathbf{D}(H') \in \mathcal{T}\big] \le c_{K,B}\,(1+M)^{2B}\,q^{3B+2}\exp\Big(q \sup_{\mathbf{D}\in\mathcal{T},\ \mu} \Xi(\mu;\mathbf{D})\Big),$$
$$c_{K,B} = e^2\,2^{B+2}\exp\big(K^{9/2}/30 + K^{3/2}/5\big).$$

By (5.12) and (5.13) the degree array enters the first moment only through

(F1) the bin densities ν_b, in g, in the constraint $\mu_b \le \nu_b$ and in the prefactor; and

(F2) the three functions $u \mapsto \widehat{\Phi}_s(u;\mathbf{D})$, through the infima.

Both are averages over agents of a function of the binned star: $d \mapsto d_b$ and $d \mapsto L_u(d)$.

The good set. For $A > 0$ let $U_A = \{u \in \mathbb{R}^{B+1} : \max u - \min u \le A\}$. For $\delta_1, \delta_2, A > 0$ define

$$\mathcal{T}(\delta_1,\delta_2;A) = \Big\{\mathbf{D}:\ \sum_b |\nu_b(\mathbf{D}) - \lambda_b| \le \delta_1 \ \text{ and } \sup_{u\in U_A} \big|\widehat{\Phi}_s(u;\mathbf{D}) - \Phi(u)\big| \le \delta_2 \ \text{ for } s = 1,2,3\Big\}, \tag{5.15}$$

$$\mathcal{G}_q(\delta_1,\delta_2;A) = \big\{\text{configurations } h:\ \mathbf{D}(h) \in \mathcal{T}(\delta_1,\delta_2;A)\big\}.$$

Lemma 5.10 (Stability)**.** *Let h, h' be configurations with $|h \triangle h'| \le k$. Then $\sum_b |n_b(h) - n_b(h')| \le k$, and for every $u \in U_A$ and $s \in \{1,2,3\}$,*

$$\big|\widehat{\Phi}_s(u;h) - \widehat{\Phi}_s(u;h')\big| \le \frac{k}{q}\ln\big(1+e^A\big) \le \frac{k}{q}\,(A + \ln 2).$$

Consequently, for every $\tau > 0$,

$$\mathcal{G}_q(\delta_1,\delta_2;A)^{(2\tau)} \subset \mathcal{G}_q\big(\delta_1 + 2\tau,\ \delta_2 + 2\tau(A+\ln 2);\ A\big). \tag{5.16}$$

Proof. Pass from h to h' by removing the labelled teams of $h \setminus h'$ and then adding those of $h' \setminus h$, one at a time; every intermediate set is a configuration. Each step changes one n_b by 1 and, in each sector, one star in one coordinate by 1. If $S = e^{u_\infty} + \sum_b d_b e^{u_b}$, then adding a team of bin b changes $\ln S$ by $\ln(1 + e^{u_b}/S) \le \ln(1 + e^{u_b - u_\infty}) \le \ln(1+e^A)$, because $S \ge e^{u_\infty}$ and $u \in U_A$; removing a team is the same step read backwards. So each step changes $\widehat{\Phi}_s(u;\cdot)$ by at most $q^{-1}\ln(1+e^A)$. For (5.16) take $k = 2\tau q$ and use the triangle inequality in both conditions of (5.15). $\square$

Lemma 5.11 (Box Legendre identity)**.** *Let $\mu \in \mathbb{R}^{B+1}$ with $\sum_b \mu_b + \varepsilon = 1$ and let $A > 0$. The function $u \mapsto \Phi(u) - \langle \mu, u\rangle$ attains its minimum over U_A at some $\bar u$, and with $\mu' = \nabla\Phi(\bar u) \in \mathcal{P}$,*

$$\Phi(\bar u) - \langle \mu, \bar u\rangle = \mathsf{H}(\mu') - \frac{A}{2}\,\|\mu - \mu'\|_1.$$

Proof. By (I2) the function $f(u) = \Phi(u) - \langle \mu, u\rangle$ is convex, differentiable and invariant under $u \mapsto u + c\mathbf{1}$. The set U_A is convex and invariant under the same shifts, and $U_A \cap \{\sum u = 0\}$ is compact. So a minimiser $\bar u$ of f over U_A exists, and $\langle \nabla f(\bar u), v - \bar u\rangle \geq 0$ for all $v \in U_A$. Here $\nabla f(\bar u) = \mu' - \mu =: z$, and the coordinates of z sum to 0. Thus $\langle z, \bar u\rangle = \min_{v\in U_A}\langle z, v\rangle$. For $v \in U_A$ and $c = (\max v + \min v)/2$, every coordinate of $v - c\mathbf{1}$ has absolute value at most $A/2$, so $\langle z, v\rangle = \langle z, v - c\mathbf{1}\rangle \geq -\frac{A}{2}\|z\|_1$, with equality for $v = -\frac{A}{2}\operatorname{sign}(z) \in U_A$. Hence $\langle \mu' - \mu, \bar u\rangle = -\frac{A}{2}\|\mu - \mu'\|_1$. Finally, by (I2), $\mathsf{H}(\mu') = \Phi(\bar u) - \langle \mu', \bar u\rangle$, and so $f(\bar u) = \mathsf{H}(\mu') + \langle \mu' - \mu, \bar u\rangle$. □

Lemma 5.12 (Entropy modulus)**.** *Let $\varphi(x) = -x\ln x$ on $[0,\infty)$, $\varphi(0) = 0$. If $a, b \in [0, M]$ and $c' \geq 1 + \ln^+ M$, then*

$$\varphi(a) - \varphi(b) \leq c'\,|a - b| + e^{-1-c'}.$$

Proof. If $a \geq b$: φ is concave with $\varphi(0) = 0$, hence subadditive, so $\varphi(a) - \varphi(b) \leq \varphi(a - b)$, and $\sup_{z\geq 0}[\varphi(z) - c'z] = e^{-1-c'}$, attained at $z = e^{-1-c'}$. If $a < b$: $\varphi(a) - \varphi(b) = \int_a^b (1 + \ln x)\,\mathrm{d}x \leq (b - a)(1 + \ln^+ b) \leq c'(b - a)$. □

The next lemma is the quantitative replacement for Lemma 4.13.

Lemma 5.13 (Explicit exponent on the good set)**.** *Let $u_0 \in \mathbb{R}^{B+1}$, $c' \geq 1$, $\delta_1', \delta_2' > 0$ and $A > 0$ satisfy*

$$A \;\geq\; \frac{8c' + 4 + 2\ell_{\max}}{3}, \qquad c' \;\geq\; 1 + \ln^+\bigl(\lambda_{\max} + \delta_1'\bigr). \tag{5.17}$$

Then for every $\mathbf{D} \in \mathcal{T}(\delta_1', \delta_2'; A)$ and every profile μ for $\mathbf{D}$,

$$\Xi(\mu; \mathbf{D}) \leq \theta\beta + \Psi_0(K, B, \theta, u_0) + 3\delta_2' + 4c'\delta_1' + (6B + 2)\,e^{-1-c'}.$$

Proof. Take $\bar u$ and μ' from Lemma 5.11 for this μ. The same vector $\bar u \in U_A$ is admissible in each of the three infima of (5.12), so by the definition of $\mathcal{T}(\delta_1', \delta_2'; A)$ and Lemma 5.11,

$$\begin{aligned}\sum_{s=1}^{3} \inf_u \bigl[\widehat\Phi_s(u; \mathbf{D}) - \langle \mu, u\rangle\bigr] &\leq 3\bigl[\Phi(\bar u) - \langle \mu, \bar u\rangle\bigr] + 3\delta_2' \\ &= 3\mathsf{H}(\mu') - \frac{3A}{2}\|\mu - \mu'\|_1 + 3\delta_2'.\end{aligned}$$

Since $\mu' \in \mathcal{P}$, (I3) gives $3\mathsf{H}(\mu') \leq \Psi_0 + 2S_0(\mu') - \langle \ell, \mu'\rangle$. Write ε' for the last coordinate of μ', so that $S_0(\mu') = \sum_b g(\mu_b', \lambda_b) + \varphi(\varepsilon') + \varepsilon'$, and put

$$\widetilde S(\mu, \nu) = \sum_b g(\mu_b, \nu_b) + \varphi(\varepsilon) + \varepsilon,$$

so that the entropy terms of (5.12) are $-2\widetilde{S}(\mu,\nu)$. Then

(5.18)
$$\Xi(\mu;\mathbf{D}) - \theta\beta - \Psi_0 - 3\delta_2' \le 2\big[S_0(\mu') - \widetilde{S}(\mu,\nu)\big] + \langle \ell, \mu - \mu'\rangle - \frac{3A}{2}\|\mu-\mu'\|_1.$$

Since $g(x,y) = \varphi(x) + \varphi(y-x) - \varphi(y)$,

$$g(\mu_b',\lambda_b) - g(\mu_b,\nu_b) = \big[\varphi(\mu_b') - \varphi(\mu_b)\big] + \big[\varphi(\lambda_b-\mu_b') - \varphi(\nu_b-\mu_b)\big] + \big[\varphi(\nu_b) - \varphi(\lambda_b)\big].$$

All six arguments lie in $[0, \lambda_{\max}+\delta_1']$, because $\mu_b' \le \lambda_b$, $\mu_b \le \nu_b$ and $\nu_b \le \lambda_b + \delta_1'$. With $z_b = |\mu_b - \mu_b'|$, $d_b = |\nu_b - \lambda_b|$ and Lemma 5.12 (with $M = \lambda_{\max}+\delta_1'$), the three brackets are at most $c'z_b + e^{-1-c'}$, $c'(z_b+d_b)+e^{-1-c'}$ and $c'd_b + e^{-1-c'}$, so

$$g(\mu_b',\lambda_b) - g(\mu_b,\nu_b) \le c'(2z_b + 2d_b) + 3e^{-1-c'}.$$

For the high coordinate, with $z_\infty = |\varepsilon - \varepsilon'|$ and $\varepsilon,\varepsilon' \in [0,1]$, Lemma 5.12 (with $M=1$) gives $\varphi(\varepsilon') + \varepsilon' - \varphi(\varepsilon) - \varepsilon \le (c'+1)z_\infty + e^{-1-c'}$. Also $\langle \ell, \mu-\mu'\rangle \le \ell_{\max}\|\mu-\mu'\|_1$, $\sum_b z_b + z_\infty = \|\mu-\mu'\|_1$ and $\sum_b d_b \le \delta_1'$. Hence the right side of (5.18) is at most

$$\Big(4c' + 2 + \ell_{\max} - \frac{3A}{2}\Big)\|\mu-\mu'\|_1 + 4c'\delta_1' + (6B+2)\,e^{-1-c'},$$

and the first coefficient is at most 0 by (5.17). □

Lemma 5.13 needs no compactness and no continuity of H at the boundary of its domain. Its error is continuous in (δ_1',δ_2') and tends to 0 as $c' \to \infty$ and then $\delta_1', \delta_2' \to 0$; this is the statement that the exponent of the first moment moves continuously with the star profile, with an explicit modulus.

In (5.17) one may use $\ell_{\max} \le \theta K$ whenever $-\theta K \le \ln J \le 0$: indeed κ_b/λ_b is the mean of $e^{-\theta x}$ under the probability measure with density $x^2/(2\lambda_b)$ on $I_b \subset [0,K]$, so $-\theta K \le \ell_b \le 0$.

Proposition 5.14 (First moment on the enlarged good set)**.** *Let $K \ge 1$, $B \ge 1$, $\theta > 0$, $u_0 \in \mathbb{R}^{B+1}$ and $\beta > 0$ be such that*

$$m := -\big(\theta\beta + \Psi_0(K,B,\theta,u_0)\big) > 0.$$

Let $c' \ge 1$, $A \ge (8c' + 4 + 2\ell_{\max})/3$ and $\delta_1,\delta_2,\tau > 0$ be such that

$$c' \ge 1 + \ln^+\big(\lambda_{\max} + \delta_1 + 2\tau\big),$$
$$\mathrm{Err} := 3\big(\delta_2 + 2\tau(A + \ln 2)\big) + 4c'(\delta_1 + 2\tau) + (6B+2)\,e^{-1-c'} < m. \tag{5.19}$$

Put $\mathcal{G}_q = \mathcal{G}_q(\delta_1,\delta_2;A)$. Then:

(i) $\mathbb{P}(H \in \mathcal{G}_q) \to 1$ *in the market;*

(ii) $\mathcal{G}_q^{(2\tau)} \subset \mathcal{G}_q\big(\delta_1 + 2\tau,\ \delta_2 + 2\tau(A+\ln 2);\ A\big)$*;*

(iii) *for $q^{2/3} \ge K$,*

$$\mathbb{E}'\big[Z_\beta;\ H' \in \mathcal{G}_q^{(2\tau)}\big] \le C\,q^{3B+2}\,e^{-(m-\mathrm{Err})\,q},$$

$$C = c_{K,B}\left(1 + \lambda_{\max} + \delta_1 + 2\tau\right)^{2B},$$

with $c_{K,B}$ as in (5.14)*. The constant C depends on K, B and $\delta_1+2\tau$, and not on q.*

For every $\beta < -\Psi_0/\theta$ such constants exist: take c' large, then A, then δ_1, δ_2, τ small.

Proof. (ii) is (5.16).

(iii) Let $\delta_1' = \delta_1 + 2\tau$ and $\delta_2' = \delta_2 + 2\tau(A + \ln 2)$. By (ii), $\{H' \in \mathcal{G}_q^{(2\tau)}\} \subset \{\mathbf{D}(H') \in \mathcal{T}(\delta_1', \delta_2'; A)\}$. On $\mathcal{T}(\delta_1', \delta_2'; A)$ every bin density is at most $\lambda_{\max} + \delta_1'$, so (5.14) applies with $M = \lambda_{\max} + \delta_1'$, and by Lemma 5.13 the supremum of Ξ is at most $\theta\beta + \Psi_0 + \mathrm{Err} = -(m - \mathrm{Err})$.

(i) For $u \in U_A$ the vector $u - (\max u)\mathbf{1}$ lies in the cube $[-A, 0]^{B+1}$, and $\widehat{\Phi}_s$ and Φ change by the same constant under $u \mapsto u + c\mathbf{1}$. So the supremum over U_A in (5.15) is at most the supremum over the cube. For u in the cube and $d \in \mathbb{N}^B$ we have $-A \le u_\infty \le L_u(d) \le \ln(1+|d|)$, hence $|L_u(d)| \le A + |d|$; moreover $\mathbb{E}L_u(D) = \Phi(u)$, and $|L_u(d) - L_{u'}(d)| \le \|u - u'\|_\infty$ for all d, because the gradient of $u \mapsto L_u(d)$ is a probability vector. Hence $\widehat{\Phi}_s(\cdot\,; \mathbf{D})$ and Φ are 1-Lipschitz in the supremum norm. Let $\mathcal{N}$ be a finite subset of the cube such that every point of the cube is within supremum distance $\delta_2/4$ of $\mathcal{N}$; it does not depend on q. Apply Proposition 5.8(c) in each of the three sectors to the finitely many functions L_u, $u \in \mathcal{N}$, and $d \mapsto d_b$, $1 \le b \le B$, whose mean under γ is λ_b and whose average over the agents of a sector is ν_b. With probability tending to 1 in the market, $|\widehat{\Phi}_s(u; H) - \Phi(u)| \le \delta_2/2$ for all $u \in \mathcal{N}$ and all s, and $|\nu_b(H) - \lambda_b| \le \delta_1/B$ for all b. On this event the supremum over the cube of $|\widehat{\Phi}_s - \Phi|$ is at most $\delta_2/2 + 2 \cdot \delta_2/4 = \delta_2$ and $\sum_b |\nu_b - \lambda_b| \le \delta_1$, that is, $H \in \mathcal{G}_q$.

For the last assertion, $\mathrm{Err} \to (6B+2)e^{-1-c'}$ as $\delta_1, \delta_2, \tau \to 0$ with c' and A fixed, and this limit is below m for c' large. □

Constants. At the certified parameters of (I4), $K = 5$, $B = 50$, $\eta = \frac{1}{10}$, $\theta = 1.6712$, one has $\lambda_{\max} = 7351/6000$, $\ln J \in [-6.10637801, -6.10637800]$ and therefore $\ell_{\max} \le \theta K = 8.356$. (In floating point $\ell_{\max} = |\ell_{50}| \approx 8.2718$.) The box size is taken as $A = (8c' + 4 + 2\theta K)/3$, which satisfies (5.17). Table 2 lists three admissible choices of the constants of Proposition 5.14, with Ψ_0 replaced by its certified upper bound -3.84751328.

TABLE 2. Constants for Proposition 5.14 at $K = 5$, $B = 50$, $\theta = 1.6712$. Here $m = -(\theta\beta + \Psi_0)$ and $A = (8c' + 4 + 2\theta K)/3$.

β	$m \ge$	c'	A	δ_1	δ_2	τ	Err $\le$	$m - \mathrm{Err} \ge$
2.30	$3.7532 \cdot 10^{-3}$	14	44.2374	10^{-5}	$2 \cdot 10^{-4}$	$5 \cdot 10^{-6}$	$3.1603 \cdot 10^{-3}$	$5.929 \cdot 10^{-4}$
2.302	$4.108 \cdot 10^{-4}$	17	52.2374	10^{-6}	$2 \cdot 10^{-5}$	$2.5 \cdot 10^{-7}$	$2.460 \cdot 10^{-4}$	$1.648 \cdot 10^{-4}$
2.30224	$9.79 \cdot 10^{-6}$	21	62.9040	$5 \cdot 10^{-9}$	$5 \cdot 10^{-7}$	$5 \cdot 10^{-10}$	$2.28 \cdot 10^{-6}$	$7.5 \cdot 10^{-6}$

Provenance of the rows. The rows $\beta = 2.30$ and $\beta = 2.302$ are evaluated by the hardened script `lemmaC/round3/final/certI/constants_rigorous.py`. The row $\beta = 2.30224$ was evaluated in exact rational arithmetic by a referee's script, `lemmaC/round3/v13/exact_budget.py`, and is not used in the sequel: every later statement uses $\beta_0 = 2.302$.

The values of A in the table are rounded up in the last digit. What the computations certify is the following. For the first two rows, the script `lemmaC/round3/final/certI/constants_rigorous.py` checks, in interval arithmetic with verified exponential and logarithm, that $|\ln J| < \theta K$, that $c' \geq 1 + \ln^+(\lambda_{\max} + \delta_1 + 2\tau)$, and that $\theta\beta - 3.84751328 + \mathrm{Err} < 0$ with the stated bounds; the constant -3.84751328 is asserted there, in exact rational arithmetic, to be at least the exact upper endpoint of the enclosure of Ψ_0 produced by `certify_rigorous.py`. For the third row (and again for the first two) the script `lemmaC/round3/v13/exact_budget.py` checks the same inequalities in exact rational arithmetic, with rational bounds for $\ln 2$ and for the exponential. Details are in Section 11. Since $\{\Phi_q \leq \beta q^{1/3}\}$ increases with β, the second row alone suffices for Theorem 5.1 as stated; the first shows a larger margin and the third shows how the admissible τ shrinks as β approaches $-\Psi_0/\theta$. For the consequence $\Phi_q \geq \beta_0 q^{1/3}$ with high probability, which is what Theorem 1.1 uses, the second row suffices.

Remark 5.15 (What the good set uses)**.** (1) In the exponent the enlargement costs $2\tau\,(3(A + \ln 2) + 4c')$, which is about $382\,\tau$ in the first row of Table 2. So the method tolerates τ up to about 10^{-5} there, and nothing near $2\tau = 1$. The explicit τ is small because Lemma 5.12 is used with the same constant in every bin; it has not been optimised, and its size plays no role in Theorem 5.1. (2) The good set uses two families of averages over the agents of a sector, of $d \mapsto d_b$ (one-team increment 1) and of $d \mapsto L_u(d)$, $u \in U_A$ (one-team increment at most $A + \ln 2$). Both are bounded by $A + |d|$ after normalisation, so Proposition 5.8(c) applies. There is no condition on the maximal degree and no truncation of stars. A statistic with unbounded one-team increments, such as the mean of the squared degree, is typical by Proposition 5.8(c) but is not stable in the sense of Lemma 5.10. (3) Z_β counts every perfect assignment with $\xi(N) < \beta q$, whatever mix of teams below and above K it uses; the independent-team model has exactly q^3 teams with the exact one-team law of the market. No Poisson number of teams is introduced: "Poisson" refers only to the limit law γ of a star.

5.10. **Proof of Theorem 5.1.** It suffices to treat $\beta = \beta_0 = 2.302$, because $\{\Phi_q \leq \beta q^{1/3}\}$ increases with β; the argument is the same for every β with $m > 0$.

Order of choices.

(1) $K = 5$, $B = 50$ (so $\eta = \frac{1}{10}$), $\theta = 1.6712$ and the weight vector $u_0 = u^*$ of (I4). This fixes $\Psi_0 \leq -3.84751328$, $\lambda_{\max} = 7351/6000$, $\ell_{\max} \leq \theta K$, and the constants of the comparison, $C_1 = 36K^5/\eta^6 = 1.125 \cdot 10^{11}$ and $W_q = C_1 q^{2/3}$.

(2) $\beta < -\Psi_0/\theta$, which gives the margin $m = -(\theta\beta + \Psi_0) > 0$.
(3) c', and then $A = (8c' + 4 + 2\theta K)/3$.
(4) $\delta_1, \delta_2, \tau > 0$ satisfying (5.19) (Table 2, second row). This fixes the good set $\mathcal{G}_q = \mathcal{G}_q(\delta_1, \delta_2; A)$.
(5) $\zeta = \frac{1}{6}$, so $s_0 = q^{5/6}$.
(6) $q \to \infty$.

The chain. By (5.1), $\mathbb{P}(\Phi_q \le \beta q^{1/3}) = \mathbb{P}(Z_\beta \ge 1)$. By Theorem 5.2 with these K, B, β, τ, ζ and $\mathcal{G} = \mathcal{G}_q$ (inequality (5.5) and the union bound preceding (5.7)), for $q \ge q_0(K, B, \tau)$,

$$\begin{aligned}(5.20)\quad \mathbb{P}(Z_\beta \ge 1) \le{}& \mathbb{P}(H \notin \mathcal{G}_q) + \mathbb{P}\big(s(H) > q^{5/6}\big) + \mathbb{P}\big(|H| > K^3 q\big) \\ &+ e^{R_q}\,\mathbb{E}'\big[Z_\beta;\ H' \in \mathcal{G}_q^{(2\tau)}\big] + 3\,e^{R_q}\,\mathbb{E}'[Z_\beta]\,e^{-(\tau/4)\,q\ln q},\end{aligned}$$

where, by (5.11),

$$R_q \le \tfrac{3}{40}K^5 q^{2/3} + 2K^6 + q^{5/6}\ln\big(C_1 q^{2/3}\big) + \tfrac{1}{2}K^3 C_1\, q^{2/3} = O\big(q^{5/6}\ln q\big).$$

The five terms are bounded as follows.

- $\mathbb{P}(H \notin \mathcal{G}_q) \to 0$, without a rate, by Proposition 5.14(i), which rests on Proposition 5.8(c).
- $\mathbb{P}(s(H) > q^{5/6}) \le \frac{3}{40}K^5 q^{-1/6}$ by Proposition 5.8(b).
- $\mathbb{P}(|H| > K^3 q) \le 1/q$ by Proposition 5.8(a).
- By Proposition 5.14(iii),

$$e^{R_q}\,\mathbb{E}'\big[Z_\beta;\ H' \in \mathcal{G}_q^{(2\tau)}\big] \le C\,q^{3B+2}\exp\big(R_q - (m - \mathrm{Err})\,q\big) \longrightarrow 0,$$

  because $R_q = o(q)$ and $m - \mathrm{Err} > 0$. All factors $e^{o(q)}$ of the first moment (the number of profiles, the binomial and factorial corrections, the cost of the simplicity conditioning and the factor $(1-p)^{-q}$) are contained in $C\,q^{3B+2}$.
- By Lemma 5.3, since $\beta \le 2.31$,

$$3\,e^{R_q}\,\mathbb{E}'[Z_\beta]\,e^{-(\tau/4)\,q\ln q} \le 3e^2(q+1)^2\exp\big(R_q + 0.22\,q - \tfrac{\tau}{4}\,q\ln q\big) \longrightarrow 0,$$

  because $\tau > 0$.

Hence $\mathbb{P}(\Phi_q \le \beta q^{1/3}) \to 0$. □

Every excluded event. In the market: $\{H \notin \mathcal{G}_q\}$, $\{s(H) > q^{5/6}\}$ and $\{|H| > K^3 q\}$, the first three terms of (5.20). In the independent-team model: for each candidate assignment N, the event $\{X_N > \tau q\}$ that more than τq near-tight teams share a pair with N; it is excluded inside Lemma 5.7 and produces the last term of (5.20). The teams above K are handled by the union bound inside (I1). The exceptional event $\{D(G) \notin \mathcal{T}_\delta\}$ of Proposition 4.3 (stars not typical in the independent-team model) does not appear: typicality is needed only in the market.

Remark 5.16 (How large q must be)**.** Theorem 5.1 is asymptotic only. The proof gives no information for any q that could be simulated, or whose decimal expansion could be written down in a paper of this length. Take the

first row of Table 2 ($\beta = 2.30$, $\tau = 5{\cdot}10^{-6}$, $m{-}\mathrm{Err} \geq 5.9{\cdot}10^{-4}$). (1) The term $\frac{1}{2}K^3C_1q^{2/3} = 7.0 \cdot 10^{12}\, q^{2/3}$ of R_q is below $(m - \mathrm{Err})\, q$ only for $q > 1.6 \cdot 10^{48}$. (2) Lemma 5.7 is stated for $q \geq (eK^3C_1/\tau)^{12}$, which is about 10^{227}. (3) The binding requirement is in the last term of (5.20). The tail bound in the proof of Lemma 5.7 is $2r_q^{\tau q} = 2\exp\bigl(-\frac{\tau}{3}\, q \ln q + \tau q \ln(eK^3C_1/(2\tau))\bigr)$, and it has to beat the plain first moment, which is bounded only by $(\beta^3 e/27)^q = e^{0.203\, q}$. This needs $\frac{\tau}{3} \ln q > 0.203$, that is $\ln q > 1.2 \cdot 10^5$, or q above about $10^{52{,}900}$. (With the rate $\tau/4$ used in the statement of the lemma the threshold is higher still.) For the second and third rows of the table the same requirement reads $\ln q > 2.4 \cdot 10^6$ and $\ln q > 1.2 \cdot 10^9$. The slow convergence is already visible in the typicality step: by (5.3), the expected number of pair-sharings per near-tight team is $2\Gamma_q/\mathbb{E}|H| \leq \frac{9}{10}K^2q^{-1/3}$, which at $K = 5$ is below 0.1 only for q above about 10^7.

6. The Poisson hypertree

Seen from one agent, and with team deficits rescaled to $\xi_e = q^{2/3}\Delta(e)$, the cheap teams of the market look like an infinite tree of teams in which every agent has infinitely many teams but only finitely many below any cost level. This section defines that object, the Poisson hypertree $\mathbb{T}$, and the one notion of invariance used in this paper. It then proves three things. Theorem 6.7: an invariant fractional perfect matching is the marginal of an invariant random perfect matching of the same cost. Lemma 6.8: weak duality between invariant fractional perfect matchings and invariant prices. Theorem 6.12: a fixed point of the recursion map Θ gives an invariant perfect matching on $\mathbb{T}$ whose cost is nine times the mean of the fixed point. Existence of a fixed point, and a bound on its mean, are the subject of Section 7. The convergence of the market to $\mathbb{T}$ is the subject of Section 8 and is not used here.

The construction follows the pattern of the recursive tree processes used for the random assignment problem [1, 2, 4]; invariance is the hypergraph form of involution invariance [3]. Two features need care: every agent has infinite degree, and the matching is not known to be a function of the tree, so it has to be built together with the tree by an extension theorem.

Standard tools. The following theorems are used by name. From measure theory: Kolmogorov's extension theorem for a sequence of standard Borel spaces, and the Ionescu-Tulcea theorem [16]. For Poisson processes: the marking, mapping and restriction theorems; the fact that a Poisson process with diffuse locally finite intensity is almost surely simple; and Mecke's formula for a Poisson process with σ-finite intensity on a standard Borel space [19].

6.1. Hypertrees and their codes.

A *hypertree* is a set of agents, each with a sector in $\mathbb{Z}/3$, and a set of teams, each made of three agents, one from each sector, such that the incidence graph (agents and teams as nodes, membership as edges) is a tree. Two agents a, b are *adjacent*, written $a \sim b$,

if they share a team; in a hypertree that team is unique and is written $e(a,b)$. The *distance* between two agents is the number of teams on the path joining them. Every team e carries a cost $\xi_e \in [0,\infty)$. The standing condition is

(R0) every agent is in infinitely many teams, their costs are pairwise distinct, and only finitely many of them lie below any level.

An *incidence* is a pair (v,g) of an agent v and a team $g \ni v$. Given an agent $b \neq v$, the incidence (v,g) *points toward* b if g is the first team on the path from v to b, and *points away from* b otherwise; every incidence at b itself points away from b. Given a team e, the incidence (v,g) *points toward* e if $g = e$ or g is the first team on the path from v to the agents of e, and away from e otherwise.

Rank coding. Let $V = \bigcup_{n\geq 0}(\mathbb{N}\times\{1,2\})^n$ be the set of addresses and $|v|$ the length of $v \in V$. For $v \in V$, $k \geq 1$ and $i \in \{1,2\}$ write (v,k,i) for the address v followed by the letter (k,i); for $v = \varnothing$ this is (k,i). In a rooted hypertree with (R0) the root o gets the address $\varnothing$ and sector s_o. Its teams, in increasing order of cost, are called $[\varnothing,1],[\varnothing,2],\dots$, and the two other agents of $[\varnothing,k]$ get the addresses $(k,1)$ and $(k,2)$, where (k,i) is the one in sector $s_o + i$. Recursively, an agent $v \neq \varnothing$ has a *parent team* $e(v)$ (the first team on the path to o), and in it a *parent agent* $p(v)$ (the agent of $e(v)$ nearest to o) and a *sibling* $\mathrm{sib}(v)$; its other teams, in increasing order of cost, are $[v,1],[v,2],\dots$, and the other agents of $[v,k]$ are $(v,k,1)$ and $(v,k,2)$, the i-th one in sector $s_v + i$. Write $\xi_{v,k}$ for the cost of $[v,k]$. The *code* of the rooted hypertree is

$$\omega = \big(s_o, (\xi_{v,k})_{v\in V,\, k\geq 1}\big) \in \Omega := \mathbb{Z}/3 \times \mathcal{S}^V,$$

where $\mathcal{S}$ is the set of strictly increasing sequences in $[0,\infty)$ that tend to infinity, a Borel subset of $[0,\infty)^{\mathbb{N}}$. Coding is a bijection between rooted hypertrees with (R0), up to root-preserving isomorphism, and the Borel subset $\Omega_0 \subset \Omega$ of codes in which no child team of an agent has the same cost as the team through which the agent was reached; the Poisson construction below takes values in Ω_0 almost surely. The set Ω carries its product Borel σ-field and is a standard Borel space.

A *decoration* is a family of marks on teams, on agents or on incidences with values in a Polish space (messages $X_{v\to g}$, weights x_e, indicators m_e, prices u_v). In the code it is a family indexed by addresses, so a decorated code is a point of a countable product of Polish spaces. A *random decorated rooted hypertree* is a random decorated code.

Remark 6.1 (truncated balls)**.** For $K > 0$ let T_K be the component of the root after the teams of cost above K are deleted (written $\mathbb{T}_K$ for the Poisson hypertree), and let $B_r(T_K, o)$ be the rooted marked hypergraph formed by the teams of T_K that contain an agent at distance at most $r-1$ from o in T_K, with sectors and costs, up to isomorphism. The product σ-field on Ω equals the σ-field generated by the maps $\omega \mapsto B_r(T_K, o)$, $r, K \in \mathbb{N}$. Indeed, a truncated ball is a measurable function of the code: it reads coordinates

selected by comparisons with K. Conversely, truncation at K removes the most expensive teams at each agent and so preserves ranks; hence $\{\xi_{v,k} \le t\}$ is the event that for all large integers K the ball $B_{|v|+1}(T_K, o)$ contains the agent with address v and that agent has at least k teams of cost at most t other than its parent team.

6.2. **Invariance.** Let $\mathfrak{n}$ be a decorated code and $b = (k, i)$ a neighbour of the root. Put $\rho(\mathfrak{n}, k, i) = (\mathfrak{n}', k', i')$, where $\mathfrak{n}'$ is the code of the same decorated hypertree rooted at b, the marks staying on their teams, agents and incidences, and (k', i') is the address of o in $\mathfrak{n}'$. Explicitly $k' = 1 + \#\{j : \xi_{b,j} < \xi_{\varnothing,k}\}$ and $i' = 3 - i$, and every coordinate of $\mathfrak{n}'$ is a coordinate of $\mathfrak{n}$ selected by finitely many such comparisons. So ρ is a measurable involution of the space of doubly rooted codes $\{(\mathfrak{n}, k, i)\}$. For a measurable function $f \ge 0$ on that space write

$$f(\mathfrak{n}; o, b) = f(\mathfrak{n}, k, i), \qquad f(\mathfrak{n}; b, o) = f(\rho(\mathfrak{n}, k, i)).$$

Definition 6.2 (invariance)**.** A random decorated rooted hypertree $\mathfrak{N}$ is *invariant* if for every measurable $f \ge 0$ on doubly rooted codes

$$\mathbb{E} \sum_{b \sim o} f(\mathfrak{N}; o, b) = \mathbb{E} \sum_{b \sim o} f(\mathfrak{N}; b, o). \tag{6.1}$$

The sums are over all neighbours $b = (k, i)$ of the root, and both sides may be $+\infty$.

Identity (6.1) is a mass-transport identity at the level of teams: mass is sent only between agents that share a team. It is the only form of unimodularity used in this paper. Its equivalence with mass transport between agents at arbitrary distance is not needed and not claimed.

Lemma 6.3 (null sets transfer)**.** *Let $\mathfrak{N}$ be invariant and let A be a measurable set of decorated rooted codes with $\mathbb{P}(\mathfrak{N} \in A) = 0$. Then almost surely no re-rooting of $\mathfrak{N}$, at any agent, lies in A.*

Proof. Fix $K > 0$ and let $f(\mathfrak{n}; a, b) = \mathbf{1}\{\xi_{e(a,b)} \le K\}\, \mathbf{1}_A(\mathfrak{n} \text{ rooted at } a)$. By (6.1),

$$\mathbb{E}\#\{b \sim o :\ \xi_{e(o,b)} \le K,\ \mathfrak{N} \text{ rooted at } b \text{ lies in } A\} \\ = \mathbb{E}\big[\mathbf{1}_A(\mathfrak{N}) \cdot 2\,\#\{e \ni o : \xi_e \le K\}\big] = 0.$$

Let $K \to \infty$: almost surely no re-rooting at a neighbour of the root lies in A. So the set A_1 of codes that have a re-rooting at a neighbour of the root in A is measurable and null. Apply the same argument to A_1 and induct on the distance from the root. □

6.3. **The Poisson hypertree and matchings on it.** Let Λ be the measure $\Lambda(\,\mathrm{d}\xi) = \frac{1}{2}\xi^2\,\mathrm{d}\xi$ on $[0, \infty)$.

Definition 6.4 (Poisson hypertree)**.** Let s_o be uniform on $\mathbb{Z}/3$ and let $(\Pi_v)_{v\in V}$ be independent Poisson processes on $[0,\infty)$ with intensity Λ, independent of s_o. Let $\xi_{v,k}$ be the k-th point of Π_v. Since $\Lambda([0,K]) = K^3/6$ is finite for every K and $\Lambda([0,\infty)) = \infty$, the code $(s_o, (\xi_{v,k}))$ lies in Ω almost surely. The random rooted hypertree with this code is the *Poisson hypertree* $\mathbb{T} = (\mathbb{T}, o, \xi)$.

Lemma 6.5. *The Poisson hypertree is invariant.*

The proof is given after Lemma 6.18, of which it is a special case.

Let H be a hypertree. A *fractional perfect matching* of H is a family of weights $x_e \in [0,1]$ on the teams with $\sum_{e\ni v} x_e = 1$ at every agent v; it is a *perfect matching* if every x_e is 0 or 1, that is, if the teams of weight one are pairwise disjoint and cover every agent. An *invariant fractional perfect matching* is an invariant random decorated rooted hypertree (T, o, ξ, x), possibly with further decorations, with (R0), in which the decoration x is almost surely a fractional perfect matching. It is *on* $\mathbb{T}$ if (T, o, ξ) is the Poisson hypertree. By Lemma 6.3 it is enough to require $x_e \in [0,1]$ for the teams at the root and $\sum_{e\ni o} x_e = 1$ almost surely. The decoration may use randomness beyond the tree and its costs. The *cost* of x is

$$c(x) = \mathbb{E}\sum_{e\ni o} \xi_e\, x_e \in [0,\infty], \tag{6.2}$$

the expected cost of the root's team when x is a perfect matching. Remark 6.23 relates c to the deficit of an assignment in the market.

6.4. **Decomposition of fractional perfect matchings.** The deterministic part comes first. Let H be a hypertree (countable degrees; (R0) and costs are not needed), x a fractional perfect matching of H and o an agent. Assign to every agent w a *state* $t(w)$, which is a team containing w or a cemetery state †, as follows.

- $t(o) = e$ with probability x_e, for $e \ni o$.
- Let $w \neq o$ have parent team f (the first team on the path to o) and parent agent p. If $t(p) = f$ then $t(w) = f$. If $t(p) \neq f$ and $x_f < 1$, then $t(w) = g$ with probability $x_g/(1-x_f)$, for $g \ni w$, $g \neq f$; this is a probability vector because $\sum_{g\ni w,\, g\neq f} x_g = 1 - x_f$. If $t(p) \neq f$ and $x_f = 1$, then $t(w) = \dagger$.
- Conditionally on the states of all agents at distance at most n from o, the states of the agents at distance $n+1$ are drawn independently.

So t is a Markov chain indexed by the rooted tree; its law on the product σ-field exists by the Ionescu-Tulcea theorem (one countable product of kernels per generation). Let $m = \{t(w) : w \text{ an agent},\ t(w) \neq \dagger\}$ and let $\mu^o_{H,x}$ be the law of m, a probability measure on sets of teams.

Lemma 6.6 (the chain)**.** (i) *$\mu^o_{H,x}$-almost surely m is a perfect matching of H.*

(ii) $\mu^o_{H,x}(t(w) = g) = x_g$ *for every incidence* (w, g)*; in particular* $\mu^o_{H,x}(e \in m) = x_e$ *for every team* e*.*
(iii) $\mu^o_{H,x}$ *does not depend on* o*; write* $\mu_{H,x}$*.*
(iv) *If* $\varphi : (H, x) \to (H', x')$ *is an isomorphism, then* $\varphi_* \mu_{H,x} = \mu_{H',x'}$*.*

Proof. Step 1 (consistency). If $t(w) = g$ for one agent w of a team g, then $t = g$ at all three agents of g. Indeed, let p be the agent of g nearest to o; the other two agents of g have parent team g and parent agent p. Such an agent has state g only if $t(p) = g$, and if $t(p) = g$ then both have state g. Hence two teams of m that share an agent w both equal $t(w)$: the teams of m are pairwise disjoint, and they cover every agent whose state is not †.

Step 2 (marginals). We show $\mathbb{P}(t(w) = g) = x_g$ by induction on the distance from o. At o it is the definition. Let w have parent team f and parent agent p. First, $\mathbb{P}(t(w) = f) = \mathbb{P}(t(p) = f) = x_f$. Let $g \ni w$, $g \neq f$. If $x_f < 1$, then $\mathbb{P}(t(w) = g) = \mathbb{P}(t(p) \neq f) \cdot x_g/(1 - x_f) = x_g$. If $x_f = 1$, then $x_g = 0$, and $\mathbb{P}(t(w) = g) = 0$ by the rule for that case. This proves (ii). Summing over $g \ni w$ gives $\mathbb{P}(t(w) = \dagger) = 0$ for every w, and there are countably many agents; with Step 1 this proves (i). (The rule for $x_f = 1$ is applied with probability $\mathbb{P}(t(p) \neq f) = 0$, so any other convention gives the same measure.)

Step 3 (root independence). Let $f = \{o, o', o''\}$ be a team at o. If $x_f = 1$ then under μ^o and under $\mu^{o'}$ the three states are (f, f, f) almost surely. Let $x_f < 1$. Under μ^o the triple $(t(o), t(o'), t(o''))$ equals (f, f, f) with probability x_f and equals (e, g, h), with e, g, h all different from f, with probability $x_e\, x_g\, x_h/(1 - x_f)^2$: the root draws e with probability x_e, and then o' and o'' draw independently from their kernels. This expression is symmetric in the roles of o, o', o'', so it is also the law of the triple under $\mu^{o'}$. Given the triple, under either root the restrictions of t to the three branches hanging from f (the branch at $a \in f$ consists of the agents separated from f by a, with a itself) are conditionally independent, and the restriction to the branch at a is the chain started from $t(a)$ at a and run with the transition rule above in which "parent" is read as "neighbour toward f". That rule is the same for the roots o and o', because inside the branch at a the directions toward o, toward o' and toward f coincide. So $\mu^o = \mu^{o'}$, and since H is connected μ^o does not depend on o.

Step 4 (equivariance). φ carries the construction with root o to the construction with root $\varphi(o)$; now use (iii). □

Measurability. In rank coding the construction is a measurable function $m = \mathfrak{m}(\omega, x, U)$ of the code ω, the weights x and a family $U = (U_v)_{v \in V}$ of independent uniform random variables, independent of (ω, x): the agent at address v uses U_v and the cumulative sums of its kernel along the rank order of its teams. (Where x is not a fractional perfect matching put $m = \emptyset$.) For a finite set of addresses and a prescribed pattern on it, $(\omega, x) \mapsto \mathbb{P}(\mathfrak{m}(\omega, x, U)$ shows the pattern) is measurable by Fubini's theorem, and such patterns generate the σ-field on sets of teams. So $(\omega, x) \mapsto$

$\mu_{(\omega,x)}$ is a probability kernel. By Lemma 6.6(iii),(iv) it commutes with re-rooting: the image of $\mu_{(\omega,x)}$ under the relabelling of addresses that ρ induces is the measure μ at the re-rooted code.

Theorem 6.7 (decomposition)**.** *Let (T,o,ξ,x) be an invariant fractional perfect matching. Conditionally on (T,o,ξ,x) draw m from $\mu_{(T,\xi,x)}$, independently of everything else, and write $m_e=\mathbf{1}\{e\in m\}$. Then (T,o,ξ,x,m) is invariant, m is almost surely a perfect matching, $\mathbb{E}[m_e\mid T,o,\xi,x]=x_e$ for every team e, and $c(m)=c(x)$ in $[0,\infty]$. In particular every invariant fractional perfect matching on $\mathbb{T}$ is the conditional expectation, given the tree, the costs and x, of an invariant random perfect matching on $\mathbb{T}$ of the same cost.*

Proof. Write $\mathfrak{n}$ for the code of (T,o,ξ,x). *Invariance.* Let $f\ge 0$ be measurable on doubly rooted codes decorated with (x,m) and put

$$F(\mathfrak{n};a,b)=\int f\big((\mathfrak{n},m');a,b\big)\,\mu_{\mathfrak{n}}(\,\mathrm{d}m').$$

F is measurable because μ is a kernel. Because the kernel commutes with re-rooting, $\int f((\mathfrak{n},m');b,o)\,\mu_{\mathfrak{n}}(\,\mathrm{d}m')$ is the value of F at the re-rooted code, that is, $F(\mathfrak{n};b,o)$ in the notation of Section 6.2. Conditioning on (T,o,ξ,x) and using Tonelli's theorem,

$$\mathbb{E}\sum_{b\sim o} f(\cdot,m;o,b)=\mathbb{E}\sum_{b\sim o}F(\cdot;o,b),\qquad \mathbb{E}\sum_{b\sim o} f(\cdot,m;b,o)=\mathbb{E}\sum_{b\sim o}F(\cdot;b,o),$$

and the two right-hand sides agree by (6.1) for (T,o,ξ,x). *Matching and marginals.* Lemma 6.6(i),(ii), applied conditionally on (T,o,ξ,x). *Cost.* By Tonelli's theorem, $c(m)=\mathbb{E}\sum_{e\ni o}\xi_e\,\mathbb{E}[m_e\mid T,o,\xi,x]=\mathbb{E}\sum_{e\ni o}\xi_e x_e=c(x)$. □

6.5. **Weak duality.**

Lemma 6.8 (weak duality)**.** *Let (T,o,ξ,x,u) be* jointly invariant*: one random decorated rooted hypertree with (R0), with weights $x_e\in[0,1]$ on teams and prices $u_v\in\mathbb{R}$ on agents, that satisfies* (6.1) *for functions of all of (T,ξ,x,u). Assume that almost surely $\sum_{e\ni o}x_e=1$ and $u_o+u_b+u_c\le\xi_e$ for every team $e=\{o,b,c\}$ at the root, and that $\mathbb{E}|u_o|<\infty$. Then*

$$3\,\mathbb{E}u_o\le c(x)=\mathbb{E}\sum_{e\ni o}\xi_e\,x_e.$$

Proof. Apply (6.1) to $f_\pm(\mathfrak{n};a,b)=x_{e(a,b)}\,(u_b)_\pm$, where $(\cdot)_\pm$ are the positive and negative parts:

$$\mathbb{E}\sum_{b\sim o}x_{e(o,b)}(u_b)_\pm=\mathbb{E}\sum_{b\sim o}x_{e(o,b)}(u_o)_\pm=\mathbb{E}\Big[(u_o)_\pm\cdot 2\sum_{e\ni o}x_e\Big]=2\,\mathbb{E}(u_o)_\pm<\infty.$$

So $\sum_{e=\{o,b,c\}\ni o}x_e(u_b+u_c)$ is integrable with mean $2\,\mathbb{E}u_o$, and, using $\sum_{e\ni o}x_e=1$,

$$3\,\mathbb{E}u_o=\mathbb{E}\sum_{e=\{o,b,c\}\ni o}x_e\,(u_o+u_b+u_c)\le\mathbb{E}\sum_{e\ni o}x_e\,\xi_e.$$ □

Corollary 6.9 (prices that are a function of the tree)**.** *Let h be a measurable function of rooted codes of (T,ξ) and put $u_v = h(T \text{ rooted at } v)$. If (T,o,ξ,x) is an invariant fractional perfect matching, the prices are almost surely dual feasible at the root ($u_o+u_b+u_c \le \xi_e$ for every team $e=\{o,b,c\}$) and $\mathbb{E}|u_o|<\infty$, then $3\,\mathbb{E}u_o \le c(x)$.*

Proof. Every function of $(T,\xi,x,u;o,b)$ is a function of $(T,\xi,x;o,b)$, so (T,o,ξ,x,u) is jointly invariant and Lemma 6.8 applies. □

Remark 6.10. Joint invariance is what the proof uses: the transported quantity $x_e u_b$ involves x and u together. If x and u are each invariant together with (T,ξ) but use separate additional randomness with an unspecified coupling, (6.1) for the pair is not given.

6.6. **The recursion map.** Throughout the rest of this section $(\xi_k)_{k\ge1}$ are the points of a Poisson process on $[0,\infty)$ with intensity Λ, and $(X'_k,X''_k)_{k\ge1}$ are independent pairs of independent variables with law μ, independent of (ξ_k). For a probability law μ on $\mathbb{R}$ such that $\inf_k(\xi_k - X'_k - X''_k)$ is almost surely finite, $\Theta(\mu)$ is the law of this infimum. A *fixed point* is a probability law μ on $\mathbb{R}$ for which the infimum is almost surely finite and $\Theta(\mu)=\mu$. For a fixed point, X,X',X'',X_1,X_2,X_3 denote independent variables with law μ, and

$$Y = X' + X'', \qquad S = X_1+X_2+X_3.$$

Lemma 6.11 (what the fixed-point equation forces)**.** *Let μ be a fixed point and put*

$$G(x) = \tfrac{1}{6}\,\mathbb{E}\big[(x+X'+X'')_+^3\big], \qquad x\in\mathbb{R}. \tag{6.3}$$

(i) *$G(x)<\infty$ for every x, and $\mathbb{P}(X>x)=\exp(-G(x))$. The function G is convex and twice continuously differentiable, with $G'(x)=\frac{1}{2}\mathbb{E}(x+Y)_+^2>0$ and $G''(x)=\mathbb{E}(x+Y)_+$, and it is an increasing bijection of $\mathbb{R}$ onto $(0,\infty)$. In particular μ has no atoms.*

(ii) *Almost surely, for every x only finitely many k have $\xi_k - X'_k - X''_k \le x$, the infimum is a minimum, and it is attained at exactly one k.*

(iii) *Fix m with $p := \mathbb{P}(Y>-m)>0$. Then $\mathbb{P}(X>x)\le \exp(-p(x-m)^3/6)$ for $x>m$, and for every integer $n\ge1$*

$$\mathbb{E}X_-^n \le \frac{\mathbb{E}Y_+^{n+3}}{(n+1)(n+2)(n+3)} < \infty.$$

Hence $\mathbb{E}|X|^n<\infty$ for every n; in particular $\mathbb{E}|X|<\infty$ and $\mathbb{E}S_+^4<\infty$.

Proof. Put $Y_k = X'_k + X''_k$. By the marking theorem the points (ξ_k,Y_k) form a Poisson process on $[0,\infty)\times\mathbb{R}$ with intensity $\Lambda\otimes\mathrm{law}(Y)$. So $N(x) := \#\{k : \xi_k - Y_k \le x\}$ is a Poisson variable with mean

$$\int_0^\infty \tfrac{1}{2}\xi^2\,\mathbb{P}(Y\ge\xi-x)\,\mathrm{d}\xi = \mathbb{E}\int_0^\infty \tfrac{1}{2}\xi^2\,\mathbf{1}\{\xi\le x+Y\}\,\mathrm{d}\xi = G(x)\in[0,\infty].$$

(i) Suppose $G(x_0) = \infty$ for some x_0. For $x < x_0$ we have $(x_0 + Y)_+ \le (x+Y)_+ + (x_0 - x)$, so $(x_0+Y)_+^3 \le 4(x+Y)_+^3 + 4(x_0-x)^3$ and $G(x) = \infty$; and G is nondecreasing, so $G \equiv \infty$. A Poisson variable with infinite mean is almost surely infinite, so $N(x) = \infty$ almost surely for every integer x and the infimum is $-\infty$ almost surely. This contradicts the hypothesis that the infimum is almost surely finite. So G is finite. Then $N(x+1) < \infty$ almost surely, so the infimum over the points with $\xi_k - Y_k \le x+1$ is a minimum, and $\{\inf_k(\xi_k - Y_k) > x\} = \{N(x) = 0\}$; hence $\mathbb{P}(X > x) = \mathbb{P}(N(x) = 0) = \exp(-G(x))$. Finiteness of G gives $\mathbb{E}Y_+^3 < \infty$, and differentiation under the expectation (the derivatives $\frac{1}{2}(x+Y)_+^2$ and $(x+Y)_+$ are dominated, for $x \le x_0$, by their values at x_0) gives the formulas for G' and G'' and their continuity. G is convex because $x \mapsto (x+y)_+^3$ is. Since $\mathbb{P}(X > x) = \exp(-G(x)) > 0$ for every x, the variable X is unbounded above; so is Y, and therefore $\mathbb{P}(Y > -x) > 0$ and $G'(x) > 0$ for every x. Finally $G(x) \to 0$ as $x \to -\infty$ by dominated convergence and $G(x) \to \infty$ as $x \to \infty$ by monotone convergence.

(ii) $N(x) < \infty$ almost surely for every integer x, and N is nondecreasing; so almost surely finitely many points lie below every level and the infimum is attained. By the mapping theorem the values $Z_k = \xi_k - Y_k$ form a Poisson process on $\mathbb{R}$ whose intensity measure has distribution function G. As G is continuous and finite, this measure is diffuse and locally finite, and such a Poisson process is almost surely simple: no two Z_k coincide. So the minimiser is unique.

(iii) For $x > m$, $G(x) \ge \frac{1}{6}\mathbb{E}[(x+Y)_+^3;\, Y > -m] \ge p(x-m)^3/6$. So X_+ has moments of every order, and so has $Y_+ \le X'_+ + X''_+$. For the lower tail, $\mathbb{P}(X \le -t) = 1 - \exp(-G(-t)) \le G(-t) = \frac{1}{6}\mathbb{E}(Y-t)_+^3$, so

$$\begin{aligned}\mathbb{E}X_-^n &= \int_0^\infty nt^{n-1}\,\mathbb{P}(X \le -t)\,\mathrm{d}t \le \mathbb{E}\int_0^{Y_+} \frac{nt^{n-1}(Y-t)^3}{6}\,\mathrm{d}t \\ &= \frac{n\,\mathrm{B}(n,4)}{6}\,\mathbb{E}Y_+^{n+3} = \frac{\mathbb{E}Y_+^{n+3}}{(n+1)(n+2)(n+3)},\end{aligned}$$

where $\mathrm{B}(n,4) = 6\,(n-1)!/(n+3)!$ is the Beta integral. □

So the tail condition that the construction below needs, $\mathbb{E}Y_+^3 < \infty$, is a consequence of $\Theta(\mu) = \mu$ and not an extra hypothesis. Equation (6.3), read as an equation for $G = -\ln \mathbb{P}(X > \cdot\,)$ in which X', X'' have survival function $\exp(-G)$, is the fixed-point equation in the form used in Section 7.

6.7. **From a fixed point to an invariant perfect matching.**

Theorem 6.12 (the tree process)**.** *Let μ be a fixed point of Θ. Then there is an invariant random perfect matching M on the Poisson hypertree: a random decorated rooted hypertree $(\mathbb{T}, o, \xi, M)$ that is invariant, whose $(\mathbb{T}, o, \xi)$-marginal is the Poisson hypertree, and in which M is almost surely a perfect*

matching. Its cost (6.2) *is*

$$c(M) = \tfrac{1}{8}\,\mathbb{E}S_+^4 = 9\,\mathbb{E}X, \qquad \textit{and moreover} \qquad \mathbb{E}S_+^3 = 6.$$

The proof occupies the rest of this subsection. The matching is defined from a family of real *messages*, one on each incidence, which solve the recursion (6.4) below in both directions. The messages are constructed first on incidences that point toward the root (Lemma 6.13), then on all incidences (Lemma 6.16); invariance is proved through a Palm formula (Lemmas 6.17 and 6.18); the matching and its cost follow. Throughout, μ is a fixed point and G is the function (6.3).

Inward messages.

Lemma 6.13 (inward messages)**.** *There is a probability space carrying the Poisson hypertree $(s_o, (\Pi_v)_{v\in V})$ and real random variables $(X_v)_{v\in V}$ such that*

(a) *for every $n \geq 0$ the variables $(X_v)_{|v|=n}$ are independent with law μ and independent of $(s_o, (\Pi_w)_{|w|<n})$;*

(b) *almost surely, for every $v \in V$, $X_v = \min_{k\geq 1}\big(\xi_{v,k} - X_{(v,k,1)} - X_{(v,k,2)}\big)$, and the minimum is attained at exactly one k.*

The joint law of $(s_o, (\Pi_v), (X_v))$ is uniquely determined by μ.

For $v \neq \varnothing$, X_v will be the message from v into its parent team. $X_\varnothing$ is an auxiliary variable, the root's own minimum over all its teams.

Proof. Depth-n laws. For $n \geq 0$ let $V_{<n}$ and V_n be the sets of addresses of length less than n and equal to n, and $\mathcal{E}_n = \mathbb{Z}/3 \times \mathcal{S}^{V_{<n}} \times \mathbb{R}^{V_n}$. Both index sets are countable, so $\mathcal{E}_n$ is a countable product of standard Borel spaces. Let P_n be the product measure on $\mathcal{E}_n$ under which s_o is uniform, each Π_w ($|w| < n$) is a Poisson process of intensity Λ and each X_v ($|v| = n$) has law μ. A countable product of probability measures needs no truncation of the cost level: this is where the infinite degree is absorbed.

Bonding maps. Define $\Phi_n : \mathcal{E}_{n+1} \to \mathcal{E}_n$ by keeping s_o and Π_w for $|w| < n$ and setting, for $|v| = n$, $X_v := \inf_k(\xi_{v,k} - X_{(v,k,1)} - X_{(v,k,2)})$, with the value 0 where the infimum is $-\infty$. Φ_n is measurable.

Consistency: $\Phi_n(P_{n+1}) = P_n$. Under P_{n+1} the blocks

$$\big(\Pi_v, (X_{(v,k,i)})_{k\geq 1,\, i=1,2}\big), \qquad |v| = n,$$

are independent of one another and of $(s_o, (\Pi_w)_{|w|<n})$, because they are built from disjoint sets of coordinates of a product measure. In each block Π_v is a Poisson process of intensity Λ and the $X_{(v,k,i)}$ are independent with law μ, independent of Π_v. So X_v, a function of the block of v, has law $\Theta(\mu) = \mu$. Hence under $\Phi_n(P_{n+1})$ the variables $(X_v)_{|v|=n}$ are independent with law μ and independent of $(s_o, (\Pi_w)_{|w|<n})$: this is P_n.

Extension. For $N \geq 0$ let Q_N be the law on $\mathcal{E}_0 \times \cdots \times \mathcal{E}_N$ of $(\Phi_0 \circ \cdots \circ \Phi_{N-1}(\eta), \ldots, \Phi_{N-1}(\eta), \eta)$ with $\eta \sim P_N$. By consistency the projection of Q_{N+1} on $\mathcal{E}_0 \times \cdots \times \mathcal{E}_N$ is Q_N. Kolmogorov's extension theorem gives a

probability measure Q on $\prod_n \mathcal{E}_n$ with these marginals. Under Q, almost surely $\eta_n = \Phi_n(\eta_{n+1})$ for every n. So the coordinates s_o and Π_w of η_n do not depend on $n > |w|$. Let X_v be the X_v-coordinate of $\eta_{|v|}$. Then (a) holds because $\eta_n \sim P_n$, and the equation in (b) holds because the X_v-coordinate of $\Phi_n(\eta_{n+1})$ is the infimum over the messages of level $n+1$. By (a) at level $|v|+1$ and Lemma 6.11(ii), that infimum is almost surely a minimum attained at a unique k; there are countably many v.

Uniqueness. By (b), for every n the family $(s_o, (\Pi_w)_{|w|<n}, (X_v)_{|v|\le n})$ is almost surely a fixed measurable function of $(s_o, (\Pi_w)_{|w|<n}, (X_v)_{|v|=n})$, whose law is P_n by (a). These families generate the σ-field. □

Branches. A *branch* is an agent with all its descendants, their costs and their inward messages. Its code is $\beta = ((\xi_{v,k})_{v\in V,\, k\ge 1}, (X_v)_{v\in V})$, without a sector, a point of the standard Borel space $\mathfrak{S} := \mathcal{S}^V \times \mathbb{R}^V$; $X(\beta) := X_\varnothing$ is its *root message*. For $u \in V$ let $\mathfrak{X}_u$ be the branch at u in the process of Lemma 6.13, with addresses read relative to u, and let L be the law of $\mathfrak{X}_\varnothing$.

Lemma 6.14 (independent branches)**.** *For every $n \ge 0$ the branches $(\mathfrak{X}_u)_{|u|=n}$ are independent with law L, and independent of $(s_o, (\Pi_w)_{|w|<n})$. In particular $\mathfrak{X}_\varnothing$ consists of $\Pi_\varnothing$ and of the branches $\mathfrak{X}_{(k,i)}$, which are independent with law L and independent of $\Pi_\varnothing$, together with $X_\varnothing = \min_k\big(\xi_{\varnothing,k} - X(\mathfrak{X}_{(k,1)}) - X(\mathfrak{X}_{(k,2)})\big)$.*

Proof. Write $w \succeq u$ if u is a prefix of w. Fix $N \ge n$. For $|u| = n$ the depth-N truncation of $\mathfrak{X}_u$ and its input are

$$\big((\Pi_w)_{w\succeq u,\,|w|<N}, (X_w)_{w\succeq u,\,|w|\le N}\big) \quad\text{and}\quad \big((\Pi_w)_{w\succeq u,\,|w|<N}, (X_w)_{w\succeq u,\,|w|=N}\big).$$

By Lemma 6.13(b) the truncation is one fixed measurable function of the input. Under P_N these inputs, for the different u of length n, are independent, independent of $(s_o, (\Pi_w)_{|w|<n})$, and each has, after shifting addresses, the law of the corresponding input at the root under P_{N-n}. So the truncations are independent, each with the law of the depth-$(N-n)$ truncation of $\mathfrak{X}_\varnothing$, and independent of $(s_o, (\Pi_w)_{|w|<n})$. The truncations increase with N and generate the σ-field of the branch; a π-system argument finishes the proof. The last sentence is the case $n = 1$ with Lemma 6.13(b). □

Independence at fixed addresses is not the whole story for messages in both directions: the k-th cheapest team at an agent is not independent of the other teams there, since those of lower rank are cheaper. The correct two-directional statement is the Palm formula of Lemma 6.17: seen from a typical team, its cost and the three branches hanging from it are independent, and the branches are independent with law L.

Messages in both directions. For a team g and $v \in g$ write $g \setminus v$ for the set of the other two agents. The target is a family $\mathfrak{M} = (X_{v\to g})$ of real numbers

indexed by all incidences such that, at every incidence (v, g),

$$X_{v\to g} = \min_{g'\ni v,\ g'\neq g}\Big(\xi_{g'} - \sum_{w\in g'\setminus v} X_{w\to g'}\Big). \tag{6.4}$$

Definition 6.15 (the message family)**.** *Inward:* $X_{v\to e(v)} := X_v$ for $v \neq \varnothing$. *Outward,* by induction on $|v|$:

$$X_{\varnothing\to[\varnothing,k]} := \min_{j\neq k}\big(\xi_{\varnothing,j} - X_{(j,1)} - X_{(j,2)}\big),$$

$$X_{v\to[v,k]} := \min\Big\{\xi_{e(v)} - X_{p(v)\to e(v)} - X_{\mathrm{sib}(v)},\ \min_{j\neq k}\big(\xi_{v,j} - X_{(v,j,1)} - X_{(v,j,2)}\big)\Big\},$$

the second line for $v \neq \varnothing$.

By Lemma 6.13(a) and Lemma 6.11(ii), almost surely at every agent only finitely many of the values $\xi_{v,j} - X_{(v,j,1)} - X_{(v,j,2)}$ lie below any level. So each of these minima is finite and attained, and each message is a measurable function of $\mathfrak{X}_\varnothing$.

Lemma 6.16 (the recursion in both directions)**.** (i) *Almost surely,* (6.4) *holds for* $\mathfrak{M}$ *at every incidence.*

(ii) *Let* b *be an agent. A family satisfying* (6.4) *at every incidence, with all minima finite, is determined by its values on the incidences that point toward* b*. The same holds with a team* e *in place of* b*.*

Proof. (i) For the incidence $(v, e(v))$ the teams $g' \neq e(v)$ at v are its child teams $[v, k]$, whose other agents send their inward messages; so (6.4) at this incidence is Lemma 6.13(b). For $(v, [v, k])$ the teams $g' \neq [v, k]$ at v are the other child teams and, if $v \neq \varnothing$, the parent team $e(v)$, into which $p(v)$ sends $X_{p(v)\to e(v)}$ and $\mathrm{sib}(v)$ sends $X_{\mathrm{sib}(v)}$; so (6.4) at this incidence is Definition 6.15.

(ii) Let (v, g) point away from b. In (6.4) the teams $g' \neq g$ at v are of two kinds. If g' is not the first team on the path from v to b, then b and v lie on the same side of g', so both incidences (w, g'), $w \in g' \setminus v$, point toward b. If g' is that first team (then $v \neq b$), $g' \setminus v$ consists of the next agent p on the path, with (p, g') pointing away from b and p strictly closer to b than v, and of a third agent s, with (s, g') pointing toward b. So $X_{v\to g}$ is a function of messages that point toward b and of at most one message that points away from b at an agent closer to b. Induct on the distance from b; at $v = b$ only the first kind occurs. For a team e the argument is the same, by induction on the distance from e: at an agent $v \in e$ and an incidence (v, g) with $g \neq e$, the team $g' = e$ contributes the two messages into e from the other agents of e, which point toward e. □

So the family $\mathfrak{M}$ is an object on the unrooted hypertree: each message is the stated function of the messages one step further out, in both directions on every incidence, and re-rooting only relabels it. Let

$$\mathfrak{N} = (\mathbb{T}, o, \xi, \mathfrak{M})$$

be the Poisson hypertree decorated with all messages. It is a measurable function of $(s_o, \mathfrak{X}_\varnothing)$.

The Palm formula and invariance. For a branch code β and $k \geq 1$, $i \in \{1,2\}$, let $\beta^{(k,i)}$ be its sub-branch at (k,i), and let $\beta^{\setminus k}$ be the branch obtained by deleting the root's k-th child team with its two sub-branches, re-indexing the remaining child teams by rank and recomputing the root message as the minimum over the remaining teams. Both are measurable in β. For the team $e = [\varnothing, k]$ at the root, with other agents $b = (k,1)$ and $c = (k,2)$, the branches hanging from e at o, b, c are $\mathfrak{X}_\varnothing^{\setminus k}, \mathfrak{X}_b, \mathfrak{X}_c$, and their root messages are $X_{o\to e}, X_{b\to e}, X_{c\to e}$ (Definition 6.15, first line).

Lemma 6.17 (Palm formula)**.** *For every measurable $F \geq 0$ on $[0,\infty)\times\mathfrak{S}^3$,*

$$\mathbb{E}\sum_{k\geq 1} F\big(\xi_{\varnothing,k}; \mathfrak{X}_\varnothing^{\setminus k}, \mathfrak{X}_{(k,1)}, \mathfrak{X}_{(k,2)}\big) = \int_0^\infty \Lambda(\,\mathrm{d}\xi) \int L(\,\mathrm{d}\beta_0) L(\,\mathrm{d}\beta_1) L(\,\mathrm{d}\beta_2)\, F(\xi;\beta_0,\beta_1,\beta_2), \tag{6.5}$$

and the same holds with s_o as an additional argument of F on the left and an independent uniform element of $\mathbb{Z}/3$ on the right.

Proof. By Lemma 6.14 and the marking theorem, $\Xi := \sum_k \delta_{(\xi_{\varnothing,k}, \mathfrak{X}_{(k,1)}, \mathfrak{X}_{(k,2)})}$ is a Poisson process on $[0,\infty)\times\mathfrak{S}^2$ with intensity $\Lambda\otimes L\otimes L$, which is σ-finite. There is a measurable map $\mathfrak{b}$ from counting measures to branch codes with $\mathfrak{X}_\varnothing = \mathfrak{b}(\Xi)$ (rank the points by their first coordinate, attach the marks, take the minimum as root message), and then $\mathfrak{X}_\varnothing^{\setminus k} = \mathfrak{b}(\Xi - \delta_{x_k})$, where x_k is the k-th point of Ξ. Mecke's formula

$$\mathbb{E}\int \tilde{F}(x, \Xi - \delta_x)\, \Xi(\,\mathrm{d}x) = \int \mathbb{E}\tilde{F}(x,\Xi)\, (\Lambda\otimes L\otimes L)(\,\mathrm{d}x),$$

applied to $\tilde{F}((\xi,\beta_1,\beta_2),\varphi) = F(\xi;\mathfrak{b}(\varphi),\beta_1,\beta_2)$, gives (6.5), because $\mathfrak{b}(\Xi) = \mathfrak{X}_\varnothing$ has law L. The sector s_o is independent of everything else. □

Lemma 6.18 (invariance of the message process)**.** *$\mathfrak{N} = (\mathbb{T}, o, \xi, \mathfrak{M})$ is invariant in the sense of Definition 6.2.*

Proof. For an ordered triple (a_1,a_2,a_3) of distinct agents forming a team e define the *view*

$$\mathfrak{v}(a_1,a_2,a_3) = (\sigma; \xi_e; \beta_1, \beta_2, \beta_3),$$

where $\sigma = (s_{a_1}, s_{a_2}, s_{a_3})$ is the triple of sectors, an ordering of $\mathbb{Z}/3$, and β_i is the branch hanging from e at a_i: the agents separated from e by a_i, together with a_i, with their costs, coded by rank from a_i outward, and with the messages on their incidences that point toward e. The branch codes do not depend on a root: inside the branch at a_i, "toward e" and "toward a_j" are the same direction for every $j \neq i$.

Step 1: the doubly rooted network is a fixed function of the view. There is one measurable map Ψ such that, almost surely, for every team e at the

root and every ordering (a_1, a_2, a_3) of its agents, $\Psi(\mathfrak{v}(a_1, a_2, a_3))$ is the code of $\mathfrak{N}$ re-rooted at a_1 together with the address of a_2 in that code. Indeed, the hypertree with its sectors and costs is obtained by gluing the three branches to a team of cost ξ_e; the messages on incidences pointing toward e are those carried by the branches, the three messages into e being the root messages $X(\beta_i)$; and the messages on incidences pointing away from e are obtained from them by the recursion (6.4), outward from e, as in the proof of Lemma 6.16(ii). This defines Ψ (with the value 0 for a message whose minimum is not attained), and each coordinate of its output is obtained from coordinates of the view by countably many comparisons and minima, so Ψ is measurable. By Lemma 6.16(i) the family $\mathfrak{M}$ satisfies (6.4) everywhere, so by Lemma 6.16(ii) the family reconstructed by Ψ is $\mathfrak{M}$, whichever agent of e comes first.

Step 2: the view measure is symmetric. For o the root, $e = [\varnothing, k]$, $b = (k, i)$ and $c = (k, 3-i)$ the third agent,

$$\mathfrak{v}(o, b, c) = \big((s_o, s_o + i, s_o - i); \xi_{\varnothing,k}; \mathfrak{X}_{\varnothing}^{\setminus k}, \mathfrak{X}_{(k,i)}, \mathfrak{X}_{(k,3-i)}\big).$$

Sum (6.5) over $i \in \{1, 2\}$, using that s_o is uniform and independent and that $L \otimes L$ is exchangeable. As s_o and i vary, $(s_o, s_o + i, s_o - i)$ runs once through the six orderings of $\mathbb{Z}/3$. So for every measurable $h \geq 0$

$$\mathbb{E} \sum_{b \sim o} h(\mathfrak{v}(o, b, c)) = \frac{1}{3} \sum_{\sigma} \int_0^\infty \Lambda(\,\mathrm{d}\xi) \int L^{\otimes 3}(\,\mathrm{d}\beta)\, h(\sigma; \xi; \beta_1, \beta_2, \beta_3),$$

the sum being over the six orderings σ of $\mathbb{Z}/3$. The right-hand side is unchanged when h is composed with the swap

$$\tau(\sigma; \xi; \beta_1, \beta_2, \beta_3) = \big((\sigma_2, \sigma_1, \sigma_3); \xi; \beta_2, \beta_1, \beta_3\big),$$

because $L^{\otimes 3}$ is exchangeable and the set of orderings is closed under the transposition.

Step 3. $\mathfrak{v}(b, o, c) = \tau(\mathfrak{v}(o, b, c))$. By Step 1, $f(\mathfrak{N}; o, b) = (f \circ \Psi)(\mathfrak{v}(o, b, c))$ and $f(\mathfrak{N}; b, o) = (f \circ \Psi)(\tau\, \mathfrak{v}(o, b, c))$. Apply Step 2 to $h = f \circ \Psi$ and to $h = f \circ \Psi \circ \tau$; the two results are equal, which is (6.1). □

The intensity Λ is σ-finite and not finite. That is harmless: (6.1) is an identity between elements of $[0, \infty]$.

Proof of Lemma 6.5. Repeat the proofs of Lemmas 6.14, 6.17 and 6.18 with the message coordinates deleted: a branch is then an agent with its descendants and their costs, its law is that of $(\Pi_v)_{v \in V}$, the branches at the agents of length n are independent with that law by construction, and Steps 1 to 3 need only the marking theorem and Mecke's formula. No law μ is involved. □

The matching.

Definition 6.19. A team $e = \{a, b, c\}$ belongs to M if and only if $\xi_e < X_{a \to e} + X_{b \to e} + X_{c \to e}$.

Lemma 6.20 (the matching)**.** *Almost surely M is a perfect matching of $\mathbb{T}$: every agent is in exactly one team of M. The decorated hypertree $(\mathbb{T}, o, \xi, M)$ is invariant and its $(\mathbb{T}, o, \xi)$-marginal is the Poisson hypertree. Moreover, almost surely, the team of M at the root is the unique minimiser e^* of $e = \{o, b, c\} \mapsto \xi_e - X_{b\to e} - X_{c\to e}$ over the teams at the root, and the minimum equals $X_\varnothing$.*

Proof. For an agent v and a team $g \ni v$ put $Z_g^v = \xi_g - \sum_{w\in g\setminus v} X_{w\to g}$. By (6.4), $X_{v\to e} = \min_{g\ni v,\, g\neq e} Z_g^v$. The defining inequality of M reads $Z_e^v < X_{v\to e}$; so for each agent v of a team e,

$$e \in M \iff Z_e^v < Z_g^v \text{ for every other team } g \text{ at } v. \tag{6.6}$$

So at most one team of M contains v, and exactly one does if and only if the minimum of $(Z_g^v)_{g\ni v}$ is attained at a single team. At the root, $Z^o_{[\varnothing,k]} = \xi_{\varnothing,k} - X_{(k,1)} - X_{(k,2)}$, and by Lemma 6.13(b) the minimum over k is almost surely attained at a single k and equals $X_\varnothing$. The event that the minimum at the root is not attained at a single team is therefore a null set of codes of $\mathfrak{N}$, and by Lemmas 6.18 and 6.3 it almost surely fails at every agent. Hence almost surely every agent is in exactly one team of M.

M is a measurable function of $\mathfrak{N}$ defined through incidences, with no reference to the root. So every function of $(\mathbb{T}, \xi, M; o, b)$ is a function of $(\mathfrak{N}; o, b)$, and (6.1) for $(\mathbb{T}, o, \xi, M)$ follows from Lemma 6.18. □

Cost.

Proof of Theorem 6.12. The matching M of Definition 6.19 is an invariant random perfect matching on $\mathbb{T}$ by Lemma 6.20. It remains to compute its cost. Apply (6.5) to a function F that depends on the branches only through their root messages, which under $L^{\otimes 3}$ are independent with law μ: for measurable $f \geq 0$, and then for f with $\int \Lambda(\mathrm{d}\xi)\, \mathbb{E}|f(\xi, X_1, X_2, X_3)| < \infty$ by splitting f into positive and negative parts,

$$\mathbb{E} \sum_{e=\{o,b,c\}\ni o} f(\xi_e, X_{o\to e}, X_{b\to e}, X_{c\to e}) = \int_0^\infty \tfrac{1}{2}\xi^2\, \mathbb{E} f(\xi, X_1, X_2, X_3)\, \mathrm{d}\xi. \tag{6.7}$$

First, $f = \mathbf{1}\{\xi < x_1 + x_2 + x_3\}$. The left-hand side of (6.7) is $\mathbb{E}\#\{e \ni o : e \in M\} = 1$, and the right-hand side is $\mathbb{E}\int_0^{S_+} \frac{1}{2}\xi^2\, \mathrm{d}\xi = \mathbb{E}S_+^3/6$. So $\mathbb{E}S_+^3 = 6$.

Second, $f = \xi\, \mathbf{1}\{\xi < x_1 + x_2 + x_3\}$. The left-hand side is $c(M)$ and the right-hand side is $\mathbb{E}\int_0^{S_+} \frac{1}{2}\xi^3\, \mathrm{d}\xi = \mathbb{E}S_+^4/8$, which is finite by Lemma 6.11(iii).

Third, $f = (x_2 + x_3)\, \mathbf{1}\{\xi < x_1 + x_2 + x_3\}$. It is integrable: $\int_0^\infty \frac{1}{2}\xi^2\, \mathbb{E}[|X_2 + X_3|\, \mathbf{1}\{\xi < S\}]\, \mathrm{d}\xi = \mathbb{E}[|X_2 + X_3|\, S_+^3]/6$, and $S_+^3 \leq 9\sum_i (X_i)_+^3$, so this is at most a multiple of $\mathbb{E}|X|\, \mathbb{E}X_+^3 + \mathbb{E}|X|^4 < \infty$. Hence the expected sum of the two partners' messages into the root's team is $\mathbb{E}[(X_2 + X_3)\, S_+^3]/6$. By exchangeability $\mathbb{E}[X_i\, S_+^3]$ does not depend on i, so $\mathbb{E}[(X_2 + X_3)\, S_+^3] = \frac{2}{3}\mathbb{E}[S\, S_+^3] = \frac{2}{3}\mathbb{E}S_+^4$, and the expectation equals $\mathbb{E}S_+^4/9$.

Fourth. Let $e^* = \{o, b^*, c^*\}$ be the root's team in M. By Lemma 6.20, $X_\varnothing = \xi_{e^*} - X_{b^*\to e^*} - X_{c^*\to e^*}$. The variable $X_\varnothing$ has law μ (Lemma 6.13(a) with $n = 0$), and the terms on the right are integrable by the second and third steps. So

$$\mathbb{E}X = \frac{\mathbb{E}S_+^4}{8} - \frac{\mathbb{E}S_+^4}{9} = \frac{\mathbb{E}S_+^4}{72}, \qquad \text{that is,} \qquad c(M) = \frac{\mathbb{E}S_+^4}{8} = 9\,\mathbb{E}X. \quad \square$$

Remark 6.21 (a check of $\mathbb{E}S_+^3 = 6$ without the tree)**.** Since $\mathbb{P}(X > x) = \exp(-G(x))$ and G is continuous, $G(X)$ is a standard exponential variable. With X_3 independent of $X_1 + X_2$, which has the law of Y, equation (6.3) gives $\mathbb{E}S_+^3/6 = \mathbb{E}\,G(X_3) = 1$.

Remark 6.22 (the matching is not claimed to be a function of the tree)**.** M is a function of the tree, the costs and the messages. The messages are constructed jointly with the tree by an extension theorem and are not claimed to be functions of $(\mathbb{T}, \xi)$; consequently M is not claimed to be a function of $(\mathbb{T}, \xi)$ either. The definition of an invariant perfect matching allows such additional randomness, and Theorem 9.1 is stated for matchings of this kind. The transfer to the market in Section 9 works with conditional expectations of M given the tree and the costs, which are fractional; accordingly Theorem 6.12 leads to a bound on the fractional optimum LP_q^* and not on T_q. In fact the additional randomness cannot be avoided: by Corollary 10.1, no invariant perfect matching on $\mathbb{T}$ of cost below β_0 is a measurable function of the tree and its costs.

Remark 6.23 (normalisation)**.** In the market a team has deficit $\Delta(e)$ and rescaled cost $\xi_e = q^{2/3}\Delta(e)$. Since $\mathbb{P}(\Delta(e) \le t) = t^3/6$ for $t \le 1$, an agent has on average $q^2 \cdot K^3/(6q^2) = K^3/6 = \Lambda([0, K])$ teams of rescaled cost at most K, for $K \le q^{2/3}$; this fixes the intensity Λ. A perfect assignment N has q teams and every one of the $3q$ agents is in exactly one of them, so its total deficit is

$$\sum_{e\in N} \Delta(e) = q^{-2/3} \sum_{e\in N} \xi_e = q^{1/3} \cdot \frac{1}{3q} \sum_{\text{agents } v} \xi_{N(v)},$$

where $N(v)$ is the team of v. So a total deficit of $c\,q^{1/3}$ means that c is the average cost per team, equivalently the average over agents of the cost of the agent's own team. The tree quantity $c(m) = \mathbb{E}\sum_{e\ni o} \xi_e m_e$ is exactly this; the cost per agent is $c/3$. The same holds for fractional x: $\sum_e \Delta(e)x_e = q^{1/3} \cdot \frac{1}{3q} \sum_v \sum_{e\ni v} \xi_e x_e$, to be compared with (6.2).

7. Existence and cost of a fixed point

Theorem 6.12 turns a fixed point μ of the recursion map Θ into an invariant perfect matching on the Poisson hypertree of cost $9\,\mathbb{E}X$, $X \sim \mu$. This section supplies the fixed point and bounds its cost.

Theorem 7.1 (existence and cost)**.** (i) *The map Θ has a fixed point: there is a probability law μ on $\mathbb{R}$ with $\Theta(\mu) = \mu$.*

(ii) *Every fixed point μ of Θ satisfies $9\,\mathbb{E}X \le \alpha_0 = 2.2519$, where $X \sim \mu$.*

Corollary 7.2. *There is an invariant random perfect matching M on the Poisson hypertree $\mathbb{T}$ with cost $c(M) \le \alpha_0 = 2.2519$.*

Proof. Theorem 7.1 and Theorem 6.12. □

Uniqueness of the fixed point is neither needed nor claimed. The proof has three parts. The map Θ preserves the class of laws with increasing failure rate and is monotone on it for the dispersive order (Theorem 7.7); the dispersive order ignores shifts, which is what makes it suitable for a map under which a shift by a becomes a shift by $-2a$. A cost functional $c(\cdot)$ that equals $9\,\mathbb{E}X$ at a fixed point is antitone for the same order (Lemma 7.9). Hence every law below a fixed point gives an upper bound on its cost, the iterates of Θ started from δ_0 are such laws, and they converge below an explicit super-solution, which gives existence. Part (ii) is then reduced to one inequality for one explicit law, $c(\tilde{Q}_{10}) \le 2.2519$, and that inequality is the computer-assisted step (Proposition 7.16).

7.1. **Increasing failure rate and the dispersive order.** Throughout, E, E_1, E_2 are independent standard exponential variables.

Definition 7.3. $\mathcal{Q}$ is the set of concave nondecreasing functions $Q : (0,\infty) \to \mathbb{R}$ such that $\mathbb{E}|Q(E)|^n < \infty$ for every $n \ge 1$. For $Q \in \mathcal{Q}$, μ_Q is the law of $Q(E)$. For $Q_0, Q_1 \in \mathcal{Q}$ write

$$Q_0 \preceq_{\mathrm{disp}} Q_1 \quad \text{if and only if} \quad Q_1 - Q_0 \text{ is nondecreasing on } (0,\infty).$$

The function $\theta \mapsto Q(\theta)$ is the quantile function of μ_Q in the exponential scale: $Q(\theta)$ is the quantile of order $1 - e^{-\theta}$. For $x \in \mathbb{R}$ put $a_Q(x) = \inf\{\theta > 0 : Q(\theta) > x\} \in [0,\infty]$. Since Q is continuous and nondecreasing, $\{Q > x\} = (a_Q(x), \infty)$ and

$$\mathbb{P}(Q(E) > x) = \exp(-a_Q(x)), \qquad x \in \mathbb{R}, \tag{7.1}$$

with the convention $e^{-\infty} = 0$. The function a_Q is convex: if $a_Q(x_0), a_Q(x_1) < \infty$, $\lambda \in (0,1)$ and $\theta_j > a_Q(x_j)$, then by concavity $Q((1-\lambda)\theta_0 + \lambda\theta_1) \ge (1-\lambda)Q(\theta_0) + \lambda Q(\theta_1) > (1-\lambda)x_0 + \lambda x_1$, so $a_Q((1-\lambda)x_0 + \lambda x_1) \le (1-\lambda)\theta_0 + \lambda\theta_1$; now let $\theta_j \downarrow a_Q(x_j)$. So the survival function of μ_Q is log-concave: μ_Q has *increasing failure rate* (IFR). The class $\mathcal{Q}$ contains $Q \equiv 0$, for which $\mu_Q = \delta_0$, and functions that are constant on a half-line $[\theta_N, \infty)$, for which μ_Q has an atom of mass $e^{-\theta_N}$ at its upper end. Conversely, a law with moments of every order and survival function $\exp(-G)$, where G is a convex increasing bijection of $\mathbb{R}$ onto $(0,\infty)$, is μ_Q for the concave increasing function $Q = G^{-1} \in \mathcal{Q}$.

The relation $\preceq_{\mathrm{disp}}$ is the dispersive order of the laws μ_{Q_0}, μ_{Q_1} [24], written on quantile functions. It is reflexive and transitive, it ignores additive constants ($Q \preceq_{\mathrm{disp}} Q + a \preceq_{\mathrm{disp}} Q$ for every $a \in \mathbb{R}$), and $0 \preceq_{\mathrm{disp}} Q$ for every $Q \in \mathcal{Q}$. The set $\mathcal{Q}$ is convex, and if $Q_0 \preceq_{\mathrm{disp}} Q_1$ then $Q_0 \preceq_{\mathrm{disp}} Q_\tau \preceq_{\mathrm{disp}} Q_1$ for $Q_\tau = Q_0 + \tau(Q_1 - Q_0)$, $\tau \in [0,1]$.

Lemma 7.4 (the map on quantile functions)**.** *Let $Q \in \mathcal{Q}$, $Y = Q(E_1) + Q(E_2)$, $t_Q = -2\sup Q \in [-\infty, \infty)$ and*

$$g_Q(t) = \tfrac{1}{6}\,\mathbb{E}\big(t + Q(E_1) + Q(E_2)\big)_+^3, \qquad t \in \mathbb{R}. \tag{7.2}$$

(i) *g_Q is finite, convex, nondecreasing and twice continuously differentiable, with $g_Q'(t) = \frac{1}{2}\mathbb{E}(t+Y)_+^2$ and $g_Q''(t) = \mathbb{E}(t+Y)_+$. It vanishes on $(-\infty, t_Q]$ and is an increasing bijection of (t_Q, ∞) onto $(0, \infty)$, with $g_Q' > 0$ there.*

(ii) *The infimum defining $\Theta(\mu_Q)$ is almost surely finite and attained, and $\Theta(\mu_Q)$ has survival function $\exp(-g_Q)$. Let*

$$\Theta(Q) := \big(g_Q|_{(t_Q,\infty)}\big)^{-1} : (0,\infty) \to (t_Q, \infty).$$

Then $\Theta(Q)$ is concave, increasing and twice continuously differentiable, $\Theta(Q) \in \mathcal{Q}$, and $\Theta(\mu_Q) = \mu_{\Theta(Q)}$.

(iii) *Shift law: $\Theta(Q+a) = \Theta(Q) - 2a$ for $a \in \mathbb{R}$. Scaling law: $\Theta(aQ)(\theta) = a\,\Theta(Q)(\theta/a^3)$ for $a > 0$.*

(iv) *Let $s_Q(\theta) = g_Q'(\Theta(Q)(\theta))$ be the slope of g_Q at level $\theta > 0$. Then s_Q is positive, continuous and nondecreasing, $s_{Q+a} = s_Q$, and $\Theta(Q)'(\theta) = 1/s_Q(\theta)$. Consequently, for $Q_0, Q_1 \in \mathcal{Q}$,*

$$\Theta(Q_0) \preceq_{\mathrm{disp}} \Theta(Q_1) \quad \text{if and only if} \quad s_{Q_1} \le s_{Q_0} \text{ on } (0,\infty).$$

Proof. (i) Y has moments of every order, so g_Q is finite and differentiation under the expectation gives the two derivatives and their continuity. Convexity follows from that of $t \mapsto (t+y)_+^3$. Since Q is nondecreasing and E has full support in $(0,\infty)$, the essential supremum of Y is $2\sup Q$; so $\mathbb{P}(Y > -t) > 0$ if and only if $t > t_Q$, which gives the vanishing of g_Q on $(-\infty, t_Q]$ and $g_Q' > 0$ on (t_Q, ∞). Finally $g_Q(t) \to 0$ as $t \downarrow t_Q$ by continuity (or dominated convergence if $t_Q = -\infty$), and $g_Q(t) \to \infty$ as $t \to \infty$ by monotone convergence.

(ii) As in the proof of Lemma 6.11, with $Y_k = X_k' + X_k''$ and (X_k', X_k'') independent with law $\mu_Q \otimes \mu_Q$, the number $N(x)$ of indices k with $\xi_k - Y_k \le x$ is a Poisson variable with mean $g_Q(x) < \infty$. So almost surely finitely many points lie below every level, the infimum is attained and finite, and $\mathbb{P}(\inf_k(\xi_k - Y_k) > x) = \mathbb{P}(N(x) = 0) = \exp(-g_Q(x))$. On the other hand, for $t > t_Q$, $\mathbb{P}(\Theta(Q)(E) > t) = \mathbb{P}(E > g_Q(t)) = \exp(-g_Q(t))$, and for $t \le t_Q$ both survival functions equal one. So $\Theta(\mu_Q) = \mu_{\Theta(Q)}$. The inverse of a convex increasing C^2 function with positive derivative is concave, increasing and C^2. For the moments, the two tail bounds in the proof of Lemma 6.11(iii) use only the formula $\exp(-g_Q)$ for the survival function of the output and the moments of Y_+: with $p = \mathbb{P}(Y > -m) > 0$ they give $\mathbb{P}(\Theta(Q)(E) > x) \le \exp(-p(x-m)^3/6)$ for $x > m$ and $\mathbb{E}\,\Theta(Q)(E)_-^n \le \mathbb{E}Y_+^{n+3}/((n+1)(n+2)(n+3))$.

(iii) $g_{Q+a}(t) = g_Q(t+2a)$, and $g_{aQ}(t) = \frac{1}{6}\mathbb{E}(t+aY)_+^3 = a^3 g_Q(t/a)$; invert.

(iv) g'_Q is positive, continuous and nondecreasing on (t_Q, ∞) and $\Theta(Q)$ is continuous and increasing, so s_Q has the three properties. The identity $s_{Q+a} = s_Q$ follows from $g_{Q+a}(t) = g_Q(t + 2a)$. The derivative of the inverse function is $\Theta(Q)'(\theta) = 1/g'_Q(\Theta(Q)(\theta))$. Both $\Theta(Q_j)$ are continuously differentiable, so $\Theta(Q_1) - \Theta(Q_0)$ is nondecreasing if and only if $1/s_{Q_1} \geq 1/s_{Q_0}$. □

We write Θ both for the map on laws and for the map $Q \mapsto \Theta(Q)$ on $\mathcal{Q}$; by (ii) they agree. For $Q \in \mathcal{Q}$ put

$$\Gamma(Q) = \mathbb{E}\,\Theta(Q)(E) + 2\,\mathbb{E}\,Q(E), \qquad c(Q) = 3\,\Gamma(Q). \tag{7.3}$$

By the shift law, $\Gamma(Q + a) = \Gamma(Q)$ for every $a \in \mathbb{R}$. (The letter c is also used for the cost (6.2) of a matching; part (iii) of the next lemma says that the two agree at a fixed point.)

Lemma 7.5 (fixed points are IFR)**.** (i) *If μ is a fixed point of Θ and G is the function* (6.3)*, then $Q^* := G^{-1}$ belongs to $\mathcal{Q}$, $\mu = \mu_{Q^*}$ and $\Theta(Q^*) = Q^*$.*

(ii) *If $Q \in \mathcal{Q}$ and $\Theta(Q) = Q$, then μ_Q is a fixed point of Θ.*

(iii) *If $Q^* \in \mathcal{Q}$, $\Theta(Q^*) = Q^*$ and $X \sim \mu_{Q^*}$, then $\Gamma(Q^*) = 3\,\mathbb{E}X$ and $c(Q^*) = 9\,\mathbb{E}X$, the cost of the matching of Theorem 6.12.*

Proof. (i) By Lemma 6.11(i), G is a convex increasing C^2 bijection of $\mathbb{R}$ onto $(0, \infty)$ and $\mathbb{P}(X > x) = \exp(-G(x))$. So $G(X)$ is a standard exponential variable, X has the law of $Q^*(E)$ with $Q^* = G^{-1}$ concave and increasing, and all moments are finite by Lemma 6.11(iii). Since X', X'' have law μ_{Q^*}, $g_{Q^*} = G$ and therefore $\Theta(Q^*) = G^{-1} = Q^*$. (ii) is Lemma 7.4(ii). (iii) $\Gamma(Q^*) = \mathbb{E}\,\Theta(Q^*)(E) + 2\,\mathbb{E}Q^*(E) = 3\,\mathbb{E}Q^*(E) = 3\,\mathbb{E}X$. □

7.2. **Monotonicity.** A function $f : \mathbb{R}^d \to [0, \infty)$ is *log-concave* if $f((1 - \lambda)u + \lambda w) \geq f(u)^{1-\lambda} f(w)^{\lambda}$ for all u, w and $\lambda \in (0, 1)$. Indicators of convex sets are log-concave and products of log-concave functions are log-concave. Prékopa's theorem [22] states that if f is log-concave on $\mathbb{R}^d \times \mathbb{R}^{d'}$ then $u \mapsto \int_{\mathbb{R}^{d'}} f(u, w)\,\mathrm{d}w$ is log-concave on $\mathbb{R}^d$ (it may take the value $+\infty$; in the applications below it is finite).

Lemma 7.6 (log-concavity of the truncated moments)**.** *Let $Q \in \mathcal{Q}$, $X = Q(E)$ and, for $k \geq 0$ and $u \in \mathbb{R}$,*

$$M_k(u) = \frac{1}{k!}\,\mathbb{E}(u + X)_+^k, \qquad M_0(u) = \mathbb{P}(X > -u).$$

Then every M_k is log-concave, and M_2/M_1 is nondecreasing on the open half-line $\{M_1 > 0\} = (-\sup Q, \infty)$.

Proof. By (7.1), $M_0(u) = \exp(-a_Q(-u))$ with a_Q convex with values in $[0, \infty]$; so M_0 is log-concave. This covers laws with an atom at the upper end and the law δ_0, for which $M_0 = \mathbf{1}_{(0,\infty)}$. For $k \geq 0$,

$$M_{k+1}(u) = \int_{-\infty}^{u} M_k(w)\,\mathrm{d}w = \int_{\mathbb{R}} M_k(w)\,\mathbf{1}\{w \leq u\}\,\mathrm{d}w,$$

because $(u+x)_+^{k+1}/(k+1)! = \int_{-\infty}^{u}(w+x)_+^k/k!\,\mathrm{d}w$. If M_k is log-concave then $(u,w) \mapsto M_k(w)\mathbf{1}\{w \le u\}$ is log-concave on $\mathbb{R}^2$, and M_{k+1} is log-concave by Prékopa's theorem. By induction all M_k are log-concave. M_1 and M_2 are positive exactly on $(-\sup Q, \infty)$. On that half-line M_2 is continuously differentiable with $M_2' = M_1$ and $\ln M_2$ is concave, so $(\ln M_2)' = M_1/M_2$ is positive and nonincreasing. Hence M_2/M_1 is nondecreasing. □

Theorem 7.7 (monotonicity)**.** *Let $Q_0, Q_1 \in \mathcal{Q}$. If $Q_0 \preceq_{\mathrm{disp}} Q_1$, then $\Theta(Q_0) \preceq_{\mathrm{disp}} \Theta(Q_1)$.*

Proof. Let $\Delta = Q_1 - Q_0$, which is nondecreasing, and $Q_\tau = Q_0 + \tau\Delta \in \mathcal{Q}$ for $\tau \in [0,1]$. By Lemma 7.4(iv) it is enough to show that $\tau \mapsto s_{Q_\tau}(\theta)$ is nonincreasing for every $\theta > 0$. Write $X^j = Q_\tau(E_j)$, $Y = X^1 + X^2$ and $g_\tau = g_{Q_\tau}$.

Derivatives in τ. The functions $y \mapsto y_+^3$ and $y \mapsto y_+^2$ are continuously differentiable, $|Q_\tau(E_j)| \le |Q_0(E_j)| + |Q_1(E_j)|$, and all these variables have moments of every order. So by dominated convergence, and by the symmetry between E_1 and E_2,

$$\partial_\tau g_\tau(t) = \mathbb{E}\big[(t+Y)_+^2\,\Delta(E_1)\big], \qquad \partial_\tau g_\tau'(t) = 2\,\mathbb{E}\big[(t+Y)_+\,\Delta(E_1)\big],$$

and these, together with $g_\tau'(t)$ and $g_\tau''(t)$, are jointly continuous in (τ,t). Thus $(\tau,t) \mapsto g_\tau(t)$ and $(\tau,t) \mapsto g_\tau'(t)$ are continuously differentiable.

Derivative of the slope at a level. Fix $\theta > 0$ and let $t = t(\tau) = \Theta(Q_\tau)(\theta)$, so that $g_\tau(t) = \theta$ and $g_\tau'(t) > 0$. By the implicit function theorem $t(\tau)$ is continuously differentiable with $\partial_\tau t = -\partial_\tau g_\tau(t)/g_\tau'(t)$. Hence $s_{Q_\tau}(\theta) = g_\tau'(t(\tau))$ is continuously differentiable in τ and

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\tau} s_{Q_\tau}(\theta) &= \partial_\tau g_\tau'(t) - g_\tau''(t)\,\frac{\partial_\tau g_\tau(t)}{g_\tau'(t)} \\ &= 2\,\mathbb{E}(t+Y)_+ \cdot \Big(\mathbb{E}^{(1)}[\Delta(E_1)] - \mathbb{E}^{(2)}[\Delta(E_1)]\Big), \qquad (7.4)\end{aligned}$$

where $\mathbb{E}^{(k)}$ is the expectation under the probability measure $\mathbb{P}^{(k)}$ with density $(t+Y)_+^k/\mathbb{E}(t+Y)_+^k$ with respect to $\mathbb{P}$; note that $\mathbb{E}(t+Y)_+^k > 0$ because $g_\tau'(t) > 0$.

Sign. Let M_k be the functions of Lemma 7.6 for the law of $X^2 = Q_\tau(E_2)$. Conditioning on E_1, $\mathbb{E}[(t+Y)_+^k \mid E_1 = e] = k!\,M_k(t + Q_\tau(e))$, so under $\mathbb{P}^{(k)}$ the variable E_1 has density

$$p_k(e) = \frac{k!\,e^{-e}\,M_k(t+Q_\tau(e))}{\mathbb{E}(t+Y)_+^k}, \qquad e > 0.$$

The densities p_1 and p_2 are positive on the same set $I = \{e > 0 : t + Q_\tau(e) > -\sup Q_\tau\}$, which is an up-set because Q_τ is nondecreasing. On I the ratio $r = p_2/p_1$ is a positive multiple of $(M_2/M_1)(t+Q_\tau(e))$, which is nondecreasing in e by Lemma 7.6. Now $\mathbb{E}^{(1)}[r(E_1)] = 1$, $\mathbb{E}^{(2)}[\Delta(E_1)] = \mathbb{E}^{(1)}[\Delta(E_1)\,r(E_1)]$, and all the expectations involved are finite because $\Delta(E_1)$

has moments of every order under $\mathbb{P}$. With E_1' an independent copy of E_1 under $\mathbb{P}^{(1)}$,

$$\mathbb{E}^{(2)}[\Delta(E_1)] - \mathbb{E}^{(1)}[\Delta(E_1)] = \tfrac{1}{2}\,\mathbb{E}^{(1)}\big[(\Delta(E_1) - \Delta(E_1'))\,(r(E_1) - r(E_1'))\big] \geq 0,$$

since Δ and r are both nondecreasing. So the bracket in (7.4) is at most 0, $\tau \mapsto s_{Q_\tau}(\theta)$ is nonincreasing, and $s_{Q_1}(\theta) \leq s_{Q_0}(\theta)$ for every $\theta > 0$. □

Remark 7.8. No preservation of the dispersive order under convolution is used. Only the slope-at-level function of the output has to be compared, and for that the log-concavity of M_2, which is what the IFR property of one summand provides, is enough. Restricting to IFR laws costs nothing: every output of Θ is IFR by Lemma 7.4(ii).

7.3. **Cost comparison.**

Lemma 7.9 (cost comparison)**.** *If $Q_0, Q_1 \in \mathcal{Q}$ and $Q_0 \preceq_{\mathrm{disp}} Q_1$, then $\Gamma(Q_0) \geq \Gamma(Q_1)$, that is, $c(Q_0) \geq c(Q_1)$.*

Proof. Γ is invariant under shifts, so we may assume $\mathbb{E}Q_0(E) = \mathbb{E}Q_1(E) = 0$. Then $Q_1(E) = Q_0(E) + \Delta(E)$ with Δ nondecreasing and $\mathbb{E}\Delta(E) = 0$. Let φ be a convex continuously differentiable function with φ' of polynomial growth, and $a \in \mathbb{R}$. By convexity,

$$\mathbb{E}\varphi(Q_1(E)+a) \geq \mathbb{E}\varphi(Q_0(E)+a) + \mathbb{E}\big[\varphi'(Q_0(E)+a)\,\Delta(E)\big] \geq \mathbb{E}\varphi(Q_0(E)+a),$$

because $\mathbb{E}[\varphi'(Q_0(E)+a)\Delta(E)]$ is the covariance of two nondecreasing functions of E, which is nonnegative. Apply this with $\varphi(y) = (t+y)_+^3/6$, first to E_1 conditionally on E_2 with $a = Q_1(E_2)$, then to E_2 conditionally on E_1 with $a = Q_0(E_1)$:

$$\begin{aligned} g_{Q_1}(t) = \mathbb{E}\varphi(Q_1(E_1) + Q_1(E_2)) &\geq \mathbb{E}\varphi(Q_0(E_1) + Q_1(E_2)) \\ &\geq \mathbb{E}\varphi(Q_0(E_1) + Q_0(E_2)) = g_{Q_0}(t). \end{aligned}$$

So $g_{Q_1} \geq g_{Q_0}$ pointwise, hence $\Theta(Q_1) \leq \Theta(Q_0)$ pointwise on $(0,\infty)$, and $\Gamma(Q_1) = \mathbb{E}\,\Theta(Q_1)(E) \leq \mathbb{E}\,\Theta(Q_0)(E) = \Gamma(Q_0)$. □

Corollary 7.10 (laws below a fixed point)**.** *Let $Q_n = \Theta^n(0)$, $n \geq 0$, be the iterates of Θ started from the law δ_0, and $c_n = c(Q_n)$.*

(i) *$Q_0 \preceq_{\mathrm{disp}} Q_1 \preceq_{\mathrm{disp}} Q_2 \preceq_{\mathrm{disp}} \cdots$ and $c_0 \geq c_1 \geq c_2 \geq \cdots$.*
(ii) *Let $Q^* \in \mathcal{Q}$ with $\Theta(Q^*) = Q^*$. For every $Q \in \mathcal{Q}$ with $Q \preceq_{\mathrm{disp}} Q^*$ we have $\Theta(Q) \preceq_{\mathrm{disp}} Q^*$ and $c(Q^*) \leq c(Q)$. In particular $Q_n \preceq_{\mathrm{disp}} Q^*$ and $c(Q^*) \leq c_n$ for every n.*

Proof. (i) $0 \preceq_{\mathrm{disp}} Q_1$; apply Theorem 7.7 repeatedly, then Lemma 7.9. (ii) $\Theta(Q) \preceq_{\mathrm{disp}} \Theta(Q^*) = Q^*$ by Theorem 7.7, and $c(Q^*) \leq c(Q)$ by Lemma 7.9. Since $0 \preceq_{\mathrm{disp}} Q^*$, induction gives $Q_n \preceq_{\mathrm{disp}} Q^*$. □

Here $g_0(t) = t_+^3/6$, so $Q_1(\theta) = (6\theta)^{1/3}$, the law with survival function $\exp(-x^3/6)$ on $[0,\infty)$, and $c_0 = 3\,\mathbb{E}(6E)^{1/3} = 3 \cdot 6^{1/3}\,\Gamma(4/3)$ with $\Gamma(\cdot)$ here the Gamma function. By Lemma 7.5 and Corollary 7.10(ii), every fixed point of Θ has $9\,\mathbb{E}X \leq c_n$ for every n.

7.4. **The scaling law and a super-solution.**

Lemma 7.11 (super-solution)**.** *Let $R(\theta) = \ln\theta$ for $0 < \theta \le 1$ and $R(\theta) = 3(\theta^{1/3} - 1)$ for $\theta \ge 1$. Then $R \in \mathcal{Q}$, and there is a finite $a \ge 1$ with $\Theta(aR) \preceq_{\text{disp}} aR$.*

The multiplier a is obtained from a supremum that is shown to be finite; no numerical value is claimed or needed.

Proof. The law of $R(E)$. R is continuously differentiable with $R'(\theta) = 1/\theta$ on $(0, 1]$ and $R'(\theta) = \theta^{-2/3}$ on $[1, \infty)$, positive and nonincreasing; so R is concave and increasing. $R(E)$ has moments of every order, and for every $\lambda > 0$

$$\mathbb{E}e^{\lambda R(E)} = \int_0^1 \theta^\lambda e^{-\theta}\,\mathrm{d}\theta + \int_1^\infty e^{3\lambda(\theta^{1/3}-1)-\theta}\,\mathrm{d}\theta < \infty. \tag{7.5}$$

Reduction by scaling. For a differentiable Q put $\ell_Q(\theta) = \theta Q'(\theta)$. Then $\ell_R(\theta) = \max(1, \theta^{1/3})$ and $\ell_{aR} = a\,\ell_R$. By the scaling law, $\Theta(aR)'(\theta) = a^{-2}\Theta(R)'(\theta/a^3)$, so $\ell_{\Theta(aR)}(\theta) = a\,\ell_{\Theta(R)}(\theta/a^3)$. Both aR and $\Theta(aR)$ are continuously differentiable, so

$$\Theta(aR) \preceq_{\text{disp}} aR \iff \ell_{\Theta(R)}(\vartheta) \le \ell_R(a^3\vartheta) \text{ for all } \vartheta > 0. \tag{7.6}$$

Log-concavity. Let $Y = R(E_1) + R(E_2)$ and $M_k(t) = \mathbb{E}(t + Y)_+^k/k!$, so that $g_R = M_3$, $g_R' = M_2$, and

$$\ell_{\Theta(R)}(\vartheta) = \frac{\vartheta}{s_R(\vartheta)} = \frac{M_3(t)}{M_2(t)} \qquad \text{at the point } t \text{ with } g_R(t) = \vartheta.$$

Since $\sup R = \infty$, t ranges over $\mathbb{R}$ and every M_k is positive on $\mathbb{R}$. Let $m_0(u) = \mathbb{P}(R(E) > -u)$, which is log-concave by Lemma 7.6. The area of the triangle $\{u_1 > -x_1,\ u_2 > -x_2,\ u_1 + u_2 \le t\}$ is $(t + x_1 + x_2)_+^2/2$, so

$$M_2(t) = \int_{\mathbb{R}^2} m_0(u_1)\, m_0(u_2)\, \mathbf{1}\{u_1 + u_2 \le t\}\,\mathrm{d}u_1\,\mathrm{d}u_2.$$

The integrand is log-concave on $\mathbb{R}^3$, so M_2 is log-concave by Prékopa's theorem [22], and then so is $M_3(t) = \int_{\mathbb{R}} M_2(w)\mathbf{1}\{w \le t\}\,\mathrm{d}w$. The law of Y has a density, so $M_0(t) = \mathbb{P}(Y > -t)$ is continuous, M_1 is C^1 with $M_1' = M_0$, M_2 is C^2 and M_3 is C^3, with $M_{k+1}' = M_k$. Log-concavity of the C^2 functions M_3 and M_2 means $M_3''M_3 \le (M_3')^2$ and $M_2''M_2 \le (M_2')^2$, that is,

$$\frac{M_3}{M_2} \le \frac{M_2}{M_1} \le \frac{M_1}{M_0} \qquad \text{on } \mathbb{R}, \tag{7.7}$$

and $M_1/M_2 = (\ln M_2)'$ is nonincreasing.

Small levels. Let $\kappa = \lim_{t\to-\infty} M_1(t)/M_2(t) \in (0, \infty]$, which exists by monotonicity. Suppose $\kappa < \infty$. Then $(\ln M_2)' \le \kappa$ on $\mathbb{R}$, so $M_2(t) \ge M_2(0)\,e^{\kappa t}$ for $t \le 0$. But for $\lambda > \kappa$, $(t + y)_+^2 \le 2\lambda^{-2}e^{\lambda(t+y)}$, so by (7.5)

$$M_2(t) \le \lambda^{-2}\big(\mathbb{E}e^{\lambda R(E)}\big)^2 e^{\lambda t},$$

which is smaller than $M_2(0)e^{\kappa t}$ for t sufficiently negative: a contradiction. So $M_2/M_1 \to 0$ as $t \to -\infty$, and by (7.7) there is t_1 with $M_3/M_2 \le 1$ on $(-\infty, t_1]$. With $\vartheta_1 = g_R(t_1) > 0$ this says $\ell_{\Theta(R)} \le 1$ on $(0, \vartheta_1]$.

Large levels. Choose $t_0 \ge \max(t_1, 0)$ with $M_0(t_0) \ge 1/2$, and put $\vartheta_0 = g_R(t_0) \ge \vartheta_1$. For $t \ge t_0$, by (7.7), $M_3/M_2 \le M_1/M_0 \le 2\,\mathbb{E}(t+Y)_+ \le 2(t + \mathbb{E}|Y|)$. By Jensen's inequality $\vartheta = g_R(t) \ge (t + \mathbb{E}Y)_+^3/6$, so $t \le (6\vartheta)^{1/3} + \mathbb{E}|Y|$. Hence $\ell_{\Theta(R)}(\vartheta) \le 2(6\vartheta)^{1/3} + 4\,\mathbb{E}|Y|$ for $\vartheta \ge \vartheta_0$. As $\ell_{\Theta(R)}$ is continuous on $[\vartheta_1, \vartheta_0]$,

$$A := \sup_{\vartheta \ge \vartheta_1} \ell_{\Theta(R)}(\vartheta)\, \vartheta^{-1/3} < \infty.$$

Conclusion. Take $a = \max(1, A)$. For $\vartheta \le \vartheta_1$, $\ell_{\Theta(R)}(\vartheta) \le 1 \le \ell_R(a^3\vartheta)$. For $\vartheta \ge \vartheta_1$, $\ell_{\Theta(R)}(\vartheta) \le A\vartheta^{1/3} \le (a^3\vartheta)^{1/3} \le \ell_R(a^3\vartheta)$. By (7.6), $\Theta(aR) \preceq_{\text{disp}} aR$. □

7.5. **Proof of existence.**

Proof of Theorem 7.1(i). Let $Q_n = \Theta^n(0)$ and let a, R be as in Lemma 7.11.

Bounds on the slopes. $Q_n \preceq_{\text{disp}} Q_{n+1}$ by Corollary 7.10(i). Also $Q_n \preceq_{\text{disp}} aR$ for every n: this holds for $Q_0 = 0$, and if $Q_n \preceq_{\text{disp}} aR$ then $Q_{n+1} = \Theta(Q_n) \preceq_{\text{disp}} \Theta(aR) \preceq_{\text{disp}} aR$ by Theorem 7.7 and Lemma 7.11. All the functions involved are continuously differentiable (Lemma 7.4(ii)), so

$$0 \le Q_n'(\theta) \le Q_{n+1}'(\theta) \le a\,R'(\theta), \qquad \theta > 0,\ n \ge 0.$$

Let $q_\infty(\theta) = \lim_n Q_n'(\theta)$, which is nonincreasing in θ and at most $aR'(\theta)$, and put

$$\bar{Q}_n(\theta) = Q_n(\theta) - Q_n(1) = \int_1^\theta Q_n'(\vartheta)\,\mathrm{d}\vartheta, \qquad Q_\infty(\theta) = \int_1^\theta q_\infty(\vartheta)\,\mathrm{d}\vartheta.$$

By monotone convergence $\bar{Q}_n \to Q_\infty$ pointwise, and since $R(1) = 0$, $|\bar{Q}_n| \le a|R|$ and $|Q_\infty| \le a|R|$. So Q_∞ is concave, nondecreasing and has moments of every order: $Q_\infty \in \mathcal{Q}$.

Passage to the limit. Write $g_n = g_{\bar{Q}_n}$ and $g_\infty = g_{Q_\infty}$. By dominated convergence, with dominating variables $(|t| + a|R(E_1)| + a|R(E_2)|)^k$, $k = 3, 2$, we get $g_n \to g_\infty$ and $g_n' \to g_\infty'$ pointwise on $\mathbb{R}$. These functions are nondecreasing and the limits are continuous, so the convergence is locally uniform. Fix $\theta > 0$ and let $t_n = \Theta(\bar{Q}_n)(\theta)$, $t_\infty = \Theta(Q_\infty)(\theta)$. For $\varepsilon > 0$, $g_\infty(t_\infty - \varepsilon) < \theta < g_\infty(t_\infty + \varepsilon)$, so $g_n(t_\infty - \varepsilon) < \theta < g_n(t_\infty + \varepsilon)$ for large n, that is, $|t_n - t_\infty| < \varepsilon$. Hence $t_n \to t_\infty$ and, by locally uniform convergence, $s_{\bar{Q}_n}(\theta) = g_n'(t_n) \to g_\infty'(t_\infty) = s_{Q_\infty}(\theta) > 0$.

The limit is a fixed point up to a shift. By Lemma 7.4(iv), $Q_{n+1}'(\theta) = 1/s_{Q_n}(\theta) = 1/s_{\bar{Q}_n}(\theta)$. Letting $n \to \infty$,

$$q_\infty(\theta) = \frac{1}{s_{Q_\infty}(\theta)} = \Theta(Q_\infty)'(\theta), \qquad \theta > 0.$$

So q_∞ is continuous, Q_∞ is continuously differentiable with $Q'_\infty = \Theta(Q_\infty)'$, and $\Theta(Q_\infty) = Q_\infty + k$ for a constant $k \in \mathbb{R}$. Let $Q^* = Q_\infty + k/3$. By the shift law, $\Theta(Q^*) = \Theta(Q_\infty) - 2k/3 = Q_\infty + k/3 = Q^*$. By Lemma 7.5(ii), μ_{Q^*} is a fixed point of Θ. □

No continuity of the cost along the iteration is needed for part (ii): the fixed point just constructed, like every other, is covered by Corollary 7.10(ii).

7.6. **One-sided rounding and the certified bound.** By Corollary 7.10, the cost of every fixed point is at most c_n for every n. The numbers c_n are explicit iterated integrals, but the iterates Q_n have no closed form for $n \geq 2$. Theorem 7.7 and Lemma 7.9 make rounding safe provided it goes one way: an iterate may be replaced by any element of $\mathcal{Q}$ below it in the dispersive order.

Definition 7.12 (rounded law)**.** Let $N \geq 0$, $0 < \theta_0 < \cdots < \theta_N$ and $0 < \hat{s}_0 \leq \cdots \leq \hat{s}_N$. The *rounded law* with these parameters is the continuous piecewise linear function $\tilde{Q} : (0, \infty) \to \mathbb{R}$ with $\tilde{Q}(\theta_N) = 0$, slope $1/\hat{s}_0$ on $(0, \theta_0]$, slope $1/\hat{s}_i$ on $(\theta_{i-1}, \theta_i]$ for $1 \leq i \leq N$, and slope 0 on (θ_N, ∞).

The slopes are nonincreasing because the $\hat{s}_i$ are nondecreasing, so $\tilde{Q}$ is concave, nondecreasing and bounded: $\tilde{Q} \in \mathcal{Q}$. In terms of laws, $X = \tilde{Q}(E)$ is supported on $[x_{\min}, 0]$, where $x_N = 0$, $x_{i-1} = x_i - (\theta_i - \theta_{i-1})/\hat{s}_i$ and $x_{\min} = x_0 - \theta_0/\hat{s}_0$; it has an atom of mass $e^{-\theta_N}$ at 0, the upper end of its support; and $\mathbb{P}(X > x) = \exp(-\tilde{G}(x))$ for $x < 0$, where $\tilde{G}$ is convex and piecewise linear with $\tilde{G}(x_i) = \theta_i$, slope $\hat{s}_i$ on $(x_{i-1}, x_i]$, slope $\hat{s}_0$ on $(x_{\min}, x_0]$ and $\tilde{G} = 0$ on $(-\infty, x_{\min}]$.

Lemma 7.13 (safety of the rounding)**.** *Let $Q \in \mathcal{Q}$ and $g = g_Q$. Let $t_0 < \cdots < t_N$ be real numbers and let $0 < \theta_0 < \cdots < \theta_N$ and $0 < \hat{s}_0 \leq \cdots \leq \hat{s}_N$ satisfy*

$$\theta_i \leq g(t_i) \qquad \text{and} \qquad \hat{s}_i \geq g'(t_i), \qquad 0 \leq i \leq N.$$

Then the rounded law $\tilde{Q}$ with parameters $(\theta_i, \hat{s}_i)$ satisfies $\tilde{Q} \in \mathcal{Q}$ and $\tilde{Q} \preceq_{\text{disp}} \Theta(Q)$.

Proof. By Lemma 7.4(iv), s_Q is nondecreasing and $\Theta(Q)' = 1/s_Q$. Since $g(t_i) \geq \theta_i > 0$ and g is increasing on $\{g > 0\}$, $\Theta(Q)(\theta_i) \leq t_i$, and therefore $s_Q(\theta_i) = g'(\Theta(Q)(\theta_i)) \leq g'(t_i) \leq \hat{s}_i$. For θ in (θ_{i-1}, θ_i), or in $(0, \theta_0)$ if $i = 0$, $s_Q(\theta) \leq s_Q(\theta_i) \leq \hat{s}_i$, so $\Theta(Q)'(\theta) = 1/s_Q(\theta) \geq 1/\hat{s}_i = \tilde{Q}'(\theta)$. For $\theta > \theta_N$, $\tilde{Q}'(\theta) = 0 \leq \Theta(Q)'(\theta)$. Both functions are locally absolutely continuous, so $\Theta(Q) - \tilde{Q}$ is nondecreasing on $(0, \infty)$. □

No tail estimate is used: below θ_0 the slope bound at θ_0 is valid for all smaller levels, and above θ_N the rounded slope is 0, which is the atom at the top. The monotonicity of the $\hat{s}_i$ is what keeps $\tilde{Q}$ concave; it can always be enforced by replacing $\hat{s}_i$ with the running maximum $\max_{j \leq i} \hat{s}_j$, which is still at least $g'(t_i)$.

Lemma 7.14 (the chain)**.** *Let $\tilde{Q}_0 = 0$ and, for $n \geq 0$, let $\tilde{Q}_{n+1}$ be a rounded law whose parameters satisfy the hypotheses of Lemma 7.13 with $Q = \tilde{Q}_n$ and some points $t_0 < \cdots < t_N$. Then for every $Q^* \in \mathcal{Q}$ with $\Theta(Q^*) = Q^*$ and every $n \geq 0$,*

$$\tilde{Q}_n \preceq_{\mathrm{disp}} Q^* \qquad \text{and} \qquad c(Q^*) \leq c(\tilde{Q}_n).$$

Proof. $\tilde{Q}_0 = 0 \preceq_{\mathrm{disp}} Q^*$. If $\tilde{Q}_n \preceq_{\mathrm{disp}} Q^*$, then $\tilde{Q}_{n+1} \preceq_{\mathrm{disp}} \Theta(\tilde{Q}_n) \preceq_{\mathrm{disp}} \Theta(Q^*) = Q^*$ by Lemma 7.13 and Theorem 7.7. The cost bound is Lemma 7.9. □

Theorem 7.7 is applied here to laws with an atom at the upper end and to δ_0; its proof covers them, through Lemma 7.6.

Enclosures for a rounded law. To use Lemma 7.13 one needs a lower bound on g and an upper bound on g' at finitely many points, and to bound $c(\tilde{Q})$ one needs in addition a lower bound on g'. For a rounded law these come from a lattice. Let $\tilde{Q}$ be a rounded law, $X = \tilde{Q}(E)$, $\tilde{\mu}$ its law, $\bar{F}(x) = \mathbb{P}(X > x)$ and $g = g_{\tilde{Q}}$. Fix $h > 0$, let $K_{\min} = \lfloor x_{\min}/h \rfloor$ and, for $K_{\min} \leq k \leq -1$,

$$C_k = [kh, (k+1)h), \qquad \bar{F}_k = \bar{F}(kh), \qquad J_k = \frac{1}{h}\int_{C_k} \bar{F}(x)\,\mathrm{d}x,$$

and $\bar{F}_0 = \bar{F}(0-) = e^{-\theta_N}$. The mass and the first moment of $\tilde{\mu}$ on a cell are

$$p_k = \tilde{\mu}(C_k) = \bar{F}_k - \bar{F}_{k+1}, \qquad D_k = \int_{C_k} (x - kh)\,\tilde{\mu}(\,\mathrm{d}x) = h\,(J_k - \bar{F}_{k+1}) \geq 0$$

(integration by parts), and for the atom put $p_0 = e^{-\theta_N}$, $D_0 = 0$. Let $\nu_j = J_{j-1} - J_j \geq 0$ for $K_{\min} \leq j \leq 0$, with the conventions $J_{K_{\min}-1} = 1$ and $J_0 = 0$. Write $\varphi_j(u) = u_+^j/j!$ and $(a * b)_m = \sum_{k+k'=m} a_k b_{k'}$ for sequences indexed by $K_{\min}, \ldots, 0$.

Lemma 7.15 (lattice enclosures)**.** *For every real t:*

(a) *(tangent planes)*

$$g(t) \geq \sum_m (p * p)_m\, \varphi_3(t + mh) + 2\sum_m (D * p)_m\, \varphi_2(t + mh),$$
$$g'(t) \geq \sum_m (p * p)_m\, \varphi_2(t + mh) + 2\sum_m (D * p)_m\, \varphi_1(t + mh);$$

(b) *(chords)* $g'(t) \leq \sum_m (\nu * \nu)_m\, \varphi_2(t + mh)$.

(c) $\mathbb{E}\tilde{Q}(E) = K_{\min}h + h\sum_{k=K_{\min}}^{-1} J_k$. *Moreover, for any points $0 < t_0 < \cdots < t_{N'}$ and numbers $\vartheta_i \leq g(t_i)$ and $0 < a_i \leq g'(t_i)$, with $d_i = t_{i+1} - t_i$,*

$$\mathbb{E}\,\Theta(\tilde{Q})(E) = \int_0^\infty e^{-g(t)}\,\mathrm{d}t \leq t_0 + \sum_{i<N'} e^{-\vartheta_i}\,\frac{1 - e^{-a_i d_i}}{a_i} + \frac{e^{-\vartheta_{N'}}}{a_{N'}}.$$

All terms in (a) and (b) are nonnegative. So (a) remains true when p and D are replaced by termwise lower bounds that are nonnegative, and (b) remains true when ν is replaced by termwise upper bounds.

Proof. (a) $g(t) = \mathbb{E}\varphi_3(t + X' + X'')$ with X', X'' independent with law $\tilde{\mu}$. The function φ_3 is convex and continuously differentiable with $\varphi_3' = \varphi_2 \geq 0$. So for $x \in C_k$ and $x' \in C_{k'}$,

$$\varphi_3(t+x+x') \geq \varphi_3(t+(k+k')h) + \varphi_2(t+(k+k')h)\big((x-kh)+(x'-k'h)\big).$$

Integrate over $C_k \times C_{k'}$ against $\tilde{\mu} \otimes \tilde{\mu}$ and sum over k, k', the atom being included as the one-point cell $k = 0$. The same argument with φ_2 and $\varphi_2' = \varphi_1$ gives the bound for $g'(t) = \mathbb{E}\varphi_2(t + X' + X'')$.

(b) φ_2 is convex, so for $x \in C_k$ and $w = (x - kh)/h \in [0, 1)$, $\varphi_2(c + x) \leq (1 - w)\varphi_2(c + kh) + w\,\varphi_2(c + (k + 1)h)$. Integrating over C_k against $\tilde{\mu}$ gives the weight $p_k - D_k/h = \bar{F}_k - J_k$ at kh and the weight $D_k/h = J_k - \bar{F}_{k+1}$ at $(k+1)h$. Adding the two contributions at a lattice point jh gives $J_{j-1} - J_j$ for $K_{\min} < j < 0$; at $j = K_{\min}$ it gives $\bar{F}_{K_{\min}} - J_{K_{\min}} = 1 - J_{K_{\min}}$, because $K_{\min}h \leq x_{\min}$; at $j = 0$ it gives $J_{-1} - \bar{F}_0$ plus the atom $\bar{F}_0$, that is, J_{-1}. So $\mathbb{E}\varphi_2(c + X) \leq \sum_j \nu_j \varphi_2(c + jh)$ for every c. Apply this in X' for fixed X'' and then in X''.

(c) $X \leq 0$ and $\bar{F} = 1$ below $x_{\min} \geq K_{\min}h$, so $\mathbb{E}X = -\int_{K_{\min}h}^{0}(1 - \bar{F}(x))\,\mathrm{d}x = K_{\min}h + h\sum_k J_k$. Since $\sup \tilde{Q} = 0$, $t_{\tilde{Q}} = 0$, so $\Theta(\tilde{Q})(E)$ is positive with survival function $\exp(-g(t))$, $t > 0$, and its mean is $\int_0^\infty e^{-g}$. On $[0, t_0]$ use $e^{-g} \leq 1$. On $[t_i, t_{i+1}]$ and on $[t_{N'}, \infty)$ use convexity: $g(t) \geq g(t_i) + g'(t_i)(t - t_i) \geq \vartheta_i + a_i(t - t_i)$, and integrate. □

The computation. The script `lemmaC/round3/cert/cert.py`, run with the arguments `12 4000 8000 10`, uses the lattice width $h = 2^{-12}$ and produces rounded laws $\tilde{Q}_1, \ldots, \tilde{Q}_{10}$ from $\tilde{Q}_0 = 0$. At stage $n = 0, 1, \ldots, 10$ it does the following with $\tilde{Q} = \tilde{Q}_n$ and $g = g_{\tilde{Q}_n}$ (for $n = 0$ the law is δ_0: there are no cells, $p_0 = 1$ and $\nu_0 = 1$).

(1) It computes double-precision numbers $p_k^{\mathrm{lo}} \in [0, p_k]$, $D_k^{\mathrm{lo}} \in [0, D_k]$ and $\nu_j^{\mathrm{hi}} \geq \nu_j$ for all cells, and an upper bound for $\mathbb{E}\tilde{Q}(E)$ from Lemma 7.15(c), all from the parameters of $\tilde{Q}$ by outward-rounded interval arithmetic.
(2) It computes the three convolutions $p^{\mathrm{lo}} * p^{\mathrm{lo}}$, $D^{\mathrm{lo}} * p^{\mathrm{lo}}$ and $\nu^{\mathrm{hi}} * \nu^{\mathrm{hi}}$, the first two rounded down and the third rounded up.
(3) It chooses points $0 < t_0 < \cdots < t_N$, multiples of 2^{-20}, about 12 000 of them. The choice is made in plain floating point and has no bearing on validity.
(4) It evaluates the three sums of Lemma 7.15(a),(b) at every t_i, the first two rounded down and the third rounded up. This gives numbers $\theta_i \leq g(t_i)$, $a_i \leq g'(t_i)$ and $\hat{s}_i \geq g'(t_i)$. Points at which the θ_i fail to be positive and strictly increasing are discarded, and the program

checks that the remaining θ_i are positive and strictly increasing and that $a_i > 0$.

(5) It evaluates the bound of Lemma 7.15(c), with the points t_i and the numbers θ_i, a_i of step 4, in interval arithmetic and prints an upper bound for $c(\tilde{Q}_n) = 3\big(\mathbb{E}\,\Theta(\tilde{Q}_n)(E) + 2\,\mathbb{E}\tilde{Q}_n(E)\big)$. The internal bound is an upper endpoint; the printed six-decimal figure is that endpoint rounded to nearest (Remark 11.1).

(6) If $n < 10$, it replaces $\hat{s}_i$ by the running maximum $\max_{j\le i}\hat{s}_j$, checks that the result is positive and nondecreasing, and takes $(\theta_i, \hat{s}_i)$ as the parameters of $\tilde{Q}_{n+1}$.

By steps 4 and 6 and Lemma 7.13, each $\tilde{Q}_{n+1}$ satisfies the hypotheses of Lemma 7.14. The arithmetic (the interval enclosures of step 1, including the exponential; the error bound for the floating-point sums of nonnegative terms in steps 2 and 4; the directed rounding) is described in Section 11.

Proposition 7.16 (computer-assisted)**.** *Let $\tilde{Q}_0 = 0$ and let $\tilde{Q}_1, \dots, \tilde{Q}_{10}$ be the rounded laws produced by the run just described. For $0 \le n \le 9$ the parameters of $\tilde{Q}_{n+1}$ satisfy the hypotheses of Lemma 7.13 with $Q = \tilde{Q}_n$, and*

$$c(\tilde{Q}_{10}) \le 2.2519.$$

What the computation certifies is exactly this: the one-sided inequalities $\theta_i \le g_{\tilde{Q}_n}(t_i)$ and $\hat{s}_i \ge g'_{\tilde{Q}_n}(t_i)$ at every retained mesh point of every stage $n \le 9$, and an upward-rounded bound $u \ge c(\tilde{Q}_{10})$ at stage 10. The log of the run (`lemmaC/round3/cert/run_B_h12.log`) prints u to six decimal places, rounded to nearest, as 2.251887; hence $u < 2.2518875 < 2.2519$, which is the form in which the bound is quoted. The parameters of $\tilde{Q}_1, \dots, \tilde{Q}_{10}$ are stored in `lemmaC/round3/v10/B_stage1.npz` to `B_stage10.npz`.

Proof of Theorem 7.1(ii). Let μ be a fixed point and $X \sim \mu$. By Lemma 7.5, $\mu = \mu_{Q^*}$ with $Q^* \in \mathcal{Q}$, $\Theta(Q^*) = Q^*$ and $9\,\mathbb{E}X = c(Q^*)$. By Proposition 7.16 the laws $\tilde{Q}_0, \dots, \tilde{Q}_{10}$ satisfy the hypotheses of Lemma 7.14, so $c(Q^*) \le c(\tilde{Q}_{10}) \le 2.2519 = \alpha_0$. □

Remark 7.17 (floating-point values; evidence only)**.** Nothing in this remark is used in a proof. In floating point, without rounding of the laws, the costs $c_n = c(Q_n)$ of the exact iterates are

n	1	2	3	4	5
c_n	3.41528213	2.66021540	2.39808802	2.30169894	2.26786005

n	6	7	8	9	10
c_n	2.25596056	2.25179199	2.25033206	2.24982093	2.24964199

and $c_0 = 3 \cdot 6^{1/3}\,\Gamma(4/3) = 4.86795\ldots$. The differences $c_n - c_{n+1}$ decrease geometrically with ratio about 0.35, and the limit is numerically 2.2495456. The upper bounds for $c(\tilde{Q}_n)$ printed by the certified run at stages $n = 0, \dots, 10$ are 4.867955, 3.415996, 2.661760, 2.400021, 2.303815, 2.270053,

2.258185, 2.254029, 2.252574, 2.252065, 2.251887; for $n \geq 1$ each exceeds the corresponding c_n by between 0.0007 and 0.0023, the loss caused by the one-sided rounding. Only the last of these bounds is used. Against the integer constant $\beta_0 = 2.302$ of Theorem 5.1 the quoted bound leaves $\beta_0 - \alpha_0 = 0.0501$.

8. Local convergence to the tree

This section compares the neighbourhood of an agent in the near-tight hypergraph $H_q(K)$ with the corresponding ball of the Poisson hypertree $\mathbb{T}$ of Section 6. The comparison is in total variation and the bound is explicit (Theorem 8.4). It is the only fact about the finite market, apart from the finiteness of C_0, that the tree bound of Section 9 uses. The leading error term is $\frac{9}{10}K^2q^{-1/3}$ for each team of the ball, and it comes from near-tight teams that share a pair.

8.1. **Balls, shapes and the two laws.** Throughout the section $q \geq 1$ and $r \geq 1$ are integers and K is a real number with $0 < K \leq q^{2/3}$. We put

$$s = Kq^{-2/3} \leq 1, \qquad \lambda = \frac{K^3}{6} = \Lambda([0,K]), \qquad S_r = \sum_{j=0}^{r-1} (2\lambda)^j.$$

A team e is *near-tight* if $\xi_e \leq K$, that is, if $\Delta(e) \leq s$. The condition $s \leq 1$ makes (2.3) available for every $t \leq s$.

Definition 8.1. Let G be a hypergraph whose vertices are agents, each with its sector, and whose edges are teams, each marked by a cost. The *distance* between two agents of G is the least number of teams in a chain joining them. For an agent v the *ball* $B_r(G,v)$ is the rooted marked hypergraph formed by the root v, all teams of G that contain an agent at distance at most $r-1$ from v, the agents of these teams, their sectors and the costs. It is taken up to bijections of the agents that preserve the root, the sectors, the teams and the costs. The agents at distance at most $r-1$ from v are the *interior* agents of the ball. For the near-tight hypergraph we write $B_r(o) = B_r(H_q(K), o)$.

Let $\mathcal{G}_r$ be the set of finite rooted marked hypergraphs of this kind with costs in $[0,K]$, up to the same bijections. An element of $\mathcal{G}_r$ is an unmarked rooted hypergraph with sectors, of which there are countably many up to isomorphism, together with a vector of costs modulo the finite automorphism group of the unmarked hypergraph. So $\mathcal{G}_r$ is a countable disjoint union of quotients of cubes $[0,K]^n$ by finite groups, and it carries the corresponding Borel σ-field. For probability measures μ, ν on a measurable space, $d_{\mathrm{TV}}(\mu,\nu) = \sup_{\mathcal{A}} |\mu(\mathcal{A}) - \nu(\mathcal{A})|$.

Tree shapes. A *tree shape* of radius r with root sector X is a finite rooted hypertree t of the following kind. The root has sector X. Every agent of depth at most $r-1$ has a finite, linearly ordered set of child teams. Every team consists of its parent agent and two further agents, one in each of the two other sectors. Agents of depth r have no child teams. There are

countably many tree shapes. For a tree shape t we write $E(t)$ for its set of teams, $|E| = |E(t)|$ for their number, $V = 1+2|E|$ for the number of agents, and

$$m = \#\{\text{agents of depth at most } r-1\},$$
$$P_2 = \#\{\text{unordered pairs of distinct teams with a common agent}\}.$$

Let $D_t \subseteq [0,K]^{E(t)}$ be the set of cost vectors with pairwise distinct coordinates that increase along the child teams of every agent.

Let $\mathcal{T}_r \subseteq \mathcal{G}_r$ be the set of balls that are hypertrees (the incidence graph of agents and teams is a tree) with pairwise distinct costs. In such a ball the distance from the root is the depth, every team contains an interior agent, and so agents of depth r have no child teams. Ordering the child teams of every agent by increasing cost, and telling the two further agents of a team apart by their sectors, we see that every element of $\mathcal{T}_r$ with root sector X has exactly one representation (t,ξ) with t a tree shape and $\xi \in D_t$. We use two consequences. First, a bijection of the agents of (t,ξ) that preserves the root, the sectors, the teams and the costs is the identity: it fixes the root, hence each child team of the root, since their costs are distinct, hence the two further agents of each such team, since their sectors differ, and so on by induction on the depth. Secondly, $\mathcal{T}_r$ carries the reference measure ϱ whose restriction to each shape t is Lebesgue measure on D_t. For a tree shape t and a Borel set $A \subseteq D_t$ we write $[t,A] \subseteq \mathcal{T}_r$ for the set of balls with representation (t,ξ), $\xi \in A$. Every measurable subset of $\mathcal{T}_r$ consisting of balls with root sector X is a countable disjoint union of such sets.

The two laws. Let X be a sector.

- $\mu = \mu_{q,K,r,X}$ is the law of $B_r(o)$ for an agent o of sector X of the market. By symmetry it does not depend on the choice of o.
- Let $\mathbb{T}_K$ be the component of the root o of the Poisson hypertree $\mathbb{T}$ (Definition 6.4) after all teams of cost above K have been deleted, as in Remark 6.1, and let $\nu = \nu_{K,r,X}$ be the law of $B_r(\mathbb{T}_K, o)$ when the root has sector X. By Definition 6.4 and the restriction theorem for Poisson processes, ν is the law of the following random marked hypertree: the root has sector X, and every agent of depth at most $r-1$ has, independently of everything else, child teams whose costs are the points of a Poisson process on $[0,K]$ with intensity $\xi^2/2\,\mathrm{d}\xi$, of total mass λ; every child team has two new agents, one in each of the two other sectors.

8.2. **The tree ball.** The facts about Poisson processes used in this section (the restriction theorem, the density of the ordered points of a Poisson process of finite total mass, Campbell's formula) are standard; see [19].

Lemma 8.2. (i) *$\nu(\mathcal{T}_r) = 1$, and ν has the density*

$$g(t,\xi) = e^{-m(t)\lambda} \prod_{e \in E(t)} \frac{\xi_e^2}{2} \tag{8.1}$$

with respect to ϱ.

(ii) *For every measurable* $\varphi\colon [0,K] \to [0,\infty)$,

$$\mathbb{E}_\nu \sum_{e\in E} \varphi(\xi_e) = S_r \int_0^K \varphi(\xi)\, \frac{\xi^2}{2}\, \mathrm{d}\xi.$$

In particular $\mathbb{E}_\nu |E| = \lambda S_r$. *Moreover* $\mathbb{E}_\nu m = S_r$.

(iii) $\mathbb{E}_\nu P_2 = \dfrac{\lambda^2}{2} + \Big(\lambda + \dfrac{\lambda^2}{2}\Big)(S_r - 1) \le \Big(\lambda + \dfrac{\lambda^2}{2}\Big) S_r$.

(iv) $Q_r := \mathbb{E}_\nu V^2 \le (r+1)^2(2r+1)\max(1, 2\lambda)^{2r}$.

Proof. (i) The points $\xi_1 < \cdots < \xi_n$ of a Poisson process on $[0,K]$ with intensity $\xi^2/2\,\mathrm{d}\xi$, listed in increasing order, have density $e^{-\lambda}\prod_i \xi_i^2/2$ on $\{n\} \times \{\xi_1 < \cdots < \xi_n\}$; there is no factorial because the points are ordered. The processes at the m agents of depth at most $r-1$ are independent, which gives (8.1) on the set where the child costs increase at every agent. The costs are almost surely pairwise distinct, so $\nu(\mathcal{T}_r) = 1$.

(ii) Let Z_j be the number of agents of depth j, so $Z_0 = 1$. The variable Z_j is a function of the Poisson processes attached to agents of depth less than j, and is independent of those attached to agents of depth j. Every agent has a Poisson(λ) number of child teams and every team has two new agents, so $\mathbb{E}Z_j = (2\lambda)^j$. By Campbell's formula each agent of depth $j \le r-1$ contributes $\int_0^K \varphi(\xi)\xi^2/2\,\mathrm{d}\xi$ to the expectation of the sum over its child teams. Summing over $j \le r-1$ gives the first formula; $\varphi \equiv 1$ gives $\mathbb{E}_\nu|E| = \lambda S_r$; and $m = \sum_{j<r} Z_j$ gives $\mathbb{E}_\nu m = S_r$.

(iii) Agents of depth r lie in one team of the ball. The root, with N child teams, lies in N teams and contributes $\binom{N}{2}$ pairs; any other agent of depth at most $r-1$ lies in its parent team and N child teams and contributes $\binom{N+1}{2}$. For a Poisson(λ) variable N, $\mathbb{E}\binom{N}{2} = \lambda^2/2$ and $\mathbb{E}\binom{N+1}{2} = \lambda + \lambda^2/2$. The number of child teams of an agent is independent of the presence of the agent, and the expected number of agents of depth between 1 and $r-1$ is $S_r - 1$.

(iv) (Z_j) is a Galton–Watson process whose offspring number is twice a Poisson(λ) variable, with mean 2λ and variance 4λ. The conditional variance formula gives $\operatorname{Var} Z_j = (2\lambda)^2 \operatorname{Var} Z_{j-1} + 4\lambda(2\lambda)^{j-1}$, hence

$$\mathbb{E}Z_j^2 = (2\lambda)^{2j} + 2\sum_{i=0}^{j-1} (2\lambda)^{j+i} \le (1+2j)\max(1,2\lambda)^{2j}.$$

Since $V = \sum_{j=0}^r Z_j$, the Cauchy–Schwarz inequality gives $\mathbb{E}V^2 \le (r+1)\sum_{j=0}^r \mathbb{E}Z_j^2 \le (r+1)^2(2r+1)\max(1,2\lambda)^{2r}$. □

8.3. **One team and its neighbours.**

Lemma 8.3. *Let* e *be a team with pair deficits* d_1, d_2, d_3.

(i) *The cost* ξ_e *has density* $\xi^2/(2q^2)$ *on* $[0,K]$, *and* $\mathbb{P}(e \text{ is near-tight}) = s^3/6 = \lambda/q^2$. *Conditionally on* e *being near-tight,* ξ_e *has density*

$3\xi^2/K^3$ *on* $[0,K]$*, the vector* (d_1,d_2,d_3) *is uniform on the solid simplex* $\{x\in[0,\infty)^3:\ x_1+x_2+x_3\le s\}$*, and* d_1 *has density* $3(s-x)^2/s^3$ *on* $[0,s]$*.*

(ii) *Let* δ',δ'' *be independent and uniform on* $[0,1]$*. For* $x\in[0,s]$*,* $\mathbb{P}(x+\delta'+\delta''\le s)=(s-x)^2/2$*. For* $x,x'\ge 0$*,* $\mathbb{P}(x+x'+\delta'\le s)\le s$*.*

(iii) $\mathbb{E}\big[(s-d_1)^2/2\ \big|\ e \text{ is near-tight}\big]=3s^2/10$*. Consequently, if* $\psi(\xi)=\mathbb{E}[(s-d_1)^2/2\mid \xi_e=\xi]$ *for* $\xi\in[0,K]$*, then*

$$\int_0^K \psi(\xi)\,\frac{3\xi^2}{K^3}\,\mathrm{d}\xi=\frac{3s^2}{10}. \tag{8.2}$$

By symmetry ψ *is the same for* d_2 *and* d_3*.*

(iv) (Pair-sharing rate.) *Given the pair deficits of* e*, the* $3(q-1)$ *teams that share a pair with* e *are near-tight independently of each other, the one through the pair* p *of* e *with probability* $(s-d(p))^2/2$ *when* e *is near-tight. Hence, if* $\pi(e)$ *denotes the number of near-tight teams other than* e *that share a pair with* e*,*

$$\mathbb{E}\big[\pi(e)\ \big|\ e \text{ is near-tight}\big]=3(q-1)\cdot\frac{3s^2}{10}=\frac{9}{10}\,K^2q^{-1/3}\Big(1-\frac{1}{q}\Big). \tag{8.3}$$

Proof. (i) Since $s\le 1$, for $t\le s$ the simplex $\{x\ge 0,\ x_1+x_2+x_3\le t\}$ lies in the unit cube and has volume $t^3/6$; this is (2.3). The density of $\xi_e=q^{2/3}\Delta(e)$ follows by differentiation, and the conditional statements because the deficits are uniform on the cube. The section of the simplex at $x_1=x$ has area $(s-x)^2/2$, and $\big((s-x)^2/2\big)/(s^3/6)=3(s-x)^2/s^3$.

(ii) The triangle $\{\delta',\delta''\ge 0,\ \delta'+\delta''\le s-x\}$ lies in the unit square and has area $(s-x)^2/2$. The second event is contained in $\{\delta'\le s\}$.

(iii) By (i), $\int_0^s \frac{(s-x)^2}{2}\cdot\frac{3(s-x)^2}{s^3}\,\mathrm{d}x=\frac{3}{2s^3}\cdot\frac{s^5}{5}=\frac{3s^2}{10}$. Formula (8.2) is the tower property, because ξ_e has density $3\xi^2/K^3$ given that e is near-tight.

(iv) A team $\alpha\ne e$ that shares a pair with e shares exactly one pair p with it, and there are $q-1$ such teams for each of the three pairs of e. The other two pairs of α are not pairs of e, and for different α these pairs are different. So, given the deficits of e, the events $\{\alpha \text{ is near-tight}\}$ are independent, with the probabilities given by (ii). Take expectations with (iii). □

8.4. **The theorem.**

Theorem 8.4. *Let* $q\ge 1$*,* $r\ge 1$*,* $0<K\le q^{2/3}$*, and let* X *be a sector. Then*

$$d_{\mathrm{TV}}\big(\mu_{q,K,r,X},\nu_{K,r,X}\big)\le\varepsilon_1(q,K,r),$$

$$\varepsilon_1(q,K,r):=\frac{9}{10}\,K^2q^{-1/3}\cdot\lambda S_r+(2\lambda+\lambda^2)\,S_r\,Kq^{-2/3}+\frac{Q_r}{q}+\frac{\lambda^2S_r}{q^2}, \tag{8.4}$$

where $\lambda=K^3/6$*,* $S_r=\sum_{j<r}(2\lambda)^j$ *and* $Q_r\le(r+1)^2(2r+1)\max(1,2\lambda)^{2r}$ *is as in Lemma 8.2. The four terms account, in this order, for a near-tight*

team that shares a pair with one team of the ball; a near-tight team that shares pairs with two teams of the ball; coincidences among the agents when the ball is placed in the market; and the replacement of a binomial count by a Poisson one. In particular, for an agent o of the market,

$$\mathbb{P}\big(B_r(o) \text{ is not a hypertree}\big) \le \varepsilon_1(q, K, r),$$

and $\varepsilon_1(q, K, r) \to 0$ as $q \to \infty$ for fixed K and r. The same bound holds for the total variation distance between the law of $B_r(H_q(K), v)$, where v is a uniformly random agent independent of the market, and the law of $B_r(\mathbb{T}_K, o)$, where the root sector of $\mathbb{T}$ is uniform.

The first term is the pair-sharing rate (8.3) for one team, multiplied by the expected number λS_r of teams in the ball of the tree.

Proof. Fix an agent o of sector X, a tree shape t with root sector X and a Borel set $A \subseteq D_t$.

Step 1: the event $\{B_r(o) \in [t, A]\}$. An *embedding* is an injective map ϕ from the agents of t to the agents of the market that preserves sectors and sends the root to o. For a team e of t, $\phi(e)$ is the team formed by the images of its agents; these are the *tree teams* of ϕ, and the pairs contained in them are its *tree pairs*. Put $\xi_\phi = (\xi_{\phi(e)})_{e \in E(t)}$, and let W_ϕ be the number of near-tight teams that are not tree teams and contain $\phi(a)$ for some agent a of t of depth at most $r-1$. We claim that

$$\{B_r(o) \in [t, A]\} = \bigsqcup_\phi \{\xi_\phi \in A\} \cap \{W_\phi = 0\}, \tag{8.5}$$

a disjoint union over all embeddings. Suppose $\xi_\phi \in A$ and $W_\phi = 0$. Then the tree teams are near-tight, and by induction on $j \le r-1$ the agents at distance j from o in $H_q(K)$ are exactly the images of the agents of depth j, because every near-tight team that contains such an image is a tree team. Hence the teams of $B_r(o)$ are exactly the tree teams, and $B_r(o)$ has the representation (t, ξ_ϕ). Conversely, if $B_r(o) \in [t, A]$, an isomorphism from (t, ξ) onto $B_r(o)$ is an embedding ϕ with $\xi_\phi = \xi \in A$, and every near-tight team containing an interior agent of $B_r(o)$ belongs to $B_r(o)$, so $W_\phi = 0$. The union is disjoint because two embeddings for which the event holds differ by a bijection of the agents of (t, ξ) that preserves the root, the sectors, the teams and the costs, and such a bijection is the identity.

Two different tree teams share at most one agent, because t is a hypertree. So the $3|E|$ tree pairs are distinct, the triples of pair deficits of the tree teams are independent, and by Lemma 8.3(i) the vector ξ_ϕ has density $\prod_e \xi_e^2/(2q^2)$ on $[0, K]^{E(t)}$. Write $\mathbf{d}_\phi$ for the family of deficits of the tree pairs. Then, by (8.5),

$$\mu([t, A]) = \sum_\phi \mathbb{E}\big[\mathbf{1}\{\xi_\phi \in A\}\, \mathbb{P}(W_\phi = 0 \mid \mathbf{d}_\phi)\big]. \tag{8.6}$$

The agents of t other than the root are placed one at a time, and at each of the $2|E|$ steps at least $q - V$ agents of the required sector are free. So the number of embeddings is at least

$$(q-V)_+^{2|E|} \;\geq\; q^{2|E|}\Big(1 - \frac{2|E|V}{q}\Big)_+, \tag{8.7}$$

by Bernoulli's inequality when $V \leq q$; when $V > q$ the right-hand side is 0.

Step 2: the other teams. Fix an embedding ϕ and let F be the set of teams that are not tree teams and contain the image of an agent of depth at most $r-1$, so that W_ϕ is the number of near-tight teams in F. Classify the teams of F by the number of tree pairs they contain.

- F_0: no tree pair. Each agent lies in q^2 teams, so $|F_0| \leq mq^2$. Given $\mathbf{d}_\phi$, a team of F_0 is near-tight with probability $s^3/6 = \lambda/q^2$.
- F_1: exactly one tree pair p, contained in the tree team $\phi(e)$. The team is one of the $q-1$ other teams through p, and its other two pairs are not tree pairs. Given $\mathbf{d}_\phi$, on the event $\{\xi_\phi \in A\}$ (where $d(p) \leq s$) it is near-tight with probability $(s - d(p))^2/2$, by Lemma 8.3(ii).
- F_2: exactly two tree pairs. Two pairs of a team have a common agent b. If both were contained in the same tree team, the team would be that tree team. So they lie in two different tree teams $\phi(e_1)$, $\phi(e_2)$ through b, and the team consists of b, one further agent of $\phi(e_1)$ and the agent of $\phi(e_2)$ in the remaining sector. There are two such teams for each pair $\{e_1, e_2\}$ of teams of t with a common agent, so $|F_2| \leq 2P_2$. The third pair of the team is not a tree pair, so given $\mathbf{d}_\phi$ the team is near-tight with probability at most s, by Lemma 8.3(ii).
- Three tree pairs: impossible. No two of them lie in the same tree team, so they would lie in three different tree teams that pairwise share an agent of the team, and this is a cycle of length six in the incidence graph of $\phi(t)$.

Step 3: Harris' inequality. Conditionally on $\mathbf{d}_\phi$, the deficits of all other pairs are independent and uniform on $[0,1]$, and for each $\alpha \in F$ the event $\{\alpha \text{ is not near-tight}\} = \{\Delta(\alpha) > s\}$ is increasing in them. By Harris' inequality [12], with $p_\alpha = \mathbb{P}(\alpha \text{ is near-tight} \mid \mathbf{d}_\phi)$,

$$\mathbb{P}(W_\phi = 0 \mid \mathbf{d}_\phi) \;\geq\; \prod_{\alpha \in F} (1 - p_\alpha).$$

On the event $\{\xi_\phi \in A\}$ put

$$Y_\phi = (q-1) \sum_{e \in E(t)} \sum_{p \subset \phi(e)} \frac{(s - d(p))^2}{2},$$

the inner sum being over the three pairs of $\phi(e)$. By Step 2, $\sum_{\alpha \in F_1} p_\alpha \leq Y_\phi$ and $\sum_{\alpha \in F_2} p_\alpha \leq 2P_2 s$. We use $\prod_i (1 - p_i) \geq 1 - \sum_i p_i$ for $p_i \in [0,1]$ on $F_1 \cup F_2$, and on F_0 the inequality $\log(1-u) \geq -u - u^2$ with $u = \lambda/q^2 \leq 1/6$

(recall $K \le q^{2/3}$), which gives $(1-\lambda/q^2)^{mq^2} \ge e^{-m\lambda}(1-m\lambda^2/q^2)$. Hence, on $\{\xi_\phi \in A\}$,

$$\mathbb{P}(W_\phi = 0 \mid \mathbf{d}_\phi) \;\ge\; e^{-m\lambda}\Big(1-\frac{m\lambda^2}{q^2}\Big)_+\big(1-Y_\phi-2P_2 s\big)_+. \tag{8.8}$$

(The positive parts may be taken because the left-hand side is nonnegative.)

Step 4: a lower bound for the density of μ on $\mathcal{T}_r$. The triples of pair deficits of the tree teams are independent, so for a pair p of $\phi(e)$ we have $\mathbb{E}[(s-d(p))^2/2 \mid \xi_\phi] = \psi(\xi_{\phi(e)})$, with ψ as in Lemma 8.3(iii). Therefore, with

$$\omega_t(\xi) = 1 - 3(q-1)\sum_{e\in E(t)} \psi(\xi_e) - 2P_2 K q^{-2/3},$$

and since $s = Kq^{-2/3}$,

$$\begin{aligned}\mathbb{E}\big[\mathbf{1}\{\xi_\phi \in A\}\,(1-Y_\phi-2P_2 s)_+\big] &\ge \mathbb{E}\big[\mathbf{1}\{\xi_\phi \in A\}\,(1-Y_\phi-2P_2 s)\big]\\ &= \int_A \omega_t(\xi)\prod_{e\in E(t)}\frac{\xi_e^2}{2q^2}\,\mathrm{d}\xi.\end{aligned}$$

Insert (8.8) into (8.6). Every term of the sum over ϕ is then bounded below by the same nonnegative quantity, so the sum is at least that quantity times the lower bound (8.7) for the number of embeddings; then apply the last display, whose left-hand side is multiplied by a nonnegative factor, and compare with (8.1). With

$$\underline{f}(t,\xi) = \Big(1-\frac{2|E|V}{q}\Big)_+\Big(1-\frac{m\lambda^2}{q^2}\Big)_+\omega_t(\xi) \tag{8.9}$$

we obtain $\mu([t,A]) \ge \int_{[t,A]} \underline{f}\,\mathrm{d}\nu$. Since the sets $[t,A]$ exhaust the measurable subsets of $\mathcal{T}_r$ with root sector X up to countable disjoint unions,

$$\mu(\mathcal{A}) \;\ge\; \int_{\mathcal{A}} \underline{f}_+\,\mathrm{d}\nu \qquad \text{for every measurable } \mathcal{A}\subseteq\mathcal{T}_r \tag{8.10}$$

(apply the bound to $\mathcal{A}\cap\{\underline{f}>0\}$). The first two factors of (8.9) lie in $[0,1]$ and the third is at most 1, so

$$(1-\underline{f})_+ \;\le\; \frac{2|E|V}{q} + \frac{m\lambda^2}{q^2} + 2P_2 K q^{-2/3} + 3(q-1)\sum_{e\in E(t)}\psi(\xi_e). \tag{8.11}$$

Indeed, if the third factor is negative then $\underline{f}$ is at least that factor; and if it is nonnegative, $(1-a)_+(1-b)_+(1-c) \ge 1-a-b-c$ for $a,b\ge 0$ and $c\in[0,1]$.

Step 5: total variation. Only the lower bound (8.10) is needed. For measurable $\mathcal{A}\subseteq\mathcal{G}_r$, since $\nu(\mathcal{T}_r)=1$,

$$\nu(\mathcal{A})-\mu(\mathcal{A}) \le \nu(\mathcal{A}\cap\mathcal{T}_r)-\mu(\mathcal{A}\cap\mathcal{T}_r) \le \int_{\mathcal{A}\cap\mathcal{T}_r}(1-\underline{f}_+)\,\mathrm{d}\nu \le \int (1-\underline{f})_+\,\mathrm{d}\nu,$$

and $d_{\mathrm{TV}}(\mu,\nu) = \sup_{\mathcal{A}}(\nu(\mathcal{A})-\mu(\mathcal{A}))$ for probability measures. It remains to integrate (8.11) against ν, with Lemma 8.2.

- By Lemma 8.2(ii) with $\varphi = \psi$ and by (8.2),
$$\mathbb{E}_\nu \sum_e \psi(\xi_e) = S_r \int_0^K \psi(\xi)\, \frac{\xi^2}{2}\, \mathrm{d}\xi = \lambda S_r \cdot \frac{3s^2}{10}.$$
So the last term has ν-mean $3(q-1)\cdot \frac{3s^2}{10}\cdot \lambda S_r \le \frac{9}{10} K^2 q^{-1/3}\cdot \lambda S_r$.
- By Lemma 8.2(iii), $\mathbb{E}_\nu\big[2P_2 K q^{-2/3}\big] \le (2\lambda + \lambda^2) S_r K q^{-2/3}$.
- $2|E|V \le V^2$, so $\mathbb{E}_\nu[2|E|V/q] \le Q_r/q$.
- $\mathbb{E}_\nu m = S_r$, so $\mathbb{E}_\nu[m\lambda^2/q^2] = \lambda^2 S_r/q^2$.

The sum is $\varepsilon_1(q,K,r)$. Every ball in $\mathcal{T}_r$ is a hypertree and $\nu(\mathcal{T}_r) = 1$, so $\mathbb{P}(B_r(o) \text{ is not a hypertree}) \le \mu(\mathcal{G}_r \setminus \mathcal{T}_r) \le d_{\mathrm{TV}}(\mu,\nu) \le \varepsilon_1$. For fixed K and r every term of ε_1 tends to 0. Finally, a uniformly random agent has a uniformly random sector, and the total variation distance between two mixtures with the same weights is at most the largest distance between the components. □

8.5. **Two roots.**

Proposition 8.5. *Let $o \neq o'$ be agents with sectors X, X', and let μ_2 be the joint law of $(B_r(o), B_r(o'))$. Under the hypotheses of Theorem 8.4,*
$$d_{\mathrm{TV}}\big(\mu_2,\ \nu_{K,r,X} \otimes \nu_{K,r,X'}\big) \ \le\ 2\,\varepsilon_1(q,K,r) + \frac{2Q_r}{q} \ \le\ 4\,\varepsilon_1(q,K,r).$$

Proof. Repeat the proof of Theorem 8.4 for a pair of tree shapes t, t' with root sectors X, X', a Borel set $A \subseteq D_t \times D_{t'}$ and pairs of embeddings (ϕ, ϕ') of t and t' with $\phi(\text{root}) = o$, $\phi'(\text{root}) = o'$ and disjoint images. Quantities of t' carry a prime. The tree teams are now those of ϕ and of ϕ', and $W_{\phi,\phi'}$ counts the near-tight teams that are not tree teams and contain the image of an agent of depth at most $r-1$ of t or of t'.

Step 1. If $(\xi_\phi, \xi_{\phi'}) \in A$ and $W_{\phi,\phi'} = 0$, then the induction of Step 1 above, started at o, stays inside the image of ϕ, because a tree team that contains an agent of the image of ϕ is a tree team of ϕ. So $B_r(o)$ has the representation (t, ξ_ϕ), and likewise $B_r(o')$ has the representation $(t', \xi_{\phi'})$. The events for different pairs (ϕ, ϕ') are disjoint as before. Hence $\{(B_r(o), B_r(o')) \in [t, t', A]\}$ *contains* the disjoint union of the events $\{(\xi_\phi, \xi_{\phi'}) \in A\} \cap \{W_{\phi,\phi'} = 0\}$. It may be larger, since the two balls may overlap; this is harmless, because only a lower bound for μ_2 is needed. The number of pairs of embeddings with disjoint images is at least $q^{2|E|+2|E'|}\big(1 - 2(|E| + |E'|)(V + V')/q\big)_+$. Tree teams of the same embedding share at most one agent and tree teams of different embeddings share none, so the $3(|E| + |E'|)$ tree pairs are distinct and the costs of the tree teams are independent.

Step 2. $|F_0| \le (m + m')q^2$. F_1 is as before for each tree team. A team with exactly two tree pairs contains their common agent b, and the two tree teams through b belong to the same embedding; so $|F_2| \le 2(P_2 + P_2')$. Three tree pairs are impossible as before.

Steps 3 to 5. Harris' inequality applies as before, and the density of μ_2 with respect to $\nu_{K,r,X} \otimes \nu_{K,r,X'}$ on $\mathcal{T}_r \times \mathcal{T}_r$ is bounded below by a function $\underline{f}_2$ with

$$(1 - \underline{f}_2)_+ \le \frac{2(|E| + |E'|)(V + V')}{q} + \frac{(m + m')\lambda^2}{q^2} + 2(P_2 + P_2')Kq^{-2/3} + 3(q-1)\Big(\sum_{e \in E(t)} \psi(\xi_e) + \sum_{e \in E(t')} \psi(\xi'_e) \Big).$$

The last three terms have mean at most twice the corresponding terms of ε_1. For the first, $2(|E|+|E'|)(V+V') \le (V+V')^2 \le 2V^2 + 2V'^2$, with mean at most $4Q_r$. □

Corollary 8.6. *Let $h \colon \mathcal{G}_r \to [-1,1]$ be measurable, let*

$$A_q = \frac{1}{3q} \sum_v h\big(B_r(H_q(K), v)\big)$$

be its average over the $3q$ agents, and let $\bar{\nu}(h)$ be the mean of $h(B_r(\mathbb{T}_K, o))$ when the root sector is uniform. Then

$$|\mathbb{E} A_q - \bar{\nu}(h)| \le 2\,\varepsilon_1(q, K, r), \qquad \operatorname{Var} A_q \le \frac{1}{3q} + 12\,\varepsilon_1(q, K, r).$$

In particular $A_q \to \bar{\nu}(h)$ in probability as $q \to \infty$ for fixed K and r.

Proof. If $|h| \le 1$ and μ, ν are probability measures, then $|\int h\,\mathrm{d}\mu - \int h\,\mathrm{d}\nu| \le 2\,d_{\mathrm{TV}}(\mu,\nu)$. This and Theorem 8.4 give the first bound. Let $o \ne o'$ be agents with sectors X, X', and write $\nu_X(h) = \int h\,\mathrm{d}\nu_{K,r,X}$, $h_o = h(B_r(o))$. By Proposition 8.5 and Theorem 8.4,

$$\big|\mathbb{E}[h_o h_{o'}] - \nu_X(h)\nu_{X'}(h)\big| \le 8\varepsilon_1, \qquad \big|\mathbb{E}h_o\,\mathbb{E}h_{o'} - \nu_X(h)\nu_{X'}(h)\big| \le 4\varepsilon_1.$$

So the covariance of h_o and $h_{o'}$ is at most $12\varepsilon_1$ in absolute value. The $3q$ diagonal terms of the variance of $\sum_v h_v$ are at most 1 each. □

The corollary is not used in the sequel: Section 9 needs only expectations, that is, Theorem 8.4. Note the rates for fixed K and r: the bias of A_q is $O(q^{-1/3})$, and its fluctuation is $O(q^{-1/6})$ in probability.

8.6. **Remarks.**

Remark 8.7 (Reading the bound)**.** For fixed K and r the first term of ε_1 is of order $q^{-1/3}$ and the other three are $O(q^{-2/3})$. For $K \ge 2$ we have $2\lambda \ge 8/3$ and $\lambda S_r \le \frac{4}{5}(2\lambda)^r$, so the first term is at most

$$0.72\,K^2\Big(\frac{K^3}{3}\Big)^r q^{-1/3}.$$

The bound is therefore informative at radius r only when $q^{1/3}$ is large compared with $K^{3r+2}/3^r$, and for a single team the requirement is that $q^{1/3}$ be large compared with K^2. The theorem is for fixed K and r. Nothing here controls K growing with q; by (8.3), when K^2 is comparable to $q^{1/3}$ a

near-tight team shares a pair with a number of near-tight teams that is of order one in expectation.

Remark 8.8 (Where pairs enter)**.** The fact that a near-tight team has small pair deficits, which favours the other teams through its pairs, enters the proof only through the classes F_1 and F_2. It produces the rate $K^2 q^{-1/3}$. Teams that share only an agent with the ball contribute nothing beyond the last term of ε_1.

Remark 8.9 (Floating point: how large q must be)**.** The numbers in this remark are floating-point evidence and are not used in any proof. The identity (8.3) is exact, and it shows that the tree regime starts late: the expected number of near-tight partners of a near-tight team through a shared pair is below $1/10$ only when q is at least about $(9K^2)^3$, that is, $3.0 \cdot 10^6$ for $K = 4$ and $3.4 \cdot 10^7$ for $K = 6$. Simulations of fresh markets with all q^3 teams enumerated (script `lemmaC/round3/localconv/pairshare.py`, output `out_pairshare.txt` in the same folder; 300, 80 and 14 markets at $q = 32, 64, 128$) agree with (8.3) to within sampling error: at $K = 4$ the mean number of partners per near-tight team was 4.41, 3.60, 2.81 against 4.39, 3.54, 2.84 from the formula. In the same runs, at $K = 4$ the fraction of near-tight teams with at least one such partner was 0.987, 0.970, 0.934, and the fraction of agents whose ball of radius 1 is not a hypertree was 0.952, 0.952, 0.945; at $K = 2$ the latter fraction was 0.235, 0.210, 0.185. So at every size that can be simulated the near-tight hypergraph is far from a hypertree, and Theorem 8.4, like every bound that rests on it, says nothing at such sizes.

9. The fractional optimum is bounded by the tree

This section proves that the cost of any invariant perfect matching on the Poisson hypertree is an asymptotic upper bound for the fractional deficit of the market. The tree and invariance are those of Section 6; balls, the truncated tree $\mathbb{T}_K$ and the bound $\varepsilon_1(q, K, r)$ are those of Section 8; C_0 is the constant of (3.1).

9.1. **Statement.** We recall the setting of Section 6. The Poisson hypertree $(\mathbb{T}, o, \xi)$ of Definition 6.4 is a random rooted hypertree with costs; its root sector is uniform; almost surely every agent lies in infinitely many teams, with pairwise distinct costs of which finitely many lie below any level. Rooted marked hypertrees are coded by ranks, and the σ-field on them is generated by the truncated balls $B_r(\mathbb{T}_K, o)$, $r, K \in \mathbb{N}$ (Remark 6.1; $\mathbb{T}_K$ is the truncation written T_K there, applied to $\mathbb{T}$, and the balls are those of Definition 8.1). An *invariant perfect matching* on $\mathbb{T}$ is a family $m = (m_e)$ of marks in $\{0, 1\}$ on the teams of $\mathbb{T}$, possibly defined with the help of additional randomness, such that almost surely every agent lies in exactly one team e with $m_e = 1$, and such that the decorated tree $(\mathbb{T}, o, \xi, m)$ is invariant in the sense of Definition 6.2. Two agents are adjacent, $a \sim b$, when they lie

in a common team, which is then unique. Invariance is the identity (6.1): for every measurable $f \geq 0$ on doubly rooted decorated hypertrees,

$$\mathbb{E} \sum_{b \sim o} f(o,b) = \mathbb{E} \sum_{b \sim o} f(b,o), \tag{9.1}$$

where $f(a,b)$ is the value of f at the decorated tree with first root a and second root b. By linearity (9.1) also holds for signed f with $\mathbb{E} \sum_{b \sim o} (|f(o,b)| + |f(b,o)|) < \infty$. The *cost* (6.2) of m is

$$c(m) = \mathbb{E} \sum_{e \ni o} \xi_e \, m_e \ \in [0, \infty],$$

the expected cost of the team of the root.

Theorem 9.1. *Let m be an invariant perfect matching on $\mathbb{T}$. Then*

$$\limsup_{q \to \infty} \frac{\mathbb{E}\Phi_q^{\mathrm{LP}}}{q^{1/3}} \ \leq \ c(m).$$

More precisely, there are numbers $c_r \leq c(m)$ and $\varepsilon_r \in [0,1]$, $r \geq 1$, depending only on the law of $(\mathbb{T}, o, \xi, m)$, with $\varepsilon_r \to 0$ as $r \to \infty$, such that for all integers $r \geq 1$ and $q \geq r^{3/2}$ and every $\eta \in (0,1)$,

$$\frac{\mathbb{E}\Phi_q^{\mathrm{LP}}}{q^{1/3}} \ \leq \ \frac{c_r + r\,\varepsilon_1(q,r,r+1)}{1-\eta} + \frac{C_0}{\eta}\big(\varepsilon_r + \varepsilon_1(q,r,r+1)\big)^{1/3}, \tag{9.2}$$

with ε_1 as in Theorem 8.4.

If $c(m) = \infty$ there is nothing to prove, and we assume $c(m) < \infty$. The proof has five steps. The conditional expectation of m given the tree is a fractional perfect matching that all three agents of a team compute alike (Lemma 9.2). Conditioning further on a ball of finite radius, and taking the minimum over the three agents of a team, gives a fractional packing of finite range whose cost is at most $c(m)$ and whose slack is small (Lemmas 9.5 and 9.6). The same rule is applied in the market, where its mean cost and mean slack are those of the tree up to the total variation distance of Theorem 8.4 (Lemma 9.7). The pair deficits are split into two independent uniform families (Lemma 9.8): the packing is built on the first, and its slack is filled by optimal assignments of the second on random sets of agents, priced by C_0 (Lemma 9.9).

9.2. **The conditional matching.** Let $\ell_1, \ell_2, \ldots$ be the teams at the root o in increasing order of cost. For an agent v and a team $e \ni v$ let $\mathrm{rk}_v(e)$ be the rank of e among the teams at v, so that $\mathrm{rk}_o(\ell_j) = j$. Fix measurable functions h_j, $j \geq 1$, of rooted marked hypertrees, with values in $[0,1]$, such that

$$h_j(\mathbb{T}, o, \xi) = \mathbb{E}\big[m_{\ell_j} \,\big|\, \mathbb{T}, o, \xi\big] \qquad \text{almost surely.}$$

For an agent v and a team $e \ni v$ put

$$x_e^v = h_{\mathrm{rk}_v(e)}(\mathbb{T}, v, \xi),$$

where $(\mathbb{T}, v, \xi)$ is the tree re-rooted at v. So x_e^v is the conditional probability that e is matched, computed from the point of view of v.

Lemma 9.2. *Almost surely $\sum_{e \ni o} x_e^o = 1$, and $x_e^b = x_e^o$ for every team $e \ni o$ and every agent $b \in e$. Moreover $\mathbb{E} \sum_{e \ni o} \xi_e \, x_e^o = c(m)$.*

Proof. By monotone convergence for conditional expectations, $\sum_j h_j = \mathbb{E}[\sum_{e \ni o} m_e \mid \mathbb{T}, o, \xi] = 1$ and $\mathbb{E} \sum_j \xi_{\ell_j} h_j = \mathbb{E} \sum_j \xi_{\ell_j} m_{\ell_j} = c(m)$, because $\xi_{\ell_j} \geq 0$ is a function of $(\mathbb{T}, o, \xi)$.

For adjacent agents a, b with common team e put

$$f(a,b) = \operatorname{sgn}\bigl(x_e^a - x_e^b\bigr)\,(m_e - x_e^b).$$

This is a measurable function of the doubly rooted decorated tree, and $|f(a,b)| \leq m_e + x_e^b$. We have $\mathbb{E} \sum_{b \sim o} m_e = 2$, because the matched team of the root has two further agents, and $\mathbb{E} \sum_{b \sim o} x_e^o = 2\,\mathbb{E} \sum_j h_j = 2$. By (9.1) for the nonnegative function $(a,b) \mapsto x_e^b$, $\mathbb{E} \sum_{b \sim o} x_e^b = \mathbb{E} \sum_{b \sim o} x_e^o = 2$. So $\mathbb{E} \sum_{b \sim o} (|f(o,b)| + |f(b,o)|) \leq 8$, and (9.1) holds for f. We compute both sides.

Right-hand side. Grouping the neighbours of o by their team,

$$\mathbb{E} \sum_{b \sim o} f(b,o) = \mathbb{E} \sum_{j \geq 1} \Psi_j \,(m_{\ell_j} - h_j), \qquad \Psi_j = \sum_{b \in \ell_j \setminus \{o\}} \operatorname{sgn}\bigl(x_{\ell_j}^b - h_j\bigr).$$

The variable Ψ_j is a function of $(\mathbb{T}, o, \xi)$, since the tree re-rooted at a neighbour of the root is a measurable function of the tree (Section 6); and $|\Psi_j| \leq 2$. Hence $\mathbb{E}[\Psi_j (m_{\ell_j} - h_j)] = 0$ for every j by the definition of h_j. The sum over j may be taken outside the expectation because $\mathbb{E} \sum_j |\Psi_j| (m_{\ell_j} + h_j) \leq 4$. So the right-hand side is 0.

Left-hand side. For $b \in \ell_j \setminus \{o\}$ the variables $x_{\ell_j}^b$ and $\operatorname{sgn}(h_j - x_{\ell_j}^b)$ are functions of $(\mathbb{T}, o, \xi)$. Conditioning on $(\mathbb{T}, o, \xi)$ replaces m_{ℓ_j} by h_j; this is justified by the integrability shown above. Hence

$$\mathbb{E} \sum_{b \sim o} f(o,b) = \mathbb{E} \sum_{j \geq 1} \sum_{b \in \ell_j \setminus \{o\}} \operatorname{sgn}\bigl(h_j - x_{\ell_j}^b\bigr)\bigl(h_j - x_{\ell_j}^b\bigr) = \mathbb{E} \sum_{e \ni o} \sum_{b \in e \setminus \{o\}} \bigl|x_e^o - x_e^b\bigr|.$$

The two sides are equal, so the last expectation is 0. □

So $x_e := x_e^o$, $e \ni o$, is a fractional perfect matching at the root that is a function of the tree and its costs alone, has the cost of m, and is seen equally by the three agents of each team. The matching m itself may use additional randomness; from here on it enters only through the functions h_j.

9.3. **Rules of finite range.** Fix an integer $r \geq 1$ and a level $K > 0$. We write $\mathcal{G}_r(K)$ for the space of balls $\mathcal{G}_r$ of Section 8.1 when the level has to be shown.

Definition 9.3. A *rule of range r at level K* is a sequence $\gamma = (\gamma_j)_{j \geq 1}$ of measurable functions $\gamma_j \colon \mathcal{G}_r(K) \to [0,1]$ such that, for every ball B: $\sum_j \gamma_j(B) \leq 1$; $\gamma_j(B) = 0$ if the root of B lies in fewer than j teams; and

$\gamma_j(B) = 0$ for all j unless B is a hypertree in which the teams at every agent have pairwise distinct costs.

Let G be a hypergraph of teams with costs in $[0, K]$ as in Definition 8.1; below G is $H_q(K)$ or $\mathbb{T}_K$. A rule γ assigns to every agent v and every team $e \ni v$ of G the number

$$y_e^v = \gamma_{\mathrm{rk}_v(e)}\big(B_r(G, v)\big), \tag{9.3}$$

where the rank is taken among the teams of G at v; when two of these have equal costs the rule vanishes and no rank is needed. The associated weights, slacks and local costs are

$$x_e^\gamma = \min_{v \in e} y_e^v, \qquad \sigma_v = 1 - \sum_{e \ni v} x_e^\gamma, \qquad \kappa_v = \sum_{e \ni v} \xi_e \, x_e^\gamma. \tag{9.4}$$

Each agent proposes a weight for each of its teams from what it sees within distance r, and a team receives the least of its three proposals.

Lemma 9.4. *Let γ be a rule of range r at level K.*

(a) (Packing.) *$x^\gamma \geq 0$ and $\sum_{e \ni v} x_e^\gamma \leq \sum_{e \ni v} y_e^v \leq 1$ for every agent v. Hence $\sigma_v \in [0, 1]$ and $\kappa_v \in [0, K]$.*

(b) (Range.) *There are measurable functions $S^\gamma \colon \mathcal{G}_{r+1}(K) \to [0, 1]$ and $C^\gamma \colon \mathcal{G}_{r+1}(K) \to [0, K]$, depending only on γ, such that for every G and every agent v of G,*

$$\sigma_v = S^\gamma\big(B_{r+1}(G, v)\big), \qquad \kappa_v = C^\gamma\big(B_{r+1}(G, v)\big).$$

Proof. (a) $\sum_{e \ni v} y_e^v = \sum_j \gamma_j(B_r(G, v)) \leq 1$, the sum being over the ranks of the teams at v.

(b) Let v be an agent and $e \ni v$ a team of G; it belongs to $B_{r+1}(G, v)$. Let $b \in e$. A team of $B_r(G, b)$ contains an agent a at distance at most $r - 1$ from b, hence at distance at most r from v, so it is a team of $B_{r+1}(G, v)$. The same holds for every team of a shortest chain from b to a. Therefore $B_r(G, b)$ is the ball of radius r around b inside the hypergraph $B_{r+1}(G, v)$, and it contains all teams of G at b, so the rank of e at b can be read in it. Thus x_e^γ, for $e \ni v$, is a function of $B_{r+1}(G, v)$. Define $S^\gamma(B')$ and $C^\gamma(B')$ for an arbitrary ball $B' \in \mathcal{G}_{r+1}(K)$ with root ρ by the formulas (9.3) and (9.4) for $v = \rho$, applied inside the hypergraph B'. The bounds follow from (a). □

9.4. **The min-packing on the tree.** Apply a rule γ of range r at level K to $G = \mathbb{T}_K$. Truncation at K removes the most expensive teams at each agent, so it does not change the ranks of the remaining teams. For a team e of $\mathbb{T}$ with $\xi_e > K$ we put $y_e^v = 0$ for $v \in e$ and $x_e^\gamma = 0$; for a team with $\xi_e \leq K$ at a neighbour b of the root, y_e^b is given by (9.3) with $G = \mathbb{T}_K$. We write

$$\hat{\gamma}_j = y_{\ell_j}^o = \gamma_j\big(B_r(\mathbb{T}_K, o)\big),$$

which is 0 when $\xi_{\ell_j} > K$.

Lemma 9.5. *For every rule γ of range r at level K,*

$$\mathbb{E}\, C^\gamma\big(B_{r+1}(\mathbb{T}_K, o)\big) \le \mathbb{E}\sum_{j\ge 1} \xi_{\ell_j}\hat\gamma_j,$$
$$\mathbb{E}\, S^\gamma\big(B_{r+1}(\mathbb{T}_K, o)\big) \le 3\,\mathbb{E}\sum_{j\ge 1} \big(h_j - \hat\gamma_j\big)_+.$$

Proof. The first inequality holds because $x_e^\gamma \le y_e^o$ for $e \ni o$. For the second, let $x_e = x_e^o$ be as in Lemma 9.2. Almost surely $\sum_{e\ni o} x_e = 1$ and $x_e^v = x_e$ for all $e \ni o$ and $v \in e$, so

$$\sigma_o = \sum_{e\ni o}\big(x_e - x_e^\gamma\big) = \sum_{e\ni o}\max_{v\in e}\big(x_e^v - y_e^v\big) \le \sum_{e\ni o}\ \sum_{v\in e}\big(x_e^v - y_e^v\big)_+.$$

The terms with $v = o$ sum to $\sum_j (h_j - \hat\gamma_j)_+$. For the others apply (9.1) to the nonnegative function $f(a,b) = (x_e^b - y_e^b)_+$, where e is the common team of a and b; it is a measurable function of the doubly rooted tree. This gives

$$\mathbb{E}\sum_{b\sim o}\big(x_e^b - y_e^b\big)_+ = \mathbb{E}\sum_{b\sim o}\big(x_e^o - y_e^o\big)_+ = 2\,\mathbb{E}\sum_{j\ge 1}\big(h_j - \hat\gamma_j\big)_+,$$

since every team at the root contains two neighbours of the root. □

From now on the level is tied to the range: $K = K_r = r$. Let

$$\mathcal{F}_r = \sigma\big(B_r(\mathbb{T}_r, o)\big).$$

Lemma 9.6. *For every $r \ge 1$ there is a rule $g_r = (g_{r,j})_{j\ge1}$ of range r at level r such that, almost surely, for every j,*

$$g_{r,j}\big(B_r(\mathbb{T}_r,o)\big) = \mathbb{E}\big[m_{\ell_j}\mathbf{1}\{\xi_{\ell_j}\le r\}\,\big|\,\mathcal{F}_r\big] = \mathbf{1}\{\xi_{\ell_j}\le r\}\,\mathbb{E}\big[h_j\,\big|\,\mathcal{F}_r\big]. \tag{9.5}$$

For these rules, with $c_r := \mathbb{E}\, C^{g_r}(B_{r+1}(\mathbb{T}_r,o))$ and $\varepsilon_r := \mathbb{E}\, S^{g_r}(B_{r+1}(\mathbb{T}_r,o))$,

(a) *$c_r \le \mathbb{E}\sum_{e\ni o}\xi_e\, m_e\mathbf{1}\{\xi_e\le r\} \le c(m)$ for every r;*
(b) *$g_{r,j}(B_r(\mathbb{T}_r,o)) \to h_j$ almost surely as $r\to\infty$, for every j, and $\varepsilon_r \to 0$.*

Proof. The event $\{\xi_{\ell_j}\le r\}$ says that the root of $B_r(\mathbb{T}_r,o)$ lies in at least j teams, so it belongs to $\mathcal{F}_r$; this gives the second equality in (9.5), because $\mathcal{F}_r$ is contained in the σ-field of $(\mathbb{T},o,\xi)$. Choose measurable functions $G_j \ge 0$ on $\mathcal{G}_r(r)$ with $G_j(B_r(\mathbb{T}_r,o)) = \mathbb{E}[m_{\ell_j}\mathbf{1}\{\xi_{\ell_j}\le r\}\mid\mathcal{F}_r]$ almost surely. Almost surely $\sum_j G_j \le 1$, because $\sum_j m_{\ell_j} = 1$; $G_j = 0$ when the root lies in fewer than j teams; and the ball is a hypertree with pairwise distinct costs. Let g_r be equal to $(G_j)_j$ on the measurable set of balls where the three requirements of Definition 9.3 hold for $(G_j)_j$, and 0 elsewhere. The change is on a null set of the law of $B_r(\mathbb{T}_r,o)$.

(a) By Lemma 9.5, $c_r \le \mathbb{E}\sum_j \xi_{\ell_j}\, g_{r,j}(B_r(\mathbb{T}_r,o))$. The variable $\xi_{\ell_j}\mathbf{1}\{\xi_{\ell_j}\le r\}$ is bounded and $\mathcal{F}_r$-measurable, so by (9.5)

$$\mathbb{E}\big[\xi_{\ell_j}\, g_{r,j}(B_r(\mathbb{T}_r,o))\big] = \mathbb{E}\big[\xi_{\ell_j}\mathbf{1}\{\xi_{\ell_j}\le r\}\, m_{\ell_j}\big].$$

Sum over j. The truncation costs nothing here: the rule ignores the teams above the level, and what that loses is slack.

(b) The σ-fields $\mathcal{F}_r$ increase with r: deleting from $B_{r+1}(\mathbb{T}_{r+1}, o)$ the teams of cost above r, keeping the component of the root and then the teams that contain an agent at distance at most $r-1$ from the root, gives $B_r(\mathbb{T}_r, o)$. They generate the σ-field of $(\mathbb{T}, o, \xi)$: by Remark 6.1 it is generated by the truncated balls $B_r(\mathbb{T}_K, o)$ with integers r, K, and such a ball is in the same way a function of $B_{r'}(\mathbb{T}_{r'}, o)$ with $r' = \max(r, K)$. By Lévy's upward theorem [16], $\mathbb{E}[h_j \mid \mathcal{F}_r] \to h_j$ almost surely, and $\mathbf{1}\{\xi_{\ell_j} \le r\} \to 1$. This is the first claim. By Lemma 9.5,

$$\varepsilon_r \le 3\,\mathbb{E}\sum_{j\ge 1}\big(h_j - g_{r,j}(B_r(\mathbb{T}_r, o))\big)_+.$$

The j-th term of the series tends to 0 almost surely and is at most h_j, and $\sum_j h_j = 1$. By dominated convergence for series the sum tends to 0 almost surely; it is at most 1, so its expectation tends to 0. □

The slack caused by the truncation at level r is included in ε_r; there is no separate limit in K.

9.5. **Transfer to the market.**

Lemma 9.7. *Let γ be a rule of range r at level $K \le q^{2/3}$. Apply it to the near-tight hypergraph $H = H_q(K)$ of the market, put $x_e^\gamma = 0$ for teams that are not near-tight, and let V be a uniformly random agent independent of the market.*

(a) *x^γ is a function of the pair deficits; $x^\gamma \ge 0$; every agent has load $\sum_{e\ni v} x_e^\gamma \le 1$; and $x_e^\gamma = 0$ unless $\xi_e \le K$.*

(b) *The total slack $\sum_v \sigma_v$ over the agents of a sector is the same for the three sectors, namely $\Sigma := q - \sum_e x_e^\gamma$.*

(c) *The expected slack and the expected deficit of x^γ are*

$$\mathbb{E}\,\Sigma = q\,\mathbb{E}\,S^\gamma\big(B_{r+1}(H,V)\big), \qquad \mathbb{E}\sum_e \Delta(e)\,x_e^\gamma = q^{1/3}\,\mathbb{E}\,C^\gamma\big(B_{r+1}(H,V)\big).$$

(d) *With the root sector of $\mathbb{T}$ uniform,*

$$\big|\mathbb{E}\,S^\gamma(B_{r+1}(H,V)) - \mathbb{E}\,S^\gamma(B_{r+1}(\mathbb{T}_K,o))\big| \le \varepsilon_1(q,K,r+1),$$
$$\big|\mathbb{E}\,C^\gamma(B_{r+1}(H,V)) - \mathbb{E}\,C^\gamma(B_{r+1}(\mathbb{T}_K,o))\big| \le K\,\varepsilon_1(q,K,r+1).$$

Proof. (a) is Lemma 9.4(a); the rule is defined on every ball, so nothing has to be assumed about the realisation of the market.

(b) Every team has exactly one agent in each sector, so the total load of the agents of a sector is $\sum_e x_e^\gamma$.

(c) Sums over v are over all $3q$ agents. By (b) and Lemma 9.4(b),

$$3\Sigma = \sum_v \sigma_v = \sum_v S^\gamma\big(B_{r+1}(H,v)\big),$$

and the expectation of the last sum is $3q\,\mathbb{E}\,S^\gamma(B_{r+1}(H,V))$. In the same way, since every team is counted at its three agents,

$$\sum_v C^\gamma\big(B_{r+1}(H,v)\big) = \sum_v \kappa_v = 3\sum_e \xi_e\, x_e^\gamma = 3q^{2/3}\sum_e \Delta(e)\, x_e^\gamma,$$

and the expectation of the first sum is $3q\,\mathbb{E}\,C^\gamma(B_{r+1}(H,V))$.

(d) For a measurable function h with values in $[0,M]$ and probability measures μ, ν, $|\int h\,\mathrm{d}\mu - \int h\,\mathrm{d}\nu| \le M\, d_{\mathrm{TV}}(\mu,\nu)$. Apply this to S^γ and C^γ and to the laws of $B_{r+1}(H,V)$ and $B_{r+1}(\mathbb{T}_K,o)$, whose distance is at most $\varepsilon_1(q,K,r+1)$ by the last statement of Theorem 8.4. □

No continuity of the rule is used: the conditional expectations g_r are only measurable, and this is the reason for proving Theorem 8.4 in total variation. Agents whose ball is not a hypertree receive slack 1; by Theorem 8.4 their expected fraction is at most ε_1, and it is included in (d).

9.6. **The split.**

Lemma 9.8. *Let $\eta \in (0,1)$ and let $u = (u_p)$ and $w = (w_p)$ be two independent families of independent uniform variables on $[0,1]$, indexed by the $3q^2$ pairs. Put*

$$d'(p) = 1 - (1-u_p)^{1/(1-\eta)}, \qquad d''(p) = 1 - (1-w_p)^{1/\eta},$$
$$d(p) = \min\big(d'(p), d''(p)\big).$$

Then the $d(p)$ are independent and uniform on $[0,1]$, and for every pair p

$$d(p) \le \frac{u_p}{1-\eta}, \qquad d(p) \le \frac{w_p}{\eta}.$$

Consequently, writing Δ_u, Δ_w, Δ_d for the team deficits computed from u, w, d, we have $\Delta_d(e) \le \Delta_u(e)/(1-\eta)$ and $\Delta_d(e) \le \Delta_w(e)/\eta$ for every team e.

Proof. For $t \in [0,1]$, $\mathbb{P}(d'(p) > t) = \mathbb{P}(1-u_p < (1-t)^{1-\eta}) = (1-t)^{1-\eta}$ and likewise $\mathbb{P}(d''(p) > t) = (1-t)^{\eta}$, so $\mathbb{P}(d(p) > t) = 1-t$ by independence. By Bernoulli's inequality $(1-y)^a \ge 1-ay$ for $a \ge 1$ and $y \in [0,1]$, so $d'(p) \le u_p/(1-\eta)$ and $d''(p) \le w_p/\eta$. □

So the pair deficits d form a market in the sense of Section 2, and so do u and w, which are independent of each other. The main solution will be built on the uniform market u, not on d', so local convergence is needed for uniform markets only; and the comparison of costs is made team by team for a fixed solution, so no monotonicity of optimal values is needed.

9.7. **The patch.** For a family d of pair deficits write $\Phi_q^{\mathrm{LP}}(d)$ and $\Phi_q(d)$ for the fractional and the integer deficit of the market with these deficits. Recall that $\mathbb{E}\Phi_n \le C_0\, n^{1/3}$ for every $n \ge 1$, by (3.1), which rests on [9, Theorem 2(b)]; we put $\Phi_0 = 0$.

Lemma 9.9. *Let u, w, η and d be as in Lemma 9.8. Let Z be a random element such that the pair (u, Z) is independent of w, and let $x = (x_e)$ be a vector of weights on the teams that is a measurable function of (u, Z), with $x \geq 0$ and all loads at most 1. Put $\Sigma = q - \sum_e x_e$. Then*

$$\mathbb{E}\,\Phi_q^{\mathrm{LP}}(d) \;\leq\; \frac{1}{1-\eta}\,\mathbb{E}\sum_e \Delta_u(e)\,x_e + \frac{C_0}{\eta}\,\bigl(\mathbb{E}\,\Sigma\bigr)^{1/3}. \tag{9.6}$$

If x takes values in $\{0,1\}$, the same bound holds for $\mathbb{E}\,\Phi_q(d)$.

Proof. Let $\sigma_v = 1 - \sum_{e\ni v} x_e \in [0,1]$ be the slack of agent v. As in Lemma 9.7(b), the slacks of each sector sum to Σ.

Rounding. Let U be uniform on $[0,1)$ and independent of (u, Z, w). In sector $i \in \{1,2,3\}$ list the agents as $v_1, \dots, v_q$, put $c_0 = 0$ and $c_k = \sigma_{v_1} + \cdots + \sigma_{v_k}$, and let

$$R_i = \bigl\{v_k :\ [c_{k-1}, c_k) \text{ contains a point of } U + \{0,1,2,\dots\}\bigr\}.$$

The interval $[c_{k-1}, c_k)$ has length $\sigma_{v_k} \leq 1$, so it contains at most one such point, and it contains one with conditional probability σ_{v_k} given (u, Z, w). The intervals partition $[0, \Sigma)$, so

$$|R_i| = N := \#\{n \geq 0 :\ U + n < \Sigma\}$$

for each of the three sectors. Thus the three sets have the same size $N \in \{\lfloor\Sigma\rfloor, \lceil\Sigma\rceil\}$, and $\mathbb{E}[N \mid u, Z, w] = \Sigma$.

The patch. The triple $R = (R_1, R_2, R_3)$ is a function of (u, Z, U), and (u, Z, U) is independent of w. So, conditionally on (u, Z, U), the variables w_p for the $3N^2$ pairs inside R are independent and uniform: they form a market with N agents in each sector. Let M_R be its optimal assignment, a set of N disjoint teams covering exactly the agents of R (empty if $N = 0$). Then

$$\mathbb{E}\Bigl[\sum_{e\in M_R} \Delta_w(e) \;\Big|\; u, Z, U\Bigr] = \mathbb{E}\Phi_n\Big|_{n=N} \leq C_0\,N^{1/3}. \tag{9.7}$$

Put $z_e = \mathbb{P}(e \in M_R \mid u, Z, w)$, the average over U. Then $z \geq 0$, and since an agent lies in a team of M_R exactly when it belongs to R, and then in one team only,

$$\sum_{e\ni v} z_e = \mathbb{P}(v \in R \mid u, Z, w) = \sigma_v \qquad \text{for every agent } v.$$

Hence $x + z \geq 0$ has all loads exactly 1: it is a fractional assignment in the sense of (2.5).

The cost. By (2.6) for the market d and by Lemma 9.8, pointwise,

$$\Phi_q^{\mathrm{LP}}(d) \leq \sum_e \Delta_d(e)\,(x_e + z_e) \leq \frac{1}{1-\eta}\sum_e \Delta_u(e)\,x_e + \frac{1}{\eta}\sum_e \Delta_w(e)\,z_e.$$

Since $\Delta_w(e)$ is a function of w, $\mathbb{E}\sum_e \Delta_w(e)z_e = \mathbb{E}\sum_{e\in M_R}\Delta_w(e)$, and by (9.7) and Jensen's inequality for the concave function $n \mapsto n^{1/3}$,

$$\mathbb{E}\sum_{e\in M_R}\Delta_w(e) \le C_0\,\mathbb{E}\,N^{1/3} \le C_0\,(\mathbb{E}N)^{1/3} = C_0\,(\mathbb{E}\,\Sigma)^{1/3}.$$

This proves (9.6).

Integral x. If x takes values in $\{0,1\}$, then x is the indicator of a set of disjoint teams, $\sigma_v \in \{0,1\}$, Σ is an integer, and $R_i = \{v \text{ in sector } i : \sigma_v = 1\}$ whatever U is. So z is the indicator of M_R and $x+z$ is the indicator of a perfect assignment of the market d. Hence $\Phi_q(d) \le \sum_e \Delta_d(e)(x_e+z_e)$, and the rest of the proof is unchanged. □

The bound (9.6) is concave in Σ, so only $\mathbb{E}\,\Sigma$ enters. This is why the convergence of expectations in Lemma 9.7(d) suffices and no concentration of the slack is needed. The fractional assignment $x+z$ depends on (u,Z,w) and not on d alone; this is harmless, since $\Phi_q^{\mathrm{LP}}(d)$ is bounded pointwise. The hypothesis on Z cannot be weakened to independence of Z and w: if $Z = (u+w) \bmod 1$ coordinatewise, then Z is independent of w, but (u,Z) determines w. In the proof of Theorem 9.1 there is no Z.

9.8. **Proof of Theorem 9.1.**

Proof of Theorem 9.1. Let c_r, ε_r and the rules g_r be as in Lemma 9.6; they depend only on the law of $(\mathbb{T},o,\xi,m)$, $c_r \le c(m)$ and $\varepsilon_r \to 0$. Fix $r \ge 1$, $\eta \in (0,1)$ and $q \ge r^{3/2}$, so that $K = r \le q^{2/3}$. Let u, w and d be as in Lemma 9.8. Since d is a market, $\mathbb{E}\Phi_q^{\mathrm{LP}} = \mathbb{E}\,\Phi_q^{\mathrm{LP}}(d)$. Apply the rule g_r to the near-tight hypergraph $H = H_q(r)$ of the market u, and let $x = x^{g_r}$. By Lemma 9.7(a) and (b), x is a function of u that satisfies the hypotheses of Lemma 9.9, with trivial Z. By (9.6) and Lemma 9.7(c),

$$\frac{\mathbb{E}\Phi_q^{\mathrm{LP}}}{q^{1/3}} \le \frac{\mathbb{E}\,C^{g_r}(B_{r+1}(H,V))}{1-\eta} + \frac{C_0}{\eta}\Big(\mathbb{E}\,S^{g_r}(B_{r+1}(H,V))\Big)^{1/3},$$

and by Lemma 9.7(d),

$$\mathbb{E}\,C^{g_r}(B_{r+1}(H,V)) \le c_r + r\,\varepsilon_1(q,r,r+1),$$
$$\mathbb{E}\,S^{g_r}(B_{r+1}(H,V)) \le \varepsilon_r + \varepsilon_1(q,r,r+1).$$

This is (9.2).

Order of limits. Let $q \to \infty$ with r and η fixed. By Theorem 8.4, $\varepsilon_1(q,r,r+1) \to 0$, so

$$\limsup_{q\to\infty}\frac{\mathbb{E}\Phi_q^{\mathrm{LP}}}{q^{1/3}} \le \frac{c_r}{1-\eta} + \frac{C_0}{\eta}\,\varepsilon_r^{1/3} \le \frac{c(m)}{1-\eta} + \frac{C_0}{\eta}\,\varepsilon_r^{1/3}.$$

The left-hand side does not depend on r or η. Let $r \to \infty$, and then $\eta \to 0$. □

Remark 9.10. The theorem is a statement about expectations, and it gives no value of $c(m)$: an invariant perfect matching of small cost is supplied by Theorems 6.12 and 7.1. The proof uses m only through the conditional expectations h_j. Unbounded costs never appear in it, because the rule g_r ignores the teams of cost above r. The bound (9.2) is explicit in q apart from the quantities c_r and ε_r of the tree, but it is of no use at any size that can be simulated; see Remark 8.9.

Remark 9.11 (Floating point)**.** The rounding and the patch of Lemma 9.9 were checked numerically on small instances by the script `patch_check.py` in the folder `lemmaC/round3/lplocal/` ($q = 5$, $\eta = 0.25$, six seeds, a random packing x, exact integration over U, and M_R found by exhaustive search). In all runs the three sector slack sums agreed to 10^{-12}, the three sets R_i had the same size in $\{\lfloor\Sigma\rfloor, \lceil\Sigma\rceil\}$, the loads of $x + z$ differed from 1 by at most $1.1 \cdot 10^{-15}$, and the chain of inequalities for the cost held. This is evidence only and is not used in the proof.

10. Proof of the main theorem

Theorem 1.1 follows from five results proved above: the variance bound (Theorem 3.3), the integer lower bound (Theorem 5.1), the tree bound for the fractional optimum (Theorem 9.1), the tree process (Theorem 6.12) and the existence and cost of a fixed point (Theorem 7.1). The section ends with a consequence of Theorems 5.1 and 9.1 for the tree itself.

10.1. **The derivation.**

Proof of Theorem 1.1. The fractional deficit. By Theorem 7.1 the map Θ has a fixed point μ, and $9\,\mathbb{E}X \le \alpha_0$ for $X \sim \mu$. By Theorem 6.12 there is an invariant perfect matching m on $\mathbb{T}$ with cost $c(m) = 9\,\mathbb{E}X \le \alpha_0$. By Theorem 9.1,

$$\mathbb{E}\Phi_q^{\mathrm{LP}} \le (\alpha_0 + o(1))\, q^{1/3}.$$

The integer deficit. By Theorem 5.1, $\mathbb{P}(\Phi_q \ge \beta_0 q^{1/3}) \to 1$. Since $\Phi_q \ge 0$,

$$\mathbb{E}\Phi_q \;\ge\; \beta_0\, q^{1/3}\, \mathbb{P}\big(\Phi_q \ge \beta_0 q^{1/3}\big) = (\beta_0 - o(1))\, q^{1/3}.$$

Part (i). By (2.7), $D_q = \Phi_q - \Phi_q^{\mathrm{LP}}$, so

$$\mathbb{E}D_q = \mathbb{E}\Phi_q - \mathbb{E}\Phi_q^{\mathrm{LP}} \;\ge\; (\beta_0 - \alpha_0 - o(1))\, q^{1/3},$$

and $\beta_0 - \alpha_0 = 2.302 - 2.2519 = 0.0501$. In particular $\mathbb{E}D_q \to \infty$.

Part (ii). By Theorem 3.3(ii) with $\varepsilon = 1/12$, $\operatorname{Var} D_q$ is bounded for large q (it is $O(q^{-1/12})$). Fix $t > 0$. By Chebyshev's inequality and part (i),

$$\mathbb{P}\big(|D_q - \mathbb{E}D_q| > t\,\mathbb{E}D_q\big) \le \frac{\operatorname{Var} D_q}{t^2(\mathbb{E}D_q)^2} \longrightarrow 0 \qquad (q \to \infty).$$

This holds for every fixed $t > 0$, which is the statement $D_q/\mathbb{E}D_q \to 1$ in probability.

Part (iii). By Proposition 2.3 the core is nonempty exactly when $D_q = 0$. For large q we have $\mathbb{E}D_q > 0$, and then the event $\{D_q = 0\}$ is contained in $\{|D_q - \mathbb{E}D_q| > \frac{1}{2}\mathbb{E}D_q\}$. Take $t = 1/2$ in part (ii). □

The argument gives $\mathbb{P}(D_q = 0) \le 4 \operatorname{Var} D_q/(\mathbb{E}D_q)^2 = O(q^{-5/6+\varepsilon})$ for every $\varepsilon > 0$, by Theorem 3.3(ii) and part (i); compare Corollary 3.13. The implied constants and thresholds are those of Theorems 3.3, 5.1 and 9.1, and none of them is effective at any size that can be simulated.

10.2. **The matching is not a function of the tree.** The proof of Theorem 9.1 transfers to the market only the conditional expectation of the matching given the tree and its costs, and for that reason it bounds Φ_q^{LP} and not Φ_q. If the matching were a function of the tree, the same construction would produce assignments, and Theorem 5.1 forbids this at cost below β_0.

Corollary 10.1. *Let m be an invariant perfect matching on $\mathbb{T}$ with $c(m) < \beta_0$. Then m is not a measurable function of the marked tree: with $\ell_1, \ell_2, \dots$ the teams at the root in increasing order of cost, there are no measurable functions h_j with values in $\{0,1\}$ such that $m_{\ell_j} = h_j(\mathbb{T}, o, \xi)$ almost surely for every j. In particular the invariant perfect matching of Theorem 6.12 built from a fixed point of Θ, whose cost is at most $\alpha_0 < \beta_0$ by Theorem 7.1, uses randomness beyond the tree and its costs.*

Proof. Suppose that $m_{\ell_j} = h_j(\mathbb{T}, o, \xi)$ almost surely with measurable h_j with values in $\{0,1\}$, and that $c(m) < \infty$. We show $c(m) \ge \beta_0$. We use the notation of Section 9; the functions h_j may be taken as the functions of Section 9.2, since $\mathbb{E}[m_{\ell_j} \mid \mathbb{T}, o, \xi] = h_j$.

A thresholded rule. Let g_r be the rule of Lemma 9.6 and put

$$\tilde{g}_{r,j} = \mathbf{1}\{g_{r,j} > \tfrac{1}{2}\}, \qquad j \ge 1.$$

Since $\sum_j g_{r,j} \le 1$, at most one $\tilde{g}_{r,j}(B)$ equals 1 for each ball B, and $\tilde{g}_{r,j}$ vanishes where $g_{r,j}$ does. So $\tilde{g}_r$ is a rule of range r at level r in the sense of Definition 9.3, with values in $\{0,1\}$, and the weights $x^{\tilde{g}_r}$ of (9.4) take values in $\{0,1\}$: a team is kept when its conditional probability exceeds $1/2$ from the point of view of each of its three agents.

On the tree. Write $G_{r,j} = g_{r,j}(B_r(\mathbb{T}_r, o))$ and $\tilde{G}_{r,j} = \mathbf{1}\{G_{r,j} > \frac{1}{2}\}$. Almost surely exactly one h_j equals 1; let J be its index, so that ξ_{ℓ_J} is the cost of the team of the root in m and $\mathbb{E}\,\xi_{\ell_J} = c(m)$. By Lemma 9.6(b), $G_{r,J} \to h_J = 1$ almost surely. So almost surely, for all large r, $\tilde{G}_{r,J} = 1$, hence $\tilde{G}_{r,j} = 0$ for $j \ne J$, that is, $\tilde{G}_{r,j} = h_j$ for every j. Two consequences follow.

First, by Lemma 9.5,

$$\tilde{\varepsilon}_r := \mathbb{E}\, S^{\tilde{g}_r}\big(B_{r+1}(\mathbb{T}_r, o)\big) \le 3\,\mathbb{E} \sum_{j \ge 1} \big(h_j - \tilde{G}_{r,j}\big)_+ \longrightarrow 0,$$

by bounded convergence: the sum is at most 1 and vanishes for large r.

Secondly, by Lemma 9.5, $\tilde{c}_r := \mathbb{E}\, C^{\tilde{g}_r}(B_{r+1}(\mathbb{T}_r, o)) \le \mathbb{E}\, Y_r$ with $Y_r = \sum_j \xi_{\ell_j} \tilde{G}_{r,j}$, and $Y_r = \xi_{\ell_J}$ for all large r, almost surely. Moreover $\tilde{G}_{r,j} \le 2G_{r,j}$, and by (9.5), since $\xi_{\ell_j}\mathbf{1}\{\xi_{\ell_j} \le r\}$ is nonnegative and $\mathcal{F}_r$-measurable,

$$Y_r \le 2\sum_{j\ge 1} \xi_{\ell_j} G_{r,j} = 2\,\mathbb{E}\Big[\sum_{j\ge 1} \xi_{\ell_j}\mathbf{1}\{\xi_{\ell_j} \le r\}\, m_{\ell_j} \;\Big|\; \mathcal{F}_r\Big] \le 2\,\mathbb{E}\big[\xi_{\ell_J} \mid \mathcal{F}_r\big].$$

The conditional expectations of the integrable variable ξ_{ℓ_J} form a uniformly integrable family. Hence $(Y_r)_r$ is uniformly integrable, and $\mathbb{E}\, Y_r \to \mathbb{E}\,\xi_{\ell_J} = c(m)$. So $\limsup_r \tilde{c}_r \le c(m)$.

In the market. Fix $r \ge 1$, $\eta \in (0,1)$ and $q \ge r^{3/2}$, and repeat the proof of Theorem 9.1 with the rule $\tilde{g}_r$ in place of g_r. The vector $x = x^{\tilde{g}_r}$, computed in the market u, takes values in $\{0,1\}$, so the last statement of Lemma 9.9 applies, and with Lemma 9.7 it gives

$$\frac{\mathbb{E}\Phi_q}{q^{1/3}} \le \frac{\tilde{c}_r + r\,\varepsilon_1(q,r,r+1)}{1-\eta} + \frac{C_0}{\eta}\big(\tilde{\varepsilon}_r + \varepsilon_1(q,r,r+1)\big)^{1/3}.$$

Let $q \to \infty$, then $r \to \infty$, then $\eta \to 0$: $\limsup_q \mathbb{E}\Phi_q/q^{1/3} \le c(m)$.

Conclusion. By Theorem 5.1 and $\Phi_q \ge 0$, as in the proof of Theorem 1.1, $\liminf_q \mathbb{E}\Phi_q/q^{1/3} \ge \beta_0$. Hence $c(m) \ge \beta_0$. □

Remark 10.2. The matching of Theorem 6.12 is a function of the tree, the costs and a family of messages that solve the recursion on the tree (Remark 6.22). By Corollary 10.1 the messages cannot all be measurable functions of the marked tree: in the usual terminology, the recursive tree process attached to a fixed point of Θ is not endogenous. No statement in this paper relies on endogeny.

11. The two computer-assisted steps

Two inequalities between explicit real numbers are established by computation: the bound on the functional Ψ_0 behind the integer constant (Theorems 4.1 and 5.1) and the bound on the cost of a fixed point of Θ (Theorem 7.1). This section states, for each, the inequality that is certified, the scheme, which parts are computed in interval arithmetic and which in floating point, how to reproduce the run, and which independent recomputations were made. File paths are relative to the project folder. The scripts are in Python 3 and use `mpmath` version 1.3.0 (pure-Python backend) and, for the second certificate, `numpy`.

No person has read the scripts. They were written and re-read by AI agents, as recorded below.

11.1. **The trusted base.** Both certificates rest on the following and on nothing else in the numerical libraries.

(B1) The interval type `mpmath.iv`: addition, subtraction, multiplication, division and comparison of intervals, and the conversion of integers

and binary floating-point numbers to intervals. In `mpmath` these are computed from exact integer arithmetic with directed rounding.

(B2) Python integers and the exact rationals of `fractions.Fraction`.

(B3) For the second certificate only: IEEE 754 binary64 arithmetic, rounding to nearest, for the additions and multiplications inside sums of products of nonnegative numbers (computed by `numpy` and the linear-algebra library behind it).

(B4) The logic of the scripts.

The following library functions are *not* used: the interval exponential `iv.exp`, the interval logarithm `iv.log`, the interval Euler constant `iv.euler`, interval powers, and the printing routine `iv.nstr`. The reason is given next. The four interval operations of (B1) were tested against exact rational arithmetic (468,775 containment checks at 80 bits and 160,000 at 120 bits, no failure). This is evidence for (B1), not a proof of it.

11.2. **The library's interval exponential and logarithm are not strict enclosures.** In `mpmath` 1.3.0 the interval exponential and logarithm can return an interval that does not contain the true value.

- At 80 bits the upper endpoint of `iv.exp` was below the true value for 21 of 100,000 random arguments, with relative shortfall up to $2.0 \cdot 10^{-28}$, and the upper endpoint of `iv.log` was below the true value for 1 of 1,000,000 arguments, with relative shortfall $3.0 \cdot 10^{-30}$. The lower endpoints did not fail in these samples. The reference values were computed at 600 bits.
- At 120 bits the upper endpoint of `iv.exp` was below a 400-bit reference value for 5 of 20,000 random arguments, by a relative 10^{-40}.
- One failure is established without any reference to the library's own elementary functions. For $x = -$`0x1.31c2c4b9c3ff0p-2` the verified exponential described below, run at 200 bits, has lower bound 0.74186013921160060481265814002 3, while `iv.exp` at 80 bits returns the upper endpoint 0.74186013921160060481265813992 7. A failure of `iv.log` at $x =$ `0x1.5a6695e45d861p+0` is established in the same way.

The cause, read in the library source (`libmp/libmpi.py`), is that the interval routines call the scalar exponential and logarithm with directed rounding, and those compute an approximation with a few guard bits and round *that* in the requested direction. When the approximation lies on the wrong side of a representable number, so does the result. The routine for Euler's constant has the same design. In addition, `iv.nstr` prints endpoints rounded to nearest, so a printed "upper bound" can be below the endpoint; on 20,000 random endpoints this happened 10,055 times.

The defect is below 10^{-27} in relative size per call and is far too small to change any digit reported here: the first versions of both certificates used the library functions, and replacing them left the logs of the second certificate unchanged and moved the bound of the first by less than $2 \cdot 10^{-12}$.

The certificates were nevertheless changed so that they do not rest on that estimate. Both now use the following replacements, built from (B1) only.

Exponential. Write $x = 2^k r$ with $|r| \le 2^{-10}$; the inequality is asserted on the interval endpoints. Then $e^r \in \sum_{j<12} r^j/j! + [-\rho, \rho]$, where ρ is a fixed bound for the Lagrange remainder $|r|^{12} e^{|r|}/12!$, namely $\rho = 2^{-119}/12!$. The sum is evaluated by Horner's rule in interval arithmetic, and e^x is obtained by k interval squarings.

Logarithm (first certificate only). For a point $p > 0$ let y_0 be any real number; the script takes the floating-point logarithm of p, which is a guess and is not trusted. Then $\ln p = y_0 + \ln(1+d)$ with $d = p\,e^{-y_0} - 1$ enclosed by the verified exponential, and $d - d^2 \le \ln(1+d) \le d$ for $|d| \le 1/2$; the script asserts $|d| \le 2^{-20}$. An interval is treated endpoint by endpoint, since the logarithm is increasing. The inequality holds because $\ln(1+d) \le d$ always, $\ln(1+d) \ge d - d^2/2$ for $d \ge 0$, and for $s = -d \in (0, 1/2]$, $-\ln(1-s) = s + s^2/2 + s^3/3 + \cdots \le s + (s^2/2)/(1-s) \le s + s^2$.

Euler's constant (first certificate only). With H_n the nth harmonic number, $\gamma = H_n - \ln(n+\frac{1}{2}) - S_n$ with $0 \le S_n \le 2/(45n^2)$, and the script takes $n = 2^{16}$ and sums H_n in interval arithmetic. Indeed $c_k = H_k - \ln(k+\frac{1}{2}) \to \gamma$, and with $y = 1/(2k)$, $c_{k-1} - c_k = \ln\frac{1+y}{1-y} - 2y = 2\sum_{j\ge1} y^{2j+1}/(2j+1)$, which lies between 0 and $(2y^3/3)/(1-y^2) \le (16/15)/(12k^3)$ for $k \ge 2$; and $\sum_{k>n} k^{-3} \le 1/(2n^2)$. The result is $\gamma \in [0.57721566490088,\ 0.57721566491124]$.

In tests at 80 bits against 600-bit reference values, the verified exponential and logarithm showed no failure in 100,000 points and 60,000 wide intervals each. These tests are evidence that the replacements are implemented as described; their correctness is the argument above together with (B1).

11.3. **Certificate I: the integer constant.**

The inequality. The functional $\Psi_0(K, B, \theta, u)$ is defined in Definition 4.2. Let $K = 5$, $B = 50$, $\theta = 1.6712$, and let $u^* \in \mathbb{R}^{51}$ be the weight vector listed in Proposition 4.4; it is stored as five-decimal numbers in `lemmaC/round3/firstmoment/w_K5_e10.json`. All inputs are exact rationals. The script `lemmaC/round3/final/certI/certify_rigorous.py` certifies the inequality of Proposition 4.4:

(11.1)

$$\Psi_0(5, 50, 1.6712, u^*) \;\le\; -\frac{11628395373505716339509 17}{2^{78}} \;\le\; -3.84751328322.$$

Consequently $-\Psi_0/\theta \ge 2.30224586$, and

$$\theta\beta + \Psi_0 \le \begin{cases} -0.0037532832, & \beta = 2.30, \\ -0.0004108832, & \beta = 2.302, \\ -0.0000766432, & \beta = 2.3022, \\ -0.0000097952, & \beta = 2.30224. \end{cases} \tag{11.2}$$

The second line is the criterion (4.3) of Proposition 4.3 for $\beta = \beta_0 = 2.302$, which gives Theorem 4.1. The certified threshold is $-\Psi_0/\theta \geq 2.30224586$; the criterion fails at $\beta = 2.30225$.

For the market, Theorem 5.1 needs the exponent to stay negative after an explicit error term Err is added, which accounts for the enlargement of the typical event in the comparison (Section 5, Proposition 5.14). The script `constants_rigorous.py` in the same folder takes the eight-decimal consequence $\Psi_0 \leq -3.84751328$ of (11.1), asserts it in exact rational arithmetic against the exported endpoint, and certifies

(11.3)
$$\theta\beta + \Psi_0 + \mathrm{Err} \leq \begin{cases} -0.0005929830 & (\mathrm{Err} \leq 0.0031602970), \quad \beta = 2.30, \\ -0.0001648848 & (\mathrm{Err} \leq 0.0002459952), \quad \beta = 2.302, \end{cases}$$

for the two parameter rows of the error budget (in the script's names: $c' = 14$, $\delta_1 = 10^{-5}$, $\delta_2 = 1/5000$, $\tau = 1/200000$ for $\beta = 2.30$, and $c' = 17$, $\delta_1 = 10^{-6}$, $\delta_2 = 1/50000$, $\tau = 1/4000000$ for $\beta = 2.302$).

The scheme. Recall that $W = w_\infty + \sum_b D_b w_b$ with independent $D_b \sim \mathrm{Poisson}(\lambda_b)$, $w_b = e^{u_b}$ and $w_\infty = e^{u_\infty}$. The Poisson expectations are carried exactly by the Laplace transform

$$\psi(t) = \mathbb{E}e^{-tW} = \exp\Big(-tw_\infty - \sum_b \lambda_b\big(1 - e^{-tw_b}\big)\Big),$$

so no truncation of the star size is needed. By Frullani's formula, with γ Euler's constant,

$$\Phi(u) = \mathbb{E}\ln W = -\gamma + \int_0^1 \big(1 - \psi(t)\big)\frac{\mathrm{d}t}{t} - \int_1^\infty \psi(t)\frac{\mathrm{d}t}{t},$$
$$\rho_b(u) = \int_0^\infty w_b e^{-tw_b}\psi(t)\,\mathrm{d}t, \qquad \varepsilon_0(u) = \int_0^\infty w_\infty \psi(t)\,\mathrm{d}t,$$

and λ_b, κ_b, J have closed forms. The functions ψ and $e^{-tw_b}\psi$ are Laplace transforms of positive variables, hence convex and decreasing. The integrals are enclosed on a geometric mesh $t_0 = 2^{-27}$, $t_i = 1.005\,t_{i-1}$ with 5,109 cells, reaching $t \approx 868$. On each cell a convex function lies between its tangent at the midpoint and its chord, which gives the midpoint and trapezoid rules as lower and upper bounds for the $\mathrm{d}t$-integrals and the corresponding closed-form bounds for the $\mathrm{d}t/t$-integrals. The head satisfies $0 \leq \int_0^{t_0}(1-\psi)\,\mathrm{d}t/t \leq t_0\,\mathbb{E}W$, and for $t \geq t_1$ (the last node) $\psi(t) \leq \psi(t_1)e^{-(t-t_1)w_\infty}$, which bounds the tails; the mesh ends where these bounds are below 10^{-12}. Finally, $r \mapsto (1-r)^3 + Cr^3$ and $e \mapsto -3e + Ce^3$ are convex, so their maximum over an enclosing interval is at an endpoint, and the script takes the larger endpoint value.

Interval arithmetic and floating point. Every quantity entering (11.1)–(11.3) is computed in interval arithmetic at 80 bits, with (B1), (B2) and the verified exponential, logarithm and Euler constant of Section 11.2. The weights are

parsed as exact decimals. Powers are written as products. Floating point occurs in three places, none of which affects validity: the mesh nodes are generated in floating point and then used as exact binary rationals (the script asserts that they increase); the guess y_0 in the verified logarithm; and the earlier search that produced the weights u, which is irrelevant because the theorem holds for every u. Printed bounds are produced from the exact binary endpoint by floor or ceiling at the stated number of decimals, and the inequality is asserted. The exact endpoint is exported as a rational, with the parameters and the SHA-256 hash of the weights file, to `psi0_export_w_K5_e10.json`; the second script reads it and checks the hash.

Reproduction. From `lemmaC/round3/final/certI/`:

```
python3 certify_rigorous.py ../../firstmoment/w_K5_e10.json 0.005   # 75 s
python3 constants_rigorous.py                                       # 1 s
python3 constants_rigorous.py --fresh                               # 80 s
python3 test_defect_and_replacements.py 100000                      # 4 min
```

The first command should print, among other lines,

```
Psi0 <= -3.84751328322   (8 decimals: Psi0 <= -3.84751328)
beta_cert = -Psi0/theta >= 2.30224586119   (8 decimals: >= 2.30224586)
  beta=2.302: theta*beta + Psi0 <= -0.0004108832  -> NEGATIVE (certified)
```

and the second, for $\beta = 2.302$,

```
   theta*beta+Psi0 <= -0.0004108799 (m >= 0.0004108799);
   theta*beta+Psi0+Err <= -0.0001648848 (m - Err >= 0.0001648848)
   -> NEGATIVE (certified)
```

(one line in the output; broken here). With `--fresh` the second script recomputes the endpoint in the same process instead of reading the export; both modes gave identical output. The last command reproduces the counts of Section 11.2.

Independent recomputations.

- A second certificate with other parameters ($K = 5$, $B = 20$, $\theta = 1.6671$, weights `w_K5_e25.json`) gives $\Psi_0 \le -3.83621828482$ and $-\Psi_0/\theta \ge 2.30113267639$.
- A checking agent wrote its own interval enclosure and obtained $\beta \ge 2.302028$, so that the exponent at $\beta = 2.30$ is at most -0.00339. This is weaker than (11.1) but independent. It still gives Theorem 4.1 with $\beta_0 = 2.302$, and in the market it covers the row $\beta = 2.30$ of (11.3) ($0.00339 > 0.0031603$) but not the row $\beta = 2.302$.
- A second checking agent read `certify_rigorous.py` line by line, ran it again, and recomputed $\Psi_0 = -3.847516394$ by two other methods (floating point).
- Two checking agents recomputed the perturbed exponent at $\beta = 2.30$ in exact rational arithmetic and obtained at most -0.0005929831.

- A reader from a different model family ran the original certificate (before the replacements) and confirmed by direct interval comparison that $\Psi_0 < -3.84751328$; it read the logic of the hardened scripts without running them.
- The original and the hardened scripts agree: the hardened bound is weaker by $1.94 \cdot 10^{-12}$, all of it from the wider enclosure of Euler's constant, which enters Ψ_0 with weight 3.

Not done: a re-implementation in a different interval library.

11.4. **Certificate II: the cost of the tree matching.**

The inequality. The mathematics of this step is in Section 7.6, and we use its notation: $\mathcal{Q}$ is the class of Definition 7.3; for $Q \in \mathcal{Q}$, $g_Q(t) = \mathbb{E}(t + X' + X'')_+^3/6$ with X', X'' independent copies of $Q(E)$; and $c(Q) = 3\,\Gamma(Q) = 3\big(\mathbb{E}\,\Theta(Q)(E) + 2\,\mathbb{E}\,Q(E)\big)$ as in (7.3), so that $c(Q^*) = 9\,\mathbb{E}X$ at a fixed point. Rounded laws are defined in Definition 7.12. The script `lemmaC/round3/cert/cert.py` certifies the statement of Proposition 7.16: it produces rounded laws $\tilde{Q}_0 = 0, \tilde{Q}_1, \dots, \tilde{Q}_{10}$ such that, for $0 \le n \le 9$ and $g = g_{\tilde{Q}_n}$, the parameters $(\theta_i, \hat{s}_i)$ of $\tilde{Q}_{n+1}$ satisfy

$$\theta_i \le g(t_i), \qquad \hat{s}_i \ge g'(t_i) \qquad \text{at every mesh point } t_i, \tag{11.4}$$

with θ_i strictly increasing and $\hat{s}_i$ nondecreasing, and it certifies

$$c(\tilde{Q}_{10}) \;\le\; 2.2519. \tag{11.5}$$

By Lemmas 7.13 and 7.14, the one-sided inequalities (11.4) give $\tilde{Q}_n \preceq_{\mathrm{disp}} Q^*$ for every fixed point Q^*, and then (11.5) gives $9\,\mathbb{E}X \le 2.2519 = \alpha_0$, which is Theorem 7.1(ii). Table 3 lists the bounds printed at all stages of the two recorded runs; only the last entry of run B is used.

The scheme. All bounds come from Lemma 7.15. Fix the lattice width $h = 2^{-H}$ ($H = 10$ in run A, $H = 12$ in run B). For the current rounded law, with cell masses p_k, cell moments D_k and chord weights ν_j as defined there:

- part (a) of the lemma, with termwise lower bounds p^{lo}, D^{lo}, gives lower bounds $\theta_i \le g(t_i)$ and $a_i \le g'(t_i)$ as sums over the lattice;
- part (b), with termwise upper bounds ν^{hi}, gives upper bounds $\hat{s}_i \ge g'(t_i)$;
- part (c) gives an upper bound on $\mathbb{E}\,\Theta(\tilde{Q})(E)$ from the θ_i and a_i, and the exact formula for $\mathbb{E}\,\tilde{Q}(E)$ in terms of the cell integrals.

The script replaces $\hat{s}_i$ by the running maximum $\max_{k \le i} \hat{s}_k$, which keeps (11.4) and makes the slopes monotone; in both runs this changed no entry. No tail estimate is needed: below θ_0 the slope bound at θ_0 is valid for all smaller levels, and above θ_N the rounded slope is 0. Both one-sided bounds are of second order in h. The mesh points t_i are multiples of 2^{-20}, placed so that $g(t_i)$ is roughly log-spaced on $[10^{-6}, 32]$. The printed number at stage n is the internal upper bound on $c(\tilde{Q}_n)$ rounded to nearest at six decimals (Remark 11.1).

TABLE 3. Certified upper bounds on $c(\tilde{Q}_n)$, hence on the cost of every fixed point, as printed (run A: $h = 2^{-10}$, 4000 levels; run B: $h = 2^{-12}$, 12000 levels), and the exact iterates $c_n = c(\Theta^n(\delta_0))$ in floating point for comparison. The floating-point limit is 2.24954561.

n	run A	run B	c_n (floating point)
0	4.867956	4.867955	4.86795
1	3.417530	3.415996	3.41528213
2	2.665068	2.661760	2.66021540
3	2.404165	2.400021	2.39808802
4	2.308352	2.303815	2.30169894
5	2.274755	2.270053	2.26786005
6	2.262954	2.258185	2.25596056
7	2.258825	2.254029	2.25179199
8	2.257381	2.252574	2.25033206
9		2.252065	2.24982093
10		2.251887	2.24964199

Interval arithmetic and floating point.

- *Interval arithmetic at* 120 *bits* ((B1) and the verified exponential): everything about a rounded law (breakpoints, the order comparisons between breakpoints and lattice points, survival values, cell integrals, the mean) and the bound of Lemma 7.15(c) with the final combination. Double-precision numbers enter as exact rationals. An order comparison that the enclosures do not decide stops the program. Conversion from an interval to a double uses the neighbouring double in the safe direction and is asserted exactly.
- *Double precision with a proved error bound* ((B3)): the convolutions and the sums of Lemma 7.15(a),(b). Every such quantity is a sum of at most 2^{17} products of nonnegative numbers; the arguments $t_i + mh$ are exact because both terms are multiples of 2^{-20} of size below 32 (asserted); each term involves at most three rounded multiplications. For such a sum, evaluated in any order and with or without fused multiply-add, $|\text{computed} - \text{exact}| \le ((1 + 2^{-53})^{n+3} - 1) \cdot \text{exact} + n\eta$ with $n \le 2^{17}$ terms and $\eta \le 2^{-1022}$ for underflow. The relative error is below $3 \cdot 10^{-11}$ and the absolute part below $3 \cdot 10^{-303}$. Every lower bound is multiplied by $1 - 10^{-9}$ and reduced by 10^{-290} (and clamped at 0); every upper bound is multiplied by $1 + 10^{-9}$ and increased by 10^{-290}. The margin also covers the rounding of these adjustments and of the divisions by 2 and 6.
- *Plain floating point*: only the choice of the mesh points t_i, which has no effect on validity, since Lemmas 7.13 and 7.15 hold for any mesh.

Remark 11.1 (The quoted digits)**.** The script prints its final bound with a round-to-nearest format, so the last printed digit of 2.251887 is not itself certified. Any internal bound that prints as 2.251887 is below 2.2518875, and the bound quoted in this paper is 2.2519. Three improvements are owed and have not been made: asserting the internal bound against an exact rational, as in Certificate I; making the script refuse lattice parameters other than the two for which the exactness argument above is written ($h = 2^{-10}$ and $h = 2^{-12}$); and saving the stage parameters of run B with the script's own output (regenerated copies are in `lemmaC/round3/v10/`). [**TODO: make these three changes to cert.py and re-run**]

Reproduction. From `lemmaC/round3/cert/`:

```
python3 cert.py 10 1500 2500 8 stages/A   # run A, under 1 min
python3 cert.py 12 4000 8000 10           # run B, about 4 min
python3 check_float.py stages/A 8         # 7 s
python3 test_small.py 4                   # 5 s
python3 test_rexp.py                      # 15 s
```

The arguments of `cert.py` are H, two mesh sizes, the number of stages and an optional prefix for saving the stage parameters. Run A should end with a line containing

```
stage 8: ... CERTIFIED c* <= 3*Gamma(Q~_8) <= 2.257381
```

and run B with

```
stage 10: ... CERTIFIED c* <= 3*Gamma(Q~_10) <= 2.251887
```

The logs of the recorded runs are `run_A_h10.log` and `run_B_h12.log`. The mesh choice is deterministic but may differ between builds of the linear-algebra library; any mesh is valid, and the parameters of run A are saved in `stages/`.

Independent recomputations.

- A checking agent wrote its own code, recomputed g and g' at all 4,000 points of all transitions of run A (finding θ_i below g and $\hat{s}_i$ above g' everywhere), obtained $c = 2.25188597$ for the last law of run B, and reproduced both runs.
- A second checking agent computed the exact iterates $c_1, \dots, c_{10}$ with its own code, in agreement with the values of Table 3 to all digits shown, and reproduced both certificate runs digit for digit.
- The script `check_float.py` is an independent code path in plain floating point (cells of width 0.0007 that are not dyadic, mass at cell midpoints, convolution by fast Fourier transform, trapezoid rule). On the saved laws of run A it gives $c(\tilde{Q}_n) = 4.867955$, 3.417527, 2.665064, 2.404162, 2.308348, 2.274751, 2.262951, 2.258822, 2.257377 for $n = 0, \dots, 8$; the certified bounds exceed these by at most $4 \cdot 10^{-6}$, and the certified parameters lie on the safe side of the floating-point values at every mesh point.

- The script `test_small.py` compares the bounds of Lemma 7.15(a),(b) with tensor Gauss–Legendre quadrature of the exact law on a 20-piece example for $h = 1/8, 1/16, 1/32$: they are one-sided as required and of second order in h.
- The certified values exceed the exact iterates c_n at every stage, by 0.0007 to 0.0022 in run B, and the loss shrinks with the mesh, as expected of a rounding that only loses dispersion. The final bound is 0.0023 above the floating-point limit 2.24954561 and 0.050 below β_0.
- A reader from a different model family audited the source of the script without running it and found the enclosure scheme sound for both runs; the remark on the quoted digits and the restriction on lattice parameters come from that audit.

An earlier version of the script used `iv.exp`; its logs are identical to those of the present version.

12. Discussion

12.1. **What the theorem says and does not say.** Theorem 1.1 is a statement about the limit $q \to \infty$. It gives no bound for any particular q, and none can be extracted from the proof at a size of practical interest.

The thresholds in the proof are astronomically large. In the comparison between the market and the independent-team model, the error term is negligible only when $\ln q$ exceeds an explicit multiple of $1/\tau$, where τ is one of the small parameters of the error budget; with the parameters certified for $\beta = 2.30$ this means q above about $10^{52,900}$, and for $\beta = 2.302$ the parameter τ is twenty times smaller (Remark 5.16). The local picture behind the fractional upper bound also sets in late: a team of rescaled cost at most K shares a pair with about $\frac{9}{10}K^2 q^{-1/3}$ other such teams, and for $K = 6$ this number stays above 1 until q is about $3 \cdot 10^4$. Below such sizes the near-tight hypergraph of the market is far from a tree.

The constants of the proof are not visible at sizes that can be simulated. In the market, the measured value of $\Phi_q^{\mathrm{LP}}/q^{1/3}$ is between 2.53 and 2.56 for $q \le 48$, and at $q = 2000$ the fractional optimum restricted to teams of rescaled cost at most 6, which is an upper bound on Φ_q^{LP}, is still $2.31\, q^{1/3}$ (Table 5; raising the cut-off to 7 changed it by about 10^{-4}). That is far from the limiting bound $\alpha_0 = 2.2519$. So the two inequalities that the proof compares, $\Phi_q \ge \beta_0 q^{1/3}$ and $\mathbb{E}\Phi_q^{\mathrm{LP}} \le (\alpha_0 + o(1))q^{1/3}$, do not describe the market at any size we could compute.

The conclusion, by contrast, is visible early. The measured probability of a nonempty core is 0.37 at $q = 16$, 0.12 at $q = 24$, 0.04 at $q = 32$ and 0.002 at $q = 48$ (Table 4), and the measured mean gap grows over the same range. The theorem therefore agrees with the data in its conclusion, while its proof describes a regime that begins many orders of magnitude later. Nothing proved here explains the behaviour at small q.

Further limits.

- The constant 0.050 in part (i) is the difference of two certified bounds and is not claimed to be sharp. In the independent-team model the measured gap is about 0.09 to 0.115 times $q^{1/3}$ for $q \le 40$ and rising.
- The proof gives only a polynomial rate in part (iii): by Chebyshev's inequality, part (i) and Theorem 3.3, $\mathbb{P}(D_q = 0) \le \operatorname{Var} D_q / (\mathbb{E} D_q)^2 = O(q^{-5/6+\varepsilon})$.
- The theorem is for independent uniform pair values, three sectors of equal size, and teams that need all three pairs.
- The proof imports Theorem 2(b) of [9], and two of its steps are computer-assisted (Section 11).
- Every part of the proof was written and checked by AI agents. In the working notes from which this text was consolidated, each component was attacked by at least two further agents of the same model family and read at least once by a model of a different family, and no error affecting a theorem was found. No person has checked any part.

12.2. **Numerical evidence.** *Everything in this subsection is floating-point simulation. None of it is used in any proof, and none of it has been checked by a person.* Linear programmes were solved in double precision; integer optima were computed by branch and bound and are reported only where the solver closed the instance. Standard errors are given where the data files record them.

An independent sample gave shares 0.353, 0.134, 0.037 and 0 of 211 at $q = 16, 24, 32, 48$. Among the curves fitted to Table 4, the stretched exponential $\exp(-0.0115\,(q-2)^{1.68})$ fits best without an extra parameter (deviance 20.3 on 13 degrees of freedom); a pure exponential in q fits badly (deviance 1505 on 14). The fits are descriptive and cover $q \le 48$ only.

In the market the fractional constant keeps falling with q. Its excess over 2.2 is between about 1 and 1.5 times $q^{-1/3}$ from $q = 16$ to $q = 2000$, which is compatible with a limit near the tree values and a correction of order $q^{-1/3}$, as the slow disappearance of pair-sharing suggests. The integer optimum of the market was not computed beyond $q = 48$. In the instances with $D_q > 0$ the optimal fractional solution found by the solver has non-integral weights at a share 0.71, 0.77, 0.83, 0.87 of the agents for $q = 16, 24, 32, 48$; so the fractional optimum is not a small perturbation of an assignment. The standard deviation of D_q is about 0.05 for q between 16 and 48, in line with Theorem 3.3.

In the independent-team model there is no pair-sharing and the constants are close to their limits already at small q. The fractional constant rises from 2.165 to 2.200 and fits give a limit between 2.20 and 2.22; a second code found 2.184 ± 0.012 at $q = 1000$, 2.202 at $q = 2000$ and 2.196 at $q = 4000$ (two and one instances). All of these are below $\alpha_0 = 2.2519$ and below the floating-point cost 2.2495 of the matching defined by the fixed point.

TABLE 4. Floating point. The market: share of instances with a nonempty core, with 95% Clopper–Pearson intervals. The core was declared nonempty when the vertex returned by the linear-programming solver was integral; for $q \le 24$ a fractional vertex was followed by an exact integer solve and a comparison of the two optima. Source: `gate1_ext/out/ext_table_and_fits.txt`.

q	instances	nonempty	share	95% interval
2	2000	2000	1.000	$[0.998, 1.000]$
3	2000	1987	0.994	$[0.989, 0.997]$
4	2000	1939	0.970	$[0.961, 0.977]$
5	2000	1873	0.936	$[0.925, 0.947]$
6	2000	1784	0.892	$[0.878, 0.905]$
8	2000	1568	0.784	$[0.765, 0.802]$
10	2000	1338	0.669	$[0.648, 0.690]$
12	1000	582	0.582	$[0.551, 0.613]$
16	1000	366	0.366	$[0.336, 0.397]$
20	1000	211	0.211	$[0.186, 0.238]$
24	500	61	0.122	$[0.095, 0.154]$
28	500	40	0.080	$[0.058, 0.107]$
32	500	19	0.038	$[0.023, 0.059]$
40	2000	16	0.008	$[0.005, 0.013]$
48	1000	2	0.002	$[0.000, 0.007]$

This is consistent with α_0 as an upper bound for the limiting fractional constant, which Theorems 9.1 and 7.1 prove for the market; the corresponding statement for the independent-team model is not proved in this paper. The integer constant rises from 2.253 to 2.311 over $q = 16, \dots, 40$, and fits give a limit of 2.32 or more; a second code found 2.32 ± 0.03 at $q = 32$. This is consistent with $\beta_0 = 2.302$ as a lower bound for the limit. The finite-size values at $q \le 32$ lie below 2.302; the theorems concern the limit only. The comparison was run as a test that could have refuted the proof's architecture, and it did not.

12.3. **The matching on the tree is not a function of the tree.** Theorems 5.1 and 9.1 together have a consequence for the tree itself, which is proved as Corollary 10.1. In words:

No invariant perfect matching on $\mathbb{T}$ of cost below β_0 is a measurable function of the marked tree. In particular the matching built from a fixed point of Θ (Theorem 6.12), whose cost is at most $\alpha_0 < \beta_0$, uses randomness beyond the tree and its costs. In the usual terminology, the recursive tree process attached to the recursion (1.1) is not endogenous.

TABLE 5. Floating point. The market: integer and fractional deficits divided by $q^{1/3}$, and the mean gap. Upper block: both optima on the same instances (`lemmaC/round3/scout/REPORT.md`). Lower block: fractional optimum only, restricted to teams of rescaled cost at most 6 or 7, which is an upper bound on Φ_q^{LP} (`lemmaC/round3/redteam2/NOTES.md`); the row $q = 160$ is from an earlier run, recorded as 2.46 and as 2.47.

q	instances	$\Phi_q/q^{1/3}$	$\Phi_q^{\mathrm{LP}}/q^{1/3}$	$\mathbb{E}D_q$	$\mathbb{P}(D_q > 0)$
16	1000	2.572	2.556	0.041	0.647
24	1000	2.586	2.561	0.072	0.866
32	400	2.598	2.565	0.104	0.963
48	211	2.579	2.535	0.161	1.000
160	8		2.46–2.47		
500	1		2.39		
1000	4		2.352 ± 0.006		
2000	3		2.313 ± 0.011		

TABLE 6. Floating point. The independent-team model, teams of rescaled cost at most 6: fractional deficit divided by $q^{1/3}$ (mean and standard error), mean gap divided by $q^{1/3}$ on the instances whose integer optimum was found (all were closed to optimality), their sum as an estimate of the integer constant, and the share of instances with an integral fractional optimum. Source: `lemmaC/round3/pscout/results.txt`.

	fractional		gap		integer	share
q	instances	constant	instances	mean	constant	integral
16	3000	2.1654 ± 0.0025	600	0.0878 ± 0.0016	2.253	0.027
24	163	2.1757 ± 0.0086	163	0.1011 ± 0.0024	2.277	0.006
32	2000	2.1819 ± 0.0020	74	0.1029 ± 0.0028	2.285	0.000
40	19	2.1960 ± 0.0189	19	0.1149 ± 0.0045	2.311	0.000
64	1000	2.1891 ± 0.0022				
128	400	2.1933 ± 0.0024				
256	200	2.1977 ± 0.0023				
512	20	2.1996 ± 0.0044				

The reason is the following. The proof of Theorem 9.1 uses fractionality at one place only: the weight given to a team is the conditional probability, given a ball of finite radius, that the team belongs to the matching m. If m were a measurable function of the marked tree, these conditional probabilities would converge to the indicator of m itself as the radius grows, and a

thresholded version of the construction (keep a team when its weight exceeds $1/2$ at all three of its agents, then patch) would give *integral* assignments of cost $(c(m) + o(1))q^{1/3}$ in the finite market. By Theorem 5.1 this forces $c(m) \geq \beta_0$.

This matches a simulation (floating point): two copies of the process run on the same tree from independent initial conditions do not merge. Their root variables settle at correlation about 0.51, and they choose the same team at the root with probability about 0.41. It also explains why the gap is invisible locally: on the tree the fractional and the integer problem have the same value (Theorem 6.7), but no cheap invariant matching of the tree is a measurable function of the tree and its costs (Corollary 10.1).

12.4. **Open problems.** *The constants.* Do $\Phi_q/q^{1/3}$ and $\Phi_q^{\mathrm{LP}}/q^{1/3}$ converge, and to what? For the integer problem we prove $\liminf \geq 2.302$ in probability; the star-conditioned first moment cannot give more than about 2.3025, and the simulations suggest a limit of 2.32 or more. For the fractional problem we prove $\limsup \mathbb{E}\Phi_q^{\mathrm{LP}}/q^{1/3} \leq 2.2519$, the fixed point costs 2.2495 numerically, and the simulations of the independent-team model suggest a limit near 2.20. If that is right, the matching built from the recursion is not the cheapest invariant matching on $\mathbb{T}$. Is the fractional limit equal to the infimum of $c(m)$ over invariant perfect matchings on $\mathbb{T}$, and what is that infimum? The plain first moment gives the lower bound $3e^{-1/3} = 2.1496$ for the integer problem only; no lower bound for the fractional constant is proved in this paper.

The rate. The proof gives $\mathbb{P}(\text{core nonempty}) = O(q^{-5/6+\varepsilon})$. The data for $q \leq 48$ decay much faster, like a stretched or plain exponential in q. What is the true rate? A Chebyshev argument cannot give more than a polynomial rate.

Other sector graphs and more sectors. Here every pair of sectors interacts. If one of the three types of pair is removed, the sector graph is a path, and for sector graphs that are trees the core is expected to be nonempty always [**TODO: cite the deterministic result for tree sector graphs**]. With four or more sectors and pair-additive values on a cyclic sector graph, nothing is known; the scaling of the optimum for the four-sector model with values on all pairs is itself not treated in [9]. Unequal sector sizes are a further case: for independent values on triples, the figures reported in [20] suggest that the core is nonempty with probability tending to 1 when one side stays small.

Other value laws. The proof uses the uniform law through the density of a pair deficit near zero and through the imported bound of [9]. Pair values with a bounded density that is positive at the top of the support should behave in the same way. Unbounded values, for which the two-sided law of one price fails [13], and values that are independent over triples [20, 21], are different problems.

Removing the computer assistance. The main theorem needs only some $\beta > \alpha$ with $\Phi_q \geq \beta q^{1/3}$ with high probability and $9\,\mathbb{E}X \leq \alpha$. The margin is small: the plain first moment gives 2.1496, the star-conditioned one 2.3022, and the bounds c_n on the tree cost fall below 2.302 only from the fourth iterate on ($c_4 = 2.3017$, $c_5 = 2.2679$ in floating point; the certified bounds are below 2.302 from stage 5 on). A proof without computation would need an analytic bound on a five-fold iterated integral, or a different argument for either constant. Short of that, the two certificates should be re-implemented in a second interval library, and the scripts read by a person.

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